

weft

A Modern Literate Programming Tool

Cuauhtemoc Pacheco
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September 11, 2026

¹Original nuweb work supported by ARPA, through ONR grant N00014-91-J-1989.

Contents

1	Introduction	1
1.1	weft	1
1.2	Writing weft	3
1.2.1	Source File Extension: <code>.weft</code>	3
1.2.2	The Major Commands	4
1.2.3	The Minor Commands	7
1.3	Sectioning commands	8
1.4	Running weft	9
1.5	Markdown Output	10
1.6	Section Markers and Language Auto-Detection	11
1.6.1	Overriding the Language with <code>@l</code>	12
1.6.2	Mid-Line Inline Expansion	12
1.6.3	The <code>@L</code> Command: Defining New Languages	13
1.6.4	The Language Table	13
1.6.5	Syntax Highlighting in Markdown Output	14
1.6.6	Output Format of Section Markers	14
1.6.7	Debugging Tangled Code	14
1.7	Restrictions	18
1.8	Project Introspection	19
1.8.1	Map (<code>-m</code>)	19
1.8.2	Extract (<code>-e</code>)	19
1.8.3	Reverse Map (<code>-R</code>)	19
1.9	Acknowledgements	20
2	The Overall Structure	21
2.1	Files	21
2.1.1	The Main Files	23
2.1.2	Support Files	25
2.2	The Main Routine	25
2.2.1	Command-Line Arguments	26
2.2.2	Help Message	30
2.2.3	Short Flags	34
2.2.4	File Names	37
2.3	Pass One	42
2.3.1	Accumulating Definitions	45
2.3.2	Block Comments	47
2.3.3	Fixing the Cross References	51
2.3.4	Dealing with fragment parameters	51
2.4	Writing the Latex File	54
2.4.1	Formatting Definitions	60
2.4.2	Generating the Indices	76
2.5	Writing the Markdown File	82

2.5.1	Formatting Definitions in Markdown	86
2.5.2	Generating the Indices in Markdown	94
2.5.3	Identifier Cross-References in Markdown	99
2.6	Writing the HTML File	101
2.6.1	HTML escaping helpers	101
2.6.2	The driver	102
2.6.3	HTML scrap definitions	104
2.6.4	HTML indices	106
2.7	Map and Extract Output	108
2.7.1	JSON Map Output	108
2.7.2	Fragment Extraction	110
2.7.3	Callers (Upward Call Chain)	112
2.7.4	Lint (Static Analysis)	114
2.7.5	Reverse Map Output	116
2.7.6	Error Annotator	120
2.8	Writing the Output Files	122
3	The Support Routines	127
3.1	Source Files	127
3.1.1	Global Declarations	127
3.1.2	Local Declarations	127
3.1.3	Reading a File	128
3.1.4	Opening a File	131
3.2	Scraps	131
3.2.1	Collecting Page Numbers	153
3.3	Names	154
3.3.1	Language Auto-Detection	160
3.3.2	User-Defined Languages	167
3.3.3	Weave Format Declaration	169
3.3.4	Collecting File and Fragment Names	170
3.3.5	Comment Styles	172
3.4	Searching for Index Entries	180
3.4.1	Retrieving scrap uses	181
3.4.2	Building the Automata	183
3.4.3	Searching the Scraps	186
3.5	Labels	191
3.6	Memory Management	193
3.6.1	Allocating Memory	193
3.6.2	Freeing Memory	194
4	Man page	196
5	Indices	201
5.1	Files	201
5.2	Fragments	201
5.3	Identifiers	206

Chapter 1

Introduction

In 1984, Knuth introduced the idea of *literate programming* and described a pair of tools to support the practise [3]. His approach was to combine Pascal code with T_EX documentation to produce a new language, WEB, that offered programmers a superior approach to programming. He wrote several programs in WEB, including `weave` and `tangle`, the programs used to support literate programming. The idea was that a programmer wrote one document, the web file, that combined documentation (written in T_EX [4]) with code (written in Pascal).

Running `tangle` on the web file would produce a complete Pascal program, ready for compilation by an ordinary Pascal compiler. The primary function of `tangle` is to allow the programmer to present elements of the program in any desired order, regardless of the restrictions imposed by the programming language. Thus, the programmer is free to present his program in a top-down fashion, bottom-up fashion, or whatever seems best in terms of promoting understanding and maintenance.

Running `weave` on the web file would produce a T_EX file, ready to be processed by T_EX. The resulting document included a variety of automatically generated indices and cross-references that made it much easier to navigate the code. Additionally, all of the code sections were automatically prettyprinted, resulting in a quite impressive document.

Knuth also wrote the programs for T_EX and METAFONT entirely in WEB, eventually publishing them in book form [5, 6]. These are probably the largest programs ever published in a readable form.

Inspired by Knuth's example, many people have experimented with WEB. Some people have even built web-like tools for their own favorite combinations of programming language and typesetting language. For example, CWEB, Knuth's current system of choice, works with a combination of C (or C++) and T_EX [8]. Another system, FunnelWeb, is independent of any programming language and only mildly dependent on T_EX [10]. Inspired by the versatility of FunnelWeb and by the daunting size of its documentation, Preston Briggs decided to write his own, very simple, tool for literate programming.¹

`weft` is a fork of `nuweb`. The original `nuweb` code, written by Preston Briggs and maintained over the years by several contributors, provided an excellent foundation: a simple, language-independent literate programming tool. `weft` builds on that foundation with a modernized codebase, a modular source structure organized by concept, the removal of the unmaintained HTML output path, and the addition of a Markdown weave output format. The goal is to continue evolving the tool while preserving the simplicity and philosophy that made `nuweb` valuable in the first place.

1.1 `weft`

The version number is defined once, as a single fragment, and threaded everywhere it is needed: the binary's `--version`, the default value of the `@v` directive (via `WEFT_VERSION`), and the generated documentation. Cutting a release is therefore a one-line edit to this scrap.

¹There is another system similar to `nuweb`, written by Norman Ramsey, called *noweb* [9]. It perhaps suffers from being overly Unix-dependent and requiring several programs to use. On the other hand, its command syntax is very nice. In any case, `nuweb` certainly owes its name and a number of features to his inspiration.

`<version 2>` $\equiv$
2.0.0
2 · used in 23b

weft works with any programming language and produces documentation in L^AT_EX [7] or Markdown format. L^AT_EX was the original choice because it supports a multi-level sectioning scheme and has facilities for drawing figures. Markdown support was added to allow literate programs to produce lightweight documentation that renders well on platforms like GitHub, GitLab, and any Markdown viewer, without requiring a T_EX installation.

Support for arbitrary programming languages was a design goal because programmers work in many languages (and sometimes combinations of several languages), *e.g.*, C, Fortran, C++, yacc, lex, Scheme, assembly, Postscript, and so forth. The need to support arbitrary programming languages has many consequences:

No prettyprinting Both WEB and CWEB are able to prettyprint the code sections of their documents because they understand the language well enough to parse it. Since weft aims to support *any* language, this feature must be abandoned. However, weft does allow particular individual formulas or fragments of L^AT_EX code to be formatted and still be parts of output files. Also, keywords in scraps can be surrounded by @_ to have them be bold in the output.

No index of identifiers Because WEB knows about Pascal, it is able to construct an index of all the identifiers occurring in the code sections (filtering out keywords and the standard type identifiers). Unfortunately, this isn't as easy for a language-independent tool. weft doesn't know what an identifier looks like in each language and certainly doesn't know all the keywords. (On the other hand, see the end of Section 1.2.3)

Of course, there must be some compensation for these losses or the whole idea would be a waste. Here are the advantages:

Simplicity The majority of the commands in WEB are concerned with control of the automatic prettyprinting. Since weft doesn't prettyprint, many commands are eliminated. A further set of commands is subsumed by L^AT_EX and may also be eliminated. As a result, the set of commands is reduced to only four members (explained in the next section). This simplicity is also reflected in the size of this tool, which is quite a bit smaller than the tools used with other approaches.

No prettyprinting Everyone disagrees about how their code should look, so automatic formatting annoys many people. One approach is to provide ways to control the formatting. weft's approach is simpler—it performs no automatic formatting and therefore allows the programmer complete control of code layout. weft does allow individual scraps to be presented in either verbatim, math, or paragraph mode in the T_EX output.

Control weft also offers the programmer complete control of the layout of his output files (the files generated during tangling). Of course, this is essential for languages that are sensitive to layout; but it is also important in many practical situations, *e.g.*, debugging.

Speed Since weft doesn't do too much, the weft tool runs quickly. By default it performs only tangling; weaving is activated with -w. When weaving is requested, both operations execute in a single pass.

Page numbers Inspired by the example of noweb, weft refers to all scraps by page number to simplify navigation. If there are multiple scraps on a page (say, page 17), they are distinguished by lower-case letters (*e.g.*, 17a, 17b, and so forth).

Multiple file output The programmer may specify more than one output file in a single weft file. This is required when constructing programs in a combination of languages (say, Fortran and C). It's also an advantage when constructing very large programs that would require a lot of compile time.

This last point is very important. By allowing the creation of multiple output files, the need for monolithic programs is avoided. This supports the creation of very large programs by groups of people.

A further reduction in compilation time is achieved by first writing each output file to a temporary location, then comparing the temporary file with the old version of the file. If there is no difference, the temporary file can be deleted. If the files differ, the old version is deleted and the temporary file renamed. This approach works well in combination with `make` (or similar tools), since `make` will avoid recompiling untouched output files.

1.2 Writing weft

The bulk of a weft file will be ordinary L^AT_EX (or Markdown, when using the `-w md` flag). In fact, any L^AT_EX file can serve as input to weft and will be simply copied through, unchanged, to the documentation file—unless a weft command is discovered. Similarly, any Markdown file can be processed with `-w md` and will be copied through to the `.md` output. All weft commands begin with an “at-sign” (`@`). Therefore, a file without at-signs will be copied unchanged. weft commands are used to specify *output files*, define *fragments*, and delimit *scraps*. These are the basic features of interest to the weft tool—all else is simply text to be copied to the documentation file.

1.2.1 Source File Extension: `.weft`

weft source files use the `.weft` extension. This single, format-agnostic extension replaces the earlier `.w` and `.wmd` conventions. The prose surrounding scraps may be written in L^AT_EX or Markdown; the at-sign commands (`@o`, `@d`, `@<`, etc.) are the same in both cases.

The weave output format is specified in one of two ways:

In the source file The `@W` directive declares the intended format: `@W tex` for L^AT_EX, `@W md` for Markdown. This should appear in the preamble, before any `@o` or `@d` directives.

On the command line `-w md` or `-w tex` overrides the `@W` directive. Using `-w` alone (without a format token) reads the format from `@W`.

If no extension is given on the command line, weft appends `.weft` automatically.

Default Behavior: Tangle Only

By default, `weft myfile.weft` performs only tangling (producing the output files). No documentation is generated unless `-w` is given:

<code>weft programa.weft</code>	tangle only (default)
<code>weft -w programa.weft</code>	weave per <code>@W</code> (error if missing)
<code>weft -w md programa.weft</code>	weave Markdown (override)
<code>weft -w tex programa.weft</code>	weave LaTeX (override)

The `-t` flag suppresses weaving even if `-w` is given. The `-o` flag suppresses tangling.

Cosmetic Fences

When writing `.weft` files with Markdown prose, it is often convenient to surround scrap definitions with Markdown fenced code blocks (triple backticks) so that the source file renders nicely in Markdown previewers such as GitHub or VS Code. For example:

Some prose in Markdown.

```
'''c
@d My fragment
@{code here
```

```
@}  
'''
```

More prose.

weft silently strips these fence lines (lines that consist of three or more backticks, optionally followed by a language identifier) during Markdown weaving. They do not appear in the generated .md file and do not affect tangling. This is purely a cosmetic convenience for making the source file readable in any Markdown viewer.

Note that fences are only stripped in Markdown mode (`-w md`). In L^AT_EX mode, backtick lines are treated as ordinary text.

1.2.2 The Major Commands

Files and fragments are defined with the following commands:

@o *file-name flags scrap* Output a file. The file name is terminated by whitespace.

@d *fragment-name scrap* Define a fragment. The fragment name is terminated by a return or the beginning of a scrap.

@q *fragment-name scrap* Define a fragment. The fragment name is terminated by a return or the beginning of a scrap. This a quoted fragment.

A specific file may be specified several times, with each definition being written out, one after the other, in the order they appear. The definitions of fragments may be similarly specified piecemeal.

Fragment Names with Embedded Parameters

Fragment names can carry *parameters*—placeholder values embedded directly in the name using the `@'value@'` syntax. Inside the fragment body, `@1`, `@2`, ... `@9` refer to the first, second, ... ninth parameter in order of appearance.

For example, a reusable validation fragment:

```
@d Validate field @'name@' type @'string@'  
{if (!data.@1) throw new Error("Missing @1");  
if (typeof data.@1 !== '@2')  
  throw new TypeError("@1 must be @2");  
@}
```

Each invocation supplies its own arguments:

```
@<Validate field @'email@' type @'string@'@>  
@<Validate field @'age@' type @'number@'@>
```

The first invocation expands to:

```
if (!data.email) throw new Error("Missing email");  
if (typeof data.email !== 'string')  
  throw new TypeError("email must be string");
```

Note three key points:

- The values in the *definition* (`@'name@'`, `@'string@'`) serve as default arguments and as documentation—they make the fragment name readable.
- At each *invocation*, different values can be substituted (`@'email@'` replaces `@1`, `@'string@'` replaces `@2`).
- If a fragment name is abbreviated with ..., any missing arguments fall back to the defaults given in the definition.

The difference between a quoted fragment (`@q`) and an ordinary one (`@d`) is that inside a quoted fragment fragments are not expanded on output. Rather, they are formatted as uses of fragments so that the output file can itself be `weft` source. This allows you to create files containing fragments which can undergo further processing before the fragments are expanded, while also describing and documenting them in one place.

You can have both quoted and unquoted fragments with the same name. They are written out in order as usual, with those introduced by `@q` being quoted and those with `@d` expanded as normal.

In quoted fragments, the `@f` filename is quoted as well, so that when it is expanded it refers to the finally produced file, not any of the intermediate ones.

Scraps

Scraps have specific begin markers and end markers to allow precise control over the contents and layout. Note that any amount of whitespace (including carriage returns) may appear between a name and the beginning of a scrap. Scraps may also appear in the running text where they are formatted (almost) identically to their use in definitions, but don't appear in any code output.

`@{anything@}` where the scrap body includes every character in *anything*—all the blanks, all the tabs, all the carriage returns. This scrap will be typeset in verbatim mode. Using the `-l` option will cause the program to typeset the scrap with the help of L^AT_EX's `listings` package.

`@[anything@ ]` where the scrap body includes every character in *anything*—all the blanks, all the tabs, all the carriage returns. This scrap will be typeset in paragraph mode, allowing sections of T_EX documents to be scraps, but still be prettyprinted in the document.

`@(anything@)` where the scrap body includes every character in *anything*—all the blanks, all the tabs, all the carriage returns. This scrap will be typeset in math mode. This allows this scrap to contain a formula which will be typeset nicely.

Inside a scrap, several special sequences are recognized.

Fragment invocation.

`@<fragment-name@>` Expands the named fragment inline. It is an error to specify recursive invocations. If the fragment name contains embedded parameters, the invocation supplies arguments at the same positions (see above).

When the reference appears *mid-line*—that is, after at least one visible character on the same line—section markers, `#line` directives, and fragment-name comments are automatically suppressed. The fragment content is spliced seamlessly into the surrounding text. At start-of-line positions (including indented lines), markers are emitted normally. See Section 1.6 for details.

Argument substitution.

`@1, @2, ..., @9` Inside a fragment body, substitutes the *n*th argument. If the argument was not supplied, the empty string is used.

At each argument position in an invocation, four forms are accepted:

`@'string@'` A literal string, copied verbatim (only `@@` is interpreted, producing a single `@`). This is the most common form.

`@<fragment-name@>` The fragment is expanded and its output becomes the argument value.

`@{text@}` The text is expanded as an anonymous fragment (inheriting the caller's arguments) and the result is passed as the argument. Useful for combining text and arguments into a single value.

`@1, @2, ..., @9` Forwards an argument from the calling fragment to the called one.

Other in-scrap commands.

@xlabel@x Marks a place inside a scrap and associates it to the label (which can be any text not containing a @). Expands to the reference number of the scrap followed by a numeric value. Outside scraps it expands to the same value. It is used so that text outside scraps can refer to particular places within scraps.

@f Inside a scrap this is replaced by the name of the current output file.

@t Inside a scrap this is replaced by the title of the fragment as it is at the point it is used, with all parameters replaced by actual arguments.

@# At the beginning of a line in a scrap this will suppress the normal indentation for that line. Use this, for example, when you have a `#ifdef` inside a nested scrap. Writing `@##ifdef` will cause it to be lined up on the left rather than indented with the rest of its code.

@s Inside a scrap suppresses indentation for the following fragment expansion.

Note that fragment names may be abbreviated, either during invocation or definition. For example, it would be very tedious to have to type, repeatedly, the fragment name

```
@d Check for terminating at-sequence and return name if found
```

Therefore, `weft` provides a mechanism (borrowed from Knuth) of indicating abbreviated names.

```
@d Check for terminating...
```

Basically, the programmer need only type enough characters to identify the fragment name uniquely, followed by three periods. An abbreviation may even occur before the full version; `weft` simply preserves the longest version of a fragment name. Note also that blanks and tabs are insignificant within a fragment name; each string of them is replaced by a single blank.

Sometimes, for instance during program testing, it is convenient to comment out a few lines of code. In C or Fortran placing `/* ... */` around the relevant code is not a robust solution, as the code itself may contain comments. `weft` provides the command

```
@%
```

only to be used inside scraps. It behaves exactly the same as `%` in the normal `LATEX` text body.

When scraps are written to a program file or a documentation file, tabs are expanded into spaces by default. Currently, `weft` assumes tab stops are set every eight characters. Furthermore, when a fragment is expanded in a scrap, the body of the fragment is indented to match the indentation of the fragment invocation. Therefore, care must be taken with languages (*e.g.*, Fortran) that are sensitive to indentation. These default behaviors may be changed for each output file (see below).

Flags

When defining an output file, the programmer has the option of using flags to control output of a particular file. The flags are intended to make life a little easier for programmers using certain languages. They introduce little language dependences; however, they do so only for a particular file. Thus it is still easy to mix languages within a single document. There are five “per-file” flags:

- d Forces the creation of `#line` directives in the output file. These are useful with C (and sometimes C++ and Fortran) on many Unix systems since they cause the compiler’s error messages to refer to the web file rather than to the output file. Similarly, they allow source debugging in terms of the web file.
- i Suppresses the indentation of fragments. That is, when a fragment is expanded within a scrap, it will *not* be indented to match the indentation of the fragment invocation. This flag would seem most useful for Fortran programmers.

- t Suppresses expansion of tabs in the output file. This feature seems important when generating `make` files.
 - cx Puts comments in generated code documenting the fragment that generated that code. The *x* selects the comment style for the language in use: `c` for C (`/* */`), `+` for C++ (`//`), and `p` for shell/Perl (`#`). If the global `-x` cross-reference flag is set the comment includes the page reference for the first scrap that generated the code.
 - s Suppresses section markers and `#line` directives in the tangled output for this file. The language is still detected (from the extension or a preceding `@l`), so the Markdown weaver can emit syntax-highlighting classes, but the tangled code is emitted without any tangle annotations. This is useful for bootstrap source files or generated output that should remain clean.
- These flags are still available for explicit control, but in most cases the automatic language detection described in Section 1.6 is more convenient.

1.2.3 The Minor Commands

There are some very low-level utility commands that may appear anywhere in the web file.

- `@@` Causes a single “at sign” to be copied into the output.
- `@_` Causes the text between it and the next `@_` to be made bold (for keywords, etc.)
- `@i file-name` Includes a file. Includes may be nested, though there is currently a limit of 10 levels. The file name should be complete (no extension will be appended) and should be terminated by a carriage return. Normally the current directory is searched for the file to be included, but this can be varied using the `-I` flag on the command line. Each such flag adds one directory to the search path and they are searched in the order given.
- `@rx` Changes the escape character ‘`@`’ to ‘*x*’. This must appear before any scrap definitions.
- `@v` Always replaced by the string established by the `-V` flag, or a default string if the flag isn’t given. This is intended to mark versions of the generated files.
- `@l tag` Sets the language for the next `@o` or `@d` directive. The tag is looked up in the language table (Section 1.6) to determine the comment style for section markers and the `class` attribute for Markdown syntax highlighting. Placed on the line before the directive it applies to.
- `@L name style [+d]` Defines a custom language mapping for the section marker system (see Section 1.6). This must appear in the document preamble, before any `@o` or `@d` directives that depend on it.
- `@W format` Declares the weave output format for this source file. Valid formats are `md` (Markdown) and `tex` (L^AT_EX). Used by `-w` when no explicit format is given on the command line. A CLI format (`-w md` or `-w tex`) overrides this directive. Should appear in the document preamble.

In the text of the document, that is outside scraps, you may include scrap-like material.

- `@{Anything@}` The included material, the *Anything*, is typeset as if it appeared inside a scrap. This is useful for referring to fragments in the text and for describing the literate programming process itself.
- `@<Fragment name@>` The fragment named is expanded in place in the text. The expansion is presented verbatim, it is not interpreted for typesetting, so any special environment must be set up before and after this is used.
- `@c` Outside of a scrap this is used to define a block comment using all the text until the next command. Inside a scrap the last block comment is expanded using the current indentation and breaking lines between words if 60 characters are exceeded.

Finally, there are three commands used to create indices to the fragment names, file definitions, and user-specified identifiers.

@f Create an index of file names.

@m Create an index of fragment names.

@u Create an index of user-specified identifiers.

These are usually placed in their own section in the L^AT_EX document; for example, see Chapter 5.

Identifiers must be explicitly specified for inclusion in the **@u** index. By convention, each identifier is marked at the point of its definition; all references to each identifier (inside scraps) will be discovered automatically. To “mark” an identifier for inclusion in the index, it must be mentioned at the end of a scrap. For example,

```
@d a scrap @{
  Suppose the variables FOO and BAR are being declared
  inside this scrap.
@| FOO BAR @}
```

Alphabetic identifiers have been used in this example, but any string of characters (not including whitespace or @ characters) will do. Therefore, it’s possible to add index entries for things like <=< if desired. An identifier may be declared in more than one scrap.

In the generated index, each identifier appears with a list of all the scraps using and defining it, where the defining scraps are distinguished by underlining. Note that the identifier doesn’t actually have to appear in the defining scrap; it just has to be in the list of definitions at the end of a scrap.

1.3 Sectioning commands

For larger documents the indexes and usage lists get rather unwieldy and problems arise in naming things so that different things in different parts of the document don’t get confused. `weft` has a sectioning command which keeps the fragment names and user identifiers separate. Thus, you can give a fragment in one section the same name as a fragment in another and the two won’t be confused or connected in any way. Nor will user identifiers defined in one section be referenced in another. Except for the fact that scraps in successive sections can go into the same output file, this is the same as if the sections came from separate input files.

However, occasionally you may really want fragments from one section to be used in another. More often, you will want to identify a user identifier in one section with the same identifier in another (as, for example, a header file defined in one section is included in code in another). Extra commands allow a fragment defined in one section to be accessible from all other sections. Similarly, you can have scraps which define user identifiers and export them so that they can be used in other sections.

@s Start a new section.

@S Close the current section and don’t start another.

@d+ *fragment-name* scrap Define a fragment which is accessible in all sections, a global fragment.

@D+ Likewise

@q+ Likewise

@Q+ Likewise

@m+ Create an index of all such fragments.

@u+ Create an index of globally accessible user identifiers.

There are two kinds of section, the base section which is where normally everything goes, and local sections which are introduced with the **@s** command. A local section comprises everything from the command which starts it to the one which ends it. A **@s** command will start a new local section. A **@S** command closes the current local section, but doesn’t open another, so what follows goes into the base section. Note that

fragments defined in the base section aren't global; they are accessible only in the base section, but they are accessible regardless of any local sections between their definition and their use.

Within a scrap:

`@<+fragment-name@>` Expand the globally accessible fragment with that name, rather than any local fragment.

`@+` Like `@|` except that the identifiers defined are exported to the global realm and are not directly referenced in any scrap in any section (not even the one where they are defined).

`@-` Like `@|` except that the identifiers are imported to the local realm. The cross-references show where the global variables are defined and defines the same names as locally accessible. Uses of the names within the section will point to this scrap.

Note that the `+` signs above are part of the commands. They are not part of the fragment names. If you want a fragment whose name begins with a plus sign, leave a space between the command and the name.

1.4 Running weft

weft is invoked using the following command:

```
weft flags file-name...
```

One or more files may be processed at a time. If a file name has no extension, `.weft` will be appended. While a file name may specify a file in another directory, the resulting documentation file will always be created in the current directory. For example,

```
weft /foo/bar/quux
```

will take as input the file `/foo/bar/quux.weft` and will produce the tangled output files only. To also generate documentation, use `-w` (see below).

By default, weft performs only tangling—it generates the output files defined by `@o` directives and nothing else. Documentation (weaving) is produced only when the `-w` flag is given. When `-w` is present, both tangling and weaving execute in a single pass. Three flags control what weft produces:

- `-t` Suppress weaving (useful to override `-w` in scripts).
- `-o` Suppress generation of the output files (weave only).
- `-c` Avoid testing output files for change before updating them.

Thus, the command

```
weft -to /foo/bar/quux
```

would simply scan the input and produce no output at all.

There are several additional command-line flags:

- `-v` For “verbose”, causes weft to write information about its progress to `stderr`.
- `-n` Forces scraps to be numbered sequentially from 1 (instead of using page numbers). This form is perhaps more desirable for small webs.
- `-s` Doesn't print list of scraps making up each file following each scrap.
- `-d` Print “dangling” identifiers – user identifiers which are never referenced, in indices, etc.
- `-p path` Prepend *path* to the filenames for all the output files.
- `-w [format]` Activate weaving (documentation generation). Without `-w`, no documentation file is produced. The optional *format* argument selects the output format:

- `-w md` — force Markdown output (`.md` file).
- `-w tex` — force L^AT_EX output (`.tex` file).
- `-w` alone — read the format from the `@W` directive in the source file. If `@W` is absent, `weft` reports an error.

A format given on the command line overrides any `@W` directive. When Markdown output is selected, sequential numbering (`-n`) is forced automatically, since Markdown has no concept of page numbers. The Markdown output uses HTML anchors (`<a id="...">`) for cross-references and `<pre><code>` blocks for scraps, allowing clickable navigation between definitions and references. Cosmetic fence lines (triple backticks) in the source are silently stripped.

- `-x` Include cross-reference numbers in scrap comments.
- `-h options` Provide options for the `hyperref` package.
- `-r` Turn on hyperlinks. You must include the `—usepackage—` options in the text.
- `-l` Use the `listings` package for formatting scraps. Use this if you want to have a pretty-printer for your scraps. In order to e.g. have pretty Perl scraps, include the following L^AT_EX commands in your document:

```
\usepackage{listings}
...
\lstset{extendedchars=true,keepspaces=true,language=perl}
```

 See the `listings` documentation for a list of formatting options. Be sure to include a `\usepackage{listings}` in your document.
- `-V string` Provide *string* as the replacement for the `@v` operation.
- `-m` Emit a JSON map of the project to stdout and exit. The map describes every output file and every fragment, their definition locations, usage relationships, and transitive dependencies. Neither weaving nor tangling is performed. See Section 2.7 for details.
- `-e name` Extract a single fragment by name and emit its code and dependency information as Markdown to stdout. The match uses the same prefix logic as `@<name...@>`. Neither weaving nor tangling is performed. See Section 2.7 for details.
- `-R file[:line]` Read section markers from a tangled output file and emit JSON mapping to stdout. If a line number is given, emit the single innermost scrap covering that line. Without a line number, emit all scrap regions in the file. No `.weft` source files are needed—the reverse map operates entirely on the tangled output. See Section 2.7 for details.

1.5 Markdown Output

When invoked with `-w md`, `weft` generates a Markdown file (`.md`) instead of a L^AT_EX file (`.tex`). This allows literate programs to be read on any platform that renders Markdown, including GitHub, GitLab, and local editors like VS Code.

The Markdown output preserves all the essential features of the L^AT_EX output—scrap definitions, cross-references, and indices—but uses a different set of formatting conventions:

Scrap headers File definitions appear as `**"filename"** **N** =` and fragment definitions as `**<Name* N>** =`, where *N* is the sequential scrap number.

Scrap bodies Code is wrapped in `<pre><code>` HTML blocks (not fenced code blocks). This allows fragment references within scraps to be rendered as clickable HTML links with italic and bold formatting, which would not be possible inside Markdown fenced blocks.

HTML escaping Characters `<`, `>`, and `&` inside scraps are escaped as `<`, `>`, and `&` respectively.

Cross-references Each scrap has an HTML anchor (`<a id="weftN">`) and cross-reference links use `<a href="#weftN">`. Footer notes (“defined by”, “referenced in”) appear as Markdown blockquotes.

Indices The `@f`, `@m`, and `@u` commands produce Markdown lists with links. In the user identifier index, defining scraps appear in bold.

Sequential numbering The `-n` flag is forced automatically, since Markdown has no page numbers. Scraps are numbered 1, 2, 3, ... without letter suffixes.

Prose Text outside scraps passes through unmodified. The author writes Markdown prose directly in the `.weft` source file; weft does not interpret or transform it.

Syntax highlighting When the language of a scrap can be determined—from a preceding `@l` tag, the `@L` command, or the file extension of an `@o` name—weft emits `<code class="language-X">` on the opening tag. This allows client-side highlighters (highlight.js, Prism) and platforms like GitHub to apply syntax colouring automatically.

Tangling (output file generation) works identically regardless of the weave format. When `-w` is given, a single invocation produces both the documentation and the tangled output files in one pass.

1.6 Section Markers and Language Auto-Detection

When debugging tangled code, it is useful to correlate lines in the generated file back to the original web source. weft can auto-detect the programming language and generate *section markers*—comments that delimit the opening and closing of each scrap in the tangled output, including the source file name and line number. For C and C++, it also generates `#line` directives automatically.

The key idea is simple:

file extension → language → comment style → section markers in tangled output

For example, `@o server.js` has the extension `.js`. weft looks up `js` in its language table, finds that JavaScript uses `//` comments, and wraps each scrap in the tangled `server.js` with `//` markers:

```
// {1: web.weft:12}
console.log("hello");
// {1: }
```

Similarly, `@o main.c` matches `c`, which uses `/* */` comments and also enables `#line` directives:

```
#line 5 "core.weft"
/* {1: core.weft:5} */
#include <stdio.h>
/* {1: } */
```

And `@o config.py` matches `py`, using `#` comments:

```
# {3: settings.weft:40}
DATABASE_URL = "postgres://localhost/app"
# {3: }
```

This happens automatically—no flags are needed. The extension alone is enough.

1.6.1 Overriding the Language with @l

Sometimes the file extension is not enough, or you want a different behaviour. The `@l` command (lowercase) placed on the line before an `@o` or `@d` directive overrides the extension:

```
@l sql
@o queries.txt
@{SELECT * FROM users;
@}
```

Here the file is `queries.txt`, whose extension `.txt` has no comment style. But `@l sql` tells weft to use `--` comments, so the tangled output gets:

```
-- {1: db.weft:5}
SELECT * FROM users;
-- {:1}
```

`@l` also works on `@d` (fragment) definitions. This is useful for fragments that will be included in files of a specific language:

```
@l python
@d Validate email
@{import re
def validate(email):
    return bool(re.match(r'^[^\@]+\@[^\@]+$', email))
@}
```

The language tag is stored on the fragment and used both for section markers in the tangled output and for syntax highlighting in the Markdown weave output.

`@l` is format-agnostic: it works identically in $\text{\LaTeX}$ and Markdown source files. Unlike the earlier approach of encoding the language in Markdown fences (`'''python`) or $\text{\LaTeX}$ markers (`%%python`), `@l` uses the same at-sign syntax as every other weft command.

Note that cosmetic fences (`'''`) may still be used around `@o/@d` definitions for readability in Markdown previewers—they are silently stripped during Markdown weaving (see Section 1.2.1). But fences no longer set the language; use `@l` for that.

Priority rules. When multiple sources of language information are available, weft resolves them in this order:

1. **Explicit per-file flags** (`-d`, `-s`, `-cx`, etc.) on the `@o` line always win for tangle behaviour.
2. **@l tag** immediately before the directive—overrides the extension.
3. **File extension** of the `@o` name—the default when no `@l` is present.

1.6.2 Mid-Line Inline Expansion

When a fragment reference `@<name>` appears in the middle of a line—after at least one visible character—weft automatically suppresses all tangle annotations for that expansion: section markers, `#line` directives, and fragment-name comments. The fragment content is spliced directly into the surrounding text, producing clean inline output.

This makes small constant fragments practical:

```
@d port @{4000@}
@d host @{"localhost"@}

@l c
```

```
@o config.c -s
@{const char *host = @<host@>;
int port = @<port@>;
@}
```

Tangling produces:

```
const char *host = "localhost";
int port = 4000;
```

If the same reference appears at the *start* of a line (even if indented), it is treated as a block expansion and markers are emitted normally. No special flags or directives are needed—the position on the line is the only thing that determines the behaviour.

This works with any language and any fragment size. When a multi-line fragment is expanded mid-line, its content is still spliced without markers—the rule is based on the *reference site*, not the fragment body.

1.6.3 The @L Command: Defining New Languages

The built-in table covers over 250 entries, but it cannot anticipate every language or every project's preferences. The @L command defines new language mappings—or overrides existing ones—directly in the document preamble.

@L *name style* [+d]

- *name* is the tag that will be recognised in @l commands and file extensions (@o *file.name*).
- *style* is the comment syntax: //, #, --, /*, or <!--.
- +d (optional) enables #line directive generation.

Examples:

```
@L wgs1 //          % WebGPU shading language: // comments
@L cedar //         % Cedar policy language
@L c //            % override built-in: use // instead of /* */
```

How @L interacts with the rest of the system. User-defined entries are searched *before* the built-in table, so @L c // effectively changes the comment style for all subsequent @o *main.c* directives (and for any @l c tag) from /* */ to //. Up to 32 user-defined languages may be registered.

@L should appear in the document preamble, before any @o or @d that depends on it. It is processed during pass 1 and silently skipped during weaving.

1.6.4 The Language Table

weft includes a built-in table of over 250 entries (languages, file extensions, and common aliases), organised by comment style. Each entry maps a tag to a comment style and optionally enables #line directives. The tag is matched against @l tags and file extensions alike—there is no distinction between them.

Comment style	Tags (representative)
<code>/* */ + #line</code>	c, h, cpp, cc, cxx, hpp, hxx, objc, objcpp, m, mm, ino, pde, cu, cuh, cilk
<code>/* */</code>	css, scss, sass, less, stylus, cl (OpenCL)
<code>//</code>	js, ts, jsx, tsx, mjs, cjs, mts, svelte, vue, java, scala, kotlin, kt, groovy, gradle, clj, cs, fs, fsx, vb, go, rust, rs, zig, swift, dart, odin, mojo, carbon, php, graphql, prisma, glsl, hlsl, wgsl, metal, vert, frag, proto, thrift, fbs, capnp, sol, reason, res, ml, sml, fut, apex, bicep, jsonnet, qml, p4
<code>#</code>	python, py, pyx, pyi, sh, bash, zsh, fish, ksh, nu, perl, pl, ruby, rb, r, tcl, awk, sed, coffee, makefile, mk, cmake, just, meson, bazel, bzl, julia, jl, nim, crystal, cr, elixir, ex, yaml, yml, toml, dockerfile, terraform, tf, hcl, nix, conf, ini, env, editorconfig, gitignore, powershell, ps1, xdc, sdc
<code>--</code>	lua, luau, sql, psql, plsql, plpgsql, mysql, sqlite, hive, haskell, hs, elm, purescript, purs, ada, adb, ads, vhdl, vhd, lean, lean4, agda, idris, idr, erlang
<code><!-- --></code>	html, htm, xml, svg, xhtml, xsl, xsd, dtd, rss, xaml, csproj, jsp, aspx, cshtml, erb, ejs, hbs, njk, twig, blade, pug, liquid, mustache, md, mdx, rst, tex, latex
(none)	json, csv, tsv, txt, diff, patch

For the full list, see the `lang_table` array in Section 3.3.1. If a tag is not found in either the user-defined (`@L`) or built-in table, weft defaults to `//` comments with no `#line` directives. Tags mapped to style 0 (e.g. `json`, `txt`) produce no markers at all.

1.6.5 Syntax Highlighting in Markdown Output

When weaving to Markdown (`-w md`), weft emits the language tag on the `<code>` element:

```
<pre><code class="language-python">def hello():
    print("hello")
</code></pre>
```

This allows client-side highlighters (highlight.js, Prism) and platforms like GitHub to apply syntax colouring automatically. The language tag comes from the same chain described above: `@l` tag or file extension.

For `@d` fragments, the tag comes from a preceding `@l` command (there is no file extension to infer from). If no language is known, weft emits a plain `<pre><code>` without a class.

1.6.6 Output Format of Section Markers

Each scrap in the tangled file is wrapped with an opening marker at the top and a closing marker at the bottom, using the comment syntax of the detected language. The opening marker includes the scrap number and the source location (file name and line number). When fragments are nested, each expansion gets its own pair of markers, producing a clear hierarchy in the output.

1.6.7 Debugging Tangled Code

When you work with literate programming, the code you execute is *tangled output*—generated files assembled from scraps scattered across `.w` files. An error at `server.js:47` does not correspond to line 47 of any file you wrote; it is line 47 of a file that weft assembled from fragments defined in `server.w`, `routes.w`, and `database.w`. Without markers, debugging means mentally reconstructing which scrap produced which line. With markers, the tangled file tells you directly.

Reading the markers. Each marker encodes three pieces of information:

Part	Meaning
{ or {:	Opening or closing delimiter
Number	Scrap ID—unique across the entire document
file:line	Source .w file and line number (opening only)

Examples by language. JavaScript (// comments):

```
// {3: routes.w:42}
app.get("/api/users", async (req, res) => {
  // {7: database.w:18}
  const users = await db.query("SELECT * FROM users");
  // {7:}
  res.json(users);
});
// {3:}
```

Python (# comments):

```
# {5: models.w:31}
class User:
    def __init__(self, name, email):
        self.name = name
        self.email = email
# {5:}
```

C (/* */ comments with #line directives):

```
#line 12 "memory.w"
/* {2: memory.w:12} */
void *safe_malloc(size_t size) {
    void *ptr = malloc(size);
    if (!ptr) {
        /* {4: errors.w:8} */
        fprintf(stderr, "out of memory\n");
        exit(1);
        /* {4:} */
    }
    return ptr;
}
/* {2:} */
```

Lua (-- comments):

```
-- {9: config.w:55}
local defaults = {
    port = 8080,
    host = "localhost",
}
-- {9:}
```

HTML (<!-- --> comments):

```
<!-- {6: layout.w:20} -->
<nav class="sidebar">
  <!-- {11: navigation.w:7} -->
```

```

    <ul>
      <li><a href="/">Home</a></li>
    </ul>
    <!-- {11} -->
  </nav>
  <!-- {6} -->

```

Debugging workflow. The following four-step procedure traces any error in tangled output back to the literate source.

Step 1: Locate the error in tangled output. You receive an error such as:

```

TypeError: Cannot read property 'id' of undefined
    at Object.<anonymous> (server.js:47)

```

Open `server.js` and go to line 47.

Step 2: Read the nearest opening marker. Scan *upward* from the error line until you find an opening marker:

```
// {3: routes.w:42}
```

This tells you: scrap 3 generated this code, defined in `routes.w` starting at line 42.

Step 3: Go to the literate source. Open `routes.w` at line 42. You will find the scrap with the bug, surrounded by prose that explains its intent:

```

We fetch the user by ID from the database. If the user
doesn't exist, we should return a 404, but currently we
assume the user always exists --- this is the bug.

```

```

@d Handle user fetch
@{const user = await getUser(req.params.id);
  res.json(user.id);
@}

```

The prose makes the bug obvious: there is no null check. Fix it in the `.w` file, re-tangle, and the error is gone.

Step 4: Nested markers = nested fragments. When fragments invoke other fragments, markers nest:

```

// {3: routes.w:42}
app.get("/api/users/:id", async (req, res) => {
  // {7: database.w:18}
  const user = await db.query(
    "SELECT * FROM users WHERE id=$1",
    [req.params.id]);
  // {7}
  if (!user) return res.status(404).json({error: "not found"});
  res.json(user);
});
// {3}

```

If the bug is inside the `{7: ... {7}}` block, go to `database.w:18`, not `routes.w:42`. The *innermost enclosing markers* point to the most specific source location. Always look at the tightest pair around the error line.

Interpreting nested hierarchies. Complex tangled output can have several nesting levels. Read them like parentheses—each opening matches its corresponding close:

```
/* {1: main.w:10} */           outermost: main file structure
#include <stdio.h>
/* {2: io.w:25} */             middle: I/O subsystem
void print_report() {
    /* {5: format.w:8} */       innermost: formatting detail
    printf("Report v%d\n", VERSION);
    /* {5} */
}
/* {2} */
int main() {
    /* {3: init.w:14} */
    init_subsystems();
    /* {3} */
    print_report();
    return 0;
}
/* {1} */
```

Scrap numbers are *document-global*—scrap 1 is the first `@{...@}` in the entire woven document, scrap 5 is the fifth. They are stable as long as you do not reorder scraps.

C/C++: #line directives. For C-family languages, weft also generates **#line** directives *above* the opening marker. These make the C compiler itself report errors against the `.w` source file:

```
#line 12 "memory.w"
/* {2: memory.w:12} */
void *safe_malloc(size_t size) { ... }
/* {2} */
```

Now gcc will say:

```
memory.w:14: error: implicit declaration of function 'malloc'
```

instead of:

```
memory.c:47: error: implicit declaration of function 'malloc'
```

The **#line** directive and the section marker are complementary: **#line** makes the compiler report errors against the `.w` file directly; the `{N: file:line}` markers are human-readable when browsing the tangled output.

Common debugging scenarios. Runtime error with stack trace.

```
Error: ENOENT: no such file or directory, open '/data/config.json'
at readConfig (server.js:123)
```

Open `server.js:123`, find the nearest marker `// {12: config.w:35}`, open `config.w:35`—the scrap that reads the config—and fix the path or add an existence check in the `.w` file.

Compiler error in C. With **#line** directives active, the compiler already reports:

```
parser.w:84: error: 'token_type' undeclared
```

Go directly to `parser.w:84`. The prose around the scrap explains what `token_type` should be and where it is defined.

Wrong output, no error. The program runs but produces wrong results. Open the tangled file and search for the relevant code. The markers tell you which `.w` file and scrap to examine:

```
# {15: transform.w:67}
result = [x * 2 for x in data]    # should be x * 3?
# {:15}
```

Go to `transform.w:67`. The prose says “triple each value” but the code says `* 2`. Bug found.

Identifying which fragment owns a line. Sometimes you know the line of code but not which `@d` definition produced it (multiple fragments with the same name accumulate). The scrap number is unique:

```
// {14: handlers.w:89}
validateInput(data);
// {:14}
// {22: handlers.w:112}
validatePermissions(user);
// {:22}
```

Scraps 14 and 22 are two different definitions (or continuations) of the same fragment. The line numbers point to each one independently.

Tips.

- *Never edit tangled files.* The markers remind you these are generated. Always trace back to the `.w` source and edit there.
- *Scrap numbers are stable* as long as you do not add or remove `@{...@}` blocks before them. If numbering shifts after edits, re-tangle and the markers update automatically.
- *Use `grep` on markers* to find all scraps from a specific file:

```
grep -n '{[0-9]*: database.w:' server.js
```
- *Closing markers as verification.* If you see `{:7}` but no `{7:` above it in your viewport, scroll up—the scrap is larger than your screen. If you see `{7:` but no `{:7}`, the scrap extends below.
- *Markers in test output.* If your test framework shows code snippets on failure, the markers point straight to the literate source.
- *Disable markers selectively.* Use explicit flags on `@o` (e.g. `@o file -t`) with no comment flags to suppress markers for a specific output file when comments would interfere—for example, binary formats, data files, or files where comments are not valid syntax.

1.7 Restrictions

Because `weft` is intended to be a simple tool, a few restrictions have been established. Over time, some of these may be eliminated; others seem fundamental.

- The handling of errors is not completely ideal. In some cases, `weft` simply warns of a problem and continues; in other cases it halts immediately. This behavior should be regularized.
- `weft` warns about references to fragments that haven’t been defined, but doesn’t halt. The name of the fragment is included in the output file surrounded by `<>` signs. This seems most convenient for development, but may change in the future.
- File names and index entries should not contain any `@` signs.
- Fragment names may be (almost) any well-formed `TeX` string. It makes sense to change fonts or use math mode; however, care should be taken to ensure matching braces, brackets, and dollar signs.
- Anything is allowed in the body of a scrap; however, very long scraps (horizontally or vertically) may not typeset well.

- Temporary files (created for comparison to the eventual output files) are placed in the current directory. Since they may be renamed to an output file name, all the output files should be on the same file system as the current directory. Alternatively, you can use the `-p` flag to specify where the files go.
- Because page numbers cannot be determined until the document has been typeset, `weft` must be rerun after `LATEX` to obtain a clean version of the document (very similar to the way it is sometimes necessary to rerun `LATEX` to obtain an up-to-date table of contents after significant edits). `weft` will warn (in most cases) when this needs to be done; in the remaining cases, `LATEX` will warn that labels may have changed. This restriction does not apply to Markdown output (`-w md`), which uses sequential numbering and requires no additional processing passes.

Very long scraps may be allowed to break across a page if declared with `@O` or `@D` (instead of `@o` and `@d`). This doesn't work very well as a default, since far too many short scraps will be broken across pages; however, as a user-controlled option, it seems very useful.

1.8 Project Introspection

The `-m` and `-e` flags support machine-readable introspection of a `weft` project. They run after pass 1 has built the name trees and do *not* invoke the weave or tangle phases.

1.8.1 Map (`-m`)

The command

```
weft -m project.weft
```

writes a JSON object to stdout with two arrays, `"files"` and `"fragments"`. For each entry the map records the name, language tag, definition scraps (with source file and line), and dependency relationships. File entries list the fragments they use; fragment entries list the scraps that invoke them and the fragments they reference in turn.

This gives an AI assistant or any external tool a complete dependency graph of the project in a single invocation, without having to parse the `weft` source files.

1.8.2 Extract (`-e`)

The command

```
weft -e "fragment name" project.weft
```

finds the named fragment (using prefix matching), then emits its scrap code and dependency information as Markdown to stdout. Referenced fragments are followed transitively, so the output contains the full closure of the requested fragment.

This allows an AI assistant to retrieve exactly the code it needs to understand or modify a specific fragment, together with all its dependencies, in one step.

1.8.3 Reverse Map (`-R`)

The `-m` and `-e` flags require the `.weft` source files. The reverse map provides the opposite direction: given a tangled output file, it parses the section markers already embedded in that file and reports which `.weft` source and line produced each region. No source files are needed and pass 1 is never invoked.

Single-line query.

```
weft -R server.js:47
```

emits a JSON object identifying the innermost scrap covering line 47:

```
{"scrap": 3, "file": "routes.weft", "line": 42}
```

If the line is outside any scrap region, an empty object `{}` is returned.

Full-file mapping.

```
weft -R server.js
```

emits a JSON object listing every scrap region in the file:

```
{ "source": "server.js", "regions": [
  { "start": 2, "end": 15, "scrap": 3,
    "file": "routes.weft", "line": 42 },
  { "start": 5, "end": 8, "scrap": 7,
    "file": "database.weft", "line": 18 }
]}
```

Regions may overlap when fragments are nested.

This enables CI/CD pipelines to translate compiler errors back to the literate source automatically, and lets AI agents resolve tangled-output locations without loading the `.weft` files at all.

1.9 Acknowledgements

Several people have contributed their time, ideas, and debugging skills. In particular, the contributions of the following people are gratefully acknowledged: Osman Buyukisik, Manuel Carriba, Adrian Clarke, Tim Harvey, Michael Lewis, Walter Ravenek, Rob Shillingsburg, Kayvan Sylvan, Dominique de Waleffe, and Scott Warren. Of course, most of these people would never have heard of nuweb (the original tool) without the efforts of George Greenwade.

Since maintenance has been taken over by Marc Mengel, Simon Wright and Keith Harwood online contributions have been made by:

- Walter Brown <wb@fnal.gov>
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- Javier Goizueta <jgoizueta@jazzfree.com>
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Chapter 2

The Overall Structure

Processing a web requires three major steps:

1. Read the source, accumulating file names, fragment names, scraps, and lists of cross-references.
2. Reread the source, copying out to the documentation file, with protection and cross-reference information for all the scraps.
3. Traverse the list of files names. For each file name:
 - (a) Dump all the defining scraps into a temporary file.
 - (b) If the file already exists and is unchanged, delete the temporary file; otherwise, rename the temporary file.

2.1 Files

I have divided the program into several files for quicker recompilation during development.

```
<global.h 21a> ≡  
<Include files 21b>  
<Type declarations 23a, ...>  
<Limits 23b>  
<Global variable declarations 27, ...>  
<Function prototypes 42a, ...>  
<Operating System Dependencies 26>
```

21a

We'll need at least five of the standard system include files. We also require `_POSIX_C_SOURCE` at level 200809L so that POSIX functions like `strdup` are visible when compiling under strict C11 mode (`-std=c11`).

```
<Include files 21b> ≡  
  
#define _POSIX_C_SOURCE 200809L  
  
#include <stdlib.h>  
#include <stdio.h>  
#include <string.h>  
#include <ctype.h>  
#include <signal.h>  
#include <locale.h>
```

21b · used in 21a · defines `exit` 25f, 29f, 36d, 37, 39b, 46d, 47a, 67b, 112ab, 117, 122b, 124b, 127ab, 128a, 129a, 134bc, 136a, 141d, 142d, 147d, 167a, 170ab, 171d, 172a, 173, 174, 178b, 183b, 192, `fclose` 55, 75a, 82d, 101, 117, 119, 122ab, 123b, 124a, 128b, 139, 150b, `FILE` 30, 54b, 55, 62b, 66c, 68a, 69cd, 70, 75ab, 76b, 78b, 80d, 82bcd, 86c, 89ab, 92b, 93, 94b, 95c, 97b, 98a, 99c, 100c, 101, 105c, 106ac, 107, 108, 110b, 117, 119, 122a, 123b, 124a, 125de, 130b, 131bc, 132a, 138c, 139, 142b, 149b, 150b, 184ef, 185d, 186a, 187cd, 192, `fopen` 55, 82d, 101, 117, 119, 122b, 123b, 124a, 127b, 129a, 150b, `fprintf` 33, 36e, 37, 39b, 42c, 44d, 46d, 47a, 49e, 53b, 55, 57a, 59c, 61a, 63ae, 65g, 66b, 67b, 71d, 73efg, 75b, 77a, 80e, 82d, 85c, 87ae, 88c, 91bc, 92a, 93, 94c, 97c, 99a, 100a, 101, 104b, 107, 108, 110b, 112ab, 114ab, 117, 122ab, 124b, 127ab, 128a, 129a, 131b, 132g, 134bc, 136a, 138c, 139, 141d, 142d, 144bc, 145a, 147cd, 149ab, 154c, 164, 166, 167ab, 168a, 170ab, 171d, 172a, 173, 174, 178b, 183b, 185b, 186c, 188abef, `fputs` 50a, 51c, 53b, 56ab, 57b, 58a, 59bc, 61ab, 62b, 63abcde, 65aefg, 66abc, 68a, 69cd, 70, 71e, 72b, 73g, 74, 75ab, 76a, 77abc, 79abc, 80ce, 85ac, 86ac, 87ae, 88bc, 89ab, 90ac, 91ab, 92ab, 93, 94acd, 95ad, 96ab, 97ac, 98b, 99abd, 100ac, 102ab, 103ac, 104abd, 105abcd, 106abc, 107, 108, 109, 110b, 111c, 117, 120, 123b, 131b, 139, 143, 147c, 148a, 185abc, 186bc, `getc` 124a, 126a, 127ab, 128ab, 129a, `getenv` 36e, `isgraph` 128a, 167a, 173, `islower` 155c, `isspace` 65a, 87b, 104c, 136a, 167ab, 170a, 172a, `isupper` never used, `malloc` 78b, 95c, 178b, 190d, `putc` 50ab, 58bc, 59c, 68a, 70, 71c, 73g, 75ab, 77ab, 79abc, 80e, 83ab, 84a, 89b, 90abc, 92a, 93, 94c, 95d, 96a, 100c, 102c, 103a, 106a, 131b, 143, 145ce, 146a, 147d, 148abc, `remove` 123ab, 124ab, `setlocale` 36e, `size_t` 75a, 115b, 120, 189d, 190bd, `stderr` 33, 36e, 37, 39b, 42c, 44d, 46d, 47a, 55, 59c, 65g, 66b, 67b, 71d, 73g, 75b, 82d, 87e, 88c, 92a, 93, 101, 112ab, 114ab, 117, 122ab, 124b, 127ab, 128a, 129a, 132g, 134bc, 136a, 141d, 142d, 147d, 154c, 164, 166, 167ab, 168a, 170ab, 171d, 172a, 173, 174, 178b, 183b, 188bef, `strlen` 50a, 113a, 117, 121e, 150c, 153c, 154a, 179d, 187a, `toupper` 155c

I like to use `TRUE` and `FALSE` in my code. I'd use an `enum` here, except that some systems seem to provide definitions of `TRUE` and `FALSE` by default. The following code seems to work on all the local systems.

< Type declarations 23a > ≡

```
#ifndef FALSE
#define FALSE 0
#endif
#ifndef TRUE
#define TRUE 1
#endif

#if defined(__GNUC__) && __GNUC__ >= 7
#define FALLTHROUGH __attribute__((fallthrough))
#elif defined(__clang__)
#define FALLTHROUGH __attribute__((fallthrough))
#else
#define FALLTHROUGH ((void)0)
#endif
```

23a · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `FALLTHROUGH` 44d, 58c, 71e, 84a, 103a, 126a, 154b, `FALSE` 27, 28a, 33, 35abc, 36abc, 39b, 58bc, 63d, 65d, 66c, 69d, 70, 75a, 79b, 80be, 83ab, 84a, 85b, 87c, 89a, 93, 96a, 97c, 99a, 100a, 103a, 111a, 124b, 127a, 129a, 147d, 150b, 155a, 157ab, 164, 167b, 168b, 184b, 185b, 186cd, 187a, `TRUE` 27, 28a, 29f, 33, 36a, 39b, 53b, 58c, 63be, 66c, 71d, 75a, 79b, 80be, 83ab, 84a, 85b, 87ab, 89a, 93, 96a, 97c, 99a, 100a, 103a, 104b, 110b, 111c, 127a, 131c, 132g, 147d, 155a, 157ab, 164, 167b, 184b, 185b, 186cd, 187a

The `FALLTHROUGH` macro marks intentional switch case fall-throughs as a statement rather than a comment. GCC 7+ and Clang recognise `__attribute__((fallthrough))` and suppress `-Wimplicit-fallthrough` warnings; other compilers see a harmless `((void)0)`. The prose surrounding each switch already explains why the fall-through is intended, so no inline comment is needed.

Here we define the maximum length for names (of fragments etc), and the version string reported by `--version`. The version is a hard-coded literal — the single source of truth for weft's own version — so that *any* build from a fresh checkout carries the correct version, whether it is compiled with `make bootstrap`, re-tangled with plain `make`, or released with `make dist`. Cutting a release is therefore a one-line edit here. (The `-V` flag and the `@v` directive remain a separate, user-facing feature for stamping a version into *other* documents woven by weft; they no longer drive weft's own version.)

< Limits 23b > ≡

```
#ifndef MAX_NAME_LEN
#define MAX_NAME_LEN 1024
#endif

#define WEFT_VERSION "< version 2 >"
```

23b · used in 21a · defines `MAX_NAME_LEN` 136a, 138c, 141d, 167a, 170a, 173, 175a, 183b, 187b, `WEFT_VERSION` 28a, 29f, 30

2.1.1 The Main Files

The code is divided into four main files (introduced here) and five support files (introduced in the next section). The file `main.c` will contain the driver for the whole program (see Section 2.2).

< main.c 23c > ≡

```
#include "global.h"
```

23c · defined at 23c, 25f

The first pass over the source file is contained in `pass1.c`. It handles collection of all the file names, fragments names, and scraps (see Section 2.3).

```
<pass1.c 23d> ≡  
#include "global.h"  
23d · defined at 23d, 42bc
```

The `.tex` file is created during a second pass over the source file. The file `latex.c` contains the code controlling the construction of the `.tex` file (see Section 2.4).

```
<latex.c 24a> ≡  
#include "global.h"  
static int scraps = 1;  
24a · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · uses scraps 130a
```

The Markdown output is created by the code in `markdown.c` (see Section 2.5).

```
<markdown.c 24b> ≡  
#include "global.h"  
static int scraps = 1;  
24b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · uses scraps 130a
```

The HTML output is created by the code in `html.c` (see Section 2.6).

```
<html.c 24c> ≡  
#include "global.h"  
static int scraps = 1;  
24c · defined at 24c, 100c, 101, 105c, 106ac · uses scraps 130a
```

The code controlling the creation of the output files is in `output.c` (see Section 2.8).

```
<output.c 24d> ≡  
#include "global.h"  
24d · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce
```

2.1.2 Support Files

The support files contain a variety of support routines used to define and manipulate the major data abstractions. The file `input.c` holds all the routines used for referring to source files (see Section 3.1).

```
< input.c 25a > ≡  
#include "global.h"
```

25a · defined at 25a, 125de, 126ac, 129a

Creation and lookup of scraps is handled by routines in `scraps.c` (see Section 3.2).

```
< scraps.c 25b > ≡  
#include "global.h"
```

25b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac

The handling of file names and fragment names is detailed in `names.c` (see Section 3.3).

```
< names.c 25c > ≡  
#include "global.h"
```

25c · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc

Memory allocation and deallocation is handled by routines in `arena.c` (see Section 3.6).

```
< arena.c 25d > ≡  
#include "global.h"
```

25d · defined at 25d, 189e, 190ab, 191

Finally, for best portability, I seem to need a file containing (useless!) definitions of all the global variables.

```
< global.c 25e > ≡  
#include "global.h"  
< Operating System Dependencies 26 >  
< Global variable definitions 28a, ... >
```

25e · defined at 25e, 189a

2.2 The Main Routine

The main routine is quite simple in structure. It wades through the optional command-line arguments, then handles any files listed on the command line.

```
< main.c 25f > + ≡  
  
#include <stdlib.h>  
int main(int argc, char **argv)  
{  
    int arg = 1;  
    < Interpret command-line arguments 29d, ... >  
    < Set locale information 36e >  
    < Handle reverse map mode 36d >  
    initialise_delimit_scrap_array();  
    < Process the remaining arguments (file names) 37 >  
    exit(0);  
}
```

We only have two major operating system dependencies; the separators for file names, and how to set environment variables. For now we assume the latter can be accomplished via `putenv()` in `stdlib.h`.

⟨ *Operating System Dependencies* 26 ⟩ ≡

```
#if defined(VMS)
#define PATH_SEP(c) (c=='|'||c==':')
#define PATH_SEP_CHAR ""
#define DEFAULT_PATH ""
#elif defined(MSDOS)
#define PATH_SEP(c) (c=='\\')
#define PATH_SEP_CHAR "\\"
#define DEFAULT_PATH "."
#else
#define PATH_SEP(c) (c=='/')
#define PATH_SEP_CHAR "/"
#define DEFAULT_PATH "."
#endif
```

26 · used in 21a, 25e

2.2.1 Command-Line Arguments

There are numerous possible command-line arguments:

- t Suppresses generation of the `.tex` file.
- o Suppresses generation of the output files.
- d list dangling identifier references in indexes.
- c Forces output files to overwrite old files of the same name without comparing for equality first.
- v The verbose flag. Forces output of progress reports.
- n Forces sequential numbering of scraps (instead of page numbers).
- s Doesn't print list of scraps making up file at end of each scrap.
- x Include cross-reference numbers in scrap comments.
- p *path* Prepend *path* to the filenames for all the output files.
- mkdirs Create output directories as needed (a recursive `mkdir -p`) before writing each tangled file — both the `-p` prefix and any subdirectory embedded in an `@o` name. Without it, the directories must already exist (the strict default, which catches mistyped paths); a missing directory is then a hard error.
- h *options* Provide options for the `hyperref` package.
- r Turn on hyperlinks. You must include the `—usepackage—` options in the text.
- l Deprecated no-op. Code scraps are always formatted with the `listings` package now, so this flag has no effect and is kept only for backward compatibility. Remember to `\usepackage{listings}` in your document preamble.

There are two ways to get hyper-links into your documentation. With one, the `-r` flag simply arranges for fragment cross-references to be hyperlinks. You have to provide all the rest of the `hyperref` material yourself. The `-h` flag does more work, but requires you provide the `hyperref` options on the command line. You would use the first if you know you will use the same tools all the time. You would use the second if you have to distribute the weft file and use the build system to provide the `hyperref` options.

The `-h` option is used as follows. If you want hyperlinks in your document give a string to this option which is a valid option for the `hyperref` package. For example, "`dvips,colorlinks=true`" sets it to use the `dvips` driver and mark the links in colour. For more information about these options, see the `hyperref` documentation.

Then in your document you put the command `\WEFTuseHyperlinks` immediately after your `usepackage` commands. If you do both of these you will get hyperlinks. If you miss out either then nothing interesting happens.

Global flags are declared for each of the arguments.

< Global variable declarations 27 > ≡

```
extern int tex_flag; /* if FALSE, don't emit the documentation file */
extern int output_flag; /* if FALSE, don't emit the output files */
extern int compare_flag; /* if FALSE, overwrite without comparison */
extern int verbose_flag; /* if TRUE, write progress information */
extern int number_flag; /* if TRUE, use a sequential numbering scheme */
extern int scrap_flag; /* if FALSE, don't print list of scraps */
extern int dangling_flag; /* if FALSE, don't print dangling identifiers */
extern int xref_flag; /* If TRUE, print cross-references in scrap comments */
extern int prepend_flag; /* If TRUE, prepend a path to the output file names */
extern char * dirpath; /* The prepended directory path */
extern char * path_sep; /* How to join path to filename */
extern int listings_flag; /* if TRUE, use listings package for scrap formatting */
extern int version_info_flag; /* If TRUE, set up version string */
extern char * version_string; /* What to print for @v */
extern int hyperref_flag; /* Are we preparing for hyperref
                           package. */
extern int hyperopt_flag; /* Are we preparing for hyperref options */
extern char * hyperoptions; /* The options to pass to the
                             hyperref package */
extern int includepath_flag; /* Do we have an include path? */
extern struct incl{char * name; struct incl * next;} * include_list;
                           /* The list of include paths */
extern int markdown_flag; /* if TRUE, generate Markdown instead of LaTeX */
extern int weave_flag; /* if TRUE, weave was requested via -w */
extern int weave_format; /* 0=none, 1=tex, 2=md, 3=html */
extern int weave_format_from_cli; /* if TRUE, CLI override of @W */
extern int map_flag; /* if TRUE, emit JSON map to stdout */
extern int extract_flag; /* if TRUE, next arg is extract name */
extern char *extract_name; /* fragment name to extract */
extern int reverse_map_flag; /* if TRUE, reverse map mode */
extern char *reverse_map_arg; /* file[:line] argument for -R */
extern int bodies_flag; /* if TRUE, include scrap bodies in -m JSON */
extern int prose_flag; /* if TRUE, include prose context in -m JSON */
extern int diff_flag; /* if TRUE, show diff instead of writing files */
extern int callers_flag; /* if TRUE, show callers with -e */
extern int lint_flag; /* if TRUE, run static analysis */
extern int errors_flag; /* if TRUE, annotate compiler errors from stdin */
extern int mkdirs_flag; /* if TRUE, create output directories as needed */
```

27 · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `bodies_flag` 28a, 29f, 108, `callers_flag` 28a, 29f, 39b, `compare_flag` 28a, 33, 123a, `dangling_flag` 28a, 33, 80e, 97c, 106c, `diff_flag` 28a, 29f, 123a, `errors_flag` 28a, 29f, 36d, `extract_flag` 28a, 33, 36b, `extract_name` 28a, 36b, 39b, `hyperoptions` 28a, 36a, 56a, 57a, `hyperopt_flag` 28a, 33, 36a, `hyperref_flag` 28a, 33, 36a, 56a, `incl` 28a, 35b, 39a, 127b, `includepath_flag` 28a, 33, 35b,

lint_flag 28a, 29f, 39b, map_flag 28a, 33, 39b, markdown_flag 28a, 33, 39b, mkdirs_flag 28a, 29f, 122a, 124b, number_flag 28a, 33, 39b, 150b, output_flag 28a, 33, 39b, prose_flag 28a, 29f, 42d, 43b, 108, reverse_map_arg 28a, 36cd, reverse_map_flag 28a, 33, 36c, scrap_flag 28a, 33, 61a, 87d, 104d, tex_flag 28a, 33, 39b, 42c, 147d, verbose_flag 28a, 33, 42c, 55, 82d, 101, 122a, version_info_flag 28a, 33, 35c, version_string 28a, 35c, 59b, 72b, 85a, 103a, weave_flag 28a, 33, 39b, weave_format 28a, 33, 39b, 166, weave_format_from_cli 28a, 33, 39b, 166, xref_flag 28a, 33, 148ac · uses FALSE 23a, next 129b, prose 129c, reverse 176c, scraps 130a, TRUE 23a

The flags are all initialized for correct default behavior.

⟨ *Global variable definitions* 28a ⟩ ≡

```
int tex_flag = TRUE;
int output_flag = TRUE;
int compare_flag = TRUE;
int verbose_flag = FALSE;
int number_flag = FALSE;
int scrap_flag = TRUE;
int dangling_flag = FALSE;
int xref_flag = FALSE;
int prepend_flag = FALSE;
char * dirpath = DEFAULT_PATH; /* Default directory path */
char * path_sep = PATH_SEP_CHAR;
int listings_flag = FALSE;
int version_info_flag = FALSE;
char default_version_string[] = WEFT_VERSION;
char * version_string = default_version_string;
int hyperref_flag = FALSE;
int hyperopt_flag = FALSE;
char * hyperoptions = "";
int includepath_flag = FALSE; /* Do we have an include path? */
struct incl * include_list = NULL;
/* The list of include paths */
int markdown_flag = FALSE;
int weave_flag = FALSE;
int weave_format = 0;
int weave_format_from_cli = FALSE;
int map_flag = FALSE;
int extract_flag = FALSE;
char *extract_name = NULL;
int reverse_map_flag = FALSE;
char *reverse_map_arg = NULL;
int bodies_flag = FALSE;
int prose_flag = FALSE;
int diff_flag = FALSE;
int callers_flag = FALSE;
int lint_flag = FALSE;
int errors_flag = FALSE;
int mkdirs_flag = FALSE;
```

28a · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses bodies_flag 27, callers_flag 27, compare_flag 27, dangling_flag 27, diff_flag 27, errors_flag 27, extract_flag 27, extract_name 27, FALSE 23a, hyperoptions 27, hyperopt_flag 27, hyperref_flag 27, incl 27, includepath_flag 27, lint_flag 27, map_flag 27, markdown_flag 27, mkdirs_flag 27, number_flag 27, output_flag 27, prose_flag 27, reverse_map_arg 27, reverse_map_flag 27, scrap_flag 27, tex_flag 27, TRUE 23a, verbose_flag 27, version_info_flag 27, version_string 27, weave_flag 27, weave_format 27, weave_format_from_cli 27, WEFT_VERSION 23b, xref_flag 27

A global variable `nw_char` will be used for the weft meta-character, which by default will be @.

⟨ *Global variable declarations* 28b ⟩ + ≡

```
extern int nw_char;
```

28b · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `nw_char` 29a, 30, 39b, 42d, 44d, 46d, 47a, 48a, 49bf, 50d, 52, 53ab, 58bc, 61a, 67a, 69c, 70, 71ae, 73ce, 77a, 80e, 83a, 84a, 89b, 90c, 91b, 102c, 103a, 126a, 127a, 129a, 134bc, 135a, 136a, 137bc, 138c, 141d, 143, 145d, 146a, 147cd, 149abd, 154c, 164, 166, 167a, 170ab, 171a, 172a, 173, 174, 175a, 183ab, 185b, 186cd, 187b

⟨ *Global variable definitions* 29a ⟩ + ≡

```
int nw_char='@';
```

29a · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · defines `nw_char` 28b, 30, 39b, 42d, 44d, 46d, 47a, 48a, 49bf, 50d, 52, 53ab, 58bc, 61a, 67a, 69c, 70, 71ae, 73ce, 77a, 80e, 83a, 84a, 89b, 90c, 91b, 102c, 103a, 126a, 127a, 129a, 134bc, 135a, 136a, 137bc, 138c, 141d, 143, 145d, 146a, 147cd, 149abd, 154c, 164, 166, 167a, 170ab, 171a, 172a, 173, 174, 175a, 183ab, 185b, 186cd, 187b

We save the invocation name of the command in a global variable `command_name` for use in error messages.

⟨ *Global variable declarations* 29b ⟩ + ≡

```
extern char *command_name;
```

29b · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `command_name` 29cd, 30, 33, 37, 39b, 44d, 46d, 47a, 55, 59c, 66b, 71d, 73g, 75b, 82d, 88c, 92a, 93, 101, 112ab, 122b, 124b, 127ab, 128a, 129a, 132g, 134bc, 136a, 141d, 142d, 147d, 154c, 164, 166, 167ab, 168a, 170ab, 171d, 172a, 173, 174, 183b

⟨ *Global variable definitions* 29c ⟩ + ≡

```
char *command_name = NULL;
```

29c · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses `command_name` 29b

The invocation name is conventionally passed in `argv[0]`.

⟨ *Interpret command-line arguments* 29d ⟩ ≡

```
command_name = argv[0];
```

29d · defined at 29de · used in 25f · uses `command_name` 29b

We need to examine the remaining entries in `argv`, looking for command-line arguments.

⟨ *Interpret command-line arguments* 29e ⟩ + ≡

```
while (arg < argc) {
    char *s = argv[arg];
    if (*s++ == '-') {
        ⟨ Check for long options 29f ⟩
        ⟨ Interpret the argument string s 33 ⟩
        arg++;
        ⟨ Perhaps get the prepend path 35a ⟩
        ⟨ Perhaps get the version info string 35c ⟩
        ⟨ Perhaps get the hyperref options 36a ⟩
        ⟨ Perhaps add an include path 35b ⟩
        ⟨ Perhaps get the extract name 36b ⟩
        ⟨ Perhaps get the reverse map argument 36c ⟩
    }
    else break;
}
```

29e · defined at 29de · used in 25f

2.2.2 Help Message

When the user passes `--help`, we display a comprehensive reference that serves both human users and AI assistants. The help text is structured in sections—usage, options grouped by purpose, examples, and an AI workflow summary—so that a language model can parse it and use `weft` correctly without additional context.

We intercept long options before the character-by-character flag parsing. Currently `--help` and `--version` are the only long options.

⟨ Check for long options 29f ⟩ ≡

```
if (*s == '-') {
    if (strcmp(s + 1, "help") == 0) {
        ⟨ Print help message 30 ⟩
        exit(0);
    }
    if (strcmp(s + 1, "version") == 0) {
        printf("weft %s\n", WEFT_VERSION);
        exit(0);
    }
    if (strcmp(s + 1, "bodies") == 0) {
        bodies_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "prose") == 0) {
        prose_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "diff") == 0) {
        diff_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "callers") == 0) {
        callers_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "lint") == 0) {
        lint_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "errors") == 0) {
        errors_flag = TRUE;
        arg++;
        continue;
    }
    if (strcmp(s + 1, "makedirs") == 0) {
        makedirs_flag = TRUE;
        arg++;
        continue;
    }
}
```

29f · used in 29e · uses `bodies_flag` 27, `callers_flag` 27, `diff_flag` 27, `errors_flag` 27, `exit` 21b, `lint_flag` 27, `makedirs_flag` 27, `prose` 129c, `prose_flag` 27, `TRUE` 23a, `WEFT_VERSION` 23b

The help message itself. We write to `stdout` (not `stderr`) because `--help` is a deliberate request, not an error condition. The output is structured so that both humans and AI assistants can use `weft` effectively from `--help` alone: options grouped by function, per-file output flags, section marker format, a complete directives reference, and a concise AI workflow section.

< Print help message 30 > ≡

```
printf("weft %s -- A Literate Programming Tool\n\n", WEFT_VERSION);
printf("USAGE\n");
printf("  %s [options] file...\n\n", command_name);
printf("WHAT IS WEFT?\n");
printf("  weft is a literate programming tool: write .weft files where\n");
printf("  documentation and code live as one narrative.  weft TANGLES them\n");
printf("  into executable output files and optionally WEAVES formatted\n");
printf("  documentation (LaTeX or Markdown).  .weft files are the ONLY\n");
printf("  source of truth; tangled outputs are GENERATED -- never edit them.\n\n");
printf("HOW IT WORKS\n");
printf("  A .weft file contains prose with embedded code chunks:\n");
printf("    @o file.c @{ ... @}      Output file: tangle writes this to file.c\n");
printf("    @d Name   @{ ... @}      Fragment: reusable code block, invoked by name\n");
printf("    @<Name@>      Invocation: inserts the named fragment here\n");
printf("    @@                Literal @: escapes the meta-character\n");
printf("    @i other.weft        Include: pulls in another .weft file\n\n");
printf("  Fragments compose hierarchically.  An @o chunk may invoke @d fragments,\n");
printf("  which invoke other fragments, forming a tree that weft flattens into\n");
printf("  the final output file.\n\n");
printf("OPTIONS\n");
printf("  Tangle control:\n");
printf("    -o          Suppress output file generation (parse only)\n");
printf("    -c          Overwrite files without comparing first\n");
printf("    -p path     Prepend path to all output file names\n");
printf("    --mkdirs    Create output directories as needed (mkdir -p)\n");
printf("    -I path     Add directory to include search path (repeatable)\n");
printf("    -x          Include cross-reference numbers in section markers\n\n");
printf("  Weave control:\n");
printf("    -w [format] Weave documentation (format: md, tex, or html)\n");
printf("                Without format, uses @W directive from source\n");
printf("    -t          Suppress weave output (even if -w given)\n");
printf("    -n          Sequential scrap numbering (auto for Markdown)\n");
printf("    -s          Omit scrap composition lists\n\n");
printf("  Analysis (AI-friendly tools):\n");
printf("    -m          Emit JSON structure map to stdout (all fragments,\n");
printf("                files, definitions, and dependency graph)\n");
printf("    -e name     Extract named fragment and all its transitive\n");
printf("                dependencies as self-contained Markdown to stdout\n");
printf("    -R file[:line] Reverse map: given a tangled output file (and\n");
printf("                optional line number), prints the .weft source\n");
printf("                location that generated it.  Needs NO .weft files\n");
printf("                -- reads section markers embedded in the output\n");
printf("    --bodies    Include raw scrap body text in -m JSON output\n");
printf("    --prose     Include preceding documentation in -m JSON output\n");
printf("    --diff      Show unified diff of what tangle would change\n");
printf("                (no files written)\n");
printf("    --lint      Static analysis: unused fragments, undefined refs,\n");
printf("                and fuzzy-match suggestions for likely typos\n");
printf("    --callers   With -e, show call chain from %co roots to fragment\n",
nw_char);
printf("    --errors    Annotate compiler errors from stdin with .weft\n");
printf("                source locations (reads section markers, no\n");
printf("                .weft files needed)\n");
```

```

printf("                                Example: make 2>&1 | weft --errors\n\n");
printf(" LaTeX:\n");
printf("  -r                                Enable hyperref cross-reference links\n");
printf("  -h options                          Pass options to hyperref package\n");
printf("  -l                                  Deprecated no-op (listings is always used)\n");
printf("  -d                                  List dangling identifier references\n\n");
printf(" General:\n");
printf("  -V string                          Set version string (expands @v in source)\n");
printf("  -v                                  Verbose progress output\n");
printf("  --help                             Show this help message\n");
printf("  --version                          Show version number\n\n");
printf("PER-FILE FLAGS (on @o)\n");
printf(" @o file [-d] [-i] [-t] [-s] [-cc|-c+|-cp] @{ ... @} \n");
printf("  -d                                Emit #line N \"file\" directives (C/C++ debuggers)\n");
printf("  -i                                Suppress auto-indentation of expanded fragments\n");
printf("  -t                                Suppress tab expansion (write tabs literally)\n");
printf("  -cc                               C block-comment markers: /* {N: file:line} */\n");
printf("  -c+                               C++ line-comment markers: // {N: file:line}\n");
printf("  -cp                               Hash-comment markers: # {N: file:line}\n");
printf("  -s                                Suppress all section markers in output\n");
printf(" Without explicit flags, weft auto-detects from file extension\n");
printf(" (250+ extensions recognized). Use @l tag before @o to override.\n\n");
printf("SECTION MARKERS\n");
printf(" When comment style is known, tangled output includes markers:\n");
printf(" /* {5: source.weft:42} */ opening -- scrap 5, from line 42\n");
printf(" /* {5} */ closing\n");
printf(" Comment adapts to language: // for C++/Go, # for Python/shell, etc.\n");
printf(" Use weft -R to parse these markers and reverse-map to source.\n\n");
printf("DIRECTIVES REFERENCE\n");
printf(" Core structure:\n");
printf(" @o file [flags] @{ code @} Output file (tangled to disk)\n");
printf(" @d Name @{ code @} Define reusable named fragment\n");
printf(" @d+ Name @{ code @} Global fragment (visible across sectors)\n");
printf(" @q Name @{ code @} Quoted fragment (no expansion on include)\n");
printf(" @Q / @D / @Q Uppercase = big-format scrap in LaTeX\n");
printf(" @<Name@> Invoke (expand) a named fragment\n");
printf(" @<+Name@> Invoke the global version of a fragment\n");
printf(" @i file Include another .weft file\n");
printf(" @@ Literal @ character\n\n");
printf(" Fragment parameters:\n");
printf(" @'text@' Pass literal text argument to a fragment\n");
printf(" @1 .. @9 Substitute nth argument inside fragment\n");
printf(" @{ code @} as argument Pass anonymous inline scrap as argument\n\n");
printf(" Inside scraps:\n");
printf(" @f Current output file name\n");
printf(" @t Current fragment title (as invoked)\n");
printf(" @v Version string (defaults to weft's version; -V overrides)\n");
printf(" @c Expand block comment at current indent\n");
printf(" @# Suppress auto-indent for this line\n");
printf(" @%% Comment to end of line (omitted)\n");
printf(" @| name Mark identifier(s) as defined here\n");
printf(" @+ @- @* Global-export / import / used markers\n");
printf(" @_ text @_ Bold (LaTeX weave); skipped in tangle\n");
printf(" @x label @x Cross-reference label (anchor position)\n\n");
printf("Scrap delimiters (LaTeX weave):\n");
printf(" @{ ... @} Verbatim code (default)\n");
printf(" @[ ... @] Paragraph mode\n");
printf(" @( ... @) Math mode\n\n");

```

```

printf(" Sections and structure:\n");
printf("  @s                               Start a new sector (section boundary)\n");
printf("  @S                               Close current sector (return to base)\n");
printf("  @W fmt                             Declare weave format: md, tex, or html\n\n");
printf(" Language and formatting:\n");
printf("  @l tag                             Set language for next @o or @d\n");
printf("  @L name style [+d]                 Define custom language (// # -- /* <!--)\n");
printf("  @r x                               Change escape character from @ to x\n\n");
printf(" Indexes (weave output only):\n");
printf("  @f @m @u                           File / fragment / identifier indexes\n");
printf("  @m+ @u+                             Global fragment / identifier indexes\n\n");
printf("EXAMPLES\n");
printf(" %s project.weft                      Tangle (default)\n", command_name);
printf(" %s -w md project.weft                Weave to Markdown\n", command_name);
printf(" %s -w tex project.weft               Weave to LaTeX\n", command_name);
printf(" %s -w html project.weft              Weave to HTML\n", command_name);
printf(" %s -m project.weft | jq '.files'     JSON structure map\n", command_name);
printf(" %s -e \"Validate\" project.weft        Extract fragment\n", command_name);
printf(" %s -R server.js:47                   Reverse map line\n", command_name);
printf(" %s -R server.js                      All regions in file\n", command_name);
printf(" %s -v -p build/ project.weft         Verbose + path prefix\n", command_name);
printf(" %s -o project.weft                  Parse only, no output\n", command_name);
);
printf("AI ASSISTANT WORKFLOW\n");
printf(" GOLDEN RULE: .weft files are the ONLY source of truth.\n");
printf(" Tangled outputs (*.c, *.go, etc.) are GENERATED -- never edit them.\n");
printf(" If you edit a tangled file, your changes WILL be lost on next tangle.\n");
printf(" Step 1 -- Understand the project structure:\n");
printf("   %s -m project.weft > map.json\n", command_name);
printf("   The JSON contains: files (all @o output files), fragments (all @d\n");
printf("   named blocks), and a dependency graph (which fragments invoke which).\n");
printf("   Useful jq queries:\n");
printf("     jq '.files'                      List output files\n");
printf("     jq '.fragments'                  List all fragments\n");
printf("     jq '.fragments[] | select(.name | test(\"Auth\"))' Search\n");
printf(" Step 2 -- Read specific code without opening .weft files:\n");
printf("   %s -e \"fragment name\" project.weft\n", command_name);
printf("   Prints the fragment AND all its transitive dependencies as\n");
printf("   self-contained Markdown to stdout. Ideal for understanding\n");
printf("   one piece of the system without reading the whole source.\n\n");
printf(" Step 3 -- Edit the .weft source (NEVER tangled output):\n");
printf("   Locate the right .weft file (use -m to find it), then edit\n");
printf("   the @d or @o scrap that contains the code you need to change.\n");
printf("   Use standard text editing -- the prose around code is LaTeX or\n");
printf("   plain text, and code lives between @{ and @}.\n\n");
printf(" Step 4 -- Regenerate and verify:\n");
printf("   %s project.weft                    Tangle (regenerate all output files)\n",
command_name);
printf("   Then build and test as usual.\n\n");
printf(" Step 5 -- Debug errors with reverse map:\n");
printf("   %s -R output.c:42                  Find .weft source for line 42\n", command_name);
printf("   %s -R output.c                     Show ALL .weft regions in file\n", command_name);
printf("   Reads section markers in the tangled output -- needs NO .weft files.\n");
printf("   Fix errors in the .weft source, then re-tangle (step 4).\n");

```

30 · used in 29f · uses [command_name](#) 29b, [FILE](#) 21b, [first](#) 190a, [Name](#) 152b, [next](#) 129b, [nw_char](#) 28b, 29a, [prose](#) 129c, [reverse](#) 176c, [scraps](#) 130a, [search](#) 180c, [sector](#) 129c, uses 129c, [WEFT_VERSION](#) 23b

2.2.3 Short Flags

Several flags can be stacked behind a single minus sign; therefore, we've got to loop through the string, handling them all. If this flag requires an argument we skip to getting its argument straight away. This allows arguments to butt up to their flags and also avoids ambiguity about which value goes with which flag.

(Interpret the argument string `s` ³³) $\equiv$

```
{
  char c = *s++;
  while (c) {
    switch (c) {
      case 'c': compare_flag = FALSE;
                break;
      case 'd': dangling_flag = TRUE;
                break;
      case 'h': hyperopt_flag = TRUE;
                goto HasValue;
      case 'I': includepath_flag = TRUE;
                goto HasValue;
      case 'l': listings_flag = TRUE;
                break;
      case 'n': number_flag = TRUE;
                break;
      case 'o': output_flag = FALSE;
                break;
      case 'p': prepend_flag = TRUE;
                goto HasValue;
      case 'r': hyperref_flag = TRUE;
                break;
      case 's': scrap_flag = FALSE;
                break;
      case 't': tex_flag = FALSE;
                break;
      case 'e': extract_flag = TRUE;
                goto HasValue;
      case 'R': reverse_map_flag = TRUE;
                goto HasValue;
      case 'm': map_flag = TRUE;
                break;
      case 'v': verbose_flag = TRUE;
                break;
      case 'V': version_info_flag = TRUE;
                goto HasValue;
      case 'w': goto HasWeaveFormat;
      case 'x': xref_flag = TRUE;
                break;
      default: fprintf(stderr, "%s: unexpected argument ignored. "
                        "Try '%s --help' for usage information.\n",
                        command_name, command_name);
                break;
    }
    c = *s++;
  }
  goto HasValue;
HasWeaveFormat:
{
  char *fmt = NULL;
  weave_flag = TRUE;
  if (*s != '\0')
```

```

    fmt = s;
else if (arg + 1 < argc && argv[arg + 1][0] != '-') {
    /* Peek: only consume next arg if it looks like a format token */
    char *candidate = argv[arg + 1];
    if (strcmp(candidate, "md") == 0 || strcmp(candidate, "tex") == 0
        || strcmp(candidate, "html") == 0) {
        fmt = candidate;
        ++arg;
    }
}
if (fmt != NULL) {
    if (strcmp(fmt, "md") == 0) {
        weave_format = 2;
        markdown_flag = TRUE;
        weave_format_from_cli = TRUE;
    } else if (strcmp(fmt, "tex") == 0) {
        weave_format = 1;
        weave_format_from_cli = TRUE;
    } else if (strcmp(fmt, "html") == 0) {
        weave_format = 3;
        weave_format_from_cli = TRUE;
    } else {
        fprintf(stderr,
            "%s: unknown weave format '%s'. Use 'md', 'tex', or 'html'.\\n",
            command_name, fmt);
    }
}
/* If fmt is NULL, -w alone: format will come from @W */
s = "";
}
HasValue::
}

33 · used in 29e · uses command_name 29b, compare_flag 27, dangling_flag 27, extract_flag 27, FALSE 23a, fprintf 21b,
hyperopt_flag 27, hyperref_flag 27, includepath_flag 27, map_flag 27, markdown_flag 27, next 129b, number_flag 27,
output_flag 27, reverse_map_flag 27, scrap_flag 27, stderr 21b, tex_flag 27, TRUE 23a, verbose_flag 27,
version_info_flag 27, weave_flag 27, weave_format 27, weave_format_from_cli 27, xref_flag 27

```

⟨ *Perhaps get the prepend path* 35a ⟩ ≡

```

if (prepend_flag)
{
    if (*s == '\\0')
        s = argv[arg++];
    dirpath = s;
    prepend_flag = FALSE;
}

```

35a · used in 29e · uses FALSE 23a

⟨ *Perhaps add an include path* 35b ⟩ ≡

```

if (includepath_flag)
{
    struct incl * le
        = (struct incl *)arena_getmem(sizeof(struct incl));
    struct incl ** p = &include_list;

    if (*s == '\\0')
        s = argv[arg++];
    le->name = save_string(s);
}

```

```

    le->next = NULL;
    while (*p != NULL)
        p = &((*p)->next);
    *p = le;
    includepath_flag = FALSE;
}

```

35b · used in 29e · uses arena_getmem 190b, FALSE 23a, incl 27, includepath_flag 27, next 129b, save_string 154a

⟨ *Perhaps get the version info string* 35c ⟩ ≡

```

if (version_info_flag)
{
    if (*s == '\0')
        s = argv[arg++];
    version_string = s;
    version_info_flag = FALSE;
}

```

35c · used in 29e · uses FALSE 23a, version_info_flag 27, version_string 27

⟨ *Perhaps get the hyperref options* 36a ⟩ ≡

```

if (hyperopt_flag)
{
    if (*s == '\0')
        s = argv[arg++];
    hyperoptions = s;
    hyperopt_flag = FALSE;
    hyperref_flag = TRUE;
}

```

36a · used in 29e · uses FALSE 23a, hyperoptions 27, hyperopt_flag 27, hyperref_flag 27, TRUE 23a

⟨ *Perhaps get the extract name* 36b ⟩ ≡

```

if (extract_flag) {
    extract_name = (*s != '\0') ? s : argv[arg++];
    extract_flag = FALSE;
}

```

36b · used in 29e · uses extract_flag 27, extract_name 27, FALSE 23a

⟨ *Perhaps get the reverse map argument* 36c ⟩ ≡

```

if (reverse_map_flag) {
    reverse_map_arg = (*s != '\0') ? s : argv[arg++];
    reverse_map_flag = FALSE;
}

```

36c · used in 29e · uses FALSE 23a, reverse_map_arg 27, reverse_map_flag 27

If the user gave `-R file[:line]`, we handle it immediately and exit. No `.weft` source files are required—only the tangled output file.

⟨ *Handle reverse map mode* 36d ⟩ ≡

```

if (errors_flag) {
    write_error_annotator();
    exit(0);
}
if (reverse_map_arg) {
    write_reverse_map(reverse_map_arg);
    exit(0);
}

```

36d · used in 25f · uses `errors_flag` 27, `exit` 21b, `reverse_map_arg` 27, `write_error_annotator` 120, `write_reverse_map` 117

In order to be able to process files in foreign languages we set the locale information. `isgraph()` and friends need this (see Section 3.3). Try to read `LC_CTYPE` from the environment. Use `LC_ALL` if that fails. Don't set the locale if reading `LC_ALL` fails, too. Print a warning if setting the program's locale fails.

⟨ *Set locale information* 36e ⟩ ≡

```
{
    /* try to get locale information */
    char *s=getenv("LC_CTYPE");
    if (s==NULL) s=getenv("LC_ALL");

    /* set it */
    if (s!=NULL)
        if(setlocale(LC_CTYPE, s)==NULL)
            fprintf(stderr, "Setting locale failed\n");
}
```

36e · used in 25f · uses `fprintf` 21b, `getenv` 21b, `setlocale` 21b, `stderr` 21b

2.2.4 File Names

We expect at least one file name. While a missing file name might be ignored without causing any problems, we take the opportunity to report the usage convention.

⟨ *Process the remaining arguments (file names)* 37 ⟩ ≡

```
{
    if (arg >= argc) {
        fprintf(stderr, "%s: expected a file name.\n"
            "Usage: %s [options] file...\n"
            "Try '%s --help' for more information.\n",
            command_name, command_name, command_name);
        exit(-1);
    }
    do {
        ⟨ Handle the file name in argv[arg] 38a ⟩
        arg++;
    } while (arg < argc);
}
```

37 · used in 25f · uses `command_name` 29b, `exit` 21b, `fprintf` 21b, `stderr` 21b

The code to handle a particular file name is rather more tedious than the actual processing of the file. A file name may be an arbitrarily complicated path name, with an optional extension. If no extension is present, we add `.w` as a default. The extended path name will be kept in a local variable `source_name`. The resulting documentation file will be written in the current directory; its name will be kept in the variable `tex_name`.

```
< Handle the file name in argv[arg] 38a > ≡
{
    char source_name[FILENAME_MAX];
    char tex_name[FILENAME_MAX];
    char aux_name[FILENAME_MAX];
    char *trim;
    char *dot;
    < Build source_name and tex_name 38b >
    < Process a file 39b >
}
```

38a · used in 37 · uses `source_name` 125b

I bump the pointer `p` through all the characters in `argv[arg]`, copying all the characters into `source_name` (via the pointer `q`).

At each slash, I update `trim` to point just past the slash in `source_name`. The effect is that `trim` will point at the file name without any leading directory specifications.

The pointer `dot` is made to point at the file name extension, if present. If there is no extension, we add `.weft` to the source name. The `tex_name` (weave output name) is computed later in “Process a file” once the weave format is known.

```
< Build source_name and tex_name 38b > ≡
{
    char *p = argv[arg];
    char *q = source_name;
    char c;
    trim = q;
    dot = NULL;
    c = *p++;
    while (c) {
        *q++ = c;
        if (PATH_SEP(c)) {
            trim = q;
            dot = NULL;
        }
        else if (c == '.')
            dot = q - 1;
        c = *p++;
    }
    < Add the source path to the include path list 39a >
    *q = '\0';
    if (!dot) {
        *q++ = '.';
        *q++ = 'w';
        *q++ = 'e';
        *q++ = 'f';
        *q++ = 't';
        *q = '\0';
    }
    /* Find trim again for tex_name computation later */
    {
        char *t = source_name;
        trim = t;
```

```

    dot = NULL;
    while (*t) {
        if (PATH_SEP(*t)) { trim = t + 1; dot = NULL; }
        else if (*t == '.') dot = t;
        t++;
    }
}
}

```

38b · used in 38a · uses [source_name 125b](#)

If the source file has a directory part we add that directory to the end of the search path so that the path of last resort is the same as the source path, not, as one person put it, something irrelevant like the directory weft was invoked from.

⟨ *Add the source path to the include path list* 39a ⟩ ≡

```

if (trim != source_name) {
    struct incl * le
        = (struct incl *)arena_getmem(sizeof(struct incl));
    struct incl ** p = &include_list;
    char sv = *trim;

    *trim = '\0';
    le->name = save_string(source_name);
    le->next = NULL;
    while (*p != NULL)
        p = &((*p)->next);
    *p = le;
    *trim = sv;
}

```

39a · used in 38b · uses [arena_getmem 190b](#), [incl 27](#), [next 129b](#), [save_string 154a](#), [source_name 125b](#)

Now that we're finally ready to process a file, it's not really too complex. We bundle most of the work into three routines `pass1` (see Section 2.3), `write_tex` (see Section 2.4), and `write_files` (see Section 2.8). After we're finished with a particular file, we must remember to release its storage (see Section 3.6).

The default behavior is to tangle only. Weaving happens only when `-w` is given. The weave format is determined by (in priority order): a CLI argument (`-w md` or `-w tex`), or the `@W` directive in the source file. If `-w` is given without a format and no `@W` is present, weft reports an error and exits. The `-t` flag suppresses weaving even if `-w` was given.

⟨ *Process a file* 39b ⟩ ≡

```

{
    int do_weave = FALSE;
    int effective_format = 0;

    /* Reset weave_format for this file unless CLI locked it */
    if (!weave_format_from_cli)
        weave_format = 0;

    pass1(source_name);
    current_sector = 1;
    prev_sector = 1;

    if (lint_flag) {
        write_lint(file_names, macro_names);
        /* skip weave and tangle */
    } else if (map_flag) {
        write_map(file_names, macro_names);
    }
}

```

```

    /* skip weave and tangle */
} else if (extract_name) {
    if (callers_flag)
        write_callers(extract_name, file_names, macro_names);
    else
        write_extract(extract_name, file_names, macro_names);
    /* skip weave and tangle */
} else {
    /* Determine whether to weave and with which format */
    if (weave_flag && !tex_flag) {
        /* -t suppresses weave: tex_flag==FALSE means user passed -t */
    } else if (weave_flag) {
        effective_format = weave_format;
        if (effective_format == 0) {
            fprintf(stderr, "%s: -w requires a weave format.\n"
                "    Use -w md, -w tex, or add %cW md/%cW tex to the source file.\n",
                command_name, nw_char, nw_char);
            exit(-1);
        }
        do_weave = TRUE;
    }

    if (do_weave) {
        /* Compute weave output name now that format is known */
        const char *weave_ext = (effective_format == 2) ? ".md"
            : (effective_format == 3) ? ".html" : ".tex";

        if (dot) {
            *dot = '\\0';
            sprintf(tex_name, "%s%s%s%s", dirpath, path_sep, trim, weave_ext);
            sprintf(aux_name, "%s%s%s.aux", dirpath, path_sep, trim);
            *dot = '.';
        } else {
            sprintf(tex_name, "%s%s%s%s", dirpath, path_sep, trim, weave_ext);
            sprintf(aux_name, "%s%s%s.aux", dirpath, path_sep, trim);
        }

        if (effective_format == 2) {
            markdown_flag = TRUE;
            number_flag = TRUE;
            collect_numbers(aux_name);
            write_md(source_name, tex_name, 0);
        } else if (effective_format == 3) {
            number_flag = TRUE;
            collect_numbers(aux_name);
            write_html(source_name, tex_name, 0);
        } else {
            collect_numbers(aux_name);
            write_tex(source_name, tex_name, 0);
        }
    }

    if (output_flag)
        write_files(file_names);
}
arena_free();
}

```

39b · used in 38a · uses [arena_free](#) 191, [callers_flag](#) 27, [collect_numbers](#) 150b, [command_name](#) 29b, [current_sector](#) 45c, [exit](#) 21b, [extract_name](#) 27, [FALSE](#) 23a, [file_names](#) 152c, [fprintf](#) 21b, [lint_flag](#) 27, [macro_names](#) 152c, [map_flag](#) 27, [markdown_flag](#) 27, [number_flag](#) 27, [nw_char](#) 28b, 29a, [output_flag](#) 27, [pass1](#) 42c, [prev_sector](#) 45c, [source_name](#) 125b,

`stderr` 21b, `tex_flag` 27, `TRUE` 23a, `weave_flag` 27, `weave_format` 27, `weave_format_from_cli` 27, `write_callers` 112a, `write_extract` 112b, `write_files` 121c, `write_html` 101, `write_lint` 114b, `write_map` 109, `write_md` 82d, `write_tex` 55

2.3 Pass One

During the first pass, we scan the file, recording the definitions of each fragment and file and accumulating all the scraps.

< Function prototypes 42a > ≡

```
extern void pass1(char *file_name);
```

42a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses file_name 129c, pass1 42c

The routine `pass1` takes a single argument, the name of the source file. It opens the file, then initializes the scrap structures (see Section 3.2) and the roots of the file-name tree, the fragment-name tree, and the tree of user-specified index entries (see Section 3.3). After completing all the necessary preparation, we make a pass over the file, filling in all our data structures. Next, we search all the scraps for references to the user-specified index entries. Finally, we must reverse all the cross-reference lists accumulated while scanning the scraps.

< pass1.c 42b > + ≡

```
static char prose_buf[4096];
static int prose_len = 0;
static char *pending_prose = NULL;
```

42b · defined at 23d, 42bc · defines pending_prose 42c, 43b, 46ab, prose_buf 43ab, prose_len 42c, 43ab

< pass1.c 42c > + ≡

```
void pass1(char *file_name)
{
    if (verbose_flag)
        fprintf(stderr, "reading %s\n", file_name);
    source_open(file_name);
    init_scraps();
    macro_names = NULL;
    file_names = NULL;
    user_names = NULL;
    prose_len = 0;
    pending_prose = NULL;
    < Scan the source file, looking for at-sequences 42d >
    if (tex_flag)
        search();
    < Reverse cross-reference lists 51a >
}
```

42c · defined at 23d, 42bc · defines pass1 39b, 42a, 71e, 90c · uses file_name 129c, file_names 152c, fprintf 21b, init_scraps 131a, macro_names 152c, pending_prose 42b, prose_len 42b, search 180c, source_open 129a, stderr 21b, tex_flag 27, user_names 152c, verbose_flag 27

The only thing we look for in the first pass are the command sequences. All ordinary text is skipped entirely.

< Scan the source file, looking for at-sequences 42d > ≡

```
{
    int c = source_get();
    while (c != EOF) {
        if (c == nw_char)
            < Scan at-sequence 44d >
        else if (prose_flag) {
            < Accumulate prose character 43a >
        }
    }
}
```

```

    c = source_get();
}
}
42d · used in 42c · uses nw_char 28b, 29a, prose_flag 27, source_get 126a

```

```

< Accumulate prose character 43a > ≡
if (prose_len < (int)sizeof(prose_buf) - 1)
    prose_buf[prose_len++] = c;

```

43a · used in 42d · uses prose_buf 42b, prose_len 42b

```

< Save pending prose 43b > ≡
if (prose_flag && prose_len > 0) {
    /* trim trailing whitespace */
    while (prose_len > 0 && (prose_buf[prose_len-1] == ' '
        || prose_buf[prose_len-1] == '\n'
        || prose_buf[prose_len-1] == '\t'
        || prose_buf[prose_len-1] == '\r'))
        prose_len--;
    prose_buf[prose_len] = '\0';
    pending_prose = save_string(prose_buf);
    prose_len = 0;
} else {
    pending_prose = NULL;
    prose_len = 0;
}

```

43b · used in 46ab · uses pending_prose 42b, prose_buf 42b, prose_flag 27, prose_len 42b, save_string 154a

The @l command (lowercase) sets the language for the next @o or @d directive by storing the tag in pending_lang. It replaces the earlier mechanism of detecting Markdown fences (‘‘‘lang) and L^AT_EX markers (%%lang) during pass1. A typical usage is:

```

@l python
@d Validate input
@{import re
...
@}

```

```

< Collect language apply in pass1 43c > ≡
{
    char lang[64];
    char *lp = lang;
    int lc = source_get();
    /* skip optional whitespace */
    while (lc == ' ' || lc == '\t')
        lc = source_get();
    /* read language tag */
    while (lc != EOF && lc != '\n' && lc != ' ' && lc != '\t'
        && (lp - lang) < 62) {
        *lp++ = lc;
        lc = source_get();
    }
    *lp = '\0';
    if (lang[0] != '\0') {
        strncpy(pending_lang, lang, 63);
        pending_lang[63] = '\0';
    }
    /* consume rest of line */
}

```

```

while (lc != EOF && lc != '\n')
    lc = source_get();
}

```

43c · used in 44d · uses [pending_lang 152c](#), [source_get 126a](#)

The @L command (Section 3.3.2) lets users register custom language mappings. During pass1, the actual parsing and registration is handled by `collect_lang_def()` in `names.c`. During the weave passes (L^AT_EX and Markdown), we simply consume the rest of the line so that the arguments do not leak into the typeset output.

⟨ Collect user language definition in pass1 44a ⟩ ≡
`collect_lang_def();`

44a · used in 44d · uses [collect_lang_def 164](#)

⟨ Collect weave format declaration in pass1 44b ⟩ ≡
`collect_weave_format();`

44b · used in 44d · uses [collect_weave_format 166](#)

⟨ Skip at-L language definition line 44c ⟩ ≡

```

{ int lc = source_get();
  while (lc != '\n' && lc != EOF)
    lc = source_get();
}
c = source_get();

```

44c · used in 58c, 84a, 103a · uses [source_get 126a](#)

Six at-sequences are interesting during the first pass: @o and @O (file definitions), @d, @D, and @q (fragment definitions), @c (block comments), @s/@S (sectors), @l (language apply), and @L (language definitions). We skip past all others immediately, warning if unexpected sequences are discovered.

⟨ Scan at-sequence 44d ⟩ ≡

```

{
    char quoted = 0;

    c = source_get();
    switch (c) {
        case 'r':
            c = source_get();
            nw_char = c;
            update_delimit_scrap();
            break;
        case 'O':
        case 'o': ⟨ Build output file definition 46a ⟩
            break;
        case 'Q':
        case 'q': quoted = 1;
            FALLTHROUGH;
        case 'D':
        case 'd': ⟨ Build fragment definition 46b ⟩
            break;
        case 's':
            ⟨ Step to next sector 45a ⟩
            break;
        case 'S':
            ⟨ Close the current sector 45b ⟩
            break;
    }
}

```

```

case '<':
case '(':
case '[':
case '{': < Skip over an in-text scrap 46d >
    break;
case 'c': < Collect a block comment 47b >
    break;
case 'l': < Collect language apply in pass1 43c >
    break;
case 'L': < Collect user language definition in pass1 44a >
    break;
case 'W': < Collect weave format declaration in pass1 44b >
    break;
case 'x':
case 'v':
case 'u':
case 'm':
case 'f': /* ignore during this pass */
    break;
default: if (c==nw_char) /* ignore during this pass */
    break;
    fprintf(stderr,
        "%s: unexpected %c sequence ignored (%s, line %d)\n",
        command_name, nw_char, source_name, source_line);
    break;
}
}

```

44d · used in 42d · uses `command_name` 29b, `FALLTHROUGH` 23a, `fprintf` 21b, `nw_char` 28b, 29a, `source_get` 126a, `source_line` 125b, `source_name` 125b, `stderr` 21b, `update_delimit_scrap` 71a

< Step to next sector 45a > ≡

```

prev_sector += 1;
current_sector = prev_sector;
c = source_get();

```

45a · used in 44d, 58c, 84a, 103a · uses `current_sector` 45c, `prev_sector` 45c, `source_get` 126a

< Close the current sector 45b > ≡

```

current_sector = 1;
c = source_get();

```

45b · used in 44d, 58c, 84a, 103a · uses `current_sector` 45c, `source_get` 126a

< Global variable declarations 45c > + ≡

```

extern unsigned char current_sector;
extern unsigned char prev_sector;

```

45c · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `current_sector` 39b, 45abd, 77c, 80c, 95a, 97a, 105d, 106b, 134a, 136a, 170a, 173, `prev_sector` 39b, 45ad

< Global variable definitions 45d > + ≡

```

unsigned char current_sector = 1;
unsigned char prev_sector = 1;

```

45d · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses `current_sector` 45c, `prev_sector` 45c

2.3.1 Accumulating Definitions

There are three steps required to handle a definition:

1. Build an entry for the name so we can look it up later.
2. Collect the scrap and save it in the table of scraps.
3. Attach the scrap to the name.

We go through the same steps for both file names and fragment names.

⟨ Build output file definition 46a ⟩ ≡

```
{
    Name *name;
    int scrap;
    ⟨ Save pending prose 43b ⟩
    name = collect_file_name(); /* returns a pointer to the name entry */
    scrap = collect_scrap();    /* returns an index to the scrap */
    if (pending_prose) set_scrap_prose(scrap, pending_prose);
    pending_prose = NULL;
    ⟨ Add scrap to name's definition list 46c ⟩
}
```

46a · used in 44d · uses `collect_file_name` 167a, `collect_scrap` 133f, `Name` 152b, `pending_prose` 42b, `set_scrap_prose` 140a

⟨ Build fragment definition 46b ⟩ ≡

```
{
    Name *name;
    int scrap;
    ⟨ Save pending prose 43b ⟩
    name = collect_macro_name();
    scrap = collect_scrap();
    if (pending_prose) set_scrap_prose(scrap, pending_prose);
    pending_prose = NULL;
    ⟨ Add scrap to name's definition list 46c ⟩
}
```

46b · used in 44d · uses `collect_macro_name` 170a, `collect_scrap` 133f, `Name` 152b, `pending_prose` 42b, `set_scrap_prose` 140a

Since a file or fragment may be defined by many scraps, we maintain them in a simple linked list. The list is actually built in reverse order, with each new definition being added to the head of the list.

⟨ Add scrap to name's definition list 46c ⟩ ≡

```
{
    Scrap_Node *def = (Scrap_Node *) arena_getmem(sizeof(Scrap_Node));
    def->scrap = scrap;
    def->quoted = quoted;
    def->next = name->defs;
    name->defs = def;
    set_scrap_owner(scrap, name);
}
```

46c · used in 46ab · uses `arena_getmem` 190b, `defs` 129c, `next` 129b, `Scrap_Node` 152a, `set_scrap_owner` 140b

⟨ Skip over an in-text scrap 46d ⟩ ≡

```
{
    int c;
    int depth = 1;
    while ((c = source_get()) != EOF) {
        if (c == nw_char)
            ⟨ Skip over at-sign or go to skipped 47a ⟩
    }
}
```

```

    fprintf(stderr, "%s: unexpected EOF in text at (%s, %d)\n",
               command_name, source_name, source_line);
    exit(-1);

skipped: ;
}

46d · used in 44d · uses command_name 29b, exit 21b, fprintf 21b, nw_char 28b, 29a, source_get 126a, source_line 125b,
source_name 125b, stderr 21b

⟨ Skip over at-sign or go to skipped 47a ⟩ ≡

{
    c = source_get();
    switch (c) {
        case '{': case '[': case '(': case '<':
            depth += 1;
            break;
        case '}': case ']': case ')': case '>':
            if (--depth == 0)
                goto skipped;
        case 'x': case '|': case ',':
        case '%': case '1': case '2':
        case '3': case '4': case '5': case '6':
        case '7': case '8': case '9': case '_':
        case 'f': case '#': case '+': case '-':
        case 'v': case '*': case 'c': case '\\':
        case 's':
            break;
        default:
            if (c != nw_char) {
                fprintf(stderr, "%s: unexpected %c%c in text at (%s, %d)\n",
                           command_name, nw_char, c, source_name, source_line);
                exit(-1);
            }
            break;
    }
}

47a · used in 46d · uses command_name 29b, exit 21b, fprintf 21b, nw_char 28b, 29a, source_get 126a, source_line 125b,
source_name 125b, stderr 21b

```

2.3.2 Block Comments

```

⟨ Collect a block comment 47b ⟩ ≡

{
    char * p = blockBuff;
    char * e = blockBuff + (sizeof(blockBuff)/sizeof(blockBuff[0])) - 1;

    ⟨ Skip whitespace 47c ⟩
    while (p < e)
    {
        ⟨ Add one char to the block buffer 48a ⟩
    }
    if (p == e)
    {
        ⟨ Skip to the next nw-char 49b ⟩
    }
    *p = '\000';
}

```

```
}
```

47b · used in 44d · uses `blockBuff` 49c

⟨ *Skip whitespace* 47c ⟩ ≡

```
while (source_peek == ' '
      || source_peek == '\t'
      || source_peek == '\n')
    (void)source_get();
```

47c · used in 47b · uses `source_get` 126a, `source_peek` 126a

⟨ *Add one char to the block buffer* 48a ⟩ ≡

```
int c = source_get();

if (c == nw_char)
{
    ⟨ Add an at character to the block or break 48b ⟩
}
else if (c == EOF)
{
    source_ungetc(&c);
    break;
}
else
{
    ⟨ Add any other character to the block 48c ⟩
}
```

48a · used in 47b · uses `nw_char` 28b, 29a, `source_get` 126a, `source_ungetc` 126c

⟨ *Add an at character to the block or break* 48b ⟩ ≡

```
int cc = source_peek;

if (cc == 'c')
{
    do
        c = source_get();
    while (c <= ' ');

    break;
}
else if (cc == 'd'
        || cc == 'D'
        || cc == 'q'
        || cc == 'Q'
        || cc == 'o'
        || cc == 'O'
        || cc == EOF)
{
    source_ungetc(&c);
    break;
}
else
{
    *p++ = c;
    *p++ = source_get();
}
```

48b · used in 48a · uses `source_get` 126a, `source_peek` 126a, `source_ungetc` 126c

⟨ *Add any other character to the block* 48c ⟩ ≡

```
⟨ Perhaps skip white-space 49a ⟩
*p++ = c;
```

48c · used in 48a

⟨ *Perhaps skip white-space* 49a ⟩ ≡

```
if (c == ' ')
{
    while (source_peek == ' ')
        c = source_get();
}
if (c == '\n')
{
    if (source_peek == '\n')
    {
        do
            c = source_get();
        while (source_peek == '\n');
    }
    else
        c = ' ';
}
```

49a · used in 48c · uses `source_get` 126a, `source_peek` 126a

⟨ *Skip to the next nw-char* 49b ⟩ ≡

```
int c;
```

```
while ((c = source_get()), c != nw_char && c != EOF) /* Skip */
    source_ungetc(&c);
```

49b · used in 47b · uses `nw_char` 28b, 29a, `source_get` 126a, `source_ungetc` 126c

⟨ *Global variable declarations* 49c ⟩ + ≡

```
extern char blockBuff[6400];
```

49c · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `blockBuff` 47b, 49df

⟨ *Global variable definitions* 49d ⟩ + ≡

```
char blockBuff[6400];
```

49d · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses `blockBuff` 49c

We don't show block comments inside scraps in the T_EX file, but we do show where they go.

⟨ *Show presence of a block comment* 49e ⟩ ≡

```
{
    begin_tex_escape(file);
    fprintf(file, "\\hbox{\\sffamily\\slshape (Comment)}");
    end_tex_escape(file);
}
```

49e · used in 71e · uses `begin_tex_escape` 69d, `end_tex_escape` 69d, `fprintf` 21b

The block comment is presently in the block buffer. Here we copy it into the scrap whence we shall copy it into the output file.

⟨ *Include block comment in a scrap* 49f ⟩ ≡

```

{
    char * p = blockBuff;

    push(nw_char, &writer);
    do
    {
        push(c, &writer);
        c = *p++;
    } while (c != '\0');
}

```

49f · used in 134c · uses [blockBuff 49c](#), [nw_char 28b, 29a](#), [push 133d](#)

⟨ *Copy block comment from scrap* 50a ⟩ ≡

```

{
    int bgn = indent + global_indent;
    int posn = bgn + strlen(comment_begin[comment_flag]);
    int i;

    ⟨ Perhaps put a delayed indent 148b ⟩
    c = pop(&reader);
    fputs(comment_begin[comment_flag], file);
    while (c != '\0')
    {
        ⟨ Move a word to the file 50b ⟩
        ⟨ If we break the line at this word 50c ⟩
        {
            putc('\n', file);
            for (i = 0; i < bgn ; i++)
                putc(' ', file);
            c = pop(&reader);
            if (c != '\0')
            {
                posn = bgn + strlen(comment_mid[comment_flag]);
                fputs(comment_mid[comment_flag], file);
            }
        }
    }
    fputs(comment_end[comment_flag], file);
}

```

50a · used in 146a · uses [comment_begin 169b](#), [comment_end 169b](#), [comment_mid 169b](#), [fputs 21b](#), [pop 138b](#), [putc 21b](#), [strlen 21b](#)

⟨ *Move a word to the file* 50b ⟩ ≡

```

do
{
    putc(c, file);
    posn += 1;
    c = pop(&reader);
} while (c > ' ');

```

50b · used in 50a · uses [pop 138b](#), [putc 21b](#)

⟨ *If we break the line at this word* 50c ⟩ ≡

```

if (c == '\n' || (c == ' ' && posn > 60))

```

50c · used in 50a

⟨ *Skip over a block comment* 50d ⟩ ≡

```

if (last == nw_char && c == 'c')
    while ((c = pop(reader.m)) != '\0')
        /* Skip */;

```

50d · used in 182 · uses `nw_char` 28b, 29a, `pop` 138b

⟨ *Begin or end a block comment* 50e ⟩ ≡

```

if (inBlock)
{
    ⟨ End block 63d ⟩
}
else
{
    ⟨ Start block 63b ⟩
}

```

50e · used in 58c

2.3.3 Fixing the Cross References

Since the definition and reference lists for each name are accumulated in reverse order, we take the time at the end of `pass1` to reverse them all so they'll be simpler to print out prettily. The code for `reverse_lists` appears in Section 3.3.

⟨ *Reverse cross-reference lists* 51a ⟩ ≡

```

{
    reverse_lists(file_names);
    reverse_lists(macro_names);
    reverse_lists(user_names);
}

```

51a · used in 42c · uses `file_names` 152c, `macro_names` 152c, `reverse_lists` 176b, `user_names` 152c

2.3.4 Dealing with fragment parameters

Fragment parameters were added on later in nuweb's development. There still is not, for example, an index of fragment parameters. We need a data type to keep track of fragment parameters.

⟨ *Type declarations* 51b ⟩ + ≡

```

typedef int *Parameters;

```

51b · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `Parameters` 53a, 85b, 130b, 141d, 142b, 143

When we are copying a scrap to the output, we check for an embedded parameter. If we find it we copy it. Otherwise it's an old-style parameter and we can then pull the *n*th string from the `Parameters` list when we see an @1 @2, etc.

⟨ *Handle macro parameter substitution* 51c ⟩ ≡

```

case '1': case '2': case '3':
case '4': case '5': case '6':
case '7': case '8': case '9':
{
    Arglist * args;
    Name * name;

    lookup(c - '1', inArgs, inParams, &name, &args);
}

```

```

if (name == (Name *)1) {
    Embed_Node * q = (Embed_Node *)args;
    indent = write_scraps(file, spelling, q->defs,
                          global_indent + indent,
                          indent_chars, debug_flag,
                          tab_flag, indent_flag,
                          0, 0,
                          q->args, inParams,
                          local_parameters, "");
}
else if (name != NULL) {
    int i, nargs;
    char * p = name->spelling;
    Arglist *q = args;

    < Perhaps comment this macro 148a >
    indent = write_scraps(file, spelling, name->defs,
                          global_indent + indent,
                          indent_chars, debug_flag,
                          tab_flag, indent_flag,
                          comment_flag, location_flag,
                          args, name->arg,
                          local_parameters, p);
}
else if (args != NULL) {
    if (delayed_indent) {
        < Put out the indent 145c >
    }
    fputs((char *)args, file);
}
else if ( parameters && parameters[c - '1'] ) {
    Scrap_Node param_defs;
    param_defs.scrap = parameters[c - '1'];
    param_defs.next = 0;
    write_scraps(file, spelling, &param_defs,
                  global_indent + indent,
                  indent_chars, debug_flag,
                  tab_flag, indent_flag,
                  comment_flag, location_flag,
                  NULL, NULL, 0, "");
} else if (delayed_indent) {
    < Put out the indent 145c >
}
}

```

51c · used in 146a · uses Arglist 172d, defs 129c, delayed_indent 142c, Embed_Node 175d, fputs 21b, lookup 140d, Name 152b, next 129b, Scrap_Node 152a, write_scraps 142b

Now onto actually parsing fragment parameters from a call. We start off checking for fragment parameters, an @ (sequence followed by parameters separated by @, sequences, and terminated by a @) sequence.

We collect separate scraps for each parameter, and write the scrap numbers down in the text. For example, if the file has:

```
@<foo @( param1 @, param2 @)@>
```

we actually make new scraps, say 10 and 11, for param1 and param2, and write in the collected scrap:

```
@<foo @(10@,11@)@>
```

< Save macro parameters 52 > ≡

```

{
    int param_scrap;
    char param_buf[10];

    push(nw_char, &writer);
    push('(', &writer);
    do {
        param_scrap = collect_scrap();
        sprintf(param_buf, "%d", param_scrap);
        pushs(param_buf, &writer);
        push(nw_char, &writer);
        push(scrap_ended_with, &writer);
        add_to_use(name, current_scrap);
    } while( scrap_ended_with == ',' );
    do
        c = source_get();
    while( ' ' == c );
    if (c == nw_char) {
        c = source_get();
    }
    if (c != '>') {
        /* ZZZ print error */;
    }
}

```

52 · used in 137b · uses [add_to_use 137d](#), [collect_scrap 133f](#), [nw_char 28b, 29a](#), [push 133d](#), [pushs 133e](#), [source_get 126a](#)

If we get inside, we have at least one parameter, which will be at the beginning of the parms buffer, and we prime the pump with the first character.

< Check for macro parameters 53a > ≡

```

if (c == '(') {
    Parameters res = arena_getmem(10 * sizeof(int));
    int *p2 = res;
    int count = 0;
    int scrapnum;

    while( c && c != ')' ) {
        scrapnum = 0;
        c = pop(manager);
        while( '0' <= c && c <= '9' ) {
            scrapnum = scrapnum * 10 + c - '0';
            c = pop(manager);
        }
        if ( c == nw_char ) {
            c = pop(manager);
        }
        *p2++ = scrapnum;
    }
    while (count < 10) {
        *p2++ = 0;
        count++;
    }
    while( c && c != nw_char ) {
        c = pop(manager);
    }
    if ( c == nw_char ) {
        c = pop(manager);
    }
}

```

```

    }
    *parameters = res;
}

```

53a · used in 142a · uses [arena_getmem](#) 190b, [nw_char](#) 28b, 29a, [Parameters](#) 51b, [pop](#) 138b

These are used in `write_tex` to output the argument list for a fragment.

⟨ *Format macro parameters* 53b ⟩ ≡

```

char sep;

sep = '(';
do {
    fputc(sep,file);

    fputs("{\\footnotesize ", file);
    write_single_scrap_ref(file, scraps + 1);
    fprintf(file, "\\label{scrap%d}", scraps + 1);
    fputs(" }", file);

    source_last = '{';
    {
        /* The parameter is rendered inside a fragment reference, i.e.
           inside a listings escape, so it must be single-line LaTeX with
           the special characters neutralised: switch copy_scrap into its
           inline mode for the duration. */
        int saved_inline = code_inline;
        code_inline = TRUE;
        copy_scrap(file, TRUE, NULL);
        code_inline = saved_inline;
    }

    ++scraps;

    sep = ',';
} while ( source_last != ')' && source_last != EOF );
fputs(" ) ",file);
do
    c = source_get();
while(c != nw_char && c != EOF);
if (c == nw_char) {
    c = source_get();
}

```

53b · used in 73g, 75b · uses [code_inline](#) 69d, [copy_scrap](#) 70, [fprintf](#) 21b, [fputs](#) 21b, [nw_char](#) 28b, 29a, [scraps](#) 130a, [source_get](#) 126a, [source_last](#) 126a, [TRUE](#) 23a, [write_single_scrap_ref](#) 131c

2.4 Writing the Latex File

The second pass (invoked via a call to `write_tex`) copies most of the text from the source file straight into a `.tex` file. Definitions are formatted slightly and cross-reference information is printed out.

Note that all the formatting is handled in this section. If you don't like the format of definitions or indices or whatever, it'll be in this section somewhere. Similarly, if someone wanted to modify `weft` to work with a different typesetting system, this would be the place to look.

⟨ *Function prototypes* 54a ⟩ + ≡

```
extern void write_tex(char *file_name, char *tex_name, unsigned char sector);
```

54a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `file_name` 129c, `sector` 129c, `write_tex` 55

We need a few local function declarations before we get into the body of `write_tex`.

```
<latex.c 54b> + ≡
static void copy_scrap(FILE *file, int prefix, Name *name);           /* formats the
    body of a scrap */
static void latex_code_putc(FILE *file, int c);                     /* escapes one
    code character for LaTeX */
static const char *latex_language(const char *lang);                /* maps a weft
    language tag to a listings name */
static void print_scrap_numbers(FILE *tex_file, Scrap_Node *scraps); /* formats a
    list of scrap numbers */
static void format_entry(Name *name, FILE *tex_file, unsigned char sector);
    /* formats an index entry */
static void format_file_entry(Name *name, FILE *tex_file);          /* formats a file index
    entry */
static void format_user_entry(Name *name, FILE *tex_file, unsigned char sector);
static void write_arg(FILE *tex_file, char *p);
static void write_literal(FILE *tex_file, char *p, int mode);
static void write_ArglistElement(FILE *file, Arglist *args, char **params);

54b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · uses Arglist 172d, copy_scrap 70,
FILE 21b, format_entry 78b, format_file_entry 76b, format_user_entry 80d, latex_code_putc 68a, latex_language 68b,
Name 152b, print_scrap_numbers 66c, scraps 130a, Scrap_Node 152a, sector 129c, write_arg 62b,
write_ArglistElement 75b, write_literal 69c
```

The routine `write_tex` takes two file names as parameters: the name of the web source file and the name of the `.tex` output file.

```
<latex.c 55> + ≡
void write_tex(char *file_name, char *tex_name, unsigned char sector)
{
    (void)sector;
    FILE *tex_file = fopen(tex_name, "w");
    if (tex_file) {
        if (verbose_flag)
            fprintf(stderr, "writing %s\n", tex_name);
        source_open(file_name);
        < Write LaTeX limbo definitions 56a >
        < Copy source_file into tex_file 58b >
        fclose(tex_file);
    }
    else
        fprintf(stderr, "%s: can't open %s\n", command_name, tex_name);
}
```

55 · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `write_tex` 39b, 54a · uses `command_name` 29b, `fclose` 21b, `FILE` 21b, `file_name` 129c, `fopen` 21b, `fprintf` 21b, `sector` 129c, `source_open` 129a, `stderr` 21b, `verbose_flag` 27

Weft's woven output relies on a family of `\WEFT...` macros. We write default definitions for all of them into the document's limbo so that source files need not define anything new. If a user wants to change any of the macros (to use `hyperref`, to write in some language other than english, or to tune the code layout) he or she can redefine the commands *after* the point where weft's definitions are read, or (better) redefine them in the preamble with `\renewcommand`.

Three groups of macros are written. The first group are the textual and cross-reference macros (`\WEFTtarget`, `\WEFTlink`, the `\WEFTtxt...` labels, and so on). The second group, the *modern decoration macros* of §2.4, give a scrap its visual identity (fragment name, scrap number, the $\equiv$ / $+$ $\equiv$ marker, the one-line footer) and route all colour through the degradable `\WEFTtint` indirection. The third group, written by the scrap-formatting fragment below, are the layout macros that frame a scrap.

Code bodies are typeset with the `listings` package: every scrap becomes an `lstlisting` environment. This gives syntax highlighting (driven by the scrap’s language, see `latex_language`), automatic line wrapping (`breaklines`), and page breaking for free. The document *must* therefore load `\usepackage{listings}` in its preamble — `weft` cannot inject it, because the `limbo` is written before `\documentclass` (a colour package, `color` or `xcolor`, is likewise the user’s to load, and enables the accent and syntax colours; without it the output still compiles in black and white). Fragment references that appear in the middle of code are typeset by briefly escaping out of the listing back to L^AT_EX; see §2.4 and the `escapeinside` option.

```
< Write LaTeX limbo definitions 56a > ≡
if (hyperref_flag) {
  fputs("\newcommand{\WEFTtarget}[2]{\hypertarget{#1}{#2}}\n", tex_file);
  fputs("\newcommand{\WEFTlink}[2]{\hyperlink{#1}{#2}}\n", tex_file);
} else {
  fputs("\newcommand{\WEFTtarget}[2]{#2}\n", tex_file);
  fputs("\newcommand{\WEFTlink}[2]{#2}\n", tex_file);
}
fputs("\newcommand{\WEFTtxtMacroDefBy}{defined at}\n", tex_file);
fputs("\newcommand{\WEFTtxtMacroRefIn}{used in}\n", tex_file);
fputs("\newcommand{\WEFTtxtMacroNoRef}{unreferenced}\n", tex_file);
fputs("\newcommand{\WEFTtxtDefBy}{defined at}\n", tex_file);
fputs("\newcommand{\WEFTtxtRefIn}{used in}\n", tex_file);
fputs("\newcommand{\WEFTtxtNoRef}{unreferenced}\n", tex_file);
fputs("\newcommand{\WEFTtxtFileDefBy}{defined at}\n", tex_file);
fputs("\newcommand{\WEFTtxtIdsentsUsed}{uses}\n", tex_file);
fputs("\newcommand{\WEFTtxtIdsentsNotUsed}{never used}\n", tex_file);
fputs("\newcommand{\WEFTtxtIdsentsDefed}{defines}\n", tex_file);
fputs("\newcommand{\WEFTsep}{}\n", tex_file);
fputs("\newcommand{\WEFTnotglobal}{(not defined globally)}\n", tex_file);
< Write the modern decoration macros 57b >
< Write the scrap formatting macros 56b >
fputs("\newcommand{\WEFTuseHyperlinks}{", tex_file);
if (hyperoptions[0] != '\0')
{
  < Write the hyperlink usage macro 57a >
}
fputs("}\n", tex_file);
< Write the AtBeginDocument setup 58a >
```

56a · used in 55 · uses `fputs` 21b, `hyperoptions` 27, `hyperref_flag` 27, `uses` 129c

The layout macros. `\WEFTbegin`/`\WEFTend` open and close the vertical block that holds a scrap’s header line, its `lstlisting` body, and its one-line metadata footer. `\WEFTbegin` sets the block ragged-right at the document’s ordinary size (the old `\small` is gone: the header now reads at full size, and the code carries its own, tighter, listings size). A fragment reference appearing in the middle of code is assembled inline and wrapped in an `\hbox` so that `listings`’ line-breaker never splits it while it sits inside an escape. (We use `\hbox` rather than `\mbox` because a fragment name may legitimately contain `\verb`, which is illegal in a macro argument such as `\mbox`’s but fine inside a primitive `\hbox`.) `\WEFTlstsetup` now carries `weft`’s *modern* `\lstset` defaults: a compact monospace with tightened leading, coloured keywords/strings/comments (through `\WEFTtint`, so colour is optional), line breaking, and *no* frame, background, or left indent. The user need not call it — the `\AtBeginDocument` hook of §2.4 applies it automatically once `listings` is loaded — but an explicit call, or a `\renewcommand`, still lets the user override the house style.

```

< Write the scrap formatting macros 56b > ≡
fputs("\\newcommand{\\WEFTbegin}{\\par\\addvspace{2.3ex plus .6ex}"
      "\\begingroup\\raggedright}\\n", tex_file);
fputs("\\newcommand{\\WEFTend}{\\par\\endgroup"
      "\\addvspace{2.3ex plus .6ex}}\\n", tex_file);
fputs("\\providecommand{\\WEFTlstsetup}{\\lstset{basicstyle=\\ttfamily"
      "\\fontsize{10pt}{10.8pt}\\selectfont,"
      "keywordstyle={\\WEFTtint{weftkw}\\bfseries},"
      "commentstyle={\\WEFTtint{weftcom}\\itshape},"
      "stringstyle={\\WEFTtint{weftstr}},"
      "breaklines=true,breakatwhitespace=false,"
      "columns=fullflexible,keepspaces=true,showstringspaces=false,"
      "tabsize=8,frame=none,xleftmargin=0pt,"
      "aboveskip=0pt,belowskip=0pt}}\\n", tex_file);

```

56b · used in 56a · uses fputs 21b

```

< Write the hyperlink usage macro 57a > ≡
fprintf(tex_file, "\\usepackage[%s]{hyperref}", hyperoptions);

```

57a · used in 56a · uses fprintf 21b, hyperoptions 27

The modern decoration macros

The visual identity of a woven scrap — the way a fragment name reads, the little scrap number beside it, the definition/continuation marker, the one-line metadata footer — is factored into a handful of small macros written here. Emitting them once, as named commands, keeps the per-scrap output that follows compact and lets a reader (or the user) re-style the whole document by redefining a single command.

The design is *sober and self-contained*: a fragment name is set in the document's ordinary italic tinted with one accent colour, its scrap number in a small grey, the enclosing `< >` in the same grey, and the modern layout carries *no* rules, boxes, or shaded backgrounds — highlighting alone distinguishes the code.

Colour is applied through one level of indirection, `\WEFTtint`. By default it is a no-op: `\WEFTtint{weftaccent}` expands to nothing, so a document that loads *no* colour package still compiles, merely in black and white. When a colour package *is* present, the `\AtBeginDocument` hook below swaps in `\WEFTtintOn`, and the same call becomes `\color{weftaccent}`. This is why every colour site is written `{\WEFTtint{name}...}` rather than `\color` directly: it degrades gracefully.

`\WEFTlangle`/`\WEFTrangle` are the grey angle brackets; `\WEFTnumsep` is the space set between a fragment name and its scrap number (wide enough that `name 41a` never reads as one token); `\WEFTeq` and `\WEFTpluseq` are the definition ($\equiv$) and additive-continuation ($+ \equiv$) markers; `\WEFTdot` is the mid-dot that separates footer items; and `\WEFTmetabegin`/`\WEFTmetaend` open and close the small grey metadata line, which opens with the scrap's own number.

```

< Write the modern decoration macros 57b > ≡
fputs("\\newcommand{\\WEFTtint}[1]{\\n", tex_file);
fputs("\\newcommand{\\WEFTtintOn}[1]{\\color{#1}}\\n", tex_file);
fputs("\\newcommand{\\WEFTlangle}{\\WEFTtint{weftcom}\\langle$\\,}\\n",
      tex_file);
fputs("\\newcommand{\\WEFTrangle}{\\,\\WEFTtint{weftcom}\\rangle$}\\n",
      tex_file);
fputs("\\newcommand{\\WEFTeq}{\\WEFTtint{weftcom}\\equiv$}\\n",
      tex_file);
fputs("\\newcommand{\\WEFTpluseq}{\\WEFTtint{weftcom}"
      "\\mathord{+}\\equiv$}\\n", tex_file);
fputs("\\newcommand{\\WEFTdot}{\\WEFTtint{weftcom}"
      "\\textperiodcentered}\\n", tex_file);
fputs("\\newcommand{\\WEFTnumsep}{\\n", tex_file);
fputs("\\newcommand{\\WEFTmetabegin}[1]{\\par\\vspace{3pt}\\begingroup"
      "\\footnotesize\\WEFTtint{weftcom}#1\\n", tex_file);

```

```
fputs("\\newcommand{\\WEFTmetaend}{\\endgroup\\par}\\n", tex_file);
```

57b · used in 56a · uses [fputs 21b](#)

The colours themselves, and the automatic application of `\WEFTlstsetup`, are deferred to `\AtBeginDocument`. This is essential: the limbo we are writing precedes `\documentclass`, where `\definecolor` and `\lstset` cannot run (no colour or listings package is loaded yet, and loading one there fails outright). The kernel hook `\AtBeginDocument` runs its body at `\begin{document}`, by which time the user's preamble has loaded its packages. We guard each half with ε -TeX's `\ifdefined` so the output still compiles when a colour package or `listings` is absent. The palette uses the `rgb` model, understood by both `color` and `xcolor`, so no specific colour package is mandated.

Write the AtBeginDocument setup 58a ≡

```
fputs("\\AtBeginDocument{%\\n", tex_file);
fputs("\\ifdefined\\definecolor\\n", tex_file);
fputs("\\definecolor{weftaccent}{rgb}{0.184,0.435,0.922}%\\n", tex_file);
fputs("\\definecolor{weftkw}{rgb}{0.020,0.314,0.682}%\\n", tex_file);
fputs("\\definecolor{weftstr}{rgb}{0.067,0.388,0.161}%\\n", tex_file);
fputs("\\definecolor{weftcom}{rgb}{0.431,0.467,0.506}%\\n", tex_file);
fputs("\\let\\WEFTtint\\WEFTtintOn\\n", tex_file);
fputs("\\fi\\n", tex_file);
fputs("\\ifdefined\\lstset\\WEFTlstsetup\\fi\\n", tex_file);
fputs("}\\n", tex_file);
```

58a · used in 56a · uses [fputs 21b](#)

We make our second (and final) pass through the source web, this time copying characters straight into the `.tex` file. However, we keep an eye peeled for `@` characters, which signal a command sequence.

Copy source_file into tex_file 58b ≡

```
{
  int inBlock = FALSE;
  int c = source_get();
  while (c != EOF) {
    if (c == nw_char)
    {
      Interpret at-sequence 58c
    }
    else {
      putc(c, tex_file);
      c = source_get();
    }
  }
}
```

58b · used in 55 · uses [FALSE 23a](#), [nw_char 28b, 29a](#), [putc 21b](#), [source_get 126a](#)

Interpret at-sequence 58c ≡

```
{
  int big_definition = FALSE;
  c = source_get();
  switch (c) {
    case 'r':
      c = source_get();
      nw_char = c;
      update_delimit_scrap();
      break;
    case '0': big_definition = TRUE;
              FALLTHROUGH;
```

```

case 'o': < Write output file definition 61a >
           break;
case 'Q':
case 'D': big_definition = TRUE;
           FALLTHROUGH;
case 'q':
case 'd': < Write macro definition 61b >
           break;
case 's':
           < Step to next sector 45a >
           break;
case 'S':
           < Close the current sector 45b >
           break;
case '{':
case '[':
case '(': < Write in-text scrap 65d >
           break;
case '<': < Expand macro into tex_file 59c >
           break;
case 'x': < Copy label from source into(59a tex_file ) 73a >
           c = source_get();
           break;
case 'c': < Begin or end a block comment 50e >
           c = source_get();
           break;
case 'f': < Write index of file names 76a >
           break;
case 'm': < Write index of macro names 77c >
           break;
case 'u': < Write index of user-specified names 80c >
           break;
case 'v': < Copy version info into tex file 59b >
           break;
case 'l':
case 'L': < Skip at-L language definition line 44c >
           break;
case 'W': < Skip at-L language definition line 44c >
           break;
default:
           if (c==nw_char)
               putc(c, tex_file);
           c = source_get();
           break;
    }
}

```

59a · used in 58b

```

< Copy version info into tex file 59b > ≡
fputs(version_string, tex_file);
c = source_get();

```

59b · used in 58c · uses fputs 21b, source_get 126a, version_string 27

A fragment reference `@<name@>` written in the *documentation* text (outside any scrap) is a *cross-reference*, not an inclusion. `weft` renders it exactly like a reference in the middle of code — the grey `< >` brackets, the accent-tinted name, and a link to the fragment’s first defining scrap — only without the `listings` escape, since prose is already ordinary L^AT_EX text. This is a deliberate change from the old `nuweb` behaviour, which *expanded* the fragment’s whole body inline into the prose: a chunk name mentioned in a sentence would

dump code into the middle of it. Referencing, not expanding, is what a literate narrative wants (“as we saw in @<Say hello@>”). To show a fragment’s *code* in the prose, define and reference it as a scrap in the usual way.

```
< Expand macro into file 59c > ≡
{
  Arglist *a = collect_scrap_name(0);
  Name *name = a->name;
  char *p = name->spelling;
  int i = 0;
  fputs("\\WEFTlangle{\\WEFTtint{\\weftaccent}\\itshape ", file);
  while (*p != '\\000') {
    if (*p == ARG_CHR) {
      write_arg(file, name->arg[i++]);
      p++;
    }
    else
      fputc(*p++, file);
  }
  fputs("}\\nobreak\\WEFTnumsep{\\footnotesize\\WEFTtint{\\weftcom}", file);
  if (name->defs) {
    Scrap_Node *q = name->defs;
    fputs("\\WEFTlink{\\weft", file);
    write_single_scrap_ref(file, q->scrap);
    fputs("}{", file);
    write_single_scrap_ref(file, q->scrap);
    fputs("}", file);
    if (q->next)
      fputs(", \\ldots\\ ", file);
  }
  else {
    putc('?', file);
    fprintf(stderr, "%s: never defined <%s>\\n",
            command_name, name->spelling);
  }
  fputs("}\\WEFTrangle", file);
  c = source_get();
}
```

59c · used in 58c · uses Arglist 172d, ARG_CHR 171b, collect_scrap_name 173, command_name 29b, defs 129c, fprintf 21b, fputs 21b, Name 152b, next 129b, putc 21b, Scrap_Node 152a, source_get 126a, stderr 21b, write_arg 62b, write_single_scrap_ref 131c

2.4.1 Formatting Definitions

We go through a fair amount of effort to format a fragment definition. The L^AT_EX for the previous fragment definition should look like this (perhaps modulo the scrap references). Note the shape of a modern header: the grey \\WEFTlangle/\\WEFTrangle brackets, the accent-tinted italic name, the small grey scrap number, and the \\WEFTeq (≡) kind marker; then a hair of space before the listing; and, after it, the one-line \\WEFTmetabegin...\\WEFTmetaend footer instead of the old bulleted list.

```
\\WEFTbegin
\\label{scrap70}
\\WEFTtarget{\\weft70}{\\WEFTlangle{\\WEFTtint{\\weftaccent}\\itshape %
  Interpret at-sequence}\\nobreak\\,{\\footnotesize\\WEFTtint{\\weftcom}}70}%
  \\WEFTrangle\\WEFTeq\\par\\nobreak\\vspace{0.6ex}\\nobreak
\\begin{lstlisting}[escapeinside={(*<}{>*)},language=C]
{
  int big_definition = FALSE;
```

```

c = source_get();
switch (c) {
  case 'o': (*<\hbox{\normalfont \WEFTlangle{\WEFTtint{weftaccent}}%
\itshape Write output file definition}\nobreak\,{\footnotesize%
\WEFTtint{weftcom}\WEFTlink{weft19a}{19a}}\WEFTrangle>*)
}
\end{lstlisting}
\WEFTmetabegin{70}\WEFTdot \WEFTtxtMacroRefIn\ \WEFTlink{weft17b}{17b}\WEFTmetaend
\WEFTend

```

The code body is now an `lstlisting`. Because `listings` reads the body verbatim, the special $\text{\LaTeX}$ characters need *no* escaping — they are printed as typed. The `escapeinside={(*<}{>*)}` option carves out the one place we *do* need $\text{\LaTeX}$: a fragment reference in the middle of code, which is emitted between `(*<` and `>*)` and wrapped in an `\hbox` so `breaklines` cannot split it. The delimiter pair `(*<...>*)` is deliberately chosen to be a sequence that never occurs in real source (weft warns if a scrap body contains it). `listings` supplies page breaking, line wrapping, and — via the `language=` option, taken from the scrap's language tag — syntax highlighting.

Fragment and file definitions are formatted nearly identically. I've factored the common parts out into separate scraps.

Both headers now carry a *kind marker* after the closing angle bracket: `\WEFTeq` ($\equiv$) for a definition, or `\WEFTpluseq` ($+\equiv$) for an *additive continuation* — a scrap that appends to a fragment (or file) already begun elsewhere. We know which is which because `name->defs` is, after pass one's `reverse_lists`, in source order: its head is the *first* scrap that defines the name. So the current scrap is a continuation exactly when `name->defs->scrap` differs from the scrap we are emitting. The file header sets the file name in monospace (it *is* a filename); the fragment header sets its name in the document's italic. Both are tinted with the accent colour and followed, on the next line after a hair of vertical space, by the code.

```

< Write output file definition 61a > ≡
{
  Name *name = collect_file_name();
  int additive = (name->defs && name->defs->scrap != scraps);
  < Begin the scrap environment 63a >
  fputs("\WEFTtarget{weft", tex_file);
  write_single_scrap_ref(tex_file, scraps);
  fputs("{}]", tex_file);
  fprintf(tex_file, "\\WEFTlangle{\\WEFTtint{weftaccent}}\\verb%c%s%c\\nobreak"
    "\\WEFTnumsep{\\footnotesize\\WEFTtint{weftcom}}", nw_char, name->spelling,
    nw_char);
  write_single_scrap_ref(tex_file, scraps);
  fputs("{}\\WEFTrangle", tex_file);
  fputs(additive ? "\\WEFTpluseq" : "\\WEFTeq", tex_file);
  fputs("\\par\\nobreak\\vspace{0.6ex}\\nobreak\\n", tex_file);
  < Fill in the middle of the scrap environment 63e >
  < Begin the cross-reference environment 65e >
  if ( scrap_flag ) {
    < Write file defs 65g >
  }
  format_defs_refs(tex_file, scraps);
  format_uses_refs(tex_file, scraps++);
  < Finish the cross-reference environment 65f >
  < Finish the scrap environment 65a >
}

```

61a · used in 58c · uses `collect_file_name` 167a, `defs` 129c, `format_defs_refs` 186a, `format_uses_refs` 184f, `fprintf` 21b, `fputs` 21b, `Name` 152b, `nw_char` 28b, 29a, `scraps` 130a, `scrap_flag` 27, `write_single_scrap_ref` 131c

The fragment name's *content* is still emitted verbatim by the *Write the macro's name* fragment, so a programmer may put italics or bold face in the midst of the name; the surrounding `\itshape` and accent

tint are only the default dress.

```

< Write macro definition 61b > ≡
{
  Name *name = collect_macro_name();
  int additive = (name->defs && name->defs->scrap != scraps);

  < Begin the scrap environment 63a >
  fputs("\\WEFTtarget{weft", tex_file);
  write_single_scrap_ref(tex_file, scraps);
  fputs("{}\\WEFTlangle{\\WEFTtint{weftaccent}\\itshape ", tex_file);
  < Write the macro's name 62a >
  fputs("\\\\nobreak\\WEFTnumsep{\\footnotesize\\WEFTtint{weftcom}", tex_file);
  write_single_scrap_ref(tex_file, scraps);
  fputs("{}\\WEFTrangle", tex_file);
  fputs(additive ? "\\WEFTpluseq" : "\\WEFTeq", tex_file);
  fputs("\\\\par\\nobreak\\vspace{0.6ex}\\nobreak\\n", tex_file);
  < Fill in the middle of the scrap environment 63e >
  < Begin the cross-reference environment 65e >
  < Write macro defs 66a >
  < Write macro refs 66b >
  format_defs_refs(tex_file, scraps);
  format_uses_refs(tex_file, scraps++);
  < Finish the cross-reference environment 65f >
  < Finish the scrap environment 65a >
}

```

61b · used in 58c · uses collect_macro_name 170a, defs 129c, format_defs_refs 186a, format_uses_refs 184f, fputs 21b, Name 152b, scraps 130a, write_single_scrap_ref 131c

```

< Write the macro's name 62a > ≡
{
  char * p = name->spelling;
  int i = 0;

  while (*p != '\\000') {
    if (*p == ARG_CHR) {
      write_arg(tex_file, name->arg[i++]);
      p++;
    }
    else
      fputc(*p++, tex_file);
  }
}

```

62a · used in 61b, 79a · uses ARG_CHR 171b, write_arg 62b

```

< latex.c 62b > + ≡
static void write_arg(FILE * tex_file, char * p)
{
  fputs("\\\\hbox{\\slshape\\sffamily ", tex_file);
  while (*p)
  {
    switch (*p)
    {
      case '$':
      case '_':
      case '~':
      case '#':

```

```

        fputc('\\', tex_file);
        break;
    default:
        break;
    }
    fputc(*p, tex_file);
    p++;
}

fputs("\\}/", tex_file);
}

```

62b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `write_arg` 54b, 59c, 62a, 71e, 75b · uses `FILE` 21b, `fputs` 21b

A scrap is opened with `\WEFTbegin` and closed with `\WEFTend` (both defined among the scrap-formatting macros above). Both big (`@0`, `@D`) and small (`@o`, `@d`) definitions use the same breakable block; the `inBlock` bookkeeping lets a run of small definitions share one open group, separated by a little vertical space.

```

⟨ Begin the scrap environment 63a ⟩ ≡
{
    if (big_definition)
    {
        if (inBlock)
        {
            ⟨ End block 63d ⟩
        }
        fputs("\\WEFTbegin", tex_file);
    }
    else
    {
        if (inBlock)
        {
            ⟨ Switch block 63c ⟩
        }
        else
        {
            ⟨ Start block 63b ⟩
        }
    }
    fprintf(tex_file, "\\label{scrap%d}\\n", scraps);
}

```

63a · used in 61ab · uses `fprintf` 21b, `fputs` 21b, `scraps` 130a

```

⟨ Start block 63b ⟩ ≡
fputs("\\WEFTbegin\\n", tex_file);
inBlock = TRUE;

```

63b · used in 50e, 63a · uses `fputs` 21b, `TRUE` 23a

```

⟨ Switch block 63c ⟩ ≡
fputs("\\par\\vspace{\\baselineskip}\\n", tex_file);

```

63c · used in 63a · uses `fputs` 21b

```

⟨ End block 63d ⟩ ≡
fputs("\\WEFTend\\n", tex_file);
inBlock = FALSE;

```

63d · used in 50e, 63a, 65a · uses `FALSE` 23a, `fputs` 21b

The body of a code scrap (`@{...@}`) is emitted as an `lstlisting`. We tag it with our fixed `escapeinside` pair and, when the scrap has a language listings understands, with a `language=` option so the code is highlighted.

Both ends of the listing must sit on their own lines. We emit a newline right after the closing `]` of the options, so the first line of code starts fresh; without it a scrap beginning with `#` (a C `#ifndef`, say) would land immediately after `]`, and `listings` reads that `#` as a macro parameter and aborts with *Illegal parameter number*. Symmetrically, `\end{lstlisting}` must start its own line, so we emit a newline before it even when the scrap body did not end in one. The modern layout draws no end-of-scrap glyph: the old diamond (`\WEFTsep`) is retired — its macro is still defined, but empty, for backward compatibility — and the metadata footer that follows delimits the scrap on its own. A non-code block scrap (the unusual `@d name @[...@]` or `@(...@)` forms) is emitted as before, without a listing.

< Fill in the middle of the scrap environment 63e > ≡

```
{
  int is_code = (source_last == '{');
  if (is_code) {
    const char *lstlang = latex_language(name->lang);
    /* the delimiter halves are split so this source line does not itself
       contain the escape sequence when weft typesets its own source */
    fputs("\\begin{lstlisting}[escapeinside={(*) "<>" "*)}", tex_file);
    if (lstlang)
      fprintf(tex_file, ",language=%s", lstlang);
    /* newline after ']' so the first code line starts fresh; a scrap
       beginning with '#' would otherwise break listings (see prose above) */
    fputs("]\n", tex_file);
  }
  extra_scrap = 0;
  copy_scrap(tex_file, TRUE, name);
  if (is_code)
    /* the closing tag is split across two literals so that this very
       line does not end a listing when weft typesets its own source
       (the same reason this comment says "the closing tag" rather than
       spelling it out); the leading newline keeps it on its own line */
    fputs("\n\\end{ls" "tlisting}\n", tex_file);
}
```

63e · used in 61ab · uses `copy_scrap` 70, `extra_scrap` 65b, `first` 190a, `fprintf` 21b, `fputs` 21b, `latex_language` 68b, `prose` 129c, `source_last` 126a, `TRUE` 23a

We've got one last spacing command, controlling the amount of white space after a scrap.

Note also the whitespace eater. I use it to remove any blank lines that appear after a scrap in the source file. This way, text following a scrap will not be indented. Again, this is a matter of personal taste.

```
< Finish the scrap environment 65a > ≡
{
    scraps += extra_scraps;
    if (big_definition)
        fputs("\\WEFTend\\n", tex_file);
    else
    {
        < End block 63d >
    }
    do
        c = source_get();
    while (isspace(c));
}
```

65a · used in 61ab · uses extra_scraps 65b, fputs 21b, isspace 21b, scraps 130a, source_get 126a

```
< Global variable declarations 65b > + ≡
extern int extra_scraps;
```

65b · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines extra_scraps 63e, 65ac, 75a, 87ab, 93, 104bc

```
< Global variable definitions 65c > + ≡
int extra_scraps = 0;
```

65c · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses extra_scraps 65b

```
< Write in-text scrap 65d > ≡
copy_scrap(tex_file, FALSE, NULL);
c = source_get();
```

65d · used in 58c · uses copy_scrap 70, FALSE 23a, source_get 126a

Formatting Cross References

```
< Begin the cross-reference environment 65e > ≡
{
    fputs("\\WEFTmetabegin{", tex_file);
    write_single_scrap_ref(tex_file, scraps);
    fputs("}", tex_file);}
}
```

65e · used in 61ab · uses fputs 21b, scraps 130a, write_single_scrap_ref 131c

The footer is one horizontal line, opened above with the scrap's own number. Every subsequent item is introduced by \WEFTdot, the grey separator, so there is never a leading or trailing dot to special-case; an item that has nothing to say simply emits nothing.

```
< Finish the cross-reference environment 65f > ≡
{
    fputs("\\WEFTmetaend\\n", tex_file);
}
```

65f · used in 61ab · uses fputs 21b

```
< Write file defs 65g > ≡
```

```

{
  if (name->defs) {
    if (name->defs->next) {
      fputs("\\WEFTdot \\WEFTtxtFileDefBy\\ ", tex_file);
      print_scrap_numbers(tex_file, name->defs);
    }
  } else {
    fprintf(stderr,
            "would have crashed in 'Write file defs' for '%s'\n",
            name->spelling);
  }
}

```

65g · used in 61a · uses [defs 129c](#), [fprintf 21b](#), [fputs 21b](#), [next 129b](#), [print_scrap_numbers 66c](#), [stderr 21b](#)

⟨ *Write macro defs* 66a ⟩ ≡

```

{
  if (name->defs->next) {
    fputs("\\WEFTdot \\WEFTtxtMacroDefBy\\ ", tex_file);
    print_scrap_numbers(tex_file, name->defs);
  }
}

```

66a · used in 61b · uses [defs 129c](#), [fputs 21b](#), [next 129b](#), [print_scrap_numbers 66c](#)

⟨ *Write macro refs* 66b ⟩ ≡

```

{
  if (name->uses) {
    if (name->uses->next) {
      fputs("\\WEFTdot \\WEFTtxtMacroRefIn\\ ", tex_file);
      print_scrap_numbers(tex_file, name->uses);
    }
    else {
      fputs("\\WEFTdot \\WEFTtxtMacroRefIn\\ ", tex_file);
      fputs("\\WEFTlink{weft", tex_file);
      write_single_scrap_ref(tex_file, name->uses->scrap);
      fputs("}{", tex_file);
      write_single_scrap_ref(tex_file, name->uses->scrap);
      fputs("}", tex_file);
    }
  }
  else {
    fputs("\\WEFTdot {\\WEFTtxtMacroNoRef}", tex_file);
    fprintf(stderr, "%s: <%s> never referenced.\n",
            command_name, name->spelling);
  }
}

```

66b · used in 61b · uses [command_name 29b](#), [fprintf 21b](#), [fputs 21b](#), [next 129b](#), [print_scrap_numbers 66c](#), [stderr 21b](#),
[uses 129c](#), [write_single_scrap_ref 131c](#)

⟨ *latex.c* 66c ⟩ + ≡

```

static void print_scrap_numbers(FILE *tex_file, Scrap_Node *scraps)
{
  int page;
  fputs("\\WEFTlink{weft", tex_file);
  write_scrap_ref(tex_file, scraps->scrap, -1, &page);
  fputs("}{", tex_file);
  write_scrap_ref(tex_file, scraps->scrap, TRUE, &page);
  fputs("}", tex_file);
}

```

```

scrap = scraps->next;
while (scraps) {
    fputs("\\WEFTlink{weft", tex_file);
    write_scrap_ref(tex_file, scraps->scrap, -1, &page);
    fputs("}{", tex_file);
    write_scrap_ref(tex_file, scraps->scrap, FALSE, &page);
    scraps = scraps->next;
    fputs("}", tex_file);
}
}

```

66c · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `print_scrap_numbers` 54b, 65g, 66ab, 77b, 79c · uses `FALSE` 23a, `FILE` 21b, `fputs` 21b, `next` 129b, `page` 129c, `scraps` 130a, `Scrap_Node` 152a, `TRUE` 23a, `write_scrap_ref` 131b

Formatting a Scrap

This is the only place we really care whether a scrap is delimited with `@{...@}`, `@[...@]`, or `@(...@)`, and we base our output sequences on that. The code (`@{...@}`) mode is now handled by `listings`, so its per-line strings are all empty: `copy_scrap` just copies the body out verbatim, one raw line per source line, and the surrounding `lstlisting` does the formatting. The `@[...@]` (verbatim `TEX`) and `@(...@)` (math) modes keep their old inline treatment.

We still keep the array `delimit_scrap`: it carries the small *begin/end* strings for the math mode (a pair of \$) and, for every mode, the single character that stands in for the weft character. Upon initialisation we copy the strings from `orig_delimit_scrap` into a writeable array; we might change the *insert nw_char* entry later, when the weft character changes.

```

<latex.c 67a> + ≡
static char *orig_delimit_scrap[3][5] = {
    /* {} mode (code): begin, end, insert nw_char, prefix, suffix.
       All empty: listings copies the body verbatim, one raw line per
       source line, so there is no per-line wrapper. */
    { "", "", "@", "", "" },
    /* [] mode: begin, end, insert nw_char, prefix, suffix */
    { "", "", "@", "", "" },
    /* () mode: begin, end, insert nw_char, prefix, suffix */
    { "$", "$", "@", "", "" },
};

static char *delimit_scrap[3][5];

```

67a · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · uses `nw_char` 28b, 29a

The function `initialise_delimit_scrap_array` just copies the originals into a writeable array.

```

<latex.c 67b> + ≡
void initialise_delimit_scrap_array(void) {
    int i,j;
    for(i = 0; i < 3; i++) {
        for(j = 0; j < 5; j++) {
            if((delimit_scrap[i][j] = strdup(orig_delimit_scrap[i][j])) == NULL) {
                fprintf(stderr, "Not enough memory for string allocation\n");
                exit(EXIT_FAILURE);
            }
        }
    }
}

```

67b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · uses `exit` 21b, `fprintf` 21b, `stderr` 21b

Copying one code character to the `.tex` file. Because the code is no longer wrapped in `\verb`, every character that is special to L^AT_EX has to be turned into something harmless in `\ttfamily` text. A literal space becomes `\WEFTsp` (a breakable, fixed-width the code body itself needs none of this — `listings` copies it verbatim. `latex_code_putc` is used only for the little bits of text that are emitted *inside* an escape (a fragment reference's name and its substituted arguments), which land in ordinary L^AT_EX text mode and so must have the L^AT_EX-special characters neutralised.

⟨`latex.c` 68a⟩ + ≡

```
static void latex_code_putc(FILE *file, int c)
{
    switch (c) {
        case '\\': fputs("\\textbackslash{}", file); break;
        case '{': fputs("\\{", file); break;
        case '}': fputs("\\}", file); break;
        case '$': fputs("\\$", file); break;
        case '&': fputs("\\&", file); break;
        case '#': fputs("\\#", file); break;
        case '%': fputs("\\%", file); break;
        case '_': fputs("\\_", file); break;
        case '^': fputs("\\textasciicircum{}", file); break;
        case '~': fputs("\\textasciitilde{}", file); break;
        default: putc(c, file); break;
    }
}
```

68a · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `latex_code_putc` 54b, 69c, 70, 75a · uses `FILE` 21b, `fputs` 21b, `putc` 21b

The language mapper turns a scrap's weft language tag ("`c`", "`python`", ...) into the exact name `listings` expects, or `NULL` when we have no safe mapping. Returning `NULL` (rather than guessing) matters: `listings` aborts the run on an unknown language, so we only ever name languages it is known to ship.

⟨`latex.c` 68b⟩ + ≡

```
static const char *latex_language(const char *lang)
{
    static const struct { const char *tag, *lst; } map[] = {
        {"c", "C"}, {"h", "C"},
        {"cpp", "C++"}, {"c++", "C++"}, {"cc", "C++"}, {"cxx", "C++"},
        {"hpp", "C++"}, {"hxx", "C++"},
        {"java", "Java"},
        {"python", "Python"}, {"py", "Python"},
        {"sql", "SQL"},
        {"html", "HTML"}, {"htm", "HTML"}, {"xml", "XML"},
        {"php", "PHP"},
        {"perl", "Perl"}, {"pl", "Perl"},
        {"ruby", "Ruby"}, {"rb", "Ruby"},
        {"pascal", "Pascal"}, {"pas", "Pascal"},
        {"lisp", "Lisp"}, {"el", "Lisp"}, {"scheme", "Lisp"},
        {"matlab", "Matlab"},
        {"tex", "[LaTeX]TeX"}, {"latex", "[LaTeX]TeX"},
        {"make", "make"}, {"makefile", "make"},
        {"awk", "Awk"},
        {"fortran", "Fortran"}, {"f", "Fortran"}, {"f90", "Fortran"},
        {"r", "R"}, {"lua", "Lua"}, {"ada", "Ada"},
        {"vhdl", "VHDL"}, {"verilog", "Verilog"},
    }
```

```

    {"erlang","erlang"}, {"cobol","COBOL"}, {"eiffel","Eiffel"},
    {"ocaml","Caml"}, {"caml","Caml"},
    {NULL,NULL}
};
int i;
if (lang == NULL || lang[0] == '\0')
    return NULL;
for (i = 0; map[i].tag; i++) {
    const char *a = lang, *b = map[i].tag;
    while (*a && *b && tolower((unsigned char)*a) == (unsigned char)*b) { a++; b++; }
    if (*a == '\0' && *b == '\0')
        return map[i].lst;
}
return NULL;
}

```

68b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `latex_language` 54b, 63e

⟨ *Function prototypes* 69a ⟩ + ≡

```
void initialise_delimit_scrap_array(void);
```

69a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a

⟨ `latex.c` 69b ⟩ + ≡

```
int scrap_type = 0;
```

69b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `scrap_type` 69d, 70, 71de, 73g

⟨ `latex.c` 69c ⟩ + ≡

```
static void write_literal(FILE * tex_file, char * p, int mode)
{
    fputs(delimit_scrap[mode][0], tex_file);
    while (*p != '\000') {
        if (*p == nw_char)
            fputs(delimit_scrap[mode][2], tex_file);
        else if (mode == 0)
            latex_code_putc(tex_file, (unsigned char)*p);
        else
            fputc(*p, tex_file);
        p++;
    }
    fputs(delimit_scrap[mode][1], tex_file);
}

```

69c · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `write_literal` 54b, 73g, 75b · uses `FILE` 21b, `fputs` 21b, `latex_code_putc` 68a, `nw_char` 28b, 29a

Two little helpers bracket every place where we must step out of the verbatim body to run L^AT_EX (a fragment reference, a bold keyword, an italic title, a substituted parameter). In code mode this opens and closes a `listings` escape; in math mode it closes and reopens the math run; in the verbatim-T_EX (`@[...@]`) mode both are empty.

The escape delimiters are written with the two halves as separate string literals ("`(*`" "`<`"). The C compiler concatenates them, so the emitted text is exactly `(*<`; but the *source* never contains the delimiter as one contiguous token, which matters because weft typesets its own source — otherwise these very lines would trip the `listings` tokeniser when they appear in a scrap.

When `code_inline` is set we are already rendering inside an escape (a parameter of a fragment reference), so we must *not* open a nested escape — both helpers do nothing in code mode.

<latex.c 69d> + ≡

```
int code_inline = FALSE;
```

```
static void begin_tex_escape(FILE *file) /* leave the verbatim body */
{
    if (scrap_type == 0) { if (!code_inline) fputs("(" "<", file); }
    else fputs(delimit_scrap[scrap_type][1], file);
}
static void end_tex_escape(FILE *file) /* re-enter the verbatim body */
{
    if (scrap_type == 0) { if (!code_inline) fputs(">" ">", file); }
    else fputs(delimit_scrap[scrap_type][0], file);
}
```

69d · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `begin_tex_escape` 49e, 71e, 73efg, `code_inline` 53b, 70, `end_tex_escape` 49e, 71e, 73efg · uses `FALSE` 23a, `FILE` 21b, `fputs` 21b, `scrap_type` 69b, 70

<latex.c 70> + ≡

```
static void copy_scrap(FILE *file, int prefix, Name *name)
{
    int indent = 0;
    int c;
    int coll_h1 = 0, coll_h2 = 0, coll_warned = FALSE;
    char ** params = name->arg;
    if (source_last == '{') scrap_type = 0;
    if (source_last == '[') scrap_type = 1;
    if (source_last == '(') scrap_type = 2;
    c = source_get();
    if (prefix) fputs(delimit_scrap[scrap_type][3], file);
    fputs(delimit_scrap[scrap_type][0], file);
    while (1) {
        switch (c) {
            case '\n': if (code_inline) {
                putc(' ', file); /* an inline parameter stays on one line */
                indent = 0;
                break;
            }
            fputs(delimit_scrap[scrap_type][1], file);
            if (prefix) fputs(delimit_scrap[scrap_type][4], file);
            fputs("\n", file);
            if (prefix) fputs(delimit_scrap[scrap_type][3], file);
            fputs(delimit_scrap[scrap_type][0], file);
            indent = 0;
            break;
            case '\t': <Expand tab into spaces 71c>
                break;
            default:
                if (c==nw_char)
                {
                    <Check at-sequence for end-of-scrap 71e>
                    break;
                }
                if (code_inline)
                    latex_code_putc(file, c); /* inside an escape: neutralise LaTeX */
                else {
                    <Note a delimiter collision 71d>
                    putc(c, file);
                }
                indent++;
        }
    }
}
```

```

        break;
    }
    c = source_get();
}
}

```

70 · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `copy_scrap` 53b, 54b, 63e, 65d, `scrap_type` 69bd, 71de, 73g · uses `code_inline` 69d, `FALSE` 23a, `FILE` 21b, `fputs` 21b, `latex_code_putc` 68a, `Name` 152b, `nw_char` 28b, 29a, `putc` 21b, `source_get` 126a, `source_last` 126a

When we encounter the command to change the ‘weft character’ we call this function. It updates the scrap formatting directives accordingly.

⟨`latex.c` 71a⟩ + ≡

```

void update_delimit_scrap(void)
{
    /* Every mode's "insert nw_char" string is the single character that
       stands in for the weft character inside a scrap; keep them in sync
       with the (possibly changed) weft character. */
    delimit_scrap[0][2][0] = nw_char;
    delimit_scrap[1][2][0] = nw_char;
    delimit_scrap[2][2][0] = nw_char;
}

```

71a · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `update_delimit_scrap` 44d, 58c, 71b, 84a, 103a · uses `nw_char` 28b, 29a

⟨*Function prototypes* 71b⟩ + ≡

```

void update_delimit_scrap(void);

```

71b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `update_delimit_scrap` 71a

⟨*Expand tab into spaces* 71c⟩ ≡

```

{
    int delta = 8 - (indent % 8);
    indent += delta;
    while (delta > 0) {
        putc(' ', file);
        delta--;
    }
}

```

71c · used in 70, 145e · uses `putc` 21b

Since the code body is copied verbatim, a scrap that literally contains our escape delimiter ((*< or >*)) would desynchronise listings’ tokeniser. This never happens in ordinary source, but we watch a two-character rolling window and warn once per scrap if it does, so a corrupted listing is never silent.

⟨*Note a delimiter collision* 71d⟩ ≡

```

if (scrap_type == 0) {
    if ((coll_h2 == '(' && coll_h1 == '*' && c == '<') ||
        (coll_h2 == '>' && coll_h1 == '*' && c == '>')) {
        if (!coll_warned) {
            fprintf(stderr,
                "%s: warning: a scrap body contains the listings escape "
                "delimiter; its typeset code may be corrupted.\n", command_name);
            coll_warned = TRUE;
        }
    }
}

```

```

    coll_h2 = coll_h1;
    coll_h1 = c;
}
71d · used in 70 · uses command_name 29b, fprintf 21b, scrap_type 69b, 70, stderr 21b, TRUE 23a

< Check at-sequence for end-of-scrap 71e > ≡
{
    c = source_get();
    switch (c) {
        case 'c': < Show presence of a block comment 49e >
            break;
        case 'x': < Copy label from source into(72a file ) 73a >
            break;
        case 'v': < Copy version info into file 72b >
        case 's':
            break;
        case '+':
        case '-':
        case '*':
        case '|': < Skip over index entries 73c >
            FALLTHROUGH;
        case ',':
        case ')':
        case ']':
        case '}': fputs(delimit_scrap[scrap_type][1], file);
            return;
        case '<': < Format macro name 73g >
            break;
        case '%': < Skip commented-out code 73d >
            break;
        case '_': < Bold Keyword 73e >
            break;
        case 't': < Italic "fragment title" 73f >
            break;
        case 'f': < Italic "file name" 73f >
            break;
        case '1': case '2': case '3':
        case '4': case '5': case '6':
        case '7': case '8': case '9':
            if (name == NULL
                || name->arg[c - '1'] == NULL) {
                fputs(delimit_scrap[scrap_type][2], file);
                fputc(c, file);
            }
            else {
                begin_tex_escape(file);
                write_arg(file, name->arg[c - '1']);
                end_tex_escape(file);
            }
            break;
        default:
            if (c==nw_char)
            {
                fputs(delimit_scrap[scrap_type][2], file);
                break;
            }
            /* ignore these since pass1 will have warned about them */
            break;
    }
}

```

```
}
}
```

72a · used in 70

⟨ *Copy version info into file* 72b ⟩ ≡
 fputs(version_string, file);

72b · used in 71e, 90c, 146a · uses fputs 21b, version_string 27

⟨ *Copy label from source into* 73a ⟩ ≡
 {
 ⟨ *Get label from*(73b source_get()) 187b ⟩
 write_label(label_name, 01);
 }

73b · used in 58c, 71e, 84a, 90c, 103a · uses source_get 126a

There's no need to check for errors here, since we will have already pointed out any during the first pass.

⟨ *Skip over index entries* 73c ⟩ ≡
 {
 do {
 do
 c = source_get();
 while (c != nw_char);
 c = source_get();
 } while (c != '}' && c != ']' && c != ')');
 }

73c · used in 71e, 90c · uses nw_char 28b, 29a, source_get 126a

⟨ *Skip commented-out code* 73d ⟩ ≡
 {
 do
 c = source_get();
 while (c != '\n');
 }

73d · used in 71e, 90c, 134c · uses source_get 126a

This scrap helps deal with bold keywords:

⟨ *Bold Keyword* 73e ⟩ ≡
 {
 begin_tex_escape(file);
 fprintf(file, "\\hbox{\\sffamily\\bfseries ");
 c = source_get();
 do {
 fputc(c, file);
 c = source_get();
 } while (c != nw_char);
 c = source_get();
 fprintf(file, "}");
 end_tex_escape(file);
 }

73e · used in 71e · uses begin_tex_escape 69d, end_tex_escape 69d, fprintf 21b, nw_char 28b, 29a, source_get 126a

⟨ *Italic "whatever"* 73f ⟩ ≡

```

{
  begin_tex_escape(file);
  fprintf(file, "\\hbox{\\sffamily\\slshape whatever}");
  end_tex_escape(file);
}

```

73f · used in 71e · uses [begin_tex_escape 69d](#), [end_tex_escape 69d](#), [fprintf 21b](#)

⟨ *Format macro name 73g* ⟩ ≡

```

{
  Arglist *args = collect_scrap_name(-1);
  Name *name = args->name;
  char * p = name->spelling;
  Arglist *q = args->args;
  int narg = 0;

  begin_tex_escape(file);
  if (prefix)
    /* \normalfont resets the listing's \ttfamily basicstyle, so the
       fragment name is set in the ordinary document font (roman italic)
       rather than as italic typewriter. */
    fputs("\\hbox{\\normalfont ", file);
  fputs("\\WEFTlangle{\\WEFTtint{weftaccent}\\itshape ", file);
  while (*p != '\\000') {
    if (*p == ARG_CHR) {
      if (q == NULL) {
        write_literal(file, name->arg[narg], scrap_type);
      }
      else {
        write_ArglistElement(file, q, params);
        q = q->next;
      }
      p++;
      narg++;
    }
    else
      fputc(*p++, file);
  }
  fputs("}\\nobreak", file);
  if (scrap_name_has_parameters) {
    ⟨ Format macro parameters 53b ⟩
  }
  fprintf(file, "\\WEFTnumsep{\\footnotesize\\WEFTtint{weftcom}");
  if (name->defs)
    ⟨ Write abbreviated definition list 74 ⟩
  else {
    putc('?', file);
    fprintf(stderr, "%s: never defined <%s>\\n",
            command_name, name->spelling);
  }
  fputs("}\\WEFTrangle", file);
  if (prefix)
    fputs("}", file);
  end_tex_escape(file);
}

```

73g · used in 71e · uses [Arglist 172d](#), [ARG_CHR 171b](#), [begin_tex_escape 69d](#), [collect_scrap_name 173](#), [command_name 29b](#), [defs 129c](#), [end_tex_escape 69d](#), [fprintf 21b](#), [fputs 21b](#), [Name 152b](#), [next 129b](#), [putc 21b](#), [scrap_type 69b](#), 70, [stderr 21b](#), [write_ArglistElement 75b](#), [write_literal 69c](#)

⟨ *Write abbreviated definition list 74* ⟩ ≡

```
{
  Scrap_Node *p = name->defs;
  fputs("\\WEFTlink{weft", file);
  write_single_scrap_ref(file, p->scrap);
  fputs("{", file);
  write_single_scrap_ref(file, p->scrap);
  fputs("}", file);
  p = p->next;
  if (p)
    fputs(", \\ldots\\ ", file);
}
```

74 · used in 73g, 75b · uses `defs 129c`, `fputs 21b`, `next 129b`, `Scrap_Node 152a`, `write_single_scrap_ref 131c`

A quoted scrap (`@{...@}`) can appear as the argument of a parameterised fragment reference, e.g. `@<Name @{code@} eman@>`. In the old `\verb`-based output that code was simply set in a verbatim run in the middle of the reference. Under `listings` we cannot leave the escape here — the reference is being built inside an `\hbox`, and closing the escape mid-box would leave the braces straddling the escape boundary (a ! Extra } error). Instead we capture the scrap's expansion into a memory buffer and re-emit it *inside* the escape, as L^AT_EX-escaped typewriter text on a single line, so the box stays balanced.

⟨ *Emit a quoted scrap argument inline 75a* ⟩ ≡

```
{
  Scrap_Node * qq = (Scrap_Node *)q;
  char *buf = NULL;
  size_t sz = 0;
  FILE *mem = open_memstream(&buf, &sz);
  qq->quoted = TRUE;
  if (mem) {
    char *s;
    write_scraps(mem, "", qq, -1, "", 0, 0, 0, 0, 0, NULL, params, 0, "");
    fclose(mem);
    fputs("\\hbox{\\ttfamily ", file);
    for (s = buf; *s != '\\0'; s++) {
      if (*s == '\\n')
        putc(' ', file);
      else
        latex_code_putc(file, (unsigned char)*s);
    }
    fputs("}", file);
    free(buf);
  }
  extra_scraps++;
  qq->quoted = FALSE;
}
```

75a · used in 75b · uses `extra_scraps 65b`, `FALSE 23a`, `fclose 21b`, `FILE 21b`, `fputs 21b`, `latex_code_putc 68a`, `putc 21b`, `Scrap_Node 152a`, `size_t 21b`, `TRUE 23a`, `write_scraps 142b`

⟨ *latex.c 75b* ⟩ + ≡

```
static void
write_ArglistElement(FILE * file, Arglist * args, char ** params)
{
  Name *name = args->name;
  Arglist *q = args->args;

  if (name == NULL) {
    char * p = (char*)q;
```

```

    if (p[0] == ARG_CHR) {
        write_arg(file, params[p[1] - '1']);
    } else {
        write_literal(file, (char *)q, 0);
    }
} else if (name == (Name *)1) {
    < Emit a quoted scrap argument inline 75a >
} else {
    char * p = name->spelling;
    fputs("$\\langle\\,${\\itshape ", file);
    while (*p != '\\000') {
        if (*p == ARG_CHR) {
            write_ArglistElement(file, q, params);
            q = q->next;
            p++;
        }
        else
            fputc(*p++, file);
    }
    fputs("\\nobreak\\ ", file);
    if (scrap_name_has_parameters) {
        int c;

        < Format macro parameters 53b >
    }
    fprintf(file, "{\\footnotesize ");
    if (name->defs)
        < Write abbreviated definition list 74 >
    else {
        putc('?', file);
        fprintf(stderr, "%s: never defined <%s>\\n",
            command_name, name->spelling);
    }
    fputs("}$\\,\\rangle$", file);
}
}

```

75b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines `write_ArglistElement` 54b, 73g · uses `Arglist` 172d, `ARG_CHR` 171b, `command_name` 29b, `defs` 129c, `FILE` 21b, `fprintf` 21b, `fputs` 21b, `Name` 152b, `next` 129b, `putc` 21b, `stderr` 21b, `write_arg` 62b, `write_literal` 69c

2.4.2 Generating the Indices

The three back-of-book indices — of file names, fragment names, and user-specified identifiers — are each a compact vertical list, one entry per line. Their entries share the scrap presentation’s visual language: a fragment name is set in the accent-tinted italic inside the grey `\WEFTlangle/\WEFTrangle` brackets, a file name and an identifier in accent and keyword-coloured monospace respectively, and each entry’s `\WEFTtxtDefBy/\WEFTtxtRefIn` metadata in the small grey used everywhere else. Colour flows through the same `\WEFTtint` indirection, so the indices too degrade to black and white when no colour package is loaded.

```

< Write index of file names 76a > ≡
{
    if (file_names) {
        fputs("\\n{\\small\\begin{list}{}{\\setlength{\\itemsep}{-\\parsep}",
            tex_file);
        fputs("\\setlength{\\itemindent}{-\\leftmargin}}\\n", tex_file);
        format_file_entry(file_names, tex_file);
    }
}

```

```

    fputs("\\end{list}}", tex_file);
}
c = source_get();
}

```

76a · used in 58c · uses file_names 152c, format_file_entry 76b, fputs 21b, source_get 126a

⟨ latex.c 76b ⟩ + ≡

```

static void format_file_entry(Name *name, FILE *tex_file)
{
    while (name) {
        format_file_entry(name->llink, tex_file);
        ⟨ Format a file index entry 77a ⟩
        name = name->rlink;
    }
}

```

76b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines format_file_entry 54b, 76a
· uses FILE 21b, Name 152b

⟨ Format a file index entry 77a ⟩ ≡

```

fputs("\\item ", tex_file);
fprintf(tex_file, "{\\WEFTtint{weftaccent}\\verb%c%s%c}\\ ",
        nw_char, name->spelling, nw_char);
⟨ Write file's defining scrap numbers 77b ⟩
putc('\\n', tex_file);

```

77a · used in 76b · uses fprintf 21b, fputs 21b, nw_char 28b, 29a, putc 21b

⟨ Write file's defining scrap numbers 77b ⟩ ≡

```

{
    Scrap_Node *p = name->defs;
    fputs("{\\footnotesize\\WEFTtint{weftcom} {\\WEFTtxtDefBy}", tex_file);
    putc(' ', tex_file);
    if (p->next) {
        print_scrap_numbers(tex_file, p);
    }
    else {
        fputs("\\WEFTlink{weft", tex_file);
        write_single_scrap_ref(tex_file, p->scrap);
        fputs("{", tex_file);
        write_single_scrap_ref(tex_file, p->scrap);
        fputs("}", tex_file);
    }
    putc('}', tex_file);
}

```

77b · used in 77a · uses defs 129c, fputs 21b, next 129b, print_scrap_numbers 66c, putc 21b, Scrap_Node 152a, write_single_scrap_ref 131c

⟨ Write index of macro names 77c ⟩ ≡

```

{
    unsigned char sector = current_sector;
    int c = source_get();
    if (c == '+')
        sector = 0;
    else
        source_ungetc(&c);
    if (has_sector(macro_names, sector)) {
        fputs("\\n{\\small\\begin{list}}{\\setlength{\\itemsep}{-\\parsep}",

```

```

        tex_file);
    fputs("\\setlength{\\itemindent}{-\\leftmargin}\\n", tex_file);
    format_entry(macro_names, tex_file, sector);
    fputs("\\end{list}\\n", tex_file);
} else {
    fputs("None.\\n", tex_file);
}
}
c = source_get();
77c · used in 58c · uses current_sector 45c, format_entry 78b, fputs 21b, has_sector 80b, macro_names 152c, sector 129c,
source_get 126a, source_ungetc 126c

```

⟨*latex.c* 78a⟩ + ≡

```

static int load_entry(Name * name, Name ** nms, int n)
{
    while (name) {
        n = load_entry(name->llink, nms, n);
        nms[n++] = name;
        name = name->rlink;
    }
    return n;
}

```

78a · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines load_entry 78b ·
uses Name 152b

⟨*latex.c* 78b⟩ + ≡

```

static void format_entry(Name *name, FILE *tex_file, unsigned char sector)
{
    Name ** nms = malloc(num_scraps()*sizeof(Name *));
    int n = load_entry(name, nms, 0);
    int i;

    ⟨Sort nms of size n for ⟨Rob's ordering 78c⟩ 78d⟩
    for (i = 0; i < n; i++)
    {
        Name * name = nms[i];

        ⟨Format an index entry 79a⟩
    }
}

```

78b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines format_entry 54b, 77c ·
uses FILE 21b, load_entry 78a, malloc 21b, Name 152b, num_scraps 130a, sector 129c

⟨*Rob's ordering* 78c⟩ ≡

```

robs_strcmp(ki->spelling, kj->spelling) < 0

```

78c · used in 78b, 95c · uses robs_strcmp 155c

⟨*Sort key of size n for ordering* 78d⟩ ≡

```

int j;
for (j = 1; j < n; j++)
{
    int i = j - 1;
    Name * kj = key[j];

    do
    {

```

```

    Name * ki = key[i];

    if (ordering)
        break;
    key[i + 1] = ki;
    i -= 1;
} while (i >= 0);
key[i + 1] = kj;
}

```

78d · used in 78b, 95c · uses Name 152b

⟨ *Format an index entry* 79a ⟩ ≡

```

if (name->sector == sector){
    fputs("\\item ", tex_file);
    fputs("\\WEFTlangle{\\WEFTtint{weftaccent}\\itshape ", tex_file);
    ⟨ Write the macro's name 62a ⟩
    fputs("\\nobreak\\WEFTnumsep{\\footnotesize\\WEFTtint{weftcom}", tex_file);
    ⟨ Write defining scrap numbers 79b ⟩
    fputs("\\WEFTrangle\\ ", tex_file);
    ⟨ Write referencing scrap numbers 79c ⟩
    putc('\\n', tex_file);
}

```

79a · used in 78b · uses fputs 21b, putc 21b, sector 129c

⟨ *Write defining scrap numbers* 79b ⟩ ≡

```

{
    Scrap_Node *p = name->defs;
    if (p) {
        int page;
        fputs("\\WEFTlink{weft", tex_file);
        write_scrap_ref(tex_file, p->scrap, -1, &page);
        fputs("{", tex_file);
        write_scrap_ref(tex_file, p->scrap, TRUE, &page);
        fputs("}", tex_file);
        p = p->next;
        while (p) {
            fputs("\\WEFTlink{weft", tex_file);
            write_scrap_ref(tex_file, p->scrap, -1, &page);
            fputs("{", tex_file);
            write_scrap_ref(tex_file, p->scrap, FALSE, &page);
            fputs("}", tex_file);
            p = p->next;
        }
    }
    else
        putc('?', tex_file);
}

```

79b · used in 79a · uses defs 129c, FALSE 23a, fputs 21b, next 129b, page 129c, putc 21b, Scrap_Node 152a, TRUE 23a, write_scrap_ref 131b

⟨ *Write referencing scrap numbers* 79c ⟩ ≡

```

{
    Scrap_Node *p = name->uses;
    fputs("{\\footnotesize\\WEFTtint{weftcom} ", tex_file);
    if (p) {
        fputs("{\\WEFTtxtRefIn}", tex_file);
        putc(' ', tex_file);
    }
}

```

```

    if (p->next) {
        print_scrap_numbers(tex_file, p);
    }
    else {
        fputs("\\WEFTlink{weft", tex_file);
        write_single_scrap_ref(tex_file, p->scrap);
        fputs("{}{", tex_file);
        write_single_scrap_ref(tex_file, p->scrap);
        fputs("}", tex_file);
    }
}
else
    fputs("{\\WEFTtxtNoRef}", tex_file);
putc('}', tex_file);
}

```

79c · used in 79a · uses fputs 21b, next 129b, print_scrap_numbers 66c, putc 21b, Scrap_Node 152a, uses 129c, write_single_scrap_ref 131c

⟨ *Function prototypes* 80a ⟩ + ≡

```
extern int has_sector(Name *, unsigned char);
```

80a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses has_sector 80b, Name 152b

⟨ *latex.c* 80b ⟩ + ≡

```

int has_sector(Name * name, unsigned char sector)
{
    while(name) {
        if (name->sector == sector)
            return TRUE;
        if (has_sector(name->llink, sector))
            return TRUE;
        name = name->rlink;
    }
    return FALSE;
}

```

80b · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines has_sector 77c, 80ac, 95a, 97a, 105d, 106b · uses FALSE 23a, Name 152b, sector 129c, TRUE 23a

⟨ *Write index of user-specified names* 80c ⟩ ≡

```

{
    unsigned char sector = current_sector;
    c = source_get();
    if (c == '+') {
        sector = 0;
        c = source_get();
    }
    if (has_sector(user_names, sector)) {
        fputs("\n{\\small\\begin{list}}{\\setlength{\\itemsep}{-\\parsep}",
            tex_file);
        fputs("\\setlength{\\itemindent}{-\\leftmargin}}\\n", tex_file);
        format_user_entry(user_names, tex_file, sector);
        fputs("\\end{list}}", tex_file);
    }
}

```

80c · used in 58c · uses current_sector 45c, format_user_entry 80d, fputs 21b, has_sector 80b, sector 129c, source_get 126a, user_names 152c

```

< latex.c 80d > + ≡
static void format_user_entry(Name *name, FILE *tex_file, unsigned char sector)
{
    while (name) {
        format_user_entry(name->llink, tex_file, sector);
        < Format a user index entry 80e >
        name = name->rlink;
    }
}

80d · defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd · defines format_user_entry 54b, 80c
· uses FILE 21b, Name 152b, sector 129c

< Format a user index entry 80e > ≡
if (name->sector == sector){
    Scrap_Node *uses = name->uses;
    if ( uses || dangling_flag ) {
        int page;
        Scrap_Node *defs = name->defs;
        fprintf(tex_file, "\\item {\\WEFTtint{weftkw}\\verb%c%s%c}: ",
            nw_char, name->spelling, nw_char);
        if (!uses) {
            fputs("\\underline{", tex_file);
            fputs("\\WEFTlink{weft", tex_file);
            write_single_scrap_ref(tex_file, defs->scrap);
            fputs("{}{", tex_file);
            write_single_scrap_ref(tex_file, defs->scrap);
            fputs("})", tex_file);
            page = -2;
            defs = defs->next;
        }
        else
            if (!defs || uses->scrap < defs->scrap) {
                fputs("\\WEFTlink{weft", tex_file);
                write_scrap_ref(tex_file, uses->scrap, -1, &page);
                fputs("{}{", tex_file);
                write_scrap_ref(tex_file, uses->scrap, TRUE, &page);
                fputs("}", tex_file);
                uses = uses->next;
            }
        else {
            if (defs->scrap == uses->scrap)
                uses = uses->next;
            fputs("\\underline{", tex_file);

            fputs("\\WEFTlink{weft", tex_file);
            write_single_scrap_ref(tex_file, defs->scrap);
            fputs("{}{", tex_file);
            write_single_scrap_ref(tex_file, defs->scrap);
            fputs("})", tex_file);
            page = -2;
            defs = defs->next;
        }
        while (uses || defs) {
            if (uses && (!defs || uses->scrap < defs->scrap)) {
                fputs("\\WEFTlink{weft", tex_file);
                write_scrap_ref(tex_file, uses->scrap, -1, &page);
                fputs("{}{", tex_file);
                write_scrap_ref(tex_file, uses->scrap, FALSE, &page);
            }
        }
    }
}

```

```

        fputs("}", tex_file);
        uses = uses->next;
    }
    else {
        if (uses && defs->scrap == uses->scrap)
            uses = uses->next;
        fputs(" ", tex_file);

        fputs("\\WEFTlink{weft", tex_file);
        write_single_scrap_ref(tex_file, defs->scrap);
        fputs("}{", tex_file);
        write_single_scrap_ref(tex_file, defs->scrap);
        fputs("}", tex_file);

        putc('}', tex_file);
        page = -2;
        defs = defs->next;
    }
}
fputs(".\\n", tex_file);
}
}

```

80e · used in 80d · uses `dangling_flag` 27, `defs` 129c, `FALSE` 23a, `fprintf` 21b, `fputs` 21b, `next` 129b, `nw_char` 28b, 29a, `page` 129c, `putc` 21b, `Scrap_Node` 152a, `sector` 129c, `TRUE` 23a, `uses` 129c, `write_scrap_ref` 131b, `write_single_scrap_ref` 131c

2.5 Writing the Markdown File

This section implements the Markdown weave output, activated by the `-w md` command-line flag. It is structurally parallel to the L^AT_EX output in Section 2.4, but emits Markdown with embedded HTML for scraps (using `<pre><code>` blocks so that fragment references inside code can be clickable links).

⟨ *Function prototypes* 82a ⟩ + ≡

```
extern void write_md(char *file_name, char *md_name, unsigned char sector);
```

82a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `file_name` 129c, `sector` 129c, `write_md` 82d

The HTML weaver (Section 2.6) reuses two format-agnostic helpers from here, so they are declared `extern` rather than `static`: `md_copy_scrap` copies a scrap body emitting HTML-escaped code and clickable inline fragment references, and `md_print_scrap_numbers` emits a comma-separated list of `<a href="#weftN">` cross-reference links. Both are pure HTML and identical to what the HTML output needs.

⟨ *Function prototypes* 82b ⟩ + ≡

```
extern void md_copy_scrap(FILE *file, int prefix, Name *name);
extern void md_print_scrap_numbers(FILE *md_file, Scrap_Node *scraps);
```

82b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `FILE` 21b, `md_copy_scrap` 89b, `md_print_scrap_numbers` 89a, `Name` 152b, `scraps` 130a, `Scrap_Node` 152a

We need a few local function declarations before we get into the body of `write_md`.

⟨ *markdown.c* 82c ⟩ + ≡

```
static void md_format_entry(Name *name, FILE *md_file, unsigned char sector);
static void md_format_file_entry(Name *name, FILE *md_file);
static void md_format_user_entry(Name *name, FILE *md_file, unsigned char sector);
```

```
static void md_write_arg(FILE *md_file, char *p);
static void md_write_literal(FILE *md_file, char *p);
static void md_write_ArglistElement(FILE *file, Arglist *args, char **params);
static void md_format_uses_refs(FILE *md_file, int scrap);
static void md_format_defs_refs(FILE *md_file, int scrap);
```

82c · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · uses Arglist 172d, FILE 21b, md_format_defs_refs 99c, md_format_entry 95c, md_format_file_entry 94b, md_format_user_entry 97b, md_format_uses_refs 98a, md_write_arg 86c, md_write_ArglistElement 93, md_write_literal 92b, Name 152b, sector 129c

The routine `write_md` takes two file names as parameters: the name of the web source file and the name of the .md output file.

```
<markdown.c 82d> + ≡
void write_md(char *file_name, char *md_name, unsigned char sector)
{
    (void)sector;
    FILE *md_file = fopen(md_name, "w");
    if (md_file) {
        if (verbose_flag)
            fprintf(stderr, "writing %s\n", md_name);
        source_open(file_name);
        <Copy source_file into md_file 83a>
        fclose(md_file);
    }
    else
        fprintf(stderr, "%s: can't open %s\n", command_name, md_name);
}
```

82d · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines write_md 39b, 82a · uses command_name 29b, fclose 21b, FILE 21b, file_name 129c, fopen 21b, fprintf 21b, sector 129c, source_open 129a, stderr 21b, verbose_flag 27

We make our second pass through the source web, copying characters straight into the .md file. We watch for @ characters which signal a command sequence. We also detect and discard cosmetic fence lines (lines that are exactly ‘‘‘ optionally followed by a language tag).

```
<Copy source_file into md_file 83a> ≡
{
    int c = source_get();
    int at_line_start = TRUE;
    while (c != EOF) {
        if (c == nw_char)
        {
            <Interpret md at-sequence 84a>
        }
        else if (c == ' ' && at_line_start) {
            <Check for cosmetic fence line 83b>
        }
        else {
            if (c == '\n') at_line_start = TRUE;
            else at_line_start = FALSE;
            putc(c, md_file);
            c = source_get();
        }
    }
}
```

83a · used in 82d · uses FALSE 23a, nw_char 28b, 29a, putc 21b, source_get 126a, TRUE 23a

Cosmetic fence lines are lines that start with ‘‘‘. We peek ahead to see if the line matches; if so, we skip to end of line. Otherwise we output what we already consumed and continue.

```

< Check for cosmetic fence line 83b > ≡
{
  int c2 = source_get();
  int c3 = (c2 == '‘') ? source_get() : 0;
  if (c2 == '‘' && c3 == '‘') {
    /* Skip rest of fence line */
    int fc = source_get();
    while (fc != '\n' && fc != EOF)
      fc = source_get();
    c = source_get();
    at_line_start = TRUE;
  } else {
    /* Not a fence, output what we consumed */
    putc('‘', md_file);
    if (c2 == '‘') {
      putc('‘', md_file);
      c = c3;
    } else {
      c = c2;
    }
    at_line_start = FALSE;
  }
}

```

83b · used in 83a · uses FALSE 23a, putc 21b, source_get 126a, TRUE 23a

```

< Interpret md at-sequence 84a > ≡
{
  int big_definition = FALSE;
  c = source_get();
  switch (c) {
    case 'r':
      c = source_get();
      nw_char = c;
      update_delimit_scrap();
      break;
    case '0': big_definition = TRUE;
              FALLTHROUGH;
    case 'o': < Write md output file definition 85c >
              break;
    case 'Q':
    case 'D': big_definition = TRUE;
              FALLTHROUGH;
    case 'q':
    case 'd': < Write md macro definition 86a >
              break;
    case 's':
              < Step to next sector 45a >
              break;
    case 'S':
              < Close the current sector 45b >
              break;
    case '{':
    case '[':
    case '(': < Write md in-text scrap 87c >
              break;

```

```

    case '<': < Expand md macro into md_file 85b >
        break;
    case 'x': < Copy label from source into(84b md_file ) 73a >
        c = source_get();
        break;
    case 'c': /* Block comments: no special handling in Markdown */
        c = source_get();
        break;
    case 'f': < Write md index of file names 94a >
        break;
    case 'm': < Write md index of macro names 95a >
        break;
    case 'u': < Write md index of user-specified names 97a >
        break;
    case 'v': < Copy version info into md file 85a >
        break;
    case 'l':
    case 'L': < Skip at-L language definition line 44c >
        break;
    case 'W': < Skip at-L language definition line 44c >
        break;
    default:
        if (c==nw_char)
            putc(c, md_file);
        c = source_get();
        break;
}
}

```

84b · used in 83a

< Copy version info into md file 85a > ≡
fputs(version_string, md_file);
c = source_get();

85a · used in 84a · uses fputs 21b, source_get 126a, version_string 27

< Expand md macro into file 85b > ≡

```

{
    Parameters local_parameters = 0;
    int changed;
    char indent_chars[MAX_INDENT];
    Arglist *a;
    Name *name;
    Arglist * args;
    a = collect_scrap_name(0);
    name = a->name;
    args = instance(a->args, NULL, NULL, &changed);
    name->mark = TRUE;
    write_scraps(file, md_name, name->defs, 0, indent_chars, 0, 0, 1, 0, 0,
        args, name->arg, local_parameters, md_name);
    name->mark = FALSE;
    c = source_get();
}

```

85b · used in 84a · uses Arglist 172d, collect_scrap_name 173, defs 129c, FALSE 23a, instance 140e, MAX_INDENT 121d, Name 152b, Parameters 51b, source_get 126a, TRUE 23a, write_scraps 142b

2.5.1 Formatting Definitions in Markdown

File and fragment definitions use HTML anchors and formatting. A file definition header looks like:

```
<a id="weft1"></a>
**"filename" ** **1** =
<pre><code>...code...</code></pre>
```

A fragment definition header looks like:

```
<a id="weft1"></a>
**&lt;*Fragment Name* 1&gt;** =
<pre><code>...code...</code></pre>
```

```
< Write md output file definition 85c > ≡
{
    Name *name = collect_file_name();
    (void)big_definition;
    fputs("\n<a id=\"weft\", md_file);
    write_single_scrap_ref(md_file, scraps);
    fputs("\n"></a>\n", md_file);
    fprintf(md_file, "**\"%s\"**", name->spelling);
    fputs(" **", md_file);
    write_single_scrap_ref(md_file, scraps);
    fputs("** =\n", md_file);
    < Fill in the md scrap 87a >
    < Write md file cross-refs 87d >
    < Finish md scrap 87b >
}
```

85c · used in 84a · uses collect_file_name 167a, fprintf 21b, fputs 21b, Name 152b, scraps 130a, write_single_scrap_ref 131c

```
< Write md macro definition 86a > ≡
{
    Name *name = collect_macro_name();
    (void)big_definition;
    fputs("\n<a id=\"weft\", md_file);
    write_single_scrap_ref(md_file, scraps);
    fputs("\n"></a>\n", md_file);
    fputs("**&lt;* ", md_file);
    < Write the md macro's name 86b >
    fputs("* ", md_file);
    write_single_scrap_ref(md_file, scraps);
    fputs("&gt;** =\n", md_file);
    < Fill in the md scrap 87a >
    < Write md macro cross-refs 88a >
    < Finish md scrap 87b >
}
```

86a · used in 84a · uses collect_macro_name 170a, fputs 21b, Name 152b, scraps 130a, write_single_scrap_ref 131c

```
< Write the md macro's name 86b > ≡
{
    char * p = name->spelling;
    int i = 0;

    while (*p != '\000') {
        if (*p == ARG_CHR) {
            md_write_arg(md_file, name->arg[i++]);
        }
    }
}
```

```

        p++;
    }
    else
        fputc(*p++, md_file);
}
}

```

86b · used in 86a, 95d · uses ARG_CHR 171b, md_write_arg 86c

⟨markdown.c 86c⟩ + ≡

```

static void md_write_arg(FILE * md_file, char * p)
{
    fputs("<i>", md_file);
    while (*p)
    {
        switch (*p)
        {
            case '<':
                fputs("&lt;", md_file);
                break;
            case '>':
                fputs("&gt;", md_file);
                break;
            case '&':
                fputs("&amp;", md_file);
                break;
            default:
                fputc(*p, md_file);
                break;
        }
        p++;
    }
    fputs("</i>", md_file);
}

```

86c · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_write_arg 82c, 86b, 90c, 93 · uses FILE 21b, fputs 21b

⟨Fill in the md scrap 87a⟩ ≡

```

{
    if (name->lang[0] != '\0')
        fprintf(md_file, "<pre><code class=\"language-%s\">", name->lang);
    else
        fputs("<pre><code>", md_file);
    extra_scraps = 0;
    md_copy_scrap(md_file, TRUE, name);
    fputs("</code></pre>\n", md_file);
}

```

87a · used in 85c, 86a · uses extra_scraps 65b, fprintf 21b, fputs 21b, md_copy_scrap 89b, TRUE 23a

⟨Finish md scrap 87b⟩ ≡

```

{
    scraps += extra_scraps;
    do
        c = source_get();
        while (isspace(c));
        at_line_start = TRUE;
    }
}

```

87b · used in 85c, 86a · uses extra_scraps 65b, isspace 21b, scraps 130a, source_get 126a, TRUE 23a

```

< Write md in-text scrap 87c > ≡
md_copy_scrap(md_file, FALSE, NULL);
c = source_get();
87c · used in 84a · uses FALSE 23a, md_copy_scrap 89b, source_get 126a

```

Cross References in Markdown

Cross-references appear as blockquotes after each scrap.

```

< Write md file cross-refs 87d > ≡
{
  if ( scrap_flag ) {
    < Write md file defs 87e >
  }
  md_format_defs_refs(md_file, scraps);
  md_format_uses_refs(md_file, scraps++);
}
87d · used in 85c · uses md_format_defs_refs 99c, md_format_uses_refs 98a, scraps 130a, scrap_flag 27

```

```

< Write md file defs 87e > ≡
{
  if (name->defs) {
    if (name->defs->next) {
      fputs("> *File defined by* ", md_file);
      md_print_scrap_numbers(md_file, name->defs);
    }
  } else {
    fprintf(stderr,
      "would have crashed in 'Write md file defs' for '%s'\n",
      name->spelling);
  }
}
87e · used in 87d · uses defs 129c, fprintf 21b, fputs 21b, md_print_scrap_numbers 89a, next 129b, stderr 21b

```

```

< Write md macro cross-refs 88a > ≡
{
  < Write md macro defs 88b >
  < Write md macro refs 88c >
  md_format_defs_refs(md_file, scraps);
  md_format_uses_refs(md_file, scraps++);
}
88a · used in 86a · uses md_format_defs_refs 99c, md_format_uses_refs 98a, scraps 130a

```

```

< Write md macro defs 88b > ≡
{
  if (name->defs->next) {
    fputs("> *Fragment defined by* ", md_file);
    md_print_scrap_numbers(md_file, name->defs);
  }
}
88b · used in 88a · uses defs 129c, fputs 21b, md_print_scrap_numbers 89a, next 129b

```

```

< Write md macro refs 88c > ≡
{
  if (name->uses) {
    if (name->uses->next) {

```

```

        fputs("> *Fragment referenced in* ", md_file);
        md_print_scrap_numbers(md_file, name->uses);
    }
    else {
        fputs("> *Fragment referenced in* ", md_file);
        fputs("<a href=\"#weft\"", md_file);
        write_single_scrap_ref(md_file, name->uses->scrap);
        fputs("\>", md_file);
        write_single_scrap_ref(md_file, name->uses->scrap);
        fputs("</a>", md_file);
        fputs(".\n", md_file);
    }
}
else {
    fputs("> *Fragment never referenced*.\\n", md_file);
    fprintf(stderr, "%s: <%s> never referenced.\\n",
            command_name, name->spelling);
}
}

```

88c · used in 88a · uses `command_name` 29b, `fprintf` 21b, `fputs` 21b, `md_print_scrap_numbers` 89a, `next` 129b, `stderr` 21b, `uses` 129c, `write_single_scrap_ref` 131c

⟨`markdown.c` 89a⟩ + ≡

```

void md_print_scrap_numbers(FILE *md_file, Scrap_Node *scraps)
{
    int page;
    fputs("<a href=\"#weft\"", md_file);
    write_scrap_ref(md_file, scraps->scrap, -1, &page);
    fputs("\>", md_file);
    write_scrap_ref(md_file, scraps->scrap, TRUE, &page);
    fputs("</a>", md_file);
    scraps = scraps->next;
    while (scraps) {
        fputs("<a href=\"#weft\"", md_file);
        write_scrap_ref(md_file, scraps->scrap, -1, &page);
        fputs("\>", md_file);
        write_scrap_ref(md_file, scraps->scrap, FALSE, &page);
        fputs("</a>", md_file);
        scraps = scraps->next;
    }
    fputs(".\n", md_file);
}

```

89a · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines `md_print_scrap_numbers` 82b, 87e, 88bc, 94d, 96b, 104d, 105ac, 106ac · uses `FALSE` 23a, `FILE` 21b, `fputs` 21b, `next` 129b, `page` 129c, `scraps` 130a, `Scrap_Node` 152a, `TRUE` 23a, `write_scrap_ref` 131b

Formatting a Scrap in Markdown

Scraps are output inside `<pre><code>` blocks. Characters `<`, `>`, and `&` are HTML-escaped. At-sequences within scraps (macro references, keywords, parameters, etc.) emit HTML markup directly.

⟨`markdown.c` 89b⟩ + ≡

```

static int md_scrap_type = 0;

void md_copy_scrap(FILE *file, int prefix, Name *name)
{
    int indent = 0;

```

```

int c;
char ** params = name ? name->arg : NULL;
if (source_last == '{') md_scrap_type = 0;
if (source_last == '[') md_scrap_type = 1;
if (source_last == '(') md_scrap_type = 2;
c = source_get();
while (1) {
    switch (c) {
        case '\n': fputs("\n", file);
                    indent = 0;
                    break;
        case '\t': < Expand md tab into spaces 90b >
                    break;
        default:
            if (c==nw_char)
            {
                < Check md at-sequence for end-of-scrap 90c >
                break;
            }
            if (md_scrap_type == 0 && prefix) {
                < HTML-escape character c 90a >
            } else {
                putc(c, file);
            }
            indent++;
            break;
    }
    c = source_get();
}
}

```

89b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_copy_scrap 82b, 87ac, 103a, 104b, md_scrap_type 90c · uses FILE 21b, fputs 21b, Name 152b, nw_char 28b, 29a, putc 21b, source_get 126a, source_last 126a

< HTML-escape character c 90a > ≡

```

switch (c) {
    case '<': fputs("<", file); break;
    case '>': fputs(">", file); break;
    case '&': fputs("&", file); break;
    default: putc(c, file); break;
}

```

90a · used in 89b · uses fputs 21b, putc 21b

< Expand md tab into spaces 90b > ≡

```

{
    int delta = 8 - (indent % 8);
    indent += delta;
    while (delta > 0) {
        putc(' ', file);
        delta--;
    }
}

```

90b · used in 89b · uses putc 21b

< Check md at-sequence for end-of-scrap 90c > ≡

```

{
    c = source_get();
    switch (c) {

```

```

case 'c': < Show md presence of a block comment 91a >
    break;
case 'x': < Copy label from source into(90d file ) 73a >
    break;
case 'v': < Copy version info into file 72b >
case 's':
    break;
case '+':
case '-':
case '*':
case '|': < Skip over index entries 73c >
case ',':
case ')':
case ']':
case '}': return;
case '<': < Format md macro name 92a >
    break;
case '%': < Skip commented-out code 73d >
    break;
case '_': < Bold md Keyword 91b >
    break;
case 't': < Italic md "fragment title" 91c >
    break;
case 'f': < Italic md "file name" 91c >
    break;
case '1': case '2': case '3':
case '4': case '5': case '6':
case '7': case '8': case '9':
    if (name == NULL
        || name->arg[c - '1'] == NULL) {
        if (md_scrap_type == 0 && prefix)
            fputs("&", file);
        else
            putc(nw_char, file);
        fputc(c, file);
    }
    else {
        md_write_arg(file, name->arg[c - '1']);
    }
    break;
default:
    if (c==nw_char)
    {
        if (md_scrap_type == 0 && prefix)
            fputs("&", file);
        else
            putc(nw_char, file);
        break;
    }
    /* ignore these since pass1 will have warned about them */
    break;
}
}

```

90d · used in [89b](#)

```

< Show md presence of a block comment 91a > ≡
{
    fputs("<i>(Comment)</i>", file);
}

```

```
}
```

91a · used in 90c · uses fputs 21b

⟨ *Bold md Keyword* 91b ⟩ ≡

```
{
    fprintf(file, "<b>");
    c = source_get();
    do {
        if (c == '<') fputs("&lt;", file);
        else if (c == '>') fputs("&gt;", file);
        else if (c == '&') fputs("&amp;", file);
        else fputc(c, file);
        c = source_get();
    } while (c != nw_char);
    c = source_get();
    fprintf(file, "</b>");
}
```

91b · used in 90c · uses fprintf 21b, fputs 21b, nw_char 28b, 29a, source_get 126a

⟨ *Italic md "whatever"* 91c ⟩ ≡

```
{
    fprintf(file, "<i>whatever</i>");
}
```

91c · used in 90c · uses fprintf 21b

⟨ *Format md macro name* 92a ⟩ ≡

```
{
    Arglist *args = collect_scrap_name(-1);
    Name *name = args->name;
    char *p = name->spelling;
    Arglist *q = args->args;
    int narg = 0;

    fputs("&lt;<a href=\"\#weft", file);
    if (name->defs) {
        write_single_scrap_ref(file, name->defs->scrap);
    } else {
        putc('?', file);
    }
    fputs("\">><i>", file);
    while (*p != '\000') {
        if (*p == ARG_CHR) {
            if (q == NULL) {
                md_write_literal(file, name->arg[narg]);
            }
            else {
                md_write_ArglistElement(file, q, params);
                q = q->next;
            }
            p++;
            narg++;
        }
        else {
            if (*p == '<') fputs("&lt;", file);
            else if (*p == '>') fputs("&gt;", file);
            else if (*p == '&') fputs("&amp;", file);
            else fputc(*p, file);
        }
    }
}
```

```

        p++;
    }
}
fputs("</i> ", file);
if (name->defs) {
    Scrap_Node *dp = name->defs;
    write_single_scrap_ref(file, dp->scrap);
    dp = dp->next;
    if (dp)
        fputs(" ...", file);
}
else {
    putc('?', file);
    fprintf(stderr, "%s: never defined <%s>\n",
            command_name, name->spelling);
}
fputs("</a>&gt;", file);
}

```

92a · used in 90c · uses Arglist 172d, ARG_CHR 171b, collect_scrap_name 173, command_name 29b, defs 129c, fprintf 21b, fputs 21b, md_write_ArglistElement 93, md_write_literal 92b, Name 152b, next 129b, putc 21b, Scrap_Node 152a, stderr 21b, write_single_scrap_ref 131c

⟨markdown.c 92b⟩ + ≡

```

static void md_write_literal(FILE * md_file, char * p)
{
    while (*p!= '\000') {
        if (*p == '<') fputs("&lt;", md_file);
        else if (*p == '>') fputs("&gt;", md_file);
        else if (*p == '&') fputs("&amp;", md_file);
        else fputc(*p, md_file);
        p++;
    }
}

```

92b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_write_literal 82c, 92a, 93 · uses FILE 21b, fputs 21b

⟨markdown.c 93⟩ + ≡

```

static void
md_write_ArglistElement(FILE * file, Arglist * args, char ** params)
{
    Name *name = args->name;
    Arglist *q = args->args;

    if (name == NULL) {
        char * p = (char*)q;

        if (p[0] == ARG_CHR) {
            md_write_arg(file, params[p[1] - '1']);
        } else {
            md_write_literal(file, (char *)q);
        }
    } else if (name == (Name *)1) {
        Scrap_Node * qq = (Scrap_Node *)q;
        qq->quoted = TRUE;
        write_scraps(file, "", qq,
                    -1, "", 0, 0, 0, 0, 0,
                    NULL, params, 0, "");
    }
}

```

```

        extra_scraps++;
        qq->quoted = FALSE;
    } else {
        char * p = name->spelling;
        fputs("<i>", file);
        while (*p != '\000') {
            if (*p == ARG_CHR) {
                md_write_ArglistElement(file, q, params);
                q = q->next;
                p++;
            }
            else {
                if (*p == '<') fputs("<i>", file);
                else if (*p == '>') fputs(">", file);
                else if (*p == '&') fputs("&", file);
                else fputc(*p, file);
                p++;
            }
        }
        fputs("</i> ", file);
        if (name->defs) {
            Scrap_Node *dp = name->defs;
            write_single_scrap_ref(file, dp->scrap);
            dp = dp->next;
            if (dp)
                fputs(" ...", file);
        }
        else {
            putc('?', file);
            fprintf(stderr, "%s: never defined <s>\n",
                    command_name, name->spelling);
        }
        fputs(">", file);
    }
}

```

93 · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines `md_write_ArglistElement` 82c, 92a · uses `Arglist` 172d, `ARG_CHR` 171b, `command_name` 29b, `defs` 129c, `extra_scraps` 65b, `FALSE` 23a, `FILE` 21b, `fprintf` 21b, `fputs` 21b, `md_write_arg` 86c, `md_write_literal` 92b, `Name` 152b, `next` 129b, `putc` 21b, `Scrap_Node` 152a, `stderr` 21b, `TRUE` 23a, `write_scraps` 142b, `write_single_scrap_ref` 131c

2.5.2 Generating the Indices in Markdown

⟨ *Write md index of file names* 94a ⟩ ≡

```

{
    if (file_names) {
        fputs("\n", md_file);
        md_format_file_entry(file_names, md_file);
        fputs("\n", md_file);
    }
    c = source_get();
}

```

94a · used in 84a · uses `file_names` 152c, `fputs` 21b, `md_format_file_entry` 94b, `source_get` 126a

⟨ *markdown.c* 94b ⟩ + ≡

```

static void md_format_file_entry(Name *name, FILE *md_file)
{
    while (name) {

```

```

    md_format_file_entry(name->llink, md_file);
    < Format a md file index entry 94c >
    name = name->rlink;
}
}

```

94b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines `md_format_file_entry` 82c, 94a · uses `FILE` 21b, `Name` 152b

```

< Format a md file index entry 94c > ≡
fputs("- ", md_file);
fprintf(md_file, "\"%s\" ", name->spelling);
< Write md file's defining scrap numbers 94d >
putc('\n', md_file);

```

94c · used in 94b · uses `fprintf` 21b, `fputs` 21b, `putc` 21b

```

< Write md file's defining scrap numbers 94d > ≡
{
    Scrap_Node *p = name->defs;
    fputs("*Defined by* ", md_file);
    if (p->next) {
        md_print_scrap_numbers(md_file, p);
    }
    else {
        fputs("<a href=\"#weft\"", md_file);
        write_single_scrap_ref(md_file, p->scrap);
        fputs(">", md_file);
        write_single_scrap_ref(md_file, p->scrap);
        fputs("</a>.", md_file);
    }
}

```

94d · used in 94c · uses `defs` 129c, `fputs` 21b, `md_print_scrap_numbers` 89a, `next` 129b, `Scrap_Node` 152a, `write_single_scrap_ref` 131c

```

< Write md index of macro names 95a > ≡
{
    unsigned char sector = current_sector;
    int c = source_get();
    if (c == '+')
        sector = 0;
    else
        source_ungetc(&c);
    if (has_sector(macro_names, sector)) {
        fputs("\n", md_file);
        md_format_entry(macro_names, md_file, sector);
        fputs("\n", md_file);
    } else {
        fputs("None.\n", md_file);
    }
}
c = source_get();

```

95a · used in 84a · uses `current_sector` 45c, `fputs` 21b, `has_sector` 80b, `macro_names` 152c, `md_format_entry` 95c, `sector` 129c, `source_get` 126a, `source_ungetc` 126c

```

< markdown.c 95b > + ≡
static int md_load_entry(Name * name, Name ** nms, int n)
{

```

```

    while (name) {
        n = md_load_entry(name->llink, nms, n);
        nms[n++] = name;
        name = name->rlink;
    }
    return n;
}

```

95b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_load_entry 95c · uses Name 152b

⟨ markdown.c 95c ⟩ + ≡

```

static void md_format_entry(Name *name, FILE *md_file, unsigned char sector)
{
    Name **nms = malloc(num_scraps()*sizeof(Name *));
    int n = md_load_entry(name, nms, 0);
    int i;

    ⟨ Sort nms of size n for ⟨ Rob's ordering 78c ⟩ 78d ⟩
    for (i = 0; i < n; i++)
    {
        Name * name = nms[i];

        ⟨ Format a md index entry 95d ⟩
    }
    free(nms);
}

```

95c · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_format_entry 82c, 95a · uses FILE 21b, malloc 21b, md_load_entry 95b, Name 152b, num_scraps 130a, sector 129c

⟨ Format a md index entry 95d ⟩ ≡

```

if (name->sector == sector){
    fputs("-", md_file);
    fputs("<*", md_file);
    ⟨ Write the md macro's name 86b ⟩
    fputs("* ", md_file);
    ⟨ Write md defining scrap numbers 96a ⟩
    fputs("> ", md_file);
    ⟨ Write md referencing scrap numbers 96b ⟩
    putc('\n', md_file);
}

```

95d · used in 95c · uses fputs 21b, putc 21b, sector 129c

⟨ Write md defining scrap numbers 96a ⟩ ≡

```

{
    Scrap_Node *p = name->defs;
    if (p) {
        int page;
        fputs("<a href=\"#weft\"", md_file);
        write_scrap_ref(md_file, p->scrap, -1, &page);
        fputs(">", md_file);
        write_scrap_ref(md_file, p->scrap, TRUE, &page);
        fputs("</a>", md_file);
        p = p->next;
        while (p) {
            fputs("<a href=\"#weft\"", md_file);
            write_scrap_ref(md_file, p->scrap, -1, &page);
            fputs(">", md_file);
            write_scrap_ref(md_file, p->scrap, FALSE, &page);
        }
    }
}

```

```

        fputs("</a>", md_file);
        p = p->next;
    }
}
else
    putc('?', md_file);
}

```

96a · used in 95d · uses defs 129c, FALSE 23a, fputs 21b, next 129b, page 129c, putc 21b, Scrap_Node 152a, TRUE 23a, write_scrap_ref 131b

⟨ *Write md referencing scrap numbers* 96b ⟩ ≡

```

{
    Scrap_Node *p = name->uses;
    if (p) {
        fputs("*Referenced in* ", md_file);
        if (p->next) {
            md_print_scrap_numbers(md_file, p);
        }
        else {
            fputs("<a href=\"#weft\"", md_file);
            write_single_scrap_ref(md_file, p->scrap);
            fputs("\">", md_file);
            write_single_scrap_ref(md_file, p->scrap);
            fputs("</a>.", md_file);
        }
    }
    else
        fputs("*Not referenced*.", md_file);
}

```

96b · used in 95d · uses fputs 21b, md_print_scrap_numbers 89a, next 129b, Scrap_Node 152a, uses 129c, write_single_scrap_ref 131c

⟨ *Write md index of user-specified names* 97a ⟩ ≡

```

{
    unsigned char sector = current_sector;
    c = source_get();
    if (c == '+') {
        sector = 0;
        c = source_get();
    }
    if (has_sector(user_names, sector)) {
        fputs("\n", md_file);
        md_format_user_entry(user_names, md_file, sector);
        fputs("\n", md_file);
    }
}

```

97a · used in 84a · uses current_sector 45c, fputs 21b, has_sector 80b, md_format_user_entry 97b, sector 129c, source_get 126a, user_names 152c

⟨ *markdown.c* 97b ⟩ + ≡

```

static void md_format_user_entry(Name *name, FILE *md_file, unsigned char sector)
{
    while (name) {
        md_format_user_entry(name->llink, md_file, sector);
        ⟨ Format a md user index entry 97c ⟩
        name = name->rlink;
    }
}

```

}

97b · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines md_format_user_entry 82c, 97a ·
uses FILE 21b, Name 152b, sector 129c

⟨ *Format a md user index entry* 97c ⟩ ≡

```
if (name->sector == sector){
    Scrap_Node *uses = name->uses;
    if ( uses || dangling_flag ) {
        int page;
        Scrap_Node *defs = name->defs;
        fprintf(md_file, "- '%s': ", name->spelling);
        if (!uses) {
            fputs("**", md_file);
            fputs("<a href=\"#weft\"", md_file);
            write_single_scrap_ref(md_file, defs->scrap);
            fputs("\>", md_file);
            write_single_scrap_ref(md_file, defs->scrap);
            fputs("</a>**", md_file);
            page = -2;
            defs = defs->next;
        }
        else
            if (!defs || uses->scrap < defs->scrap) {
                fputs("<a href=\"#weft\"", md_file);
                write_scrap_ref(md_file, uses->scrap, -1, &page);
                fputs("\>", md_file);
                write_scrap_ref(md_file, uses->scrap, TRUE, &page);
                fputs("</a>", md_file);
                uses = uses->next;
            }
        else {
            if (defs->scrap == uses->scrap)
                uses = uses->next;
            fputs("**", md_file);

            fputs("<a href=\"#weft\"", md_file);
            write_single_scrap_ref(md_file, defs->scrap);
            fputs("\>", md_file);
            write_single_scrap_ref(md_file, defs->scrap);
            fputs("</a>**", md_file);
            page = -2;
            defs = defs->next;
        }
    }
    while (uses || defs) {
        if (uses && (!defs || uses->scrap < defs->scrap)) {
            fputs("<a href=\"#weft\"", md_file);
            write_scrap_ref(md_file, uses->scrap, -1, &page);
            fputs("\>", md_file);
            write_scrap_ref(md_file, uses->scrap, FALSE, &page);
            fputs("</a>", md_file);
            uses = uses->next;
        }
        else {
            if (uses && defs->scrap == uses->scrap)
                uses = uses->next;
            fputs(", **", md_file);

            fputs("<a href=\"#weft\"", md_file);
```

```

        write_single_scrap_ref(md_file, defs->scrap);
        fputs(">", md_file);
        write_single_scrap_ref(md_file, defs->scrap);
        fputs("</a>**", md_file);
        page = -2;
        defs = defs->next;
    }
}
fputs(".\n", md_file);
}

```

97c · used in 97b · uses [dangling_flag](#) 27, [defs](#) 129c, [FALSE](#) 23a, [fprintf](#) 21b, [fputs](#) 21b, [next](#) 129b, [page](#) 129c, [Scrap_Node](#) 152a, [sector](#) 129c, [TRUE](#) 23a, [uses](#) 129c, [write_scrap_ref](#) 131b, [write_single_scrap_ref](#) 131c

2.5.3 Identifier Cross-References in Markdown

These functions are the Markdown equivalents of `format_uses_refs` and `format_defs_refs` from `scraps.c`, but emit Markdown blockquote format.

[markdown.c](#) 98a) + ≡

```

static void
md_format_uses_refs(FILE * md_file, int scrap)
{
    Uses * p = get_scrap_uses(scrap);
    if (p != NULL)
        Write md uses references 98b)
}

```

98a · defined at [24b](#), [82cd](#), [86c](#), [89ab](#), [92b](#), [93](#), [94b](#), [95bc](#), [97b](#), [98a](#), [99c](#) · defines [md_format_uses_refs](#) 82c, [87d](#), [88a](#) · uses [FILE](#) 21b, [get_scrap_uses](#) 131d, [Uses](#) 184d

[Write md uses references](#) 98b) ≡

```

{
    char join = ' ';
    fputs("> *Uses:*", md_file);
    do {
        Write one md use reference 99a)
        join = ',';
        p = p->next;
    }while (p != NULL);
    fputs(".\n", md_file);
}

```

98b · used in 98a · uses [fputs](#) 21b, [next](#) 129b, [Uses](#) 184d

[Write one md use reference](#) 99a) ≡

```

Name * name = p->defn;
Scrap_Node *defs = name->defs;
int first = TRUE, page = -1;
fprintf(md_file,
        "%c '%s' ",
        join, name->spelling);
if (defs)
{
    do {
        Write one md referenced scrap 99b)
        first = FALSE;
        defs = defs->next;
    }
}

```

```

    }while (defs!= NULL);
}
else
{
    fputs("(not defined globally)", md_file);
}

```

99a · used in 98b · uses [defs 129c](#), [FALSE 23a](#), [first 190a](#), [fprintf 21b](#), [fputs 21b](#), [Name 152b](#), [next 129b](#), [page 129c](#), [Scrap_Node 152a](#), [TRUE 23a](#)

⟨ *Write one md referenced scrap* 99b ⟩ ≡

```

fputs("<a href=\"#weft\"", md_file);
write_scrap_ref(md_file, defs->scrap, -1, &page);
fputs(">", md_file);
write_scrap_ref(md_file, defs->scrap, first, &page);
fputs("</a>", md_file);

```

99b · used in 99a, 100a · uses [defs 129c](#), [first 190a](#), [fputs 21b](#), [page 129c](#), [write_scrap_ref 131b](#)

⟨ [markdown.c](#) 99c ⟩ + ≡

```

static void
md_format_defs_refs(FILE * md_file, int scrap)
{
    Uses * p = get_scrap_defs(scrap);
    if (p != NULL)
        ⟨ Write md defs references 99d ⟩
}

```

99c · defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c · defines [md_format_defs_refs 82c](#), 87d, 88a · uses [FILE 21b](#), [get_scrap_defs 131e](#), [Uses 184d](#)

⟨ *Write md defs references* 99d ⟩ ≡

```

{
    char join = ' ';
    fputs("> *Defines:* ", md_file);
    do {
        ⟨ Write one md def reference 100a ⟩
        join = ',';
        p = p->next;
    }while (p != NULL);
    fputs(".\n", md_file);
}

```

99d · used in 99c · uses [fputs 21b](#), [next 129b](#)

⟨ *Write one md def reference* 100a ⟩ ≡

```

Name * name = p->defn;
Scrap_Node *defs = name->uses;
int first = TRUE, page = -1;
fprintf(md_file,
        "%c '%s' ",
        join, name->spelling);
if (defs == NULL
    || (defs->scrap == scrap && defs->next == NULL)) {
    fputs("*Never used*", md_file);
}
else {
    do {
        if (defs->scrap != scrap) {

```

```

    < Write one md referenced scrap 99b >
    first = FALSE;
}
defs = defs->next;
}while (defs!= NULL);
}
100a · used in 99d · uses defs 129c, FALSE 23a, first 190a, fprintf 21b, fputs 21b, Name 152b, next 129b, page 129c,
Scrap_Node 152a, TRUE 23a, uses 129c

```

2.6 Writing the HTML File

This section implements the native HTML weave, activated by `-w html` (or `@W html`). It produces a complete, self-contained `.html` document — doctype, an embedded style sheet, and body — so a literate web becomes a browsable page with clickable cross-references and no external tooling.

It is a close sibling of the Markdown weave (Section 2.5): the two share the parts that are already HTML — scrap bodies, inline fragment references, and the `<a href="#weftN">` cross-reference links — through the reused `md_copy_scrap` and `md_print_scrap_numbers`. What this file adds is the document wrapper, the HTML scrap headers (with the `≡/+≡` markers), the one-line metadata footers, and the indices. The prose is copied verbatim: an HTML weave expects HTML prose, just as the $\text{\LaTeX}$ weave expects $\text{\LaTeX}$ and the Markdown weave expects Markdown.

```

< Function prototypes 100b > + ≡
extern void write_html(char *file_name, char *html_name, unsigned char sector);
100b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b,
184e, 185d, 187d, 189d · used in 21a · uses file_name 129c, sector 129c, write_html 101

```

2.6.1 HTML escaping helpers

A handful of small helpers escape the three HTML-special characters and render a fragment name (expanding embedded parameters to their default arguments) and a linked fragment reference.

```

< html.c 100c > + ≡
static void html_esc(FILE *f, int c)
{
    if (c == '<') fputs("&lt;", f);
    else if (c == '>') fputs("&gt;", f);
    else if (c == '&') fputs("&amp;", f);
    else putc(c, f);
}

static void html_puts(FILE *f, const char *s)
{
    while (s && *s)
        html_esc(f, (unsigned char)*s++);
}

static void html_emit_name(FILE *f, Name *name)
{
    char *p = name->spelling;
    int i = 0;
    while (*p != '\0') {
        if (*p == ARG_CHR) { html_puts(f, name->arg[i++]); p++; }
        else html_esc(f, (unsigned char)*p++);
    }
}

```

```
static void html_ref(FILE *f, Name *name)
{
    fputs("<a href=\"#weft\", f);
    if (name->defs) write_single_scrap_ref(f, name->defs->scrap);
    else putc('?', f);
    fputs("><i>", f);
    html_emit_name(f, name);
    fputs("</i> ", f);
    if (name->defs) {
        write_single_scrap_ref(f, name->defs->scrap);
        if (name->defs->next) fputs(", ...", f);
    } else putc('?', f);
    fputs("</a>&gt;", f);
}
```

100c · defined at 24c, 100c, 101, 105c, 106ac · defines `html_emit_name` 104a, 106a, `html_esc` never used, `html_puts` 103c, 105c, 106c, `html_ref` 103a · uses `ARG_CHR` 171b, `defs` 129c, `FILE` 21b, `fputs` 21b, `Name` 152b, `next` 129b, `putc` 21b, `write_single_scrap_ref` 131c

2.6.2 The driver

```
<html.c 101> + ≡
static void html_format_file_entry(Name *name, FILE *f);
static void html_format_entry(Name *name, FILE *f, unsigned char sector);
static void html_format_user_entry(Name *name, FILE *f, unsigned char sector);

void write_html(char *file_name, char *html_name, unsigned char sector)
{
    (void)sector;
    FILE *html_file = fopen(html_name, "w");
    if (html_file) {
        if (verbose_flag)
            fprintf(stderr, "writing %s\n", html_name);
        < Write the HTML document head 102a >
        source_open(file_name);
        < Copy source into html_file 102c >
        < Write the HTML document tail 102b >
        fclose(html_file);
    }
    else
        fprintf(stderr, "%s: can't open %s\n", command_name, html_name);
}
```

101 · defined at 24c, 100c, 101, 105c, 106ac · defines `write_html` 39b, 100b · uses `command_name` 29b, `fclose` 21b, `FILE` 21b, `file_name` 129c, `fopen` 21b, `fprintf` 21b, `html_format_entry` 106a, `html_format_file_entry` 105c, `html_format_user_entry` 106c, `Name` 152b, `sector` 129c, `source_open` 129a, `stderr` 21b, `verbose_flag` 27

The document skeleton carries an embedded style sheet so the page needs no external files. The look mirrors the modern L^AT_EX weave: grey angle-bracketed fragment names in an accent colour, soft code blocks, accent links, and a small grey metadata line — and it adapts to the reader’s light or dark theme.

```
< Write the HTML document head 102a > ≡
fputs(
"<!doctype html>\n<html lang=\"en\">\n<head>\n<meta charset=\"utf-8\">\n"
"<meta name=\"viewport\" content=\"width=device-width,initial-scale=1\">\n"
"<title>weft</title>\n<style>\n"
":root{--acc:#2f6feb;--kw:#0550ae;--com:#6e7781;--bg:#f6f8fa;--rule:#d0d7de}\n"
"body{max-width:820px;margin:2rem auto;padding:0 1.2rem;color:#1f2328;\n"
```

```
" font:16px/1.6 -apple-system,Segoe UI,Roboto,Helvetica,Arial,sans-serif}\n"
"h1,h2,h3{font-weight:700;line-height:1.25;margin:1.8rem 0 .6rem}\n"
"a{color:var(--acc);text-decoration:none} a:hover{text-decoration:underline}\n"
".weft-name{margin:1.4rem 0 .1rem;font-size:1.02rem}\n"
".weft-name .b{color:var(--com)} .weft-name i{color:var(--acc);font-style:italic}\n"
".weft-name .f{color:var(--acc);font-family:ui-monospace,Menlo,Consolas,monospace}\n"
".weft-name .n{color:var(--com);font-size:.85em} .weft-name .eq{color:var(--com)}\n"


```
{background:var(--bg);border:1px solid var(--rule);border-radius:8px;\n"
" padding:.8rem 1rem;overflow:auto;margin:.2rem 0}\n"
pre code{font:0.86em/1.45 ui-monospace,SFMono-Regular,Menlo,Consolas,monospace}\n"
pre code i{color:var(--acc);font-style:italic} pre code a{color:var(--acc)}\n"
".weft-meta{color:var(--com);font-size:.82rem;margin:.15rem 0 .6rem}\n"
".weft-meta a{color:var(--acc)} .weft-meta code{color:var(--kw)}\n"
"code{background:var(--bg);padding:.1em .35em;border-radius:5px;\n"
" font:0.9em ui-monospace,Menlo,Consolas,monospace}\n"
"@media(prefers-color-scheme:dark){body{background:#0d1117;color:#e6edf3}\n"
":root{--bg:#161b22;--rule:#30363d;--com:#8b949e}}\n"
"</style>\n</head>\n<body>\n", html_file);
```


```

102a · used in 101 · uses [fputs 21b](#), [root 177c](#)

⟨ *Write the HTML document tail* 102b ⟩ ≡

```
fputs("</body>\n</html>\n", html_file);
```

102b · used in 101 · uses [fputs 21b](#)

We make our second pass over the source, copying prose straight through (it is already HTML) and acting on at-sequences.

⟨ *Copy source into html_file* 102c ⟩ ≡

```
{
  int c = source_get();
  while (c != EOF) {
    if (c == nw_char)
    {
      ⟨ Interpret html at-sequence 103a ⟩
    }
    else {
      putc(c, html_file);
      c = source_get();
    }
  }
}
```

102c · used in 101 · uses [nw_char 28b](#), [29a](#), [putc 21b](#), [source_get 126a](#)

⟨ *Interpret html at-sequence* 103a ⟩ ≡

```
{
  int big_definition = FALSE;
  c = source_get();
  switch (c) {
    case 'r': c = source_get(); nw_char = c; update_delimit_scrap(); break;
    case '0': big_definition = TRUE; FALLTHROUGH;
    case 'o': ⟨ Write html output file definition 103c ⟩ break;
    case 'Q':
    case 'D': big_definition = TRUE; FALLTHROUGH;
    case 'q':
    case 'd': ⟨ Write html macro definition 104a ⟩ break;
    case 's': ⟨ Step to next sector 45a ⟩ break;
    case 'S': ⟨ Close the current sector 45b ⟩ break;
```

```

case '{':
case '[':
case '(': md_copy_scrap(html_file, FALSE, NULL); c = source_get(); break;
case '<': { Arglist *a = collect_scrap_name(0);
          html_ref(html_file, a->name); c = source_get(); } break;
case 'x': < Copy label from source into(103b html_file ) 73a> c = source_get(); break;
case 'c': c = source_get(); break;
case 'f': < Write html index of file names 105b> break;
case 'm': < Write html index of macro names 105d> break;
case 'u': < Write html index of user names 106b> break;
case 'v': fputs(version_string, html_file); c = source_get(); break;
case 'l':
case 'L': < Skip at-L language definition line 44c> break;
case 'W': < Skip at-L language definition line 44c> break;
default:
    if (c==nw_char) putc(c, html_file);
    c = source_get();
    break;
}
}

```

103b · used in 102c

2.6.3 HTML scrap definitions

A file or fragment definition writes an anchor, a header line naming the scrap (with the definition/continuation marker), the code in a `<pre><code>` block, and a one-line metadata footer.

```

< Write html output file definition 103c> ≡
{
    Name *name = collect_file_name();
    int additive = (name->defs && name->defs->scrap != scraps);
    (void)big_definition;
    fputs("<div class=\"weft-name\"><a id=\"weft\", html_file);
    write_single_scrap_ref(html_file, scraps);
    fputs("\></a><span class=\"b\">&lt;</span> <span class=\"f\">", html_file);
    html_puts(html_file, name->spelling);
    fputs("</span> <span class=\"n\">", html_file);
    write_single_scrap_ref(html_file, scraps);
    fputs("</span> <span class=\"b\">&gt;</span> ", html_file);
    fputs(additive ? "<span class=\"eq\">+&equiv;</span>"
                : "<span class=\"eq\">&equiv;</span>", html_file);
    fputs("</div>\n", html_file);
    < Fill in the html scrap 104b>
    < Write html file footer 104d>
    < Finish html scrap 104c>
}

```

103c · used in 103a · uses `collect_file_name` 167a, `defs` 129c, `fputs` 21b, `html_puts` 100c, `Name` 152b, `scraps` 130a, `write_single_scrap_ref` 131c

```

< Write html macro definition 104a> ≡
{
    Name *name = collect_macro_name();
    int additive = (name->defs && name->defs->scrap != scraps);
    (void)big_definition;
    fputs("<div class=\"weft-name\"><a id=\"weft\", html_file);
    write_single_scrap_ref(html_file, scraps);
    fputs("\></a><span class=\"b\">&lt;</span> <i>", html_file);

```

```

html_emit_name(html_file, name);
fputs("</i> <span class=\"n\">", html_file);
write_single_scrap_ref(html_file, scraps);
fputs("</span> <span class=\"b\">&gt;</span> ", html_file);
fputs(additive ? "<span class=\"eq\">+&equiv;</span>"
        : "<span class=\"eq\">&equiv;</span>", html_file);
fputs("</div>\n", html_file);
< Fill in the html scrap 104b >
< Write html macro footer 105a >
< Finish html scrap 104c >
}

```

104a · used in 103a · uses [collect_macro_name 170a](#), [defs 129c](#), [fputs 21b](#), [html_emit_name 100c](#), [Name 152b](#), [scraps 130a](#), [write_single_scrap_ref 131c](#)

```

< Fill in the html scrap 104b > ≡
{
    if (name->lang[0] != '\0')
        fprintf(html_file, "<pre><code class=\"language-%s\">", name->lang);
    else
        fputs("<pre><code>", html_file);
    extra_scraps = 0;
    md_copy_scrap(html_file, TRUE, name);
    fputs("</code></pre>\n", html_file);
}

```

104b · used in 103c, 104a · uses [extra_scraps 65b](#), [fprintf 21b](#), [fputs 21b](#), [md_copy_scrap 89b](#), [TRUE 23a](#)

```

< Finish html scrap 104c > ≡
{
    scraps += extra_scraps;
    do c = source_get(); while (isspace(c));
}

```

104c · used in 103c, 104a · uses [extra_scraps 65b](#), [isspace 21b](#), [scraps 130a](#), [source_get 126a](#)

The footer is a single grey line: the scrap number, then where else the fragment is defined (for a multi-part definition) and where it is used, each a clickable link.

```

< Write html file footer 104d > ≡
{
    fputs("<div class=\"weft-meta\">", html_file);
    write_single_scrap_ref(html_file, scraps);
    if (scrap_flag && name->defs && name->defs->next) {
        fputs(" &middot; also at ", html_file);
        md_print_scrap_numbers(html_file, name->defs);
    }
    fputs("</div>\n", html_file);
    scraps++;
}

```

104d · used in 103c · uses [defs 129c](#), [fputs 21b](#), [md_print_scrap_numbers 89a](#), [next 129b](#), [scraps 130a](#), [scrap_flag 27](#), [write_single_scrap_ref 131c](#)

```

< Write html macro footer 105a > ≡
{
    fputs("<div class=\"weft-meta\">", html_file);
    write_single_scrap_ref(html_file, scraps);
    if (name->defs && name->defs->next) {
        fputs(" &middot; also at ", html_file);
    }
}

```

```

    md_print_scrap_numbers(html_file, name->defs);
}
if (name->uses) {
    fputs(" &middot; used in ", html_file);
    md_print_scrap_numbers(html_file, name->uses);
} else {
    fputs(" &middot; unreferenced", html_file);
}
fputs("</div>\n", html_file);
scraps++;
}

```

105a · used in 104a · uses [defs 129c](#), [fputs 21b](#), [md_print_scrap_numbers 89a](#), [next 129b](#), [scraps 130a](#), [uses 129c](#), [write_single_scrap_ref 131c](#)

2.6.4 HTML indices

The @f, @m, and @u indices become HTML lists with links to the defining and referencing scraps.

```

< Write html index of file names 105b > ≡
{
    if (file_names) {
        fputs("<ul class=\"weft-index\">\n", html_file);
        html_format_file_entry(file_names, html_file);
        fputs("</ul>\n", html_file);
    }
    c = source_get();
}

```

105b · used in 103a · uses [file_names 152c](#), [fputs 21b](#), [html_format_file_entry 105c](#), [source_get 126a](#)

```

< html.c 105c > + ≡
static void html_format_file_entry(Name *name, FILE *f)
{
    while (name) {
        html_format_file_entry(name->llink, f);
        fputs("<li><span class=\"f\">", f);
        html_puts(f, name->spelling);
        fputs("</span> ", f);
        if (name->defs) md_print_scrap_numbers(f, name->defs);
        fputs("</li>\n", f);
        name = name->rlink;
    }
}

```

105c · defined at 24c, 100c, 101, 105c, 106ac · defines [html_format_file_entry 101](#), 105b · uses [defs 129c](#), [FILE 21b](#), [fputs 21b](#), [html_puts 100c](#), [md_print_scrap_numbers 89a](#), [Name 152b](#)

```

< Write html index of macro names 105d > ≡
{
    unsigned char sector = current_sector;
    int cc = source_get();
    if (cc == '+') sector = 0; else source_ungetc(&cc);
    if (has_sector(macro_names, sector)) {
        fputs("<ul class=\"weft-index\">\n", html_file);
        html_format_entry(macro_names, html_file, sector);
        fputs("</ul>\n", html_file);
    }
    c = source_get();
}

```

105d · used in 103a · uses [current_sector](#) 45c, [fputs](#) 21b, [has_sector](#) 80b, [html_format_entry](#) 106a, [macro_names](#) 152c, [sector](#) 129c, [source_get](#) 126a, [source_ungetc](#) 126c

```
<html.c 106a> + ≡
static void html_format_entry(Name *name, FILE *f, unsigned char sector)
{
    while (name) {
        html_format_entry(name->llink, f, sector);
        if (name->sector == sector) {
            fputs("<li><span class=\"b\">&lt;</span> <i>", f);
            html_emit_name(f, name);
            fputs("</i> <span class=\"b\">&gt;</span> ", f);
            if (name->defs) md_print_scrap_numbers(f, name->defs);
            else putc('?', f);
            fputs("</li>\n", f);
        }
        name = name->rlink;
    }
}
```

106a · defined at 24c, 100c, 101, 105c, 106ac · defines [html_format_entry](#) 101, 105d · uses [defs](#) 129c, [FILE](#) 21b, [fputs](#) 21b, [html_emit_name](#) 100c, [md_print_scrap_numbers](#) 89a, [Name](#) 152b, [putc](#) 21b, [sector](#) 129c

```
<Write html index of user names 106b> ≡
{
    unsigned char sector = current_sector;
    c = source_get();
    if (c == '+') { sector = 0; c = source_get(); }
    if (has_sector(user_names, sector)) {
        fputs("<ul class=\"weft-index\">\n", html_file);
        html_format_user_entry(user_names, html_file, sector);
        fputs("</ul>\n", html_file);
    }
}
```

106b · used in 103a · uses [current_sector](#) 45c, [fputs](#) 21b, [has_sector](#) 80b, [html_format_user_entry](#) 106c, [sector](#) 129c, [source_get](#) 126a, [user_names](#) 152c

```
<html.c 106c> + ≡
static void html_format_user_entry(Name *name, FILE *f, unsigned char sector)
{
    while (name) {
        html_format_user_entry(name->llink, f, sector);
        if (name->sector == sector && (name->uses || dangling_flag)) {
            fputs("<li><code>", f);
            html_puts(f, name->spelling);
            fputs("</code>: ", f);
            if (name->uses) md_print_scrap_numbers(f, name->uses);
            fputs("</li>\n", f);
        }
        name = name->rlink;
    }
}
```

106c · defined at 24c, 100c, 101, 105c, 106ac · defines [html_format_user_entry](#) 101, 106b · uses [dangling_flag](#) 27, [FILE](#) 21b, [fputs](#) 21b, [html_puts](#) 100c, [md_print_scrap_numbers](#) 89a, [Name](#) 152b, [sector](#) 129c, [uses](#) 129c

2.7 Map and Extract Output

Two modes support AI-assisted navigation of weft projects:

- m (map) emits a JSON object to stdout describing every output file and every fragment, their definition locations, usage relationships, and transitive dependencies.
- e *name* (extract) emits the scrap code and dependency information for a single fragment (matched by prefix) as Markdown to stdout.

Both modes run after `pass1` has built the name trees and do *not* invoke the weave or tangle phases.

2.7.1 JSON Map Output

The `write_map` function performs an in-order traversal of the `file_names` and `macro_names` binary search trees, emitting a JSON object with two arrays: "files" and "fragments".

For each `Name` node, it records:

- The spelling (name of the file or fragment).
- The sector number.
- The language tag (if any).
- Each definition scrap with its source file and line.
- Each usage scrap (scraps that reference this name).
- For definition scraps, the list of fragments referenced inside (via `get_scrap_uses`).

A helper function escapes strings for JSON output.

```
<output.c 107> + ≡
static void json_puts(FILE *f, const char *s)
{
    fputc('\"', f);
    while (*s) {
        switch (*s) {
            case '\"': fputs("\\\"", f); break;
            case '\\': fputs("\\\\", f); break;
            case '\n': fputs("\\n", f); break;
            case '\r': fputs("\\r", f); break;
            case '\t': fputs("\\t", f); break;
            default:
                if ((unsigned char)*s < 0x20)
                    fprintf(f, "\\u%04x", (unsigned char)*s);
                else
                    fputc(*s, f);
        }
        s++;
    }
    fputc('\"', f);
}
```

107 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `json_puts` 108, 117 · uses `FILE` 21b, `fprintf` 21b, `fputs` 21b

The `emit_name_json` helper emits a single name entry as a JSON object. The `is_file` flag distinguishes file names (which use "uses_fragments" for fragments referenced inside their definition scraps) from macro fragments (which use "used_by" for the scraps that invoke them, and "uses" for fragments they invoke).

```

<output.c 108> + ≡
static void emit_name_json(FILE *f, Name *name, int is_file, int *first)
{
    Scrap_Node *sn;
    int first_inner;

    if (!name) return;
    emit_name_json(f, name->llink, is_file, first);

    if (!*first) fputs("\n", f);
    *first = 0;

    fputs("    {\n", f);
    fputs("        \"name\": ", f); json_puts(f, name->spelling);
    if (name->lang[0]) {
        fputs(",\n        \"lang\": ", f); json_puts(f, name->lang);
    }
    fprintf(f, ",\n        \"sector\": %d", name->sector);

    /* defs array */
    fputs(",\n        \"defs\": [", f);
    first_inner = 1;
    for (sn = name->defs; sn; sn = sn->next) {
        if (!first_inner) fputs(", ", f);
        first_inner = 0;
        fprintf(f, "{\"scrap\": %d, \"file\": ", sn->scrap);
        json_puts(f, scrap_file_name(sn->scrap));
        fprintf(f, ", \"line\": %d, \"end_line\": %d",
                scrap_file_line(sn->scrap), scrap_end_line(sn->scrap));
        if (bodies_flag) {
            fputs(", \"body\": ", f);
            dump_scrap_json(f, sn->scrap);
        }
        if (prose_flag && scrap_prose(sn->scrap)) {
            fputs(", \"prose\": ", f);
            json_puts(f, scrap_prose(sn->scrap));
        }
        fputc('}', f);
    }
    fputs("]", f);

    if (is_file) {
        /* uses_fragments: fragments referenced inside this file's scraps */
        fputs(",\n        \"uses_fragments\": [", f);
        first_inner = 1;
        for (sn = name->defs; sn; sn = sn->next) {
            Uses *u;
            for (u = get_scrap_uses(sn->scrap); u; u = u->next) {
                if (!first_inner) fputs(", ", f);
                first_inner = 0;
                json_puts(f, u->defn->spelling);
            }
        }
        fputs("]", f);
    } else {
        /* used_by: scraps that invoke this fragment */
        fputs(",\n        \"used_by\": [", f);
        first_inner = 1;
        for (sn = name->uses; sn; sn = sn->next) {

```

```

    if (!first_inner) fputs(" ", f);
    first_inner = 0;
    fprintf(f, "{\"scrap\": %d, \"file\": ", sn->scrap);
    json_puts(f, scrap_file_name(sn->scrap));
    fprintf(f, ", \"line\": %d}", scrap_file_line(sn->scrap));
}
fputs("]", f);

/* uses: fragments referenced inside this fragment's def scraps */
fputs("\n    \"uses\": [", f);
first_inner = 1;
for (sn = name->defs; sn; sn = sn->next) {
    Uses *u;
    for (u = get_scrap_uses(sn->scrap); u; u = u->next) {
        if (!first_inner) fputs(" ", f);
        first_inner = 0;
        json_puts(f, u->defn->spelling);
    }
}
fputs("]", f);

fputs("\n    ]", f);

emit_name_json(f, name->rlink, is_file, first);
}

108 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines emit_name_json 109 · uses bodies_flag 27, defs 129c, dump_scrap_json 139, end_line 129c, FILE 21b, first 190a,
fprintf 21b, fputs 21b, get_scrap_uses 131d, json_puts 107, Name 152b, next 129b, prose 129c, prose_flag 27,
scraps 130a, scrap_end_line 132d, scrap_file_line 132c, scrap_file_name 132b, Scrap_Node 152a, scrap_prose 132e,
sector 129c, Uses 184d, uses 129c

⟨output.c 109⟩ + ≡
void write_map(Name *files, Name *macros)
{
    int first;

    fputs("{\n    \"files\": [\n", stdout);
    first = 1;
    emit_name_json(stdout, files, 1, &first);
    fputs("\n ],\n    \"fragments\": [\n", stdout);
    first = 1;
    emit_name_json(stdout, macros, 0, &first);
    fputs("\n ]\n}\n", stdout);
}

109 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines write_map 39b, 121b · uses emit_name_json 108, first 190a, fputs 21b, Name 152b

```

2.7.2 Fragment Extraction

The `write_extract` function finds a fragment by name (using prefix matching, like `@<name...@>` in the source), then emits the scrap code and dependency information as Markdown. It recurses into referenced fragments to provide the full transitive closure of the requested fragment.

To find a name in the BST without creating a new entry, we use a read-only search function:

⟨output.c 110a⟩ + ≡

```

static Name *find_name(Name *node, const char *spelling)
{
    Name *found;
    if (!node) return NULL;
    if (robs_strcmp(node->spelling, (char *)spelling) == 0)
        return node;
    found = find_name(node->llink, spelling);
    if (found) return found;
    return find_name(node->rlink, spelling);
}

```

110a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `find_name` 112ab · uses `Name` 152b, `robs_strcmp` 155c

A recursive helper emits one fragment and all its transitive dependencies. A visited array (indexed by Name pointer cast to a simple mark) prevents infinite recursion on cyclic references.

<output.c 110b> + ≡

```

static void extract_fragment(FILE *f, Name *name, Name *files, Name *macros)
{
    Scrap_Node *sn;

    if (!name || name->mark) return;
    name->mark = TRUE;

    for (sn = name->defs; sn; sn = sn->next) {
        fprintf(f, "\n## \"%s\" --- %s:%d\n\n",
            name->spelling,
            scrap_file_name(sn->scrap),
            scrap_file_line(sn->scrap));
        if (name->lang[0])
            fprintf(f, "'\"%s\n", name->lang);
        else
            fputs("'\n", f);
        dump_scrap_text(f, sn->scrap);
        fputs("\n'\n", f);

        /* Show dependencies */
        {
            Uses *u = get_scrap_uses(sn->scrap);
            if (u) {
                fputs("\n### Uses\n", f);
                for (; u; u = u->next)
                    fprintf(f, "- %s\n", u->defn->spelling);
            }
        }

        /* Show used_by */
        if (name->uses) {
            Scrap_Node *usn;
            fputs("\n### Used by\n", f);
            for (usn = name->uses; usn; usn = usn->next) {
                fprintf(f, "- scrap %d (%s:%d)\n",
                    usn->scrap,
                    scrap_file_name(usn->scrap),
                    scrap_file_line(usn->scrap));
            }
        }
    }
}

```

```

/* Recurse into dependencies */
for (sn = name->defs; sn; sn = sn->next) {
    Uses *u;
    for (u = get_scrap_uses(sn->scrap); u; u = u->next) {
        extract_fragment(f, u->defn, files, macros);
    }
}
}

```

110b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `extract_fragment` 112b · uses `defs` 129c, `dump_scrap_text` 138c, `FILE` 21b, `fprintf` 21b, `fputs` 21b,
`get_scrap_uses` 131d, `Name` 152b, `next` 129b, `scrap_file_line` 132c, `scrap_file_name` 132b, `Scrap_Node` 152a, `TRUE` 23a,
`Uses` 184d, `uses` 129c

Clear marks after extraction so the name trees remain usable:

```

<output.c 111a> + ≡
static void clear_marks(Name *node)
{
    if (!node) return;
    clear_marks(node->llink);
    node->mark = FALSE;
    clear_marks(node->rlink);
}

```

111a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `clear_marks` 112ab · uses `FALSE` 23a, `Name` 152b

2.7.3 Callers (Upward Call Chain)

The `--callers` feature, used with `-e`, traces the upward call chain from a given fragment back to its @o file roots. Each scrap records its owning `Name` (set in `pass1` when the scrap is attached to a definition list), so we can walk upward through the `uses` list to find all callers.

```

<output.c 111b> + ≡
static int is_file_name(Name *root, Name *target)
{
    if (!root) return 0;
    if (root == target) return 1;
    if (is_file_name(root->llink, target)) return 1;
    return is_file_name(root->rlink, target);
}

```

111b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `is_file_name` 111c · uses `Name` 152b, `root` 177c

```

<output.c 111c> + ≡
static void print_callers(Name *name, Name *file_names, int depth)
{
    Scrap_Node *usn;
    int i;

    if (!name || name->mark) return;
    name->mark = TRUE;

    for (i = 0; i < depth; i++) fputs("  ", stdout);
    if (is_file_name(file_names, name))
        printf("[file] %s\n", name->spelling);
}

```

```

else
    printf("%s\n", name->spelling);

for (usn = name->uses; usn; usn = usn->next) {
    Name *owner = scrap_owner(usn->scrap);
    if (owner && !owner->mark)
        print_callers(owner, file_names, depth + 1);
}
}
111c · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines print_callers 112a · uses file_names 152c, fputs 21b, is_file_name 111b, Name 152b, next 129b, owner 129c,
Scrap_Node 152a, scrap_owner 132f, TRUE 23a, uses 129c

```

```

<output.c 112a> + ≡
void write_callers(char *name, Name *files, Name *macros)
{
    Name *found = find_name(macros, name);
    if (!found)
        found = find_name(files, name);
    if (!found) {
        fprintf(stderr, "%s: fragment \"%s\" not found\n", command_name, name);
        exit(1);
    }
    printf("Callers of \"%s\":\n\n", found->spelling);
    printf("%s\n", found->spelling);
    {
        Scrap_Node *usn;
        for (usn = found->uses; usn; usn = usn->next) {
            Name *owner = scrap_owner(usn->scrap);
            if (owner)
                print_callers(owner, files, 1);
        }
    }
    clear_marks(files);
    clear_marks(macros);
}

```

112a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines write_callers 39b, 121b · uses clear_marks 111a, command_name 29b, exit 21b, find_name 110a, fprintf 21b,
Name 152b, next 129b, owner 129c, print_callers 111c, Scrap_Node 152a, scrap_owner 132f, stderr 21b, uses 129c

```

<output.c 112b> + ≡
void write_extract(char *name, Name *files, Name *macros)
{
    Name *found = NULL;

    /* Search in macro_names first, then file_names */
    found = find_name(macros, name);
    if (!found)
        found = find_name(files, name);

    if (!found) {
        fprintf(stderr, "%s: fragment \"%s\" not found\n", command_name, name);
        exit(1);
    }

    fprintf(stdout, "# Extract: \"%s\"\n", found->spelling);
    extract_fragment(stdout, found, files, macros);
}

```

```

/* Clean up marks */
clear_marks(files);
clear_marks(macros);
}

```

112b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `write_extract` 39b, 121b · uses `clear_marks` 111a, `command_name` 29b, `exit` 21b, `extract_fragment` 110b,
`file_names` 152c, `find_name` 110a, `first` 190a, `fprintf` 21b, `macro_names` 152c, `Name` 152b, `stderr` 21b

2.7.4 Lint (Static Analysis)

The `--lint` flag performs a post-pass1 static analysis of the fragment namespace. It reports three kinds of issues:

1. **Undefined references:** fragments that are used (`@<Name@>`) but never defined (`@d Name`).
2. **Unused fragments:** fragments defined but never referenced by any other scrap or output file.
3. **Fuzzy-match suggestions:** for undefined references, the closest defined name within Levenshtein distance 3.

`<output.c 113a> + ≡`

```

static int levenshtein(const char *a, const char *b)
{
    int la = (int)strlen(a), lb = (int)strlen(b);
    int prev[256], curr[256];
    int i, j;
    if (la > 255 || lb > 255) return 999;
    for (j = 0; j <= lb; j++) prev[j] = j;
    for (i = 1; i <= la; i++) {
        curr[0] = i;
        for (j = 1; j <= lb; j++) {
            int cost = (a[i-1] != b[j-1]) ? 1 : 0;
            int del = prev[j] + 1;
            int ins = curr[j-1] + 1;
            int sub = prev[j-1] + cost;
            curr[j] = del < ins ? (del < sub ? del : sub) : (ins < sub ? ins : sub);
        }
        for (j = 0; j <= lb; j++) prev[j] = curr[j];
    }
    return prev[lb];
}

```

113a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `levenshtein` 113b · uses `strlen` 21b

`<output.c 113b> + ≡`

```

#define MAX_LINT_NAMES 4096
static Name *lint_names[MAX_LINT_NAMES];
static int lint_name_count;

static void collect_lint_names(Name *node)
{
    if (!node) return;
    collect_lint_names(node->llink);
    if (node->defs && lint_name_count < MAX_LINT_NAMES)
        lint_names[lint_name_count++] = node;
    collect_lint_names(node->rlink);
}

```

```
}
```

```
static const char *find_closest(const char *target)
{
    int i, best_dist = 4;
    const char *best = NULL;
    for (i = 0; i < lint_name_count; i++) {
        int d = levenshtein(target, lint_names[i]->spelling);
        if (d < best_dist) {
            best_dist = d;
            best = lint_names[i]->spelling;
        }
    }
    return best;
}
```

113b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
 defines collect_lint_names 114b, find_closest 114a, lint_names never used, lint_name_count 114b, MAX_LINT_NAMES never
 used · uses defs 129c, levenshtein 113a, Name 152b

<output.c 114a> + ≡

```
static int lint_name(Name *node, int check_unused, int *issues)
{
    if (!node) return 0;
    lint_name(node->llink, check_unused, issues);

    if (node->defs == NULL && node->uses != NULL) {
        /* undefined reference */
        const char *suggestion = find_closest(node->spelling);
        fprintf(stderr, "lint: undefined fragment \"%s\"", node->spelling);
        if (node->uses) {
            fprintf(stderr, " (used at %s:%d)",
                    scrap_file_name(node->uses->scrap),
                    scrap_file_line(node->uses->scrap));
        }
        if (suggestion)
            fprintf(stderr, " -- did you mean \"%s\"?", suggestion);
        fputc('\n', stderr);
        (*issues)++;
    }
    else if (node->defs != NULL && node->uses == NULL && check_unused) {
        fprintf(stderr, "lint: unused fragment \"%s\" (defined at %s:%d)\n",
                node->spelling,
                scrap_file_name(node->defs->scrap),
                scrap_file_line(node->defs->scrap));
        (*issues)++;
    }

    lint_name(node->rlink, check_unused, issues);
    return 0;
}
```

114a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
 defines lint_name 114b · uses defs 129c, find_closest 113b, fprintf 21b, Name 152b, scrap_file_line 132c,
 scrap_file_name 132b, stderr 21b, uses 129c

<output.c 114b> + ≡

```
void write_lint(Name *file_names, Name *macro_names)
{
```

```

int issues = 0;
lint_name_count = 0;
collect_lint_names(file_names);
collect_lint_names(macro_names);

/* check files for undefined refs (not unused -- files are roots) */
lint_name(file_names, 0, &issues);
/* check macros for both undefined and unused */
lint_name(macro_names, 1, &issues);

if (issues)
    fprintf(stderr, "lint: %d issue(s) found\n", issues);
else
    fprintf(stderr, "lint: no issues found\n");
}

```

114b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `write_lint` 39b, 121b · uses `collect_lint_names` 113b, `file_names` 152c, `fprintf` 21b, `lint_name` 114a,
`lint_name_count` 113b, `macro_names` 152c, `Name` 152b, `stderr` 21b

2.7.5 Reverse Map Output

The reverse map feature provides programmatic navigation from tangled output back to the `.weft` source. Given a tangled output file (and optionally a line number), it parses the section markers embedded in that file and emits JSON mapping each region to its originating scrap, source file, and source line.

This operates entirely on the tangled output—no `.weft` files are needed and `pass1` is never invoked. The section markers have the form:

```

COMMENT_BEGIN{N: file:line}COMMENT_END      (opening)
COMMENT_BEGIN{:N}COMMENT_END                (closing)

```

where `COMMENT_BEGIN` and `COMMENT_END` vary by language (`/* ... */`, `// ...`, `# ...`, etc.).

A `MarkerFrame` records the data extracted from one opening marker. We maintain a stack of these to handle nested fragment expansions.

```

<output.c 115a> + ≡
typedef struct {
    int scrap;
    char file[FILENAME_MAX];
    int line;
    int start_line; /* line in the tangled file where this marker opened */
} MarkerFrame;

```

115a · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `MarkerFrame` 115b, 117, 119, 120

The `parse_open_marker` function scans a line for the pattern `{N: file:line}`. It returns 1 on success, filling in the supplied `MarkerFrame`. The scan is intentionally loose about what precedes the opening brace so that any comment style works.

```

<output.c 115b> + ≡
static int parse_open_marker(const char *line, MarkerFrame *mf)
{
    const char *p = line;
    while (*p) {
        if (*p == '{') {
            const char *start = p + 1;
            /* scrap number */

```

```

if (*start < '0' || *start > '9') { p++; continue; }
int scrap = 0;
while (*start >= '0' && *start <= '9')
    scrap = scrap * 10 + (*start++ - '0');
if (*start != ':' || *(start+1) != ')') { p++; continue; }
start += 2; /* skip ": " */
/* file:line -- find the closing brace, then work backwards for :line */
const char *end = strchr(start, '}');
if (!end) { p++; continue; }
/* find last ':' between start and end */
const char *colon = NULL;
for (const char *q = end - 1; q >= start; q--) {
    if (*q == ':') { colon = q; break; }
}
if (!colon) { p++; continue; }
/* digits after colon? */
int src_line = 0;
const char *d = colon + 1;
if (d == end) { p++; continue; }
while (d < end) {
    if (*d < '0' || *d > '9') break;
    src_line = src_line * 10 + (*d++ - '0');
}
if (d != end) { p++; continue; }
/* success */
mf->scrap = scrap;
mf->line = src_line;
{
    size_t len = (size_t)(colon - start);
    if (len >= FILENAME_MAX) len = FILENAME_MAX - 1;
    memcpy(mf->file, start, len);
    mf->file[len] = '\0';
}
return 1;
}
p++;
}
return 0;
}

```

115b · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `parse_open_marker` 117, 119 · uses `MarkerFrame` 115a, `size_t` 21b

The `parse_close_marker` function scans for `{:N}` and returns the scrap number, or `-1` on failure.

```

<output.c 116> + ≡
static int parse_close_marker(const char *line)
{
    const char *p = line;
    while (*p) {
        if (*p == '{' && *(p+1) == ':') {
            const char *start = p + 2;
            if (*start < '0' || *start > '9') { p++; continue; }
            int scrap = 0;
            while (*start >= '0' && *start <= '9')
                scrap = scrap * 10 + (*start++ - '0');
            if (*start == '}')
                return scrap;
        }
    }
}

```

```

    p++;
}
return -1;
}

```

116 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `parse_close_marker` 117, 119

The main function `write_reverse_map` parses its argument to split an optional line number from the filename, opens the tangled file, and reads it line by line. A stack of `MarkerFrame` entries tracks nested scrap regions.

In *line mode* (a specific line was requested), as soon as `current_line` reaches the target the innermost stack frame is emitted and the function returns.

In *full mode* (no line number), every time a close marker is encountered the region from the matching open to the current line is emitted.

```

<output.c 117> +≡
void write_reverse_map(const char *arg)
{
    char filename[FILENAME_MAX];
    int target_line = -1;
    FILE *fp;
    char buf[8192];
    MarkerFrame stack[256];
    int depth = 0;
    int current_line = 0;
    int first_region = 1;

    /* Parse arg: find last ':' where suffix is all digits */
    {
        const char *colon = NULL;
        const char *p;
        for (p = arg + strlen(arg) - 1; p > arg; p--) {
            if (*p == ':') { colon = p; break; }
        }
        if (colon) {
            const char *d = colon + 1;
            int all_digits = (*d != '\0');
            while (*d) {
                if (*d < '0' || *d > '9') { all_digits = 0; break; }
                d++;
            }
            if (all_digits) {
                target_line = atoi(colon + 1);
                snprintf(filename, FILENAME_MAX, "%.s", (int)(colon - arg), arg);
            } else {
                snprintf(filename, FILENAME_MAX, "%s", arg);
            }
        } else {
            snprintf(filename, FILENAME_MAX, "%s", arg);
        }
    }

    fp = fopen(filename, "r");
    if (!fp) {
        fprintf(stderr, "weft: cannot open %s: ", filename);
        perror(NULL);
        exit(1);
    }
}

```

```

if (target_line < 0) {
    /* Full mode header */
    fputs("{\"source\": ", stdout);
    json_puts(stdout, filename);
    fputs(", \"regions\": [\n", stdout);
}

while (fgets(buf, sizeof(buf), fp)) {
    MarkerFrame mf;
    int close_scrap;
    current_line++;

    if (parse_open_marker(buf, &mf)) {
        mf.start_line = current_line;
        if (depth < 256)
            stack[depth++] = mf;
        continue;
    }

    close_scrap = parse_close_marker(buf);
    if (close_scrap >= 0) {
        /* Find matching frame on stack */
        int i;
        for (i = depth - 1; i >= 0; i--) {
            if (stack[i].scrap == close_scrap) {
                if (target_line < 0) {
                    /* Full mode: emit region */
                    if (!first_region) fputs(",\n", stdout);
                    first_region = 0;
                    fprintf(stdout,
                        "  {\"start\": %d, \"end\": %d, \"scrap\": %d, \"file\": ",
                        stack[i].start_line + 1, current_line - 1, stack[i].scrap);
                    json_puts(stdout, stack[i].file);
                    fprintf(stdout, ", \"line\": %d}\n", stack[i].line);
                }
                /* Remove frame by shifting */
                depth--;
                for (; i < depth; i++)
                    stack[i] = stack[i + 1];
                break;
            }
        }
        continue;
    }

    /* Regular line -- check for line mode match */
    if (target_line > 0 && current_line == target_line) {
        if (depth > 0) {
            MarkerFrame *inner = &stack[depth - 1];
            fprintf(stdout, "{\"scrap\": %d, \"file\": ", inner->scrap);
            json_puts(stdout, inner->file);
            fprintf(stdout, ", \"line\": %d}\n", inner->line);
        } else {
            fputs("{}\n", stdout);
        }
        fclose(fp);
        return;
    }
}

```

```

}

fclose(fp);

if (target_line > 0) {
    /* Line was beyond end of file or not inside a scrap */
    fputs("{}\n", stdout);
} else {
    fputs("\n[]}\n", stdout);
}
}

```

117 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines write_reverse_map 36d, 121b · uses exit 21b, fclose 21b, FILE 21b, fopen 21b, fprintf 21b, fputs 21b,
json_puts 107, MarkerFrame 115a, parse_close_marker 116, parse_open_marker 115b, stack 125e, stderr 21b, strlen 21b

2.7.6 Error Annotator

The `--errors` flag reads compiler output from stdin and annotates lines matching `file:line:` patterns with the corresponding `.weft` source location. It works by opening each referenced file, parsing its section markers, and finding which scrap region contains the error line. Lines that don't match the pattern are passed through unchanged.

```

<output.c 119> + ≡
static int reverse_lookup_in_file(const char *filename, int target_line,
                                MarkerFrame *result)
{
    FILE *fp = fopen(filename, "r");
    char buf[8192];
    MarkerFrame stack[256];
    int depth = 0, current_line = 0;

    if (!fp) return 0;

    while (fgets(buf, sizeof(buf), fp)) {
        MarkerFrame mf;
        int close_scrap;
        current_line++;

        if (parse_open_marker(buf, &mf)) {
            mf.start_line = current_line;
            if (depth < 256)
                stack[depth++] = mf;
            continue;
        }

        close_scrap = parse_close_marker(buf);
        if (close_scrap >= 0) {
            int i;
            for (i = depth - 1; i >= 0; i--) {
                if (stack[i].scrap == close_scrap) {
                    depth--;
                    for (; i < depth; i++)
                        stack[i] = stack[i + 1];
                    break;
                }
            }
            continue;
        }
    }
}

```

```

    if (current_line == target_line && depth > 0) {
        *result = stack[depth - 1];
        fclose(fp);
        return 1;
    }
}
fclose(fp);
return 0;
}

```

119 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines `reverse_lookup_in_file` 120 · uses `fclose` 21b, `FILE` 21b, `fopen` 21b, `MarkerFrame` 115a, `parse_close_marker` 116,
`parse_open_marker` 115b, `stack` 125e

<output.c 120> + ≡

```

void write_error_annotator(void)
{
    char line[8192];

    while (fgets(line, sizeof(line), stdin)) {
        const char *p = line;
        char err_file[FILENAME_MAX];
        int err_line = -1;
        int parsed = 0;

        /* Try to parse file:digits: pattern */
        /* Skip optional drive letter on Windows (e.g. C:) */
        if (p[0] && p[1] == ':' && ((p[0] >= 'A' && p[0] <= 'Z') ||
            (p[0] >= 'a' && p[0] <= 'z'))) {
            p += 2;
        }
        {
            const char *colon1 = strchr(p, ':');
            if (colon1 && colon1 > p) {
                const char *d = colon1 + 1;
                int num = 0;
                int has_digits = 0;
                while (*d >= '0' && *d <= '9') {
                    num = num * 10 + (*d - '0');
                    has_digits = 1;
                    d++;
                }
                if (has_digits && *d == ':') {
                    size_t flen = (size_t)(colon1 - line);
                    if (flen < FILENAME_MAX) {
                        memcpy(err_file, line, flen);
                        err_file[flen] = '\0';
                        err_line = num;
                        parsed = 1;
                    }
                }
            }
        }

        fputs(line, stdout);

        if (parsed) {
            MarkerFrame mf;

```

```

    if (reverse_lookup_in_file(err_file, err_line, &mf)) {
        int offset = err_line - mf.start_line;
        printf("    [weft: %s:%d, scrap %d",
            mf.file, mf.line + offset, mf.scrap);
        printf("\n");
    }
}
}
}
}

```

120 · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines [write_error_annotator](#) 36d, 121b · uses [fputs](#) 21b, [letter](#) 129c, [MarkerFrame](#) 115a, [reverse_lookup_in_file](#) 119,
[size_t](#) 21b

2.8 Writing the Output Files

⟨ *Function prototypes* 121a ⟩ + ≡

```
extern void write_files(Name *files);
```

121a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b,
184e, 185d, 187d, 189d · used in 21a · uses [Name](#) 152b, [write_files](#) 121c

⟨ *Function prototypes* 121b ⟩ + ≡

```

extern void write_map(Name *file_names, Name *macro_names);
extern void write_extract(char *name, Name *file_names, Name *macro_names);
extern void write_reverse_map(const char *arg);
extern void write_callers(char *name, Name *file_names, Name *macro_names);
extern void write_lint(Name *file_names, Name *macro_names);
extern void write_error_annotator(void);

```

121b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b,
184e, 185d, 187d, 189d · used in 21a · uses [file_names](#) 152c, [macro_names](#) 152c, [Name](#) 152b, [write_callers](#) 112a,
[write_error_annotator](#) 120, [write_extract](#) 112b, [write_lint](#) 114b, [write_map](#) 109, [write_reverse_map](#) 117

⟨ *output.c* 121c ⟩ + ≡

```
static void make_parent_dirs(const char *path);    /* defined below */
```

```

void write_files(Name *files)
{
    while (files) {
        write_files(files->llink);
        ⟨ Write out files->spelling 122a ⟩
        files = files->rlink;
    }
}

```

121c · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce ·
defines [write_files](#) 39b, 121a · uses [make_parent_dirs](#) 121e, [Name](#) 152b

MAX_INDENT defines the maximum number of characters tracked per line for indentation purposes. When a fragment is expanded mid-line, the characters preceding it are recorded so that subsequent lines of the fragment can reproduce the same indentation. Languages like HTML, XML, and SVG routinely contain lines longer than 500 characters, so we set this to 8192. If a line exceeds this limit, indentation beyond the recorded portion falls back to spaces instead of causing a fatal error.

⟨ *Type declarations* 121d ⟩ + ≡

```
#define MAX_INDENT 8192
```

121d · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines MAX_INDENT 85b, 122a, 144a, 145c

When `--mkdirs` is given, `weft` creates the output directories itself before writing — both the `-p` prefix path and any subdirectories embedded in an `@o` name (`@o src/gen/foo.c`). Without the flag, the directories must already exist (the strict, typo-catching default). The helper is a portable recursive `mkdir -p`: it walks the path and creates each parent component, ignoring errors such as `EEXIST` (a real problem, like a permission error, surfaces later at the temporary-file or rename step). The final component — the file itself — is never created.

< output.c 121e > + ≡

```
#include <sys/stat.h>
#ifdef _WIN32
#include <direct.h>
#define WEFT_MKDIR(p) _mkdir(p)
#else
#define WEFT_MKDIR(p) mkdir((p), 0777)
#endif
```

```
static void make_parent_dirs(const char *path)
{
    char buf[FILENAME_MAX];
    char *p;
    if (strlen(path) >= sizeof(buf))
        return;
    strcpy(buf, path);
    for (p = buf + 1; *p != '\0'; p++) {
        if (*p == path_sep[0]) {
            *p = '\0';
            if (buf[0] != '\0')
                (void) WEFT_MKDIR(buf);
            *p = path_sep[0];
        }
    }
}
```

121e · defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce · defines `make_parent_dirs` 121c, 122a · uses `strlen` 21b

< Write out files->spelling 122a > ≡

```
{
    static char temp_name[FILENAME_MAX];
    static char real_name[FILENAME_MAX];
    static int temp_name_count = 0;
    char indent_chars[MAX_INDENT];
    FILE *temp_file;

    sprintf(real_name, "%s%s%s", dirpath, path_sep, files->spelling);
    if (mkdirs_flag)
        make_parent_dirs(real_name);

    < Find a free temporary file 122b >

    if (verbose_flag)
        fprintf(stderr, "writing %s [%s]\n", files->spelling, temp_name);
    write_scraps(temp_file, files->spelling, files->defs, 0, indent_chars,
        files->debug_flag, files->tab_flag, files->indent_flag,
```

```

        files->comment_flag, files->location_flag,
        NULL, NULL, 0, files->spelling);
fclose(temp_file);

< Move the temporary file to the target, if required 123a >
}

122a · used in 121c · uses defs 129c, fclose 21b, FILE 21b, fprintf 21b, make_parent_dirs 121e, MAX_INDENT 121d,
mkdirs_flag 27, stderr 21b, verbose_flag 27, write_scraps 142b

```

```

< Find a free temporary file 122b > ≡
for( temp_name_count = 0; temp_name_count < 10000; temp_name_count++) {
    sprintf(temp_name,"%s%snw%06d", dirpath, path_sep, temp_name_count);
#ifdef O_EXCL
    if (-1 != (temp_file_fd = open(temp_name, O_CREAT|O_WRONLY|O_EXCL))) {
        temp_file = fdopen(temp_file_fd, "w");
        break;
    }
#else
    if (0 != (temp_file = fopen(temp_name, "a"))) {
        if ( 0L == ftell(temp_file)) {
            break;
        } else {
            fclose(temp_file);
            temp_file = 0;
        }
    }
#endif
}
if (!temp_file) {
    fprintf(stderr, "%s: can't create %s for a temporary file\n",
            command_name, temp_name);
    exit(-1);
}

```

122b · used in 122a · uses command_name 29b, exit 21b, fclose 21b, fopen 21b, fprintf 21b, stderr 21b

Note the call to `remove` before `rename` – the ANSI/ISO C standard does *not* guarantee that renaming a file to an existing filename will overwrite the file.

```

< Move the temporary file to the target, if required 123a > ≡
if (diff_flag) {
    < Show diff of temp file vs real file 123b >
} else if (compare_flag)
    < Compare the temp file and the old file 124a >
else {
    remove(real_name);
    < Rename the temporary file to the target 124b >
}

```

123a · used in 122a · uses compare_flag 27, diff_flag 27, remove 21b

```

< Show diff of temp file vs real file 123b > ≡
{
    FILE *old_file = fopen(real_name, "r");
    if (!old_file) {
        /* New file: show all lines with + prefix */
        FILE *tf = fopen(temp_name, "r");
        if (tf) {
            char dbuf[8192];

```

```

        printf("new file: %s\n", real_name);
        while (fgets(dbuf, sizeof(dbuf), tf))
            printf("+%s", dbuf);
        fclose(tf);
    }
} else {
    char cmd[FILENAME_MAX * 2 + 32];
    FILE *dp;
    fclose(old_file);
    sprintf(cmd, "diff -u \"%s\" \"%s\"", real_name, temp_name);
    dp = popen(cmd, "r");
    if (dp) {
        char dbuf[8192];
        while (fgets(dbuf, sizeof(dbuf), dp))
            fputs(dbuf, stdout);
        pclose(dp);
    }
}
remove(temp_name);
}

```

123b · used in 123a · uses `fclose 21b`, `FILE 21b`, `fopen 21b`, `fputs 21b`, `remove 21b`

Again, we use a call to `remove` before `rename`.

```

< Compare the temp file and the old file 124a > ≡
{
    FILE *old_file = fopen(real_name, "r");
    if (old_file) {
        int x, y;
        temp_file = fopen(temp_name, "r");
        do {
            x = getc(old_file);
            y = getc(temp_file);
        } while (x == y && x != EOF);
        fclose(old_file);
        fclose(temp_file);
        if (x == y)
            remove(temp_name);
        else {
            remove(real_name);
            < Rename the temporary file to the target 124b >
        }
    }
    else
        < Rename the temporary file to the target 124b >
}

```

124a · used in 123a · uses `fclose 21b`, `FILE 21b`, `fopen 21b`, `getc 21b`, `remove 21b`

```

< Rename the temporary file to the target 124b > ≡
if (0 != rename(temp_name, real_name)) {
    fprintf(stderr, "%s: can't rename output file to %s\n",
            command_name, real_name);
    if (mkdirs_flag == FALSE)
        fprintf(stderr, " (does its directory exist? try --mkdirs)\n");
    remove(temp_name);
    exit(-1);
}

```

124b · used in 123a, 124a · uses `command_name` 29b, `exit` 21b, `FALSE` 23a, `fprintf` 21b, `makedirs_flag` 27, `remove` 21b, `stderr` 21b

Chapter 3

The Support Routines

3.1 Source Files

3.1.1 Global Declarations

We need two routines to handle reading the source files.

⟨ Function prototypes 125a ⟩ + ≡

```
extern void source_open(char *name); /* pass in the name of the source file */
extern int source_get(void); /* no args; returns the next char or EOF */
extern int source_last; /* what last source_get() returned. */
extern int source_peek; /* The next character to get */
```

125a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses next 129b, source_get 126a, source_last 126a, source_open 129a, source_peek 126a

There are also two global variables maintained for use in error messages and such.

⟨ Global variable declarations 125b ⟩ + ≡

```
extern char *source_name; /* name of the current file */
extern int source_line; /* current line in the source file */
```

125b · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines source_line 44d, 46d, 47a, 125c, 126ac, 127ab, 128ab, 129a, 134ac, 136a, 154c, 164, 166, 167ab, 168a, 170a, 171d, 173, 174, source_name 38ab, 39ab, 44d, 46d, 47a, 125c, 127ab, 128ab, 129a, 134ac, 136a, 154c, 164, 166, 167ab, 168a, 170ab, 171d, 172a, 173, 174

⟨ Global variable definitions 125c ⟩ + ≡

```
char *source_name = NULL;
int source_line = 0;
```

125c · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses source_line 125b, source_name 125b

3.1.2 Local Declarations

⟨ input.c 125d ⟩ + ≡

```
static FILE *source_file; /* the current input file */
static int double_at;
static int include_depth;
```

125d · defined at 25a, 125de, 126ac, 129a · defines double_at 127a, 129a, include_depth 127b, 128b, 129a, source_file 126ac, 127ab, 128ab, 129a · uses FILE 21b

```

<input.c 125e> + ≡
static struct {
    FILE *file;
    char *name;
    int line;
} stack[10];

```

125e · defined at 25a, 125de, 126ac, 129a · defines `stack` 117, 119, 127b, 128b · uses `FILE` 21b

3.1.3 Reading a File

The routine `source_get` returns the next character from the current source file. It notices newlines and keeps the line counter `source_line` up to date. It also catches EOF and watches for @ characters. All other characters are immediately returned. We define `source_last` to let us tell which type of scrap we are defining.

```

<input.c 126a> + ≡

```

```

int source_peek;
int source_last;
int source_get(void)
{
    int c;
    source_last = c = source_peek;
    switch (c) {
        case EOF: <Handle EOF 128b>
            return c;
        case '\n': source_line++;
            FALLTHROUGH;
        default:
            if (c==nw_char)
            {
                <Handle an "at" character 127a>
                return c;
            }
            source_peek = getc(source_file);
            return c;
    }
}

```

126a · defined at 25a, 125de, 126ac, 129a · defines `source_get` 42d, 43c, 44cd, 45ab, 46d, 47ac, 48ab, 49ab, 52, 53b, 58bc, 59bc, 65ad, 70, 71e, 73bcde, 76a, 77c, 80c, 83ab, 84a, 85ab, 87bc, 89b, 90c, 91b, 94a, 95a, 97a, 102c, 103a, 104c, 105bd, 106b, 125a, 127b, 128b, 134bc, 135b, 136a, 137b, 164, 166, 167ab, 168a, 170ab, 171a, 172a, 173, 174, 175a, `source_last` 53b, 63e, 70, 89b, 125a, `source_peek` 47c, 48b, 49a, 125a, 126c, 127ab, 128b, 129a · uses `FALLTHROUGH` 23a, `getc` 21b, `nw_char` 28b, 29a, `source_file` 125d, `source_line` 125b

`source_ungetc` pushes a read character back to the `source_file`.

```

<Function prototypes 126b> + ≡

```

```

extern void source_ungetc(int*);

```

126b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `source_ungetc` 126c

```

<input.c 126c> + ≡

```

```

void source_ungetc(int *c)
{
    ungetc(source_peek, source_file);
    if(*c == '\n')

```

```

    source_line--;
    source_peek=*c;
}

```

126c · defined at 25a, 125de, 126ac, 129a · defines `source_ungetc` 48ab, 49b, 77c, 95a, 105d, 126b · uses `source_file` 125d, `source_line` 125b, `source_peek` 126a

This whole @ character handling mess is pretty annoying. I want to recognize @i so I can handle include files correctly. At the same time, it makes sense to recognize illegal @ sequences and complain; this avoids ever having to check anywhere else. Unfortunately, I need to avoid tripping over the @@ sequence; hence this whole unsatisfactory `double_at` business.

⟨ *Handle an “at” character* 127a ⟩ ≡

```

{
    c = getc(source_file);
    if (double_at) {
        source_peek = c;
        double_at = FALSE;
        c = nw_char;
    }
    else
        switch (c) {
            case 'i': ⟨ Open an include file 127b ⟩
                break;
            case '#': case 'f': case 'm': case 'u': case 'v':
            case 'd': case 'o': case 'D': case 'O': case 's':
            case 'q': case 'Q': case 'S': case 'l': case 'L': case 't':
            case 'W':
            case '+':
            case '-':
            case '*':
            case '\\':
            case '{': case '}': case '<': case '>': case '|':
            case '(': case ')': case '[': case ']':
            case '%': case '_':
            case '.': case ',': case 'x': case 'c':
            case '1': case '2': case '3': case '4': case '5':
            case '6': case '7': case '8': case '9':
            case 'r':
                source_peek = c;
                c = nw_char;
                break;
            default:
                if (c==nw_char)
                {
                    source_peek = c;
                    double_at = TRUE;
                    break;
                }
                fprintf(stderr, "%s: bad %c sequence %c[%d] (%s, line %d)\n",
                    command_name, nw_char, c, c, source_name, source_line);
                exit(-1);
        }
}

```

127a · used in 126a · uses `command_name` 29b, `double_at` 125d, `exit` 21b, `FALSE` 23a, `fprintf` 21b, `getc` 21b, `nw_char` 28b, 29a, `source_file` 125d, `source_line` 125b, `source_name` 125b, `source_peek` 126a, `stderr` 21b, `TRUE` 23a

⟨ *Open an include file* 127b ⟩ ≡

```

{
    char name[FILENAME_MAX];
    char fullname[FILENAME_MAX];
    struct incl * p = include_list;

    if (include_depth >= 10) {
        fprintf(stderr, "%s: include nesting too deep (%s, %d)\n",
            command_name, source_name, source_line);
        exit(-1);
    }
    < Collect include-file name 128a >
    stack[include_depth].file = source_file;
    fullname[0] = '\0';
    for (;;) {
        strcat(fullname, name);
        source_file = fopen(fullname, "r");
        if (source_file || !p)
            break;
        strcpy(fullname, p->name);
        strcat(fullname, "/");
        p = p->next;
    }
    if (!source_file) {
        fprintf(stderr, "%s: can't open include file %s\n",
            command_name, name);
        source_file = stack[include_depth].file;
    }
    else
    {
        stack[include_depth].name = source_name;
        stack[include_depth].line = source_line + 1;
        include_depth++;
        source_line = 1;
        source_name = save_string(fullname);
    }
    source_peek = getc(source_file);
    c = source_get();
}

```

127b · used in 127a · uses [command_name 29b](#), [exit 21b](#), [fopen 21b](#), [fprintf 21b](#), [getc 21b](#), [incl 27](#), [include_depth 125d](#), [next 129b](#), [save_string 154a](#), [source_file 125d](#), [source_get 126a](#), [source_line 125b](#), [source_name 125b](#), [source_peek 126a](#), [stack 125e](#), [stderr 21b](#)

< Collect include-file name 128a > ≡

```

{
    char *p = name;
    do
        c = getc(source_file);
    while (c == ' ' || c == '\t');
    while (isgraph(c)) {
        *p++ = c;
        c = getc(source_file);
    }
    *p = '\0';
    if (c != '\n') {
        fprintf(stderr, "%s: unexpected characters after file name (%s, %d)\n",
            command_name, source_name, source_line);
        exit(-1);
    }
}

```

```
}
```

128a · used in 127b · uses `command_name` 29b, `exit` 21b, `fprintf` 21b, `getc` 21b, `isgraph` 21b, `source_file` 125d, `source_line` 125b, `source_name` 125b, `stderr` 21b

If an EOF is discovered, the current file must be closed and input from the next stacked file must be resumed. If no more files are on the stack, the EOF is returned.

⟨ *Handle EOF* 128b ⟩ ≡

```
{
    fclose(source_file);
    if (include_depth) {
        include_depth--;
        source_file = stack[include_depth].file;
        source_line = stack[include_depth].line;
        source_name = stack[include_depth].name;
        source_peek = getc(source_file);
        c = source_get();
    }
}
```

128b · used in 126a · uses `fclose` 21b, `getc` 21b, `include_depth` 125d, `source_file` 125d, `source_get` 126a, `source_line` 125b, `source_name` 125b, `source_peek` 126a, `stack` 125e

3.1.4 Opening a File

The routine `source_open` takes a file name and tries to open the file. If unsuccessful, it complains and halts. Otherwise, it sets `source_name`, `source_line`, and `double_at`.

⟨ *input.c* 129a ⟩ + ≡

```
void source_open(char *name)
{
    source_file = fopen(name, "r");
    if (!source_file) {
        fprintf(stderr, "%s: couldn't open %s\n", command_name, name);
        exit(-1);
    }
    nw_char = '@';
    source_name = name;
    source_line = 1;
    source_peek = getc(source_file);
    double_at = FALSE;
    include_depth = 0;
}
```

129a · defined at 25a, 125de, 126ac, 129a · defines `source_open` 42c, 55, 82d, 101, 125a · uses `command_name` 29b, `double_at` 125d, `exit` 21b, `FALSE` 23a, `fopen` 21b, `fprintf` 21b, `getc` 21b, `include_depth` 125d, `nw_char` 28b, 29a, `source_file` 125d, `source_line` 125b, `source_name` 125b, `source_peek` 126a, `stderr` 21b

3.2 Scraps

⟨ *scraps.c* 129b ⟩ + ≡

```
#define SLAB_SIZE 1024

typedef struct slab {
    struct slab *next;
    char chars[SLAB_SIZE];
}
```

} Slab;

129b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines next 27, 30, 33, 35b, 39a, 46c, 51c, 59c, 65g, 66abc, 73g, 74, 75b, 77b, 79bc, 80e, 87e, 88bc, 89a, 92a, 93, 94d, 96ab, 97c, 98b, 99ad, 100ac, 104d, 105a, 108, 110b, 111c, 112a, 125a, 127b, 133d, 136b, 137ad, 138b, 140de, 141bc, 142b, 149ad, 152a, 172de, 175ce, 176acd, 177abd, 179a, 181ab, 182, 184bcd, 185ab, 186bc, 189e, 190cd, 192, Slab 129c, 133cd, 134a, 138b, SLAB_SIZE 133d, 138b, 140c, 180a · uses slab 129c

⟨ scraps.c 129c ⟩ + ≡

```
typedef struct {
    char *file_name;
    Slab *slab;
    struct uses *uses;
    struct uses *defs;
    int file_line;
    int end_line;
    int page;
    char letter;
    unsigned char sector;
    char *prose;
    struct name *owner;
} ScrapEntry;
```

129c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines defs 46c, 51c, 59c, 61ab, 65g, 66a, 73g, 74, 75b, 77b, 79b, 80e, 85b, 87e, 88b, 92a, 93, 94d, 96a, 97c, 99ab, 100ac, 103c, 104ad, 105ac, 106a, 108, 110b, 113b, 114a, 122a, 130b, 131e, 134a, 137a, 141b, 142b, 143, 144bc, 145a, 147cd, 148ac, 149ac, 152b, 157a, 175d, 176b, 182, 185bc, 186ac, end_line 108, 132d, 134ac, file_line 132c, 134ab, 143, file_name 42ac, 54a, 55, 82ad, 100b, 101, 132b, 134ab, 144bc, letter 120, 131b, 134a, 151d, owner 111c, 112a, 132f, 134a, 140b, page 66c, 79b, 80e, 89a, 96a, 97c, 99ab, 100a, 130b, 131bc, 134a, 150bc, 151bd, 185bc, 186c, 192, prose 27, 29f, 30, 63e, 108, 132e, 134a, 140a, 192, ScrapEntry 130a, 131a, 134a, sector 30, 54ab, 55, 77c, 78b, 79a, 80bcde, 82acd, 95acd, 97abc, 100b, 101, 105d, 106abc, 108, 134a, 136ab, 137ac, 138c, 147d, 149d, 152b, 153b, 154bc, 155a, 156b, 157a, 170a, 171d, 173, 182, slab 129b, 134a, 138c, 143, 182, uses 30, 56a, 66b, 79c, 80e, 88c, 96b, 97c, 100a, 105a, 106c, 108, 110b, 111c, 112a, 114a, 131d, 134a, 136b, 137d, 152b, 157ab, 176b, 182, 184df, 186c · uses Slab 129b

There's some issue with reading the auxiliary file if it has more than 255 lines, so (as a workaround only) increase the size of the array of ScrapEntry's held in SCRAP[n].

⟨ scraps.c 130a ⟩ + ≡

```
#define SCRAP_BITS 10
#define SCRAP_SIZE (1<<SCRAP_BITS)
#define SCRAP_MASK (SCRAP_SIZE - 1)
#define SCRAP_SHIFT SCRAP_BITS
static ScrapEntry *SCRAP[SCRAP_SIZE];

#define scrap_array(i) SCRAP[(i)>> SCRAP_SHIFT][(i) & SCRAP_MASK]

static int scraps;
int num_scraps(void)
{
    return scraps;
}
⟨ Forward declarations for scraps.c 142c, ... ⟩
```

130a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines num_scraps 78b, 95c, 130b, SCRAP 131a, 134a, scraps 24abc, 27, 30, 53b, 54b, 61ab, 63a, 65ae, 66c, 82b, 85c, 86a, 87bd, 88a, 89a, 103c, 104acd, 105a, 108, 131a, 134a, 150bc, 151bc, 182, 192, scrap_array 131bde, 132bcdef, 134abc, 136b, 137a, 138c, 140ab, 143, 144bc, 150bc, 151bd, 182, 184f, 186a, SCRAP_BITS never

used, [SCRAP_MASK 134a](#), [SCRAP_SHIFT 134a](#), [SCRAP_SIZE 131a, 134a](#) · uses [ScrapEntry 129c](#)

< Function prototypes 130b > + ≡

```
extern void init_scraps(void);
extern int collect_scrap(void);
extern int write_scraps(FILE *file, char *spelling, Scrap_Node *defs, int global_indent,
    char *indent_chars, char debug_flag, char tab_flag, char indent_flag, unsigned char
    comment_flag, char location_flag, Arglist *inArgs, char *inParams[9], Parameters
    parameters, char *title);
extern void write_scrap_ref(FILE *file, int num, int first, int *page);
extern void write_single_scrap_ref(FILE *file, int num);
extern int num_scraps(void);
extern Uses * get_scrap_uses(int scrap);
extern Uses * get_scrap_defs(int scrap);
```

130b · defined at [42a](#), [54a](#), [69a](#), [71b](#), [80a](#), [82ab](#), [100b](#), [121ab](#), [125a](#), [126b](#), [130b](#), [132a](#), [138a](#), [141a](#), [150a](#), [153b](#), [155b](#), [172b](#), [180b](#), [184e](#), [185d](#), [187d](#), [189d](#) · used in [21a](#) · uses [Arglist 172d](#), [collect_scrap 133f](#), [defs 129c](#), [FILE 21b](#), [first 190a](#), [get_scrap_defs 131e](#), [get_scrap_uses 131d](#), [init_scraps 131a](#), [num_scraps 130a](#), [page 129c](#), [Parameters 51b](#), [Scrap_Node 152a](#), [Uses 184d](#), [write_scraps 142b](#), [write_scrap_ref 131b](#), [write_single_scrap_ref 131c](#)

< scraps.c 131a > + ≡

```
void init_scraps(void)
{
    scraps = 1;
    SCRAP[0] = (ScrapEntry *) arena_getmem(SCRAP_SIZE * sizeof(ScrapEntry));
}
```

131a · defined at [25b](#), [129bc](#), [130a](#), [131abcde](#), [132bcdef](#), [133cdef](#), [137d](#), [138bc](#), [139](#), [140abcde](#), [141d](#), [142b](#), [150b](#), [176d](#), [177abcde](#), [178abc](#), [179c](#), [180cd](#), [184bcf](#), [186ad](#), [187ac](#) · defines [init_scraps 42c](#), [130b](#) · uses [arena_getmem 190b](#), [SCRAP 130a](#), [ScrapEntry 129c](#), [scraps 130a](#), [SCRAP_SIZE 130a](#)

< scraps.c 131b > + ≡

```
void write_scrap_ref(FILE *file, int num, int first, int *page)
{
    if (scrap_array(num).page >= 0) {
        if (first!=0)
            fprintf(file, "%d", scrap_array(num).page);
        else if (scrap_array(num).page != *page)
            fprintf(file, ", %d", scrap_array(num).page);
        if (scrap_array(num).letter > 0)
            fputc(scrap_array(num).letter, file);
    }
    else {
        if (first!=0)
            putc('?', file);
        else
            fputs(" ? ", file);
        < Warn (only once) about needing to rerun after Latex 132g >
    }
    if (first>=0)
        *page = scrap_array(num).page;
}
```

131b · defined at [25b](#), [129bc](#), [130a](#), [131abcde](#), [132bcdef](#), [133cdef](#), [137d](#), [138bc](#), [139](#), [140abcde](#), [141d](#), [142b](#), [150b](#), [176d](#), [177abcde](#), [178abc](#), [179c](#), [180cd](#), [184bcf](#), [186ad](#), [187ac](#) · defines [write_scrap_ref 66c](#), [79b](#), [80e](#), [89a](#), [96a](#), [97c](#), [99b](#), [130b](#), [131c](#), [185c](#) · uses [FILE 21b](#), [first 190a](#), [fprintf 21b](#), [fputs 21b](#), [letter 129c](#), [page 129c](#), [putc 21b](#), [scrap_array 130a](#)

< scraps.c 131c > + ≡

```
void write_single_scrap_ref(FILE *file, int num)
```

```

{
    int page;
    write_scrap_ref(file, num, TRUE, &page);
}

131c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines write_single_scrap_ref 53b, 59c, 61ab, 65e, 66b, 74, 77b, 79c,
80e, 85c, 86a, 88c, 92a, 93, 94d, 96b, 97c, 100c, 103c, 104ad, 105a, 130b, 148ac, 188a · uses FILE 21b, page 129c, TRUE 23a,
write_scrap_ref 131b

⟨ scraps.c 131d ⟩ + ≡
Uses * get_scrap_uses(int scrap)
{
    return scrap_array(scrap).uses;
}

131d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines get_scrap_uses 98a, 108, 110b, 130b · uses scrap_array 130a,
Uses 184d, uses 129c

⟨ scraps.c 131e ⟩ + ≡
Uses * get_scrap_defs(int scrap)
{
    return scrap_array(scrap).defs;
}

131e · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde,
178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines get_scrap_defs 99c, 130b · uses defs 129c, scrap_array 130a, Uses 184d

These accessors expose scrap metadata that is needed by the --map and --extract modes (Section 2.7)
without leaking the internal ScrapEntry or Slab types.

⟨ Function prototypes 132a ⟩ + ≡
extern const char *scrap_file_name(int i);
extern int scrap_file_line(int i);
extern int scrap_end_line(int i);
extern const char *scrap_prose(int i);
extern Name *scrap_owner(int i);
extern void dump_scrap_text(FILE *file, int scrap_idx);
extern void dump_scrap_json(FILE *f, int scrap_idx);
extern void set_scrap_prose(int i, char *p);
extern void set_scrap_owner(int i, Name *n);

132a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b,
184e, 185d, 187d, 189d · used in 21a · uses dump_scrap_json 139, dump_scrap_text 138c, FILE 21b, Name 152b,
scrap_end_line 132d, scrap_file_line 132c, scrap_file_name 132b, scrap_owner 132f, scrap_prose 132e,
set_scrap_owner 140b, set_scrap_prose 140a

⟨ scraps.c 132b ⟩ + ≡
const char *scrap_file_name(int i) { return scrap_array(i).file_name; }

132b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines scrap_file_name 108, 110b, 114a, 132a · uses file_name 129c,
scrap_array 130a

⟨ scraps.c 132c ⟩ + ≡
int scrap_file_line(int i) { return scrap_array(i).file_line; }

```

132c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `scrap_file_line` 108, 110b, 114a, 132a · uses `file_line` 129c, `scrap_array` 130a

⟨ `scraps.c` 132d ⟩ + ≡
`int scrap_end_line(int i) { return scrap_array(i).end_line; }`

132d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `scrap_end_line` 108, 132a · uses `end_line` 129c, `scrap_array` 130a

⟨ `scraps.c` 132e ⟩ + ≡
`const char *scrap_prose(int i) { return scrap_array(i).prose; }`

132e · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `scrap_prose` 108, 132a · uses `prose` 129c, `scrap_array` 130a

⟨ `scraps.c` 132f ⟩ + ≡
`Name *scrap_owner(int i) { return scrap_array(i).owner; }`

132f · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `scrap_owner` 111c, 112a, 132a · uses `Name` 152b, `owner` 129c, `scrap_array` 130a

⟨ *Warn (only once) about needing to rerun after Latex* 132g ⟩ ≡

```
{
  if (!already_warned) {
    fprintf(stderr, "%s: you'll need to rerun weft after running latex\n",
              command_name);
    already_warned = TRUE;
  }
}
```

132g · used in 131b, 150c · uses `already_warned` 133a, `command_name` 29b, `fprintf` 21b, `stderr` 21b, `TRUE` 23a

⟨ *Global variable declarations* 133a ⟩ + ≡
`extern int already_warned;`

133a · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `already_warned` 132g, 133b, 150b

⟨ *Global variable definitions* 133b ⟩ + ≡
`int already_warned = 0;`

133b · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses `already_warned` 133a

⟨ `scraps.c` 133c ⟩ + ≡

```
typedef struct {
  Slab *scrap;
  Slab *prev;
  int index;
} Manager;
```

133c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `Manager` 133def, 138bc, 140c, 141d, 143, 178a, 179c, 182 · uses `Slab` 129b

⟨ `scraps.c` 133d ⟩ + ≡

```
static void push(char c, Manager *manager)
{
  Slab *scrap = manager->scrap;
  int index = manager->index;
```

```

    scrap->chars[index++] = c;
    if (index == SLAB_SIZE) {
        Slab *new_slab = (Slab *) arena_getmem(sizeof(Slab));
        scrap->next = new_slab;
        manager->scrap = new_slab;
        index = 0;
    }
    manager->index = index;
}

```

133d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines push 49f, 52, 133e, 134bc, 135a, 137bc · uses arena_getmem 190b, Manager 133c, next 129b, Slab 129b, SLAB_SIZE 129b

< scraps.c 133e > + ≡

```

static void pushes(char *s, Manager *manager)
{
    while (*s)
        push(*s++, manager);
}

```

133e · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines pushes 52, 135a, 137c · uses Manager 133c, push 133d

< scraps.c 133f > + ≡

```

int collect_scrap(void)
{
    int current_scrap, lblseq = 0;
    int depth = 1;
    Manager writer;
    < Create new scrap, managed by writer 134a >
    < Accumulate scrap and return scraps++ 134b >
}

```

133f · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines collect_scrap 46ab, 52, 130b, 175c · uses Manager 133c

< Create new scrap, managed by writer 134a > ≡

```

{
    Slab *scrap = (Slab *) arena_getmem(sizeof(Slab));
    if ((scraps & SCRAP_MASK) == 0)
        SCRAP[scraps >> SCRAP_SHIFT] = (ScrapEntry *) arena_getmem(SCRAP_SIZE * sizeof(
            ScrapEntry));
    scrap_array(scraps).slab = scrap;
    scrap_array(scraps).file_name = save_string(source_name);
    scrap_array(scraps).file_line = source_line;
    scrap_array(scraps).page = -1;
    scrap_array(scraps).letter = 0;
    scrap_array(scraps).uses = NULL;
    scrap_array(scraps).defs = NULL;
    scrap_array(scraps).sector = current_sector;
    scrap_array(scraps).end_line = source_line;
    scrap_array(scraps).prose = NULL;
    scrap_array(scraps).owner = NULL;
    writer.scrap = scrap;
    writer.index = 0;
    current_scrap = scraps++;
}

```

134a · used in 133f · uses arena_getmem 190b, current_sector 45c, defs 129c, end_line 129c, file_line 129c,

file_name 129c, letter 129c, owner 129c, page 129c, prose 129c, save_string 154a, SCRAP 130a, ScrapEntry 129c, scraps 130a, scrap_array 130a, SCRAP_MASK 130a, SCRAP_SHIFT 130a, SCRAP_SIZE 130a, sector 129c, Slab 129b, slab 129c, source_line 125b, source_name 125b, uses 129c

< Accumulate scrap and return scraps++ 134b > ≡

```
{
  int c = source_get();
  while (1) {
    switch (c) {
      case EOF: fprintf(stderr, "%s: unexpect EOF in (%s, %d)\n",
                          command_name, scrap_array(current_scrap).file_name,
                          scrap_array(current_scrap).file_line);
                          exit(-1);
      default:
        if (c==nw_char)
        {
          < Handle at-sign during scrap accumulation 134c >
          break;
        }
        push(c, &writer);
        c = source_get();
        break;
    }
  }
}
```

134b · used in 133f · uses command_name 29b, exit 21b, file_line 129c, file_name 129c, fprintf 21b, nw_char 28b, 29a, push 133d, scrap_array 130a, source_get 126a, stderr 21b

< Handle at-sign during scrap accumulation 134c > ≡

```
{
  c = source_get();
  switch (c) {
    case '(':
    case '[':
    case '{': depth++;
              break;
    case '+':
    case '-':
    case '*':
    case '|': < Collect user-specified index entries 136a >
              /* Fall through */
    case ')':
    case ']':
    case '}': if (--depth > 0)
              break;
              /* else fall through */
    case ',':
              push('\0', &writer);
              scrap_ended_with = c;
              scrap_array(current_scrap).end_line = source_line;
              return current_scrap;
    case '<': < Handle macro invocation in scrap 137b >
              break;
    case '%': < Skip commented-out code 73d >
              /* emit line break to the output file to keep #line in sync. */
              push('\n', &writer);
              c = source_get();
              break;
  }
```

```

case 'x': < Get label while collecting scrap 135a>
    break;
case 'c': < Include block comment in a scrap 49f>
    break;
case '1': case '2': case '3':
case '4': case '5': case '6':
case '7': case '8': case '9':
case 'f': case '#': case 'v':
case 't': case 's':
    push(nw_char, &writer);
    break;
case '_': c = source_get();
    break;
default :
    if (c==nw_char)
    {
        push(nw_char, &writer);
        push(nw_char, &writer);
        c = source_get();
        break;
    }
    fprintf(stderr, "%s: unexpected %c%c in scrap (%s, %d)\n",
        command_name, nw_char, c, source_name, source_line);
    exit(-1);
}
}

```

134c · used in 134b · uses [command_name 29b](#), [end_line 129c](#), [exit 21b](#), [fprintf 21b](#), [nw_char 28b](#), [29a](#), [push 133d](#), [scrap_array 130a](#), [source_get 126a](#), [source_line 125b](#), [source_name 125b](#), [stderr 21b](#)

```

< Get label while collecting scrap 135a> ≡
{
    < Get label from(135b source_get() ) 187b>
    < Save label to label store 188c>
    push(nw_char, &writer);
    push('x', &writer);
    pushs(label_name, &writer);
    push(nw_char, &writer);
}

```

135b · used in 134c · uses [source_get 126a](#)

```

< Collect user-specified index entries 136a> ≡
{
    do {
        int type = c;
        do {
            char new_name[MAX_NAME_LEN];
            char *p = new_name;
            unsigned int sector = 0;
            do
                c = source_get();
            while (isspace(c));
            if (c != nw_char) {
                Name *name;
                do {
                    *p++ = c;
                    c = source_get();
                } while (c != nw_char && !isspace(c));
                *p = '\0';
            }
        }
    }
}

```

```

switch (type) {
case '*':
    sector = current_sector;
    < Add user identifier use 136b >
    break;
case '-':
    < Add user identifier use 136b >
    /* Fall through */
case '|':
    sector = current_sector;
    /* Fall through */
case '+':
    < Add user identifier definition 137a >
    break;
}
}
} while (c != nw_char);
c = source_get();
}while (c == '|' || c == '*' || c == '-' || c == '+');
if (c != '}' && c != ']' && c != ')') {
    fprintf(stderr, "%s: unexpected %c%c in index entry (%s, %d)\n",
        command_name, nw_char, c, source_name, source_line);
    exit(-1);
}
}
}

```

136a · used in 134c · uses [command_name 29b](#), [current_sector 45c](#), [exit 21b](#), [fprintf 21b](#), [isspace 21b](#), [MAX_NAME_LEN 23b](#), [Name 152b](#), [nw_char 28b](#), [29a](#), [sector 129c](#), [source_get 126a](#), [source_line 125b](#), [source_name 125b](#), [stderr 21b](#)

< Add user identifier use 136b > ≡

```

name = name_add(&user_names, new_name, sector);
if (!name->uses || name->uses->scrap != current_scrap) {
    Scrap_Node *use = (Scrap_Node *) arena_getmem(sizeof(Scrap_Node));
    use->scrap = current_scrap;
    use->next = name->uses;
    name->uses = use;
    add_uses(&(scrap_array(current_scrap).uses), name);
}

```

136b · used in 136a · uses [add_uses 184c](#), [arena_getmem 190b](#), [name_add 156b](#), [next 129b](#), [scrap_array 130a](#), [Scrap_Node 152a](#), [sector 129c](#), [user_names 152c](#), [uses 129c](#)

< Add user identifier definition 137a > ≡

```

name = name_add(&user_names, new_name, sector);
if (!name->defs || name->defs->scrap != current_scrap) {
    Scrap_Node *def = (Scrap_Node *) arena_getmem(sizeof(Scrap_Node));
    def->scrap = current_scrap;
    def->next = name->defs;
    name->defs = def;
    add_uses(&(scrap_array(current_scrap).defs), name);
}

```

137a · used in 136a · uses [add_uses 184c](#), [arena_getmem 190b](#), [defs 129c](#), [name_add 156b](#), [next 129b](#), [scrap_array 130a](#), [Scrap_Node 152a](#), [sector 129c](#), [user_names 152c](#)

< Handle macro invocation in scrap 137b > ≡

```

{
    Arglist * args = collect_scrap_name(current_scrap);
    Name *name = args->name;
    < Save macro name 137c >

```

```

    add_to_use(name, current_scrap);
    if (scrap_name_has_parameters) {
        < Save macro parameters 52 >
    }
    push(nw_char, &writer);
    push('>', &writer);
    c = source_get();
}

```

137b · used in 134c · uses [add_to_use 137d](#), [Arglist 172d](#), [collect_scrap_name 173](#), [Name 152b](#), [nw_char 28b, 29a](#), [push 133d](#), [source_get 126a](#)

```

< Save macro name 137c > ≡
{
    char buff[24];

    push(nw_char, &writer);
    push('<', &writer);
    push(name->sector, &writer);
    sprintf(buff, "%d", arglist_register(args));
    pushs(buff, &writer);
}

```

137c · used in 137b · uses [arglist_register 142d](#), [nw_char 28b, 29a](#), [push 133d](#), [pushs 133e](#), [sector 129c](#)

```

< scraps.c 137d > + ≡
void
add_to_use(Name * name, int current_scrap)
{
    if (!name->uses || name->uses->scrap != current_scrap) {
        Scrap_Node *use = (Scrap_Node *) arena_getmem(sizeof(Scrap_Node));
        use->scrap = current_scrap;
        use->next = name->uses;
        name->uses = use;
    }
}

```

137d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines [add_to_use 52, 137b, 138a, 175e](#) · uses [arena_getmem 190b](#), [Name 152b](#), [next 129b](#), [Scrap_Node 152a](#), [uses 129c](#)

```

< Function prototypes 138a > + ≡
extern void add_to_use(Name * name, int current_scrap);

```

138a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses [add_to_use 137d](#), [Name 152b](#)

```

< scraps.c 138b > + ≡
static char pop(Manager *manager)
{
    Slab *scrap = manager->scrap;
    int index = manager->index;
    char c = scrap->chars[index++];
    if (index == SLAB_SIZE) {
        manager->prev = scrap;
        manager->scrap = scrap->next;
        index = 0;
    }
    manager->index = index;
    return c;
}

```

```
}
```

138b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines pop 50abd, 53a, 138c, 141d, 142a, 143, 145b, 146a, 147b, 178c · uses Manager 133c, next 129b, Slab 129b, SLAB_SIZE 129b

The function `dump_scrap_text` walks the slab chain for a given scrap and writes every printable character (plus newlines and tabs) to the supplied file handle. The internal meta-character sequences that encode fragment references are decoded back to their `@<name@>` form; other control sequences are silently skipped.

(`scraps.c` 138c) + ≡

```
void dump_scrap_text(FILE *file, int scrap_idx)
{
    Manager reader;
    char c;
    reader.scrap = scrap_array(scrap_idx).slab;
    reader.prev = NULL;
    reader.index = 0;
    c = pop(&reader);
    while (c) {
        if (c == nw_char) {
            /* handle internal control sequences */
            c = pop(&reader);
            if (c == '<') {
                /* fragment reference encoded as: sector_byte + decimal_index
                 followed by nw_char + '>' */
                char idx_buf[MAX_NAME_LEN];
                char *p = idx_buf;
                int idx;
                (void)pop(&reader); /* sector byte, discard */
                c = pop(&reader);
                while (c != nw_char && c != '\0') {
                    *p++ = c;
                    c = pop(&reader);
                }
                *p = '\0';
                if (sscanf(idx_buf, "%d", &idx) == 1) {
                    Arglist *a = arglist_lookup(idx);
                    if (a && a->name) {
                        fprintf(file, "%c<%s%c>", nw_char, a->name->spelling, nw_char);
                    }
                }
                /* skip past closing '>' marker */
                if (c == nw_char)
                    c = pop(&reader);
                /* skip optional parameter block */
                if (c == '(') {
                    while (c && c != nw_char)
                        c = pop(&reader);
                    if (c == nw_char)
                        c = pop(&reader);
                }
            } else {
                /* other control seq: skip */
            }
            c = pop(&reader);
            continue;
        }
        fputc(c, file);
    }
}
```

```

    c = pop(&reader);
}
}

```

138c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `dump_scrap_text` 110b, 132a, 139 · uses `Arglist` 172d, `arglist_lookup` 142d, `FILE` 21b, `fprintf` 21b, `Manager` 133c, `MAX_NAME_LEN` 23b, `nw_char` 28b, 29a, `pop` 138b, `scrap_array` 130a, `sector` 129c, `slab` 129c

The function `dump_scrap_json` is similar to `dump_scrap_text` but wraps the output in a JSON string, escaping special characters inline. It is used by `--bodies` to embed scrap code in the `-m` JSON output.

```

⟨ scraps.c 139 ⟩ + ≡
void dump_scrap_json(FILE *f, int scrap_idx)
{
    FILE *tmp = tmpfile();
    int ch;
    if (!tmp) { fputs("\\""", f); return; }
    dump_scrap_text(tmp, scrap_idx);
    fflush(tmp);
    fseek(tmp, 0, SEEK_SET);
    fputc('"', f);
    while ((ch = fgetc(tmp)) != EOF) {
        switch (ch) {
            case '\\': fputs("\\\\", f); break;
            case '\n': fputs("\\n", f); break;
            case '\r': fputs("\\r", f); break;
            case '\t': fputs("\\t", f); break;
            default:
                if (ch < 0x20)
                    fprintf(f, "\\u%04x", ch);
                else
                    fputc(ch, f);
        }
    }
    fputc('"', f);
    fclose(tmp);
}

```

139 · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `dump_scrap_json` 108, 132a · uses `dump_scrap_text` 138c, `fclose` 21b, `FILE` 21b, `fprintf` 21b, `fputs` 21b

```

⟨ scraps.c 140a ⟩ + ≡
void set_scrap_prose(int i, char *p) { scrap_array(i).prose = p; }

```

140a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `set_scrap_prose` 46ab, 132a · uses `prose` 129c, `scrap_array` 130a

```

⟨ scraps.c 140b ⟩ + ≡
void set_scrap_owner(int i, Name *n) { scrap_array(i).owner = n; }

```

140b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `set_scrap_owner` 46c, 132a · uses `Name` 152b, `owner` 129c, `scrap_array` 130a

```

⟨ scraps.c 140c ⟩ + ≡
static void backup(int n, Manager *manager)

```

```

{
    int index = manager->index;
    if (n > index
        && manager->prev != NULL)
    {
        manager->scrap = manager->prev;
        manager->prev = NULL;
        index += SLAB_SIZE;
    }
    manager->index = (n <= index ? index - n : 0);
}

```

140c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `backup` 145b · uses `Manager` 133c, `SLAB_SIZE` 129b

⟨ `scraps.c` 140d ⟩ + ≡

```

void
lookup(int n, Arglist * par, char * arg[9], Name **name, Arglist ** args)
{
    int i;
    Arglist * p = par;

    for (i = 0; i < n && p != NULL; i++)
        p = p->next;
    if (p == NULL) {
        char * a = arg[n];

        *name = NULL;
        *args = (Arglist *)a;
    }
    else {
        *name = p->name;
        *args = p->args;
    }
}

```

140d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `lookup` 51c, 141b · uses `Arglist` 172d, `Name` 152b, `next` 129b

⟨ `scraps.c` 140e ⟩ + ≡

```

Arglist * instance(Arglist * a, Arglist * par, char * arg[9], int * ch)
{
    if (a != NULL) {
        int changed = 0;
        Arglist *args, *next;
        Name* name;
        ⟨ Set up name, args and next 141b ⟩
        if (changed){
            ⟨ Build a new arglist 141c ⟩
            *ch = 1;
        }
    }

    return a;
}

```

140e · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `instance` 85b, 141ab, 147d · uses `Arglist` 172d, `Name` 152b, `next` 129b

⟨ *Function prototypes* 141a ⟩ + ≡

```
Arglist *instance(Arglist *a, Arglist *par, char *arg[9], int *ch);
```

141a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses Arglist 172d, instance 140e

⟨ *Set up name, args and next* 141b ⟩ ≡

```
next = instance(a->next, par, arg, &changed);
name = a->name;
if (name == (Name *)1) {
    Embed_Node * q = (Embed_Node *)arena_getmem(sizeof(Embed_Node));
    q->defs = (Scrap_Node *)a->args;
    q->args = par;
    args = (Arglist *)q;
    changed = 1;
} else if (name != NULL)
    args = instance(a->args, par, arg, &changed);
else {
    char * p = (char *)a->args;
    if (p[0] == ARG_CHR) {
        lookup(p[1] - '1', par, arg, &name, &args);
        changed = 1;
    }
    else {
        args = a->args;
    }
}
```

141b · used in 140e · uses arena_getmem 190b, Arglist 172d, ARG_CHR 171b, defs 129c, Embed_Node 175d, instance 140e, lookup 140d, Name 152b, next 129b, Scrap_Node 152a

⟨ *Build a new arglist* 141c ⟩ ≡

```
a = (Arglist *)arena_getmem(sizeof(Arglist));
a->name = name;
a->args = args;
a->next = next;
```

141c · used in 140e · uses arena_getmem 190b, Arglist 172d, next 129b

⟨ *scraps.c* 141d ⟩ + ≡

```
static Arglist *pop_scrap_name(Manager *manager, Parameters *parameters)
{
    char name[MAX_NAME_LEN];
    char *p = name;
    Arglist * args;
    int c;
    int idx;

    (void)pop(manager); /* not sure why we have to pop twice */
    c = pop(manager);

    while (c != nw_char) {
        *p++ = c;
        c = pop(manager);
    }
    *p = '\000';
    if (sscanf(name, "%d", &idx) != 1)
    {
        fprintf(stderr, "%s: found an internal problem (2)\n", command_name);
        exit(-1);
    }
}
```

```

    args = arglist_lookup(idx);
    < Check for end of scrap name 142a >
    return args;
}

141d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines pop_scrap_name 147d · uses Arglist 172d,
arglist_lookup 142d, command_name 29b, exit 21b, fprintf 21b, Manager 133c, MAX_NAME_LEN 23b, nw_char 28b, 29a,
Parameters 51b, pop 138b, stderr 21b

< Check for end of scrap name 142a > ≡
{
    c = pop(manager);
    < Check for macro parameters 53a >
}

142a · used in 141d · uses pop 138b

< scraps.c 142b > + ≡
int write_scraps(FILE *file, char *spelling, Scrap_Node *defs,
                int global_indent, char *indent_chars,
                char debug_flag, char tab_flag, char indent_flag,
                unsigned char comment_flag, char location_flag,
                Arglist *inArgs,
                char *inParams[9], Parameters parameters, char *title)
{
    /* This is in file file name */
    int indent = 0;
    int newline = 1;
    while (defs) {
        < Copy defs->scrap to file 143 >
        defs = defs->next;
    }
    return indent + global_indent;
}

142b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines write_scraps 51c, 75a, 85b, 93, 122a, 130b, 147d, 149c ·
uses Arglist 172d, defs 129c, FILE 21b, next 129b, Parameters 51b, Scrap_Node 152a

< Forward declarations for scraps.c 142c > ≡
int delayed_indent = 0;

142c · defined at 142cd, 149b, 169b, 184a · used in 130a · defines delayed_indent 51c, 143, 145d, 147d, 148ab

< Forward declarations for scraps.c 142d > + ≡

#define MAX_ARGLISTS 4096
static Arglist *arglist_table[MAX_ARGLISTS];
static int arglist_count = 0;

void arglist_reset(void)
{
    arglist_count = 0;
}

static int arglist_register(Arglist *a)
{
    if (arglist_count >= MAX_ARGLISTS) {
        fprintf(stderr, "%s: too many macro argument lists (max %d)\n",

```

```

        command_name, MAX_ARGLISTS);
    exit(EXIT_FAILURE);
}
arglist_table[arglist_count] = a;
return arglist_count++;
}

```

```

static Arglist *arglist_lookup(int idx)
{
    if (idx < 0 || idx >= arglist_count) {
        fprintf(stderr, "%s: invalid arglist index %d\n", command_name, idx);
        exit(EXIT_FAILURE);
    }
    return arglist_table[idx];
}

```

142d · defined at 142cd, 149b, 169b, 184a · used in 130a · defines `arglist_count` never used, `arglist_lookup` 138c, 141d, 183b, `arglist_register` 137c, `arglist_reset` 189d, 191, `arglist_table` never used, `MAX_ARGLISTS` never used · uses `Arglist` 172d, `command_name` 29b, `exit` 21b, `fprintf` 21b, `stderr` 21b

⟨ *Copy defs->scrap to file* 143 ⟩ ≡

```

{
    char c;
    Manager reader;
    Parameters local_parameters = 0;
    int line_number = scrap_array(defs->scrap).file_line;
    reader.scrap = scrap_array(defs->scrap).slab;
    reader.prev = NULL;
    reader.index = 0;
    ⟨ Insert debugging information if required 144b ⟩
    ⟨ Insert section opening marker 144c ⟩
    if (delayed_indent)
    {
        ⟨ Insert appropriate indentation 145b ⟩
    }
    c = pop(&reader);
    while (c) {
        switch (c) {
            case '\n':
                if (global_indent >= 0) {
                    putc(c, file);
                    line_number++;
                    newline = 1;
                    delayed_indent = 0;
                    ⟨ Insert appropriate indentation 145b ⟩
                    break;
                } else {
                    /* Don't show newlines in embedded fragments */
                    fputs(". . .", file);
                    return 0;
                }
            case '\t': ⟨ Handle tab characters on output 145e ⟩
                delayed_indent = 0;
                break;
            default:
                if (c==nw_char)
                {
                    ⟨ Check for macro invocation in scrap 146a ⟩
                    break;
                }
        }
    }
}

```

```

    }
    putc(c, file);
    if (global_indent >= 0) {
        < Add more indentation ' ' 144a >
    }
    indent++;
    if (c > ' ') newline = 0;
    delayed_indent = 0;
    break;
}
c = pop(&reader);
}
< Insert section closing marker 145a >
}

```

143 · used in 142b · uses `defs 129c`, `delayed_indent 142c`, `file_line 129c`, `fputs 21b`, `Manager 133c`, `nw_char 28b`, 29a, `Parameters 51b`, `pop 138b`, `putc 21b`, `scrap_array 130a`, `slab 129c`

We need to make sure that we don't overflow `indent_chars[]`. If the line is longer than `MAX_INDENT`, we simply stop recording; the Put out the indent scrap will use spaces for any positions beyond the recorded range.

```

< Add more indentation char 144a > ≡
{
    if (global_indent + indent < MAX_INDENT)
        indent_chars[global_indent + indent] = char;
}

```

144a · used in 143, 145e, 146a · uses `MAX_INDENT 121d`

```

< Insert debugging information if required 144b > ≡
if (debug_flag) {
    fprintf(file, "\n#line %d \"%s\"\n",
            line_number, scrap_array(defs->scrap).file_name);
    < Insert appropriate indentation 145b >
}

```

144b · used in 143, 146b · uses `defs 129c`, `file_name 129c`, `fprintf 21b`, `scrap_array 130a`

Section markers in the Knuth style delimit each scrap with opening and closing comments that include the scrap number and source location. For example, in C: `/* {5: weft.w:28} */` and `/* {5:} */`.

```

< Insert section opening marker 144c > ≡
if (location_flag && comment_flag) {
    fprintf(file, "%s{%d: %s:%d}%s\n",
            comment_begin[comment_flag],
            defs->scrap,
            scrap_array(defs->scrap).file_name,
            line_number,
            comment_end[comment_flag]);
}

```

144c · used in 143 · uses `comment_begin 169b`, `comment_end 169b`, `defs 129c`, `file_name 129c`, `fprintf 21b`, `scrap_array 130a`

```

< Insert section closing marker 145a > ≡
if (location_flag && comment_flag) {
    fprintf(file, "%s{: %d}%s\n",
            comment_begin[comment_flag],
            defs->scrap,
            comment_end[comment_flag]);
}

```

```

}
145a · used in 143 · uses comment_begin 169b, comment_end 169b, defs 129c, fprintf 21b

```

```

⟨ Insert appropriate indentation 145b ⟩ ≡
{
    char c1 = pop(&reader);
    char c2 = pop(&reader);

    if (indent_flag && !(⟨ Indent suppressed 145d ⟩)) {
        ⟨ Put out the indent 145c ⟩
    }
    indent = 0;
    backup(2, &reader);
}

```

145b · used in 143, 144b · uses backup 140c, pop 138b

```

⟨ Put out the indent 145c ⟩ ≡
if (tab_flag)
    for (indent=0; indent<global_indent; indent++)
        putc(' ', file);
else
    for (indent=0; indent<global_indent; indent++)
        putc(indent < MAX_INDENT ? indent_chars[indent] : ' ', file);

```

145c · used in 51c, 145b · uses MAX_INDENT 121d, putc 21b

Indent will be suppressed if the next character is a newline or if the next two characters are @#. If the next two characters are @<, we suppress the indent for now but mark that it may be needed when the next fragment is started.

```

⟨ Indent suppressed 145d ⟩ ≡
c1 == '\n'
|| (c1 == nw_char && (c2 == '#' || (delayed_indent != (c2 == '<'))))

```

145d · used in 145b · uses delayed_indent 142c, nw_char 28b, 29a

```

⟨ Handle tab characters on output 145e ⟩ ≡
{
    if (tab_flag)
        ⟨ Expand tab into spaces 71c ⟩
    else {
        putc('\t', file);
        if (global_indent >= 0) {
            ⟨ Add more indentation ^\t' 144a ⟩
        }
        indent++;
    }
}

```

145e · used in 143 · uses putc 21b

```

⟨ Check for macro invocation in scrap 146a ⟩ ≡
{
    int oldin = indent;
    char oldcf = comment_flag;
    c = pop(&reader);
    switch (c) {
        case 't': ⟨ Copy fragment title into file 148c ⟩
            break;
    }
}

```

```

case 'c': < Copy block comment from scrap 50a >
    break;
case 'f': < Copy file name into file 147c >
    break;
case 'x': < Copy label from scrap into file 147a >
case '_': break;
case 'v': < Copy version info into file 72b >
    break;
case 's': indent = -global_indent;
    comment_flag = 0;
    break;
case '<': < Copy macro into file 147d >
    < Resync the line number after a fragment reference 146b >
    indent = oldin;
    comment_flag = oldcf;
    break;
< Handle macro parameter substitution 51c >
    indent = oldin;
    break;
default:
    if(c==nw_char)
    {
        putc(c, file);
        if (global_indent >= 0) {
            < Add more indentation ' ' 144a >
        }
        indent++;
        break;
    }
    /* ignore, since we should already have a warning */
    break;
}
}

```

146a · used in 143 · uses `nw_char` 28b, 29a, `pop` 138b, `putc` 21b

The `#line` marker that keeps the C preprocessor's line count in step is emitted *after* a fragment reference expands, to resync to the parent scrap's current line. That is right when the reference sat on a line of its own, but for a *mid-line* reference — `int port = @<port>;` or `#define V "@<version>"` — an unconditional resync would inject a stray `\n#line\n` into the middle of the line, splitting the expression or string literal across two lines (and breaking the C it produces). So we resync only when the reference *began* its line (`newline` is still set): a mid-line reference then splices its fragment's value in place and the surrounding text stays on one line, exactly as `@<port>` becoming 4000 should. The child expansion already suppresses its own markers mid-line (see `Copy macro into file`); this closes the remaining gap left by the trailing resync.

```

< Resync the line number after a fragment reference 146b > ≡
if (newline) {
    < Insert debugging information if required 144b >
}

```

146b · used in 146a

```

< Copy label from scrap into file 147a > ≡
{
    < Get label from(147b pop(&reader) ) 187b >
    write_label(label_name, file);
}

```

147b · used in 146a · uses `pop` 138b

< Copy file name into file 147c > ≡

```
if (defs->quoted)
    fprintf(file, "%cf", nw_char);
else
    fputs(spelling, file);
```

147c · used in 146a · uses defs 129c, fprintf 21b, fputs 21b, nw_char 28b, 29a

< Copy macro into file 147d > ≡

```
{
    Arglist *a = pop_scrap_name(&reader, &local_parameters);
    Name *name = a->name;
    int changed;
    Arglist * args = instance(a->args, inArgs, inParams, &changed);
    int i, nargs;
    char * p = name->spelling;
    char * * inParams = name->arg;
    Arglist *q = args;

    if (name->mark) {
        fprintf(stderr, "%s: recursive macro discovered involving <%s>\n",
            command_name, name->spelling);
        exit(-1);
    }
    if (name->defs && !defs->quoted) {
        unsigned char saved_comment = comment_flag;
        char saved_location = location_flag;
        char saved_debug = debug_flag;
        if (name->lang[0] != '\0') {
            comment_flag = name->comment_flag;
            location_flag = name->location_flag;
            debug_flag = name->debug_flag;
        }
        if (!newline) {
            comment_flag = 0;
            location_flag = FALSE;
            debug_flag = FALSE;
        }
        < Perhaps comment this macro 148a >
        name->mark = TRUE;
        indent = write_scraps(file, spelling, name->defs, global_indent + indent,
            indent_chars, debug_flag, tab_flag, indent_flag,
            comment_flag, location_flag, args, name->arg,
            local_parameters, name->spelling);
        comment_flag = saved_comment;
        location_flag = saved_location;
        debug_flag = saved_debug;
        indent -= global_indent;
        name->mark = FALSE;
    }
    else
    {
        if (delayed_indent)
        {
            for (i = indent + global_indent; --i >= 0; )
                putc(' ', file);
        }

        fprintf(file, "%c<", nw_char);
```

```

    if (name->sector == 0)
        fputc(' ', file);
    < Comment this macro use 149a >
    fprintf(file, "%c>", nw_char);
    if (!defs->quoted && !tex_flag)
        fprintf(stderr, "%s: macro never defined <%s>\n",
            command_name, name->spelling);
}

```

147d · used in 146a · uses Arglist 172d, command_name 29b, defs 129c, delayed_indent 142c, exit 21b, FALSE 23a, fprintf 21b, instance 140e, Name 152b, nw_char 28b, 29a, pop_scrap_name 141d, putc 21b, sector 129c, stderr 21b, tex_flag 27, TRUE 23a, write_scraps 142b

```

< Perhaps comment this macro 148a > ≡
if (comment_flag && newline) {
    < Perhaps put a delayed indent 148b >
    fputs(comment_begin[comment_flag], file);
    < Comment this macro use 149a >
    if (xref_flag) {
        putc(' ', file);
        write_single_scrap_ref(file, name->defs->scrap);
    }
    fputs(comment_end[comment_flag], file);
    putc('\n', file);
    if (!delayed_indent)
        for (i = indent + global_indent; --i >= 0; )
            putc(' ', file);
}

```

148a · used in 51c, 147d · uses comment_begin 169b, comment_end 169b, defs 129c, delayed_indent 142c, fputs 21b, putc 21b, write_single_scrap_ref 131c, xref_flag 27

```

< Perhaps put a delayed indent 148b > ≡
if (delayed_indent)
    for (i = indent + global_indent; --i >= 0; )
        putc(' ', file);

```

148b · used in 50a, 148a · uses delayed_indent 142c, putc 21b

```

< Copy fragment title into file 148c > ≡
{
    char * p = title;
    Arglist *q = inArgs;
    int narg;

    < Comment this macro use 149a >
    if (xref_flag) {
        putc(' ', file);
        write_single_scrap_ref(file, defs->scrap);
    }
}

```

148c · used in 146a · uses Arglist 172d, defs 129c, putc 21b, write_single_scrap_ref 131c, xref_flag 27

```

< Comment this macro use 149a > ≡
narg = 0;
while (*p != '\000') {
    if (*p == ARG_CHR) {
        if (q == NULL) {

```

```

        if (defs->quoted)
            fprintf(file, "%c'%s%c'", nw_char, inParams[narg], nw_char);
        else
            fprintf(file, "'%s'", inParams[narg]);
    }
    else {
        comment_ArglistElement(file, q, defs->quoted);
        q = q->next;
    }
    p++;
    narg++;
}
else
    fputc(*p++, file);
}

```

149a · used in 147d, 148ac · uses ARG_CHR 171b, comment_ArglistElement 149b, defs 129c, fprintf 21b, next 129b, nw_char 28b, 29a

⟨ *Forward declarations for scraps.c* 149b ⟩ + ≡

```

static void
comment_ArglistElement(FILE * file, Arglist * args, int quote)
{
    Name *name = args->name;
    Arglist *q = args->args;

    if (name == NULL) {
        if (quote)
            fprintf(file, "%c'%s%c'", nw_char, (char *)q, nw_char);
        else
            fprintf(file, "'%s'", (char *)q);
    } else if (name == (Name *)1) {
        ⟨ Include an embedded scrap in comment 149c ⟩
    } else {
        ⟨ Include a fragment use in comment 149d ⟩
    }
}

```

149b · defined at 142cd, 149b, 169b, 184a · used in 130a · defines comment_ArglistElement 149ad · uses Arglist 172d, FILE 21b, fprintf 21b, Name 152b, nw_char 28b, 29a

⟨ *Include an embedded scrap in comment* 149c ⟩ ≡

```

Embed_Node * e = (Embed_Node *)q;
fputc('{', file);
write_scraps(file, "", e->defs, -1, "", 0, 0, 0, 0, 0, e->args, NULL, NULL, "");
fputc('}', file);

```

149c · used in 149b · uses defs 129c, Embed_Node 175d, write_scraps 142b

⟨ *Include a fragment use in comment* 149d ⟩ ≡

```

char * p = name->spelling;
if (quote)
    fputc(nw_char, file);
fputc('<', file);
if (quote && name->sector == 0)
    fputc('++', file);
while (*p != '\000') {
    if (*p == ARG_CHR) {
        comment_ArglistElement(file, q, quote);
        q = q->next;
    }
    p++;
}

```

```

    p++;
}
else
    fputc(*p++, file);
}
if (quote)
    fputc(nw_char, file);
fputc('>', file);

```

149d · used in 149b · uses ARG_CHR 171b, comment_ArglistElement 149b, next 129b, nw_char 28b, 29a, sector 129c

3.2.1 Collecting Page Numbers

⟨ *Function prototypes* 150a ⟩ + ≡

```
extern void collect_numbers(char *aux_name);
```

150a · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses collect_numbers 150b

⟨ *scraps.c* 150b ⟩ + ≡

```

void collect_numbers(char *aux_name)
{
    if (number_flag) {
        int i;
        for (i=1; i<scraps; i++)
            scrap_array(i).page = i;
    }
    else {
        FILE *aux_file = fopen(aux_name, "r");
        already_warned = FALSE;
        if (aux_file) {
            char aux_line[500];
            while (fgets(aux_line, 500, aux_file)) {
                ⟨ Read line in .aux file 150c ⟩
            }
            fclose(aux_file);
            ⟨ Add letters to scraps with duplicate page numbers 151a ⟩
        }
    }
}

```

150b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines collect_numbers 39b, 150a · uses already_warned 133a, FALSE 23a, fclose 21b, FILE 21b, fopen 21b, number_flag 27, page 129c, scraps 130a, scrap_array 130a

The *memoir* document class creates references in the .aux file with nested braces, so after each open brace we have to wait for the matching closing brace.

⟨ *Read line in .aux file* 150c ⟩ ≡

```

int scrap_number;
int page_number;
int i;
int dummy_idx;
int bracket_depth = 1;
if (1 == sscanf(aux_line,
                "\\newlabel{scrap%d}{%n",
                &scrap_number,
                &dummy_idx)) {
    for (i = dummy_idx; i < (int)strlen(aux_line) && bracket_depth > 0; i++) {

```

```

    if (aux_line[i] == '{') bracket_depth++;
    else if (aux_line[i] == '}') bracket_depth--;
}
if (i > dummy_idx
    && i < (int)strlen(aux_line)
    && 1 == sscanf(aux_line+i, "%d" ,&page_number)) {
    if (scrap_number < scraps)
        scrap_array(scrap_number).page = page_number;
    else
        < Warn (only once) about needing to rerun after Latex 132g >
}
}

```

150c · used in 150b · uses [page 129c](#), [scraps 130a](#), [scrap_array 130a](#), [strlen 21b](#)

```

< Add letters to scraps with duplicate page numbers 151a > ≡
{
    int i = 0;

    < Step i to the next valid scrap 151b >
    < For all remaining scraps 151c > {
        int j = i;
        < Step j to the next valid scrap 151b >
        < Perhaps add letters to the page numbers 151d >
        i = j;
    }
}

```

151a · used in 150b

```

< Step i to the next valid scrap 151b > ≡
do
    i++;
while (i < scraps && scrap_array(i).page == -1);

```

151b · used in 151a · uses [page 129c](#), [scraps 130a](#), [scrap_array 130a](#)

```

< For all remaining scraps 151c > ≡
while (i < scraps)

```

151c · used in 151a · uses [scraps 130a](#)

```

< Perhaps add letters to the page numbers 151d > ≡
if (scrap_array(i).page == scrap_array(j).page) {
    if (scrap_array(i).letter == 0)
        scrap_array(i).letter = 'a';
    scrap_array(j).letter = scrap_array(i).letter + 1;
}

```

151d · used in 151a · uses [letter 129c](#), [page 129c](#), [scrap_array 130a](#)

3.3 Names

Each scrap is linked through a simple node:

```

< Type declarations 152a > + ≡
typedef struct scrap_node {
    struct scrap_node *next;
    int scrap;
    char quoted;
}

```

```
} Scrap_Node;
```

152a · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `Scrap_Node` 46c, 51c, 54b, 59c, 66c, 74, 75a, 77b, 79bc, 80e, 82b, 89a, 92a, 93, 94d, 96ab, 97c, 99a, 100a, 108, 110b, 111c, 112a, 130b, 136b, 137ad, 141b, 142b, 152b, 175cd, 176bc, 182, 184ab, 185b, 186c · uses `next` 129b

A `Name` represents a file or fragment definition. Beyond the spelling and the binary tree links, several flags control how the tangled output is formatted:

- `debug_flag` — if `TRUE`, emit `#line` directives (set by `-d` or auto-detected for C/C++).
- `comment_flag` — selects the comment style for section markers and block comments: 0=none, 1=C, 2=C++, 3=shell, 4=Lua/SQL, 5=HTML/XML (see Section 3.3.5).
- `location_flag` — if `TRUE`, emit Knuth-style opening and closing section markers around each scrap in the tangled output. Activated automatically by `@l` auto-detection or the `@L` command.
- `lang` — the language tag (e.g. "c", "python") used for syntax highlighting in the Markdown weave output. Set from a preceding `@l` command or inferred from the file extension of an `@o` name. Empty if unknown.

⟨ *Type declarations* 152b ⟩ + ≡

```
typedef struct name {
    char *spelling;
    struct name *llink;
    struct name *rlink;
    Scrap_Node *defs;
    Scrap_Node *uses;
    char * arg[9];
    int mark;
    char tab_flag;
    char indent_flag;
    char debug_flag;
    char location_flag;
    char suppress_markers;
    unsigned char comment_flag;
    char lang[64];
    unsigned char sector;
} Name;
```

152b · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `Name` 30, 46ab, 51c, 54b, 59c, 61ab, 70, 73g, 75b, 76b, 78abd, 80abd, 82bc, 85bc, 86a, 89b, 92a, 93, 94b, 95bc, 97b, 99a, 100ac, 101, 103c, 104a, 105c, 106ac, 108, 109, 110ab, 111abc, 112ab, 113b, 114ab, 121abc, 132af, 136a, 137bd, 138a, 140bde, 141b, 147d, 149b, 152c, 153ab, 154b, 155a, 156b, 157a, 167a, 170a, 172bcde, 174, 175c, 176bd, 179a, 180cd, 182, 184acd, 185b, 186c, 187a · uses `defs` 129c, `Scrap_Node` 152a, `sector` 129c, `uses` 129c

The variable `pending_lang` acts as a one-shot channel between the `@l` command in `pass1` (which fills it with a language tag like "js" or "python") and `collect_file_name` or `collect_macro_name` (which consume it to set the flags and the `lang` field on the next `@o` or `@d` name).

⟨ *Global variable declarations* 152c ⟩ + ≡

```
extern Name *file_names;
extern Name *macro_names;
extern Name *user_names;
extern int scrap_name_has_parameters;
extern int scrap_ended_with;
extern char pending_lang[64];
```

152c · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · defines `file_names` 39b, 42c, 51a, 76a, 94a, 105b, 111c, 112b, 114b, 121b, 153a, 167a, `macro_names` 39b, 42c, 51a, 77c, 95a, 105d, 112b, 114b, 121b, 153a, 171d, `pending_lang` 43c, 153a, 168b, 169a, `user_names` 42c, 51a, 80c, 97a, 106b, 136b, 137a, 153a, 180c · uses `Name` 152b

⟨ *Global variable definitions* 153a ⟩ + ≡

```
Name *file_names = NULL;
Name *macro_names = NULL;
Name *user_names = NULL;
int scrap_name_has_parameters;
int scrap_ended_with;
char pending_lang[64] = "";
```

153a · defined at 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a · used in 25e · uses `file_names` 152c, `macro_names` 152c, `Name` 152b, `pending_lang` 152c, `user_names` 152c

⟨ *Function prototypes* 153b ⟩ + ≡

```
extern Name *collect_file_name(void);
extern Name *collect_macro_name(void);
extern Arglist *collect_scrap_name(int current_scrap);
extern Name *name_add(Name **rt, char *spelling, unsigned char sector);
extern Name *prefix_add(Name **rt, char *spelling, unsigned char sector);
extern char *save_string(char *s);
extern void reverse_lists(Name *names);
extern void collect_lang_def(void);
extern void collect_weave_format(void);
```

153b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `Arglist` 172d, `collect_file_name` 167a, `collect_lang_def` 164, `collect_macro_name` 170a, `collect_scrap_name` 173, `collect_weave_format` 166, `Name` 152b, `name_add` 156b, `prefix_add` 154b, `reverse_lists` 176b, `save_string` 154a, `sector` 129c

⟨ *names.c* 153c ⟩ + ≡

```
enum { LESS, GREATER, EQUAL, PREFIX, EXTENSION };
```

```
static int compare(char *x, char *y)
{
    int len, result;
    int x1 = strlen(x);
    int y1 = strlen(y);
    int xp = x[x1 - 1] == ' ';
    int yp = y[y1 - 1] == ' ';
    if (xp) x1--;
    if (yp) y1--;
    len = x1 < y1 ? x1 : y1;
    result = strncmp(x, y, len);
    if (result < 0) return GREATER;
    else if (result > 0) return LESS;
    else if (x1 < y1) {
        if (xp) return EXTENSION;
        else return LESS;
    }
    else if (x1 > y1) {
        if (yp) return PREFIX;
        else return GREATER;
    }
    else return EQUAL;
}
```

153c · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `compare` 154b, 155a,

EQUAL 154b, 155a, EXTENSION 154b, 155a, GREATER 154b, 155a, LESS 154b, 155a, PREFIX 154b, 155a · uses `strlen` 21b

< names.c 154a > + ≡

```
char *save_string(char *s)
{
    char *new_str = (char *) arena_getmem((strlen(s) + 1) * sizeof(char));
    strcpy(new_str, s);
    return new_str;
}
```

154a · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `save_string` 35b, 39a, 43b, 127b, 134a, 153b, 154b, 157a, 172c, 176a · uses `arena_getmem` 190b, `strlen` 21b

< names.c 154b > + ≡

```
static int ambiguous_prefix(Name *node, char *spelling, unsigned char sector);
```

```
static char * found_name = NULL;
```

```
Name *prefix_add(Name **rt, char *spelling, unsigned char sector)
{
```

```
    Name *node = *rt;
    int cmp;

    while (node) {
        switch ((cmp = compare(node->spelling, spelling))) {
            case GREATER:    rt = &node->rlink;
                            break;
            case LESS:      rt = &node->llink;
                            break;
            case EQUAL:
                            found_name = node->spelling;
                            FALLTHROUGH;
            case EXTENSION: if (node->sector > sector) {
                            rt = &node->rlink;
                            break;
                        }
                            else if (node->sector < sector) {
                            rt = &node->llink;
                            break;
                        }
                            if (cmp == EXTENSION)
                                node->spelling = save_string(spelling);
                            return node;
            case PREFIX:    < Check for ambiguous prefix 154c >
                            return node;
        }
        node = *rt;
    }
```

< Create new name entry 157a >

```
}
```

154b · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `prefix_add` 153b, 171d · uses `compare` 153c, EQUAL 153c, EXTENSION 153c, FALLTHROUGH 23a, GREATER 153c, LESS 153c, `Name` 152b, PREFIX 153c, `save_string` 154a, `sector` 129c

Since a very short prefix might match more than one fragment name, I need to check for other matches to avoid mistakes. Basically, I simply continue the search down *both* branches of the tree.

< Check for ambiguous prefix 154c > ≡

```

{
    if (ambiguous_prefix(node->llink, spelling, sector) ||
        ambiguous_prefix(node->rlink, spelling, sector))
        fprintf(stderr,
            "%s: ambiguous prefix %c<%s...%c> (%s, line %d)\n",
            command_name, nw_char, spelling, nw_char, source_name, source_line);
}

```

154c · used in 154b · uses `command_name` 29b, `fprintf` 21b, `nw_char` 28b, 29a, `sector` 129c, `source_line` 125b, `source_name` 125b, `stderr` 21b

⟨`names.c` 155a⟩ + ≡

```

static int ambiguous_prefix(Name *node, char *spelling, unsigned char sector)
{
    while (node) {
        switch (compare(node->spelling, spelling)) {
            case GREATER:    node = node->rlink;
                            break;
            case LESS:       node = node->llink;
                            break;
            case EXTENSION:
            case PREFIX:
            case EQUAL:      if (node->sector > sector) {
                            node = node->rlink;
                            break;
                        }
                            else if (node->sector < sector) {
                            node = node->llink;
                            break;
                        }
                            return TRUE;
        }
    }
    return FALSE;
}

```

155a · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · uses `compare` 153c, `EQUAL` 153c, `EXTENSION` 153c, `FALSE` 23a, `GREATER` 153c, `LESS` 153c, `Name` 152b, `PREFIX` 153c, `sector` 129c, `TRUE` 23a

Rob Shillingsburg suggested that I organize the index of user-specified identifiers more traditionally; that is, not relying on strict ASCII comparisons via `strcmp`. Ideally, we'd like to see the index ordered like this:

```

aardvark
Adam
atom
Atomic
atoms

```

The function `robs_strcmp` implements the desired predicate. It returns -2 for alphabetically less-than, -1 for less-than but only differing in case, zero for equal, 1 for greater-than but only in case and 2 for alphabetically greater-than.

⟨*Function prototypes* 155b⟩ + ≡

```

extern int robs_strcmp(char*, char*);

```

155b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `robs_strcmp` 155c

⟨`names.c` 155c⟩ + ≡

```

int robs_strcmp(char* x, char* y)
{
    int cmp = 0;

    for (; *x && *y; x++, y++)
    {
        < Skip invisibles on x 156a >
        < Skip invisibles on y 156a >
        if (*x == *y)
            continue;
        if (islower(*x) && toupper(*x) == *y)
        {
            if (!cmp) cmp = 1;
            continue;
        }
        if (islower(*y) && *x == toupper(*y))
        {
            if (!cmp) cmp = -1;
            continue;
        }
        return 2*(toupper(*x) - toupper(*y));
    }
    if (*x)
        return 2;
    if (*y)
        return -2;
    return cmp;
}

```

155c · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `robs_strcmp` 78c, 110a, 155b, 156b, 157a, 184c · uses `islower` 21b, `toupper` 21b

Certain character sequences are invisible when printed. We don't want them to be considered for the alphabetical ordering.

< Skip invisibles on p 156a > $\equiv$

```

if (*p == '|'')
    p++;

```

156a · used in 155c

< names.c 156b > + $\equiv$

Name *name_add(Name **rt, char *spelling, unsigned char sector)

```

{
    Name *node = *rt;
    while (node) {
        int result = robs_strcmp(node->spelling, spelling);
        if (result > 0)
            rt = &node->llink;
        else if (result < 0)
            rt = &node->rlink;
        else
        {
            found_name = node->spelling;
            if (node->sector > sector)
                rt = &node->llink;
            else if (node->sector < sector)
                rt = &node->rlink;
            else
                return node;
        }
    }
}

```

```

    }
    node = *rt;
}
⟨ Create new name entry 157a ⟩
}

```

156b · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `name_add` 136b, 137a, 153b, 167a · uses `Name` 152b, `robs_strcmp` 155c, `sector` 129c

```

⟨ Create new name entry 157a ⟩ ≡
{
    node = (Name *) arena_getmem(sizeof(Name));
    if (found_name && robs_strcmp(found_name, spelling) == 0)
        node->spelling = found_name;
    else
        node->spelling = save_string(spelling);
    node->mark = FALSE;
    node->llink = NULL;
    node->rlink = NULL;
    node->uses = NULL;
    node->defs = NULL;
    node->arg[0] =
    node->arg[1] =
    node->arg[2] =
    node->arg[3] =
    node->arg[4] =
    node->arg[5] =
    node->arg[6] =
    node->arg[7] =
    node->arg[8] = NULL;
    node->tab_flag = TRUE;
    node->indent_flag = TRUE;
    node->debug_flag = FALSE;
    node->location_flag = FALSE;
    node->suppress_markers = FALSE;
    node->comment_flag = 0;
    node->lang[0] = '\0';
    node->sector = sector;
    *rt = node;
    return node;
}

```

157a · used in 154b, 156b · uses `arena_getmem` 190b, `defs` 129c, `FALSE` 23a, `Name` 152b, `robs_strcmp` 155c, `save_string` 154a, `sector` 129c, `TRUE` 23a, `uses` 129c

3.3.1 Language Auto-Detection

When a Markdown fence (‘‘‘c, ‘‘‘python) or a L^AT_EX marker (%%c) precedes an @o directive, weft auto-detects the language’s comment style and, for C/C++, enables #line directives. This is driven by a data table that maps fence tags to one of the six comment styles defined in Section 3.3.5.

Each entry records:

- **name** — the fence tag (lowercase, e.g. "js", "python", "lua").
- **comment** — index into the `comment_begin/comment_end` arrays (1–5).
- **line_directive** — TRUE only for languages whose toolchains honour #line (C family).

The built-in table covers over 250 entries (languages, file extensions, and aliases) organised by comment style. For any language not listed, the @L command (Section 3.3.2) lets users add entries at the document

level. User-defined entries are searched first, so they can also override built-in defaults. If a fence tag matches neither table, weft defaults to C++ line comments (//) with no #line.

```
<names.c 157b> + ≡
typedef struct {
    const char *name;
    unsigned char comment;
    char line_directive;
} LangEntry;

static const LangEntry lang_table[] = {
    /* ---- Style 1: C block comments WITH #line ---- */
    {"c",          1, TRUE},
    {"h",          1, TRUE},
    {"hpp",        2, TRUE},
    {"hxx",        2, TRUE},
    {"hh",         2, TRUE},
    {"cpp",        2, TRUE},
    {"c++",        2, TRUE},
    {"cxx",        2, TRUE},
    {"cc",         2, TRUE},
    {"objc",       1, TRUE},
    {"objcpp",     2, TRUE},
    {"m",          1, TRUE},
    {"mm",         2, TRUE},
    {"ino",        2, TRUE},    /* Arduino */
    {"pde",        2, TRUE},    /* Processing (C-like) */
    /* ---- Style 1: C block comments WITHOUT #line ---- */
    {"css",        1, FALSE},
    {"scss",       1, FALSE},
    {"sass",       1, FALSE},
    {"less",       1, FALSE},
    {"stylus",     1, FALSE},
    {"styl",       1, FALSE},
    /* ---- Style 2: C++ line comments (//) ---- */
    /* --- JavaScript / TypeScript --- */
    {"js",         2, FALSE},
    {"javascript", 2, FALSE},
    {"jsx",        2, FALSE},
    {"mjs",        2, FALSE},
    {"cjs",        2, FALSE},
    {"ts",         2, FALSE},
    {"typescript", 2, FALSE},
    {"tsx",        2, FALSE},
    {"mts",        2, FALSE},
    {"cts",        2, FALSE},
    {"svelte",     2, FALSE},
    {"vue",        2, FALSE},
    /* --- JVM languages --- */
    {"java",       2, FALSE},
    {"scala",      2, FALSE},
    {"sc",         2, FALSE},
    {"kotlin",     2, FALSE},
    {"kt",         2, FALSE},
    {"kts",        2, FALSE},
    {"groovy",     2, FALSE},
    {"gradle",     2, FALSE},
    {"clj",        2, FALSE},    /* Clojure (;; but // is close enough) */
    {"cljs",      2, FALSE},
```

```

{"cljc",      2, FALSE},
/* --- .NET --- */
{"cs",        2, FALSE},
{"csharp",    2, FALSE},
{"fs",        2, FALSE},
{"fsharp",    2, FALSE},
{"fsx",       2, FALSE},
{"vb",        2, FALSE}, /* VB.NET uses ' but // is a safe fallback */
/* --- Systems languages --- */
{"go",        2, FALSE},
{"rust",      2, FALSE},
{"rs",        2, FALSE},
{"zig",       2, FALSE},
{"d",         2, FALSE},
{"v",         2, FALSE},
{"vlang",     2, FALSE},
{"swift",     2, FALSE},
{"dart",      2, FALSE},
{"odin",      2, FALSE},
{"jai",       2, FALSE},
{"carbon",    2, FALSE},
{"mojo",      2, FALSE},
/* --- Web / general --- */
{"php",       2, FALSE},
{"hack",      2, FALSE},
{"jsonc",     2, FALSE},
{"json5",     2, FALSE},
{"graphql",   2, FALSE},
{"gql",       2, FALSE},
{"prisma",    2, FALSE},
{"wgs1",      2, FALSE}, /* WebGPU shading language */
/* --- GPU / shaders --- */
{"glsl",     2, FALSE},
{"hlsl",     2, FALSE},
{"vert",     2, FALSE},
{"frag",     2, FALSE},
{"comp",     2, FALSE},
{"geom",     2, FALSE},
{"tesc",     2, FALSE},
{"tese",     2, FALSE},
{"metal",    2, FALSE},
{"cu",       2, TRUE}, /* CUDA: supports #line */
{"cuh",      2, TRUE},
{"cl",       1, FALSE}, /* OpenCL: C-style block comments */
/* --- Protocol / IDL / blockchain --- */
{"proto",    2, FALSE},
{"protobuf", 2, FALSE},
{"thrift",   2, FALSE},
{"flatbuffers", 2, FALSE},
{"fbs",      2, FALSE},
{"capnp",    2, FALSE},
{"avdl",     2, FALSE}, /* Avro IDL */
{"solidity", 2, FALSE},
{"sol",      2, FALSE},
{"vyper",    3, FALSE},
{"vy",       3, FALSE},
{"move",     2, FALSE},
/* --- Functional (// style) --- */
{"reason",   2, FALSE},

```

```

{"re",          2, FALSE},
{"res",         2, FALSE}, /* ReScript */
{"ml",          2, FALSE}, /* OCaml also uses (**) but // in comments */
{"mli",         2, FALSE},
{"sml",         2, FALSE},
{"sig",         2, FALSE},
{"fut",         2, FALSE}, /* Futhark */
/* --- Other // languages --- */
{"actionscript", 2, FALSE},
{"as",          2, FALSE},
{"apex",        2, FALSE},
{"bicep",        2, FALSE},
{"ceu",         2, FALSE},
{"chapel",       2, FALSE},
{"chpl",        2, FALSE},
{"cilk",        2, TRUE},
{"jsonnet",     2, FALSE},
{"libsonnet",   2, FALSE},
{"p4",          2, FALSE},
{"qml",         2, FALSE},
{"slang",       2, FALSE},
/* ---- Style 3: Shell / hash comments (#) ---- */
/* --- Python --- */
{"py",          3, FALSE},
{"python",      3, FALSE},
{"pyx",         3, FALSE}, /* Cython */
{"pxd",         3, FALSE},
{"pyi",         3, FALSE}, /* type stubs */
{"sage",        3, FALSE}, /* SageMath */
/* --- Shell --- */
{"sh",          3, FALSE},
{"bash",        3, FALSE},
{"zsh",         3, FALSE},
{"fish",        3, FALSE},
{"ksh",         3, FALSE},
{"csh",         3, FALSE},
{"tcsh",        3, FALSE},
{"dash",        3, FALSE},
{"nu",          3, FALSE}, /* Nushell */
/* --- Classic scripting --- */
{"perl",        3, FALSE},
{"pl",          3, FALSE},
{"pm",          3, FALSE},
{"ruby",        3, FALSE},
{"rb",          3, FALSE},
{"rake",        3, FALSE},
{"gemspec",     3, FALSE},
{"r",           3, FALSE},
{"rmd",         3, FALSE},
{"tcl",         3, FALSE},
{"awk",         3, FALSE},
{"sed",         3, FALSE},
/* --- Build / make --- */
{"makefile",    3, FALSE},
{"make",        3, FALSE},
{"mk",          3, FALSE},
{"cmake",       3, FALSE},
{"just",        3, FALSE},
{"justfile",    3, FALSE},

```

```

{"meson",          3, FALSE},
{"scons",          3, FALSE},
{"bazel",          3, FALSE},
{"bzl",            3, FALSE},
{"buck",           3, FALSE},
{"pants",          3, FALSE},
{"gnumakefile",    3, FALSE},
/* --- Scientific / modern --- */
{"julia",          3, FALSE},
{"jl",             3, FALSE},
{"nim",            3, FALSE},
{"nimble",         3, FALSE},
{"crystal",        3, FALSE},
{"cr",             3, FALSE},
{"elixir",         3, FALSE},
{"ex",             3, FALSE},
{"exs",            3, FALSE},
{"coco",           3, FALSE}, /* Coconut (Python superset) */
{"coffee",        3, FALSE}, /* CoffeeScript */
{"litcoffee",     3, FALSE},
/* --- Config / DevOps --- */
{"yaml",           3, FALSE},
{"yml",            3, FALSE},
{"toml",           3, FALSE},
{"dockerfile",     3, FALSE},
{"docker",         3, FALSE},
{"containerfile",  3, FALSE},
{"terraform",      3, FALSE},
{"tf",             3, FALSE},
{"tfvars",         3, FALSE},
{"hcl",            3, FALSE},
{"nix",            3, FALSE},
{"conf",           3, FALSE},
{"ini",            3, FALSE},
{"properties",     3, FALSE},
{"env",            3, FALSE},
{"editorconfig",   3, FALSE},
{"gitignore",      3, FALSE},
{"dockerignore",   3, FALSE},
{"flake8",         3, FALSE},
{"pylint",         3, FALSE},
{"powershell",     3, FALSE},
{"ps1",            3, FALSE},
{"psm1",           3, FALSE},
{"psd1",           3, FALSE},
/* --- Hardware description (# style) --- */
{"tcl",            3, FALSE},
{"xdc",            3, FALSE}, /* Xilinx constraints */
{"sdc",            3, FALSE}, /* Synopsys constraints */
/* --- Data / query (# style) --- */
{"gn",             3, FALSE}, /* GN build */
{"cfg",            3, FALSE},
{"pip",            3, FALSE},
{"cask",           3, FALSE}, /* Homebrew */
{"gemfile",        3, FALSE},
{"vagrantfile",    3, FALSE},
{"procfile",       3, FALSE},
/* ---- Style 4: double-dash comments (--) ---- */
{"lua",            4, FALSE},

```

```

{"lua",          4, FALSE},  /* Roblox Lua */
{"sql",         4, FALSE},
{"psql",        4, FALSE},
{"plsql",        4, FALSE},
{"plpgsql",      4, FALSE},
{"mysql",        4, FALSE},
{"sqlite",       4, FALSE},
{"hive",         4, FALSE},
{"presto",       4, FALSE},
{"sparksql",     4, FALSE},
{"hs",           4, FALSE},
{"haskell",      4, FALSE},
{"lhs",          4, FALSE},
{"elm",          4, FALSE},
{"purescript",   4, FALSE},
{"purs",         4, FALSE},
{"ada",          4, FALSE},
{"adb",          4, FALSE},
{"ads",          4, FALSE},
{"vhdl",         4, FALSE},
{"vhd",          4, FALSE},
{"lean",         4, FALSE},
{"lean4",        4, FALSE},
{"agda",         4, FALSE},
{"idr",          4, FALSE},  /* Idris */
{"idris",        4, FALSE},
{"erl",          4, FALSE},  /* Erlang uses % but -- is common in embedded SQL */
{"erlang",       4, FALSE},
/* ---- Style 5: HTML comments (
```

```

{"cshtml",      5, FALSE}, /* Razor */
{"vbhtml",     5, FALSE},
{"erb",        5, FALSE},
{"ejs",        5, FALSE},
{"njk",        5, FALSE}, /* Nunjucks */
{"hbs",        5, FALSE}, /* Handlebars */
{"mustache",   5, FALSE},
{"liquid",     5, FALSE},
{"twig",       5, FALSE},
{"blade",      5, FALSE}, /* Laravel Blade */
{"pug",        5, FALSE}, /* Pug/Jade */
{"slim",       5, FALSE},
{"haml",       5, FALSE},
{"md",         5, FALSE}, /* Markdown: HTML comments are valid */
{"markdown",   5, FALSE},
{"mdx",        5, FALSE},
{"rst",        5, FALSE}, /* reStructuredText (.. is native but <!-- works) */
{"tex",        5, FALSE}, /* LaTeX: % is native but no style for it */
{"latex",     5, FALSE},
/* ---- Style 0: no comments (data formats) ---- */
{"json",       0, FALSE},
{"json5",      0, FALSE},
{"csv",        0, FALSE},
{"tsv",        0, FALSE},
{"txt",        0, FALSE},
{"text",       0, FALSE},
{"log",        0, FALSE},
{"diff",       0, FALSE},
{"patch",      0, FALSE},
/* Sentinel */
{NULL,        0, FALSE}
};

/* User-defined languages via @L command */
#define MAX_USER_LANGS 32
static LangEntry user_langs[MAX_USER_LANGS];
static char user_lang_names[MAX_USER_LANGS][64];
static int user_lang_count = 0;

static const LangEntry *find_lang(const char *name)
{
    int i;
    const LangEntry *p;
    /* User-defined languages override built-in */
    for (i = 0; i < user_lang_count; i++)
        if (strcmp(user_langs[i].name, name) == 0) return &user_langs[i];
    for (p = lang_table; p->name; p++)
        if (strcmp(p->name, name) == 0) return p;
    return NULL;
}

static void lang_to_flags(const char *lang,
                        unsigned char *comment_flag,
                        char *debug_flag,
                        char *location_flag)
{
    const LangEntry *e = find_lang(lang);
    if (e) {
        *comment_flag = e->comment;
    }
}

```

```

    *debug_flag = e->line_directive;
    *location_flag = (e->comment != 0);
} else {
    /* Unknown language: default to // comments, no #line */
    *comment_flag = 2;
    *debug_flag = FALSE;
    *location_flag = TRUE;
}
}

```

157b · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `find_lang` 168b, `LangEntry` 168b, `lang_table` never used, `lang_to_flags` 168b, 169a, `user_langs` 164, `user_lang_count` 164, `user_lang_names` 164 · uses `FALSE` 23a, `TRUE` 23a, `uses` 129c

3.3.2 User-Defined Languages

The `@L` command lets users register new language mappings (or override built-in ones) directly in the document preamble. This is useful for languages that are not in the built-in table, or when the default comment style for a language is not what the user wants.

Syntax: `@L name style [+d]`

- *name* — the fence tag that will trigger this mapping (e.g. `nim`, `gls1`, `wgs1`).
- *style* — one of `//`, `#`, `--`, `/*`, or `<!--`, selecting the comment delimiters.
- `+d` — optional; if present, enables `#line` directive generation for this language.

Examples:

```

@L nim #           % Nim uses # comments
@L glsl // +d      % GLSL uses // and supports #line
@L c //           % override: use // instead of /* */ for C

```

Up to 32 user-defined languages may be registered. They are searched before the built-in table, so a definition like `@L c //` effectively overrides the built-in mapping for C.

The `@L` command is processed during `pass1` and silently skipped during weaving. It should appear in the document preamble (before any `@o` or `@d` directives that depend on it).

```

⟨names.c 164⟩ + ≡
void collect_lang_def(void)
{
    char name[64];
    char *p = name;
    char style[16];
    char *sp;
    int c;
    unsigned char comment = 0;
    char line_dir = FALSE;

    /* Skip whitespace after @L */
    c = source_get();
    while (c == ' ' || c == '\t')
        c = source_get();

    /* Read language name */
    while (c != EOF && c != ' ' && c != '\t' && c != '\n'
           && (p - name) < 62) {
        *p++ = c;
    }
}

```

```

    c = source_get();
}
*p = '\0';

if (name[0] == '\0') {
    fprintf(stderr, "%s: expected language name after %cL (%s, %d)\n",
        command_name, nw_char, source_name, source_line);
    while (c != EOF && c != '\n') c = source_get();
    return;
}

/* Skip whitespace */
while (c == ' ' || c == '\t')
    c = source_get();

/* Read comment style token */
sp = style;
while (c != EOF && c != ' ' && c != '\t' && c != '\n'
    && (sp - style) < 14) {
    *sp++ = c;
    c = source_get();
}
*sp = '\0';

/* Map style string to comment index */
if (strcmp(style, "//") == 0)    comment = 2;
else if (strcmp(style, "#") == 0) comment = 3;
else if (strcmp(style, "--") == 0) comment = 4;
else if (strcmp(style, "/*") == 0) comment = 1;
else if (strcmp(style, "<!--") == 0) comment = 5;
else {
    fprintf(stderr,
        "%s: unknown comment style '%s' for %cL (%s, %d)\n",
        command_name, style, nw_char, source_name, source_line);
    fprintf(stderr, "  Valid styles: // # -- /* <!--\n");
    while (c != EOF && c != '\n') c = source_get();
    return;
}

/* Check for optional +d flag (#line directives) */
while (c == ' ' || c == '\t')
    c = source_get();
if (c == '+') {
    c = source_get();
    if (c == 'd') line_dir = TRUE;
}

/* Consume rest of line */
while (c != EOF && c != '\n') c = source_get();

/* Register the user language */
if (user_lang_count < MAX_USER_LANGS) {
    snprintf(user_lang_names[user_lang_count],
        sizeof(user_lang_names[user_lang_count]), "%s", name);
    user_langs[user_lang_count].name = user_lang_names[user_lang_count];
    user_langs[user_lang_count].comment = comment;
    user_langs[user_lang_count].line_directive = line_dir;
    user_lang_count++;
} else {

```

```

        fprintf(stderr, "%s: too many %cL definitions (max %d) (%s, %d)\n",
                command_name, nw_char, MAX_USER_LANGS,
                source_name, source_line);
    }
}

```

164 · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines `collect_lang_def` 44a, 153b · uses `command_name` 29b, `FALSE` 23a, `fprintf` 21b, `nw_char` 28b, 29a, `source_get` 126a, `source_line` 125b, `source_name` 125b, `stderr` 21b, `TRUE` 23a, `user_langs` 157b, `user_lang_count` 157b, `user_lang_names` 157b

3.3.3 Weave Format Declaration

The `@W` command declares the weave output format in the source file. Valid tokens are `md` (Markdown) and `tex` (L^AT_EX). If the user has already specified a format on the command line (`-w md` or `-w tex`), the declaration is silently ignored.

```

⟨names.c 166⟩ + ≡
void collect_weave_format(void)
{
    char token[64];
    char *p = token;
    int c;

    /* Skip whitespace after @W */
    c = source_get();
    while (c == ' ' || c == '\t')
        c = source_get();

    /* Read format token */
    while (c != EOF && c != ' ' && c != '\t' && c != '\n'
           && (p - token) < 62) {
        *p++ = c;
        c = source_get();
    }
    *p = '\0';

    /* Consume rest of line */
    while (c != EOF && c != '\n') c = source_get();

    if (token[0] == '\0') {
        fprintf(stderr, "%s: expected format after %cW (md or tex) (%s, %d)\n",
                command_name, nw_char, source_name, source_line);
        return;
    }

    /* If CLI already set the format, skip silently */
    if (weave_format_from_cli)
        return;

    if (strcmp(token, "md") == 0)
        weave_format = 2;
    else if (strcmp(token, "tex") == 0)
        weave_format = 1;
    else if (strcmp(token, "html") == 0)
        weave_format = 3;
    else {
        fprintf(stderr, "%s: unknown weave format '%s' for %cW (%s, %d)\n",
                command_name, token, nw_char, source_name, source_line);
        fprintf(stderr, "  Valid formats: md tex html\n");
    }
}

```

```
}
}
```

166 · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc ·
 defines `collect_weave_format` 44b, 153b · uses `command_name` 29b, `fprintf` 21b, `nw_char` 28b, 29a, `source_get` 126a,
`source_line` 125b, `source_name` 125b, `stderr` 21b, `weave_format` 27, `weave_format_from_cli` 27

3.3.4 Collecting File and Fragment Names

An output file name (@o) is terminated by whitespace. After reading the name we check for optional per-file flags (-d, -t, -i, -c) and then, if no explicit comment or debug flags were given, apply any pending language defaults from a preceding fence, %%% marker, or @L definition.

⟨`names.c` 167a⟩ + ≡

```
Name *collect_file_name(void)
{
    Name *new_name;
    char name[MAX_NAME_LEN];
    char *p = name;
    int start_line = source_line;
    int c = source_get(), c2;
    while (isspace(c))
        c = source_get();
    while (isgraph(c)) {
        *p++ = c;
        c = source_get();
    }
    if (p == name) {
        fprintf(stderr, "%s: expected file name (%s, %d)\n",
            command_name, source_name, start_line);
        exit(-1);
    }
    *p = '\0';
    /* File names are always global. */
    new_name = name_add(&file_names, name, 0);
    ⟨Handle optional per-file flags 167b⟩
    ⟨Apply pending language if no explicit flags 168b⟩
    c2 = source_get();
    if (c != nw_char || (c2 != '{' && c2 != '(' && c2 != '[')) {
        fprintf(stderr, "%s: expected %c{, %c[, or %c( after file name (%s, %d)\n",
            command_name, nw_char, nw_char, nw_char, source_name, start_line);
        exit(-1);
    }
    return new_name;
}
```

167a · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc ·
 defines `collect_file_name` 46a, 61a, 85c, 103c, 153b · uses `command_name` 29b, `exit` 21b, `file_names` 152c, `fprintf` 21b,
`isgraph` 21b, `isspace` 21b, `MAX_NAME_LEN` 23b, `Name` 152b, `name_add` 156b, `nw_char` 28b, 29a, `source_get` 126a,
`source_line` 125b, `source_name` 125b, `stderr` 21b

⟨`Handle optional per-file flags` 167b⟩ ≡

```
{
    while (1) {
        while (isspace(c))
            c = source_get();
        if (c == '-') {
            c = source_get();
            do {
```

```

switch (c) {
    case 't': new_name->tab_flag = FALSE;
                break;
    case 'd': new_name->debug_flag = TRUE;
                break;
    case 'i': new_name->indent_flag = FALSE;
                break;
    case 's': new_name->suppress_markers = TRUE;
                break;
    case 'c': < Get comment delimiters 168a >
                break;
    default : fprintf(stderr, "%s: unexpected per-file flag (%s, %d)\n",
                    command_name, source_name, source_line);
                break;
}
c = source_get();
} while (!isspace(c));
}
else break;
}
}

```

167b · used in 167a · uses `command_name` 29b, `FALSE` 23a, `fprintf` 21b, `isspace` 21b, `source_get` 126a, `source_line` 125b, `source_name` 125b, `stderr` 21b, `TRUE` 23a

The explicit per-file comment flags (`-cc`, `-c+`, `-cp`) select among the first three comment styles. These flags predate the auto-detection system; they still work and take precedence over any fence-based or `@L`-based setting.

The `-s` flag suppresses `#line` directives and section markers in the tangled output for that file, while still allowing language auto-detection to set the comment style and the `lang` tag (used by the Markdown weaver for syntax highlighting classes). This is useful for bootstrap source files or any output where tangle markers are unwanted.

```

< Get comment delimiters 168a > ≡
c = source_get();
if (c == 'c')
    new_name->comment_flag = 1;
else if (c == '+')
    new_name->comment_flag = 2;
else if (c == 'p')
    new_name->comment_flag = 3;
else
    fprintf(stderr, "%s: Unrecognised comment flag (%s, %d)\n",
            command_name, source_name, source_line);

```

168a · used in 167b · uses `command_name` 29b, `fprintf` 21b, `source_get` 126a, `source_line` 125b, `source_name` 125b, `stderr` 21b

If there is a pending language tag from a fence or `%%` marker, and the user has not specified explicit comment or debug flags, we apply the language defaults automatically.

```

< Apply pending language if no explicit flags 168b > ≡
if (pending_lang[0] != '\0') {
    strncpy(new_name->lang, pending_lang, 63);
    new_name->lang[63] = '\0';
    if (new_name->comment_flag == 0 && !new_name->debug_flag) {
        lang_to_flags(pending_lang,
                    &new_name->comment_flag,
                    &new_name->debug_flag,
                    &new_name->location_flag);
    }
}

```

```

    }
    pending_lang[0] = '\0';
} else if (new_name->lang[0] == '\0') {
    const char *dot = strrchr(new_name->spelling, '.');
    if (dot && dot[1]) {
        const LangEntry *e = find_lang(dot + 1);
        if (e) {
            strncpy(new_name->lang, dot + 1, 63);
            new_name->lang[63] = '\0';
            if (new_name->comment_flag == 0 && !new_name->debug_flag) {
                lang_to_flags(new_name->lang,
                             &new_name->comment_flag,
                             &new_name->debug_flag,
                             &new_name->location_flag);
            }
        }
    }
}
}
if (new_name->suppress_markers) {
    new_name->debug_flag = FALSE;
    new_name->location_flag = FALSE;
}

```

168b · used in 167a · uses [FALSE 23a](#), [find_lang 157b](#), [LangEntry 157b](#), [lang_to_flags 157b](#), [pending_lang 152c](#)

For `@d` (macro) definitions, the language tag is stored and flags are inferred from `@l`, just as for output files. At expansion time, if a fragment carries an explicit `@l` tag its own flags override the parent’s—so `@l txt` before `@d` suppresses markers and newlines, enabling true inline expansion.

⟨ *Apply pending language to macro* 169a ⟩ ≡

```

if (pending_lang[0] != '\0') {
    strncpy(node->lang, pending_lang, 63);
    node->lang[63] = '\0';
    if (node->comment_flag == 0 && !node->debug_flag) {
        lang_to_flags(pending_lang,
                     &node->comment_flag,
                     &node->debug_flag,
                     &node->location_flag);
    }
    pending_lang[0] = '\0';
}

```

169a · used in 170b, 172a · uses [lang_to_flags 157b](#), [pending_lang 152c](#)

3.3.5 Comment Styles

weft supports six comment styles, indexed 0–5. The arrays below provide the opening, continuation, and closing delimiters for each. Style 0 means “no comments”; the remaining five cover the vast majority of programming languages:

Index	Style	Begin	Mid	End
0	none	(empty)	(empty)	(empty)
1	C	/*	*	*/
2	C++	//	//	(empty)
3	Shell	#	#	(empty)
4	Lua/SQL	--	--	(empty)
5	HTML	<!--	(spaces)	-->

These are used both for block comments (@c) and for the Knuth-style section markers generated by the language auto-detection system.

< Forward declarations for scraps.c 169b > + ≡

```
char * comment_begin[6] = { "", "/* ", "// ", "# ", "-- ", "<!-- "};
char * comment_mid[6] = { "", " * ", "// ", "# ", "-- ", "      "};
char * comment_end[6] = { "", " */", "", "", "", " -->"};
```

169b · defined at 142cd, 149b, 169b, 184a · used in 130a · defines comment_begin 50a, 144c, 145a, 148a, comment_end 50a, 144c, 145a, 148a, comment_mid 50a

Name terminated by \n or @{; but keep skipping until @{

< names.c 170a > + ≡

```
Name *collect_macro_name(void)
```

```
{
    char name[MAX_NAME_LEN];
    char args[1000];
    char * arg[9];
    char * argp = args;
    int argc = 0;
    char *p = name;
    int start_line = source_line;
    int c = source_get(), c2;
    unsigned char sector = current_sector;

    if (c == '+') {
        sector = 0;
        c = source_get();
    }
    while (isspace(c))
        c = source_get();
    while (c != EOF) {
        Name * node;
        switch (c) {
            case '\t':
            case ' ': *p++ = ' ';
                        do
                            c = source_get();
                        while (c == ' ' || c == '\t');
                        break;
            case '\n': < Skip until scrap begins, then return name 172a >
            default:
                if (c==nw_char)
                {
                    < Check for terminating at-sequence and return name 170b >
                    break;
                }
                *p++ = c;
                c = source_get();
                break;
        }
    }
    fprintf(stderr, "%s: expected fragment name (%s, %d)\n",
            command_name, source_name, start_line);
    exit(-1);
    return NULL; /* unreachable return to avoid warnings on some compilers */
}
```

170a · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc ·

defines `collect_macro_name` 46b, 61b, 86a, 104a, 153b · uses `command_name` 29b, `current_sector` 45c, `exit` 21b, `fprintf` 21b, `isspace` 21b, `MAX_NAME_LEN` 23b, `Name` 152b, `nw_char` 28b, 29a, `sector` 129c, `source_get` 126a, `source_line` 125b, `source_name` 125b, `stderr` 21b

⟨ *Check for terminating at-sequence and return name* 170b ⟩ ≡

```
{
    c = source_get();
    switch (c) {
        case '(':
        case '[':
        case '{': ⟨ Cleanup and install name 171d ⟩
                ⟨ Apply pending language to macro 169a ⟩
                return install_args(node, argc, arg);
        case '\\': ⟨ Enter the next argument 171a ⟩
                break;
        default:
            if (c==nw_char)
            {
                *p++ = c;
                break;
            }
            fprintf(stderr,
                "%s: unexpected %c%c in fragment definition name (%s, %d)\n",
                command_name, nw_char, c, source_name, start_line);
            exit(-1);
    }
}
```

170b · used in 170a · uses `command_name` 29b, `exit` 21b, `fprintf` 21b, `nw_char` 28b, 29a, `source_get` 126a, `source_name` 125b, `stderr` 21b

⟨ *Enter the next argument* 171a ⟩ ≡

```
arg[argc] = argp;
while ((c = source_get()) != EOF) {
    if (c==nw_char) {
        c2 = source_get();
        if (c2=='\\') {
            ⟨ Make this argument 171c ⟩
            c = source_get();
            break;
        }
        else
            *argp++ = c2;
    }
    else
        *argp++ = c;
}
*p++ = ARG_CHR;
```

171a · used in 170b · uses `ARG_CHR` 171b, `nw_char` 28b, 29a, `source_get` 126a

⟨ *Type declarations* 171b ⟩ + ≡

```
#define ARG_CHR '\\001'
```

171b · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `ARG_CHR` 59c, 62a, 73g, 75b, 86b, 92a, 93, 100c, 141b, 149ad, 171a, 174, 175b

⟨ *Make this argument* 171c ⟩ ≡

```

if (argc < 9) {
    *argp++ = '\000';
    argc += 1;
}

```

171c · used in 171a

⟨ *Cleanup and install name* 171d ⟩ ≡

```

{
    if (p > name && p[-1] == ' ')
        p--;
    if (p - name > 3 && p[-1] == '.' && p[-2] == '.' && p[-3] == '.') {
        p[-3] = ' ';
        p -= 2;
    }
    if (p == name || name[0] == ' ') {
        fprintf(stderr, "%s: empty name (%s, %d)\n",
            command_name, source_name, source_line);
        exit(-1);
    }
    *p = '\0';
    node = prefix_add(&macro_names, name, sector);
}

```

171d · used in 170b, 172a, 174 · uses `command_name` 29b, `exit` 21b, `fprintf` 21b, `macro_names` 152c, `prefix_add` 154b, `sector` 129c, `source_line` 125b, `source_name` 125b, `stderr` 21b

⟨ *Skip until scrap begins, then return name* 172a ⟩ ≡

```

{
    do
        c = source_get();
    while (isspace(c));
    c2 = source_get();
    if (c != nw_char || (c2 != '{' && c2 != '(' && c2 != '[')) {
        fprintf(stderr, "%s: expected %c{ after fragment name (%s, %d)\n",
            command_name, nw_char, source_name, start_line);
        exit(-1);
    }
    ⟨ Cleanup and install name 171d ⟩
    ⟨ Apply pending language to macro 169a ⟩
    return install_args(node, argc, arg);
}

```

172a · used in 170a · uses `command_name` 29b, `exit` 21b, `fprintf` 21b, `isspace` 21b, `nw_char` 28b, 29a, `source_get` 126a, `source_name` 125b, `stderr` 21b

⟨ *Function prototypes* 172b ⟩ + ≡

```

extern Name *install_args(Name *name, int argc, char *arg[9]);

```

172b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `Name` 152b

⟨ *names.c* 172c ⟩ + ≡

```

Name *install_args(Name * name, int argc, char *arg[9])
{
    int i;

    for (i = 0; i < argc; i++) {
        if (name->arg[i] == NULL)
            name->arg[i] = save_string(arg[i]);
    }
}

```

```

    }
    return name;
}

```

172c · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · uses Name 152b, save_string 154a

⟨ *Type declarations* 172d ⟩ + ≡

```

typedef struct arglist
{Name * name;
struct arglist * args;
struct arglist * next;
} Arglist;

```

172d · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines Arglist 51c, 54b, 59c, 73g, 75b, 82c, 85b, 92a, 93, 103a, 130b, 137b, 138c, 140de, 141abcd, 142bd, 147d, 148c, 149b, 153b, 172e, 173, 175cd, 176a, 177e, 178b, 179a, 183b · uses Name 152b, next 129b

⟨ *names.c* 172e ⟩ + ≡

```

Arglist * buildArglist(Name * name, Arglist * a)
{
    Arglist * args = (Arglist *)arena_getmem(sizeof(Arglist));

    args->args = a;
    args->next = NULL;
    args->name = name;
    return args;
}

```

172e · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines buildArglist 174, 175c, 176a · uses arena_getmem 190b, Arglist 172d, Name 152b, next 129b

Terminated by @>

⟨ *names.c* 173 ⟩ + ≡

```

Arglist * collect_scrap_name(int current_scrap)
{
    char name[MAX_NAME_LEN];
    char *p = name;
    int c = source_get();
    unsigned char sector = current_sector;
    Arglist * head = NULL;
    Arglist ** tail = &head;

    if (c == '+')
    {
        sector = 0;
        c = source_get();
    }
    while (c == ' ' || c == '\t')
        c = source_get();
    while (c != EOF) {
        switch (c) {
            case '\t':
            case ' ': *p++ = ' ';
                        do
                            c = source_get();
                        while (c == ' ' || c == '\t');
                        break;
            default:

```

```

    if (c==nw_char)
    {
        < Look for end of scrap name and return 174 >
        break;
    }
    if (!isgraph(c)) {
        fprintf(stderr,
            "%s: unexpected character in fragment name (%s, %d)\n",
            command_name, source_name, source_line);
        exit(-1);
    }
    *p++ = c;
    c = source_get();
    break;
}
}
fprintf(stderr, "%s: unexpected end of file (%s, %d)\n",
    command_name, source_name, source_line);
exit(-1);
return NULL; /* unreachable return to avoid warnings on some compilers */
}

```

173 · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc ·

defines collect_scrap_name 59c, 73g, 85b, 92a, 103a, 137b, 153b, 175e · uses Arglist 172d, command_name 29b, current_sector 45c, exit 21b, fprintf 21b, isgraph 21b, MAX_NAME_LEN 23b, nw_char 28b, 29a, sector 129c, source_get 126a, source_line 125b, source_name 125b, stderr 21b

< Look for end of scrap name and return 174 > ≡

```

{
    Name * node;

    c = source_get();
    switch (c) {

        case '\': {
            < Add plain string argument 175a >
        }
        *p++ = ARG_CHR;
        c = source_get();
        break;

        case '1': case '2': case '3':
        case '4': case '5': case '6':
        case '7': case '8': case '9': {
            < Add a propagated argument 175b >
        }
        *p++ = ARG_CHR;
        c = source_get();
        break;

        case '{': {
            < Add an inline scrap argument 175c >
        }
        *p++ = ARG_CHR;
        c = source_get();
        break;

        case '<':
            < Add macro call argument 175e >
            *p++ = ARG_CHR;
            c = source_get();
            break;
    }
}

```

```

case '(':
    scrap_name_has_parameters = 1;
    < Cleanup and install name 171d >
    return buildArglist(node, head);
case '>':
    scrap_name_has_parameters = 0;
    < Cleanup and install name 171d >
    return buildArglist(node, head);

default:
    if (c==nw_char)
    {
        *p++ = c;
        c = source_get();
        break;
    }
    fprintf(stderr,
        "%s: unexpected %c%c in fragment invocation name (%s, %d)\n",
        command_name, nw_char, c, source_name, source_line);
    exit(-1);
}
}

```

174 · used in 173 · uses ARG_CHR 171b, buildArglist 172e, command_name 29b, exit 21b, fprintf 21b, Name 152b, nw_char 28b, 29a, source_get 126a, source_line 125b, source_name 125b, stderr 21b

A plain string argument has no name and a string as a value.

< Add plain string argument 175a > ≡

```

char buff[MAX_NAME_LEN];
char * s = buff;
int c, c2;

while ((c = source_get()) != EOF) {
    if (c==nw_char) {
        c2 = source_get();
        if (c2=='\''')
            break;
        *s++ = c2;
    }
    else
        *s++ = c;
}
*s = '\000';
< Add buff to current arg list 176a >

```

175a · used in 174 · uses MAX_NAME_LEN 23b, nw_char 28b, 29a, source_get 126a

A parameter argument propagated into an interior fragment has no name and a string of ARG_CHAR followed by a digit as value.

< Add a propagated argument 175b > ≡

```

char buff[3];
buff[0] = ARG_CHAR;
buff[1] = c;
buff[2] = '\000';
< Add buff to current arg list 176a >

```

175b · used in 174 · uses ARG_CHAR 171b

⟨ *Add an inline scrap argument* 175c ⟩ ≡

```
int s = collect_scrap();
Scrap_Node * d = (Scrap_Node *)arena_getmem(sizeof(Scrap_Node));
d->scrap = s;
d->quoted = 0;
d->next = NULL;
*tail = buildArglist((Name *)1, (Arglist *)d);
tail = &(*tail)->next;
```

175c · used in 174 · uses [arena_getmem](#) 190b, [Arglist](#) 172d, [buildArglist](#) 172e, [collect_scrap](#) 133f, [Name](#) 152b, [next](#) 129b, [Scrap_Node](#) 152a

⟨ *Type declarations* 175d ⟩ + ≡

```
typedef struct embed {
    Scrap_Node * defs;
    Arglist * args;
} Embed_Node;
```

175d · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines [Embed_Node](#) 51c, 141b, 149c · uses [Arglist](#) 172d, [defs](#) 129c, [Scrap_Node](#) 152a

⟨ *Add macro call argument* 175e ⟩ ≡

```
*tail = collect_scrap_name(current_scrap);
if (current_scrap >= 0)
    add_to_use((*tail)->name, current_scrap);
tail = &(*tail)->next;
```

175e · used in 174 · uses [add_to_use](#) 137d, [collect_scrap_name](#) 173, [next](#) 129b

⟨ *Add buff to current arg list* 176a ⟩ ≡

```
*tail = buildArglist(NULL, (Arglist *)save_string(buff));
tail = &(*tail)->next;
```

176a · used in 175ab · uses [Arglist](#) 172d, [buildArglist](#) 172e, [next](#) 129b, [save_string](#) 154a

⟨ *names.c* 176b ⟩ + ≡

```
static Scrap_Node *reverse(Scrap_Node *a); /* a forward declaration */
```

```
void reverse_lists(Name *names)
{
    while (names) {
        reverse_lists(names->llink);
        names->defs = reverse(names->defs);
        names->uses = reverse(names->uses);
        names = names->rlink;
    }
}
```

176b · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines [reverse_lists](#) 51a, 153b · uses [defs](#) 129c, [Name](#) 152b, [reverse](#) 176c, [Scrap_Node](#) 152a, [uses](#) 129c

Just for fun, here's a non-recursive version of the traditional list reversal code. Note that it reverses the list in place; that is, it does no new allocations.

⟨ *names.c* 176c ⟩ + ≡

```
static Scrap_Node *reverse(Scrap_Node *a)
{
    if (a) {
        Scrap_Node *b = a->next;
        a->next = NULL;
```

```

    while (b) {
        Scrap_Node *c = b->next;
        b->next = a;
        a = b;
        b = c;
    }
}
return a;
}

```

176c · defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc · defines **reverse** 27, 30, 176b · uses **next** 129b, **Scrap_Node** 152a

3.4 Searching for Index Entries

Given the array of scraps and a set of index entries, we need to search all the scraps for occurrences of each entry. The obvious approach to this problem would be quite expensive for large documents; however, there is an interesting paper describing an efficient solution [1].

```

< scraps.c 176d > + ≡
typedef struct name_node {
    struct name_node *next;
    Name *name;
} Name_Node;

```

176d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines **Name_Node** 177a, 181ab, 182 · uses **Name** 152b, **next** 129b

```

< scraps.c 177a > + ≡
typedef struct goto_node {
    Name_Node *output;           /* list of words ending in this state */
    struct move_node *moves;     /* list of possible moves */
    struct goto_node *fail;      /* and where to go when no move fits */
    struct goto_node *next;      /* next goto node with same depth */
} Goto_Node;

```

177a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines **Goto_Node** 177bcd, 180c, 181ab, 182 · uses **Name_Node** 176d, **next** 129b

```

< scraps.c 177b > + ≡
typedef struct move_node {
    struct move_node *next;
    Goto_Node *state;
    char c;
} Move_Node;

```

177b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines **Move_Node** 177d, 181ab · uses **Goto_Node** 177a, **next** 129b

```

< scraps.c 177c > + ≡
static Goto_Node *root[256];
static int max_depth;
static Goto_Node **depths;

```

177c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines **depths** 180c, 181ab, **max_depth** 180c, 181ab, **root** 102a, 111b, 180c, 181ab, 182, 184ac · uses **Goto_Node** 177a

```

< scraps.c 177d > + ≡

```

```
static Goto_Node *goto_lookup(char c, Goto_Node *g)
{
    Move_Node *m = g->moves;
    while (m && m->c != c)
        m = m->next;
    if (m)
        return m->state;
    else
        return NULL;
}
```

177d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines goto_lookup 181ab, 182 · uses Goto_Node 177a, Move_Node 177b, next 129b

3.4.1 Retrieving scrap uses

```
<scraps.c 177e> + ≡
typedef struct ArgMgr_s
{
    char * pv;
    char * bgn;
    Arglist * arg;
    struct ArgMgr_s * old;
} ArgMgr;
```

177e · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines ArgMgr 178abc, 179c · uses Arglist 172d

```
<scraps.c 178a> + ≡
typedef struct ArgManager_s
{
    Manager * m;
    ArgMgr * a;
} ArgManager;
```

178a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines ArgManager 178bc, 179c, 180c, 182, 187a · uses ArgMgr 177e, Manager 133c

```
<scraps.c 178b> + ≡
static void
pushArglist(ArgManager * mgr, Arglist * a)
{
    ArgMgr * b = malloc(sizeof(ArgMgr));

    if (b == NULL)
    {
        fprintf(stderr, "Can't allocate space for an argument manager\n");
        exit(EXIT_FAILURE);
    }
    b->pv = b->bgn = NULL;
    b->arg = a;
    b->old = mgr->a;
    mgr->a = b;
}
```

178b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines pushArglist 179a, 183b · uses Arglist 172d, ArgManager 178a, ArgMgr 177e, exit 21b, fprintf 21b, malloc 21b, stderr 21b

```

< scraps.c 178c > + ≡
static char argpop(ArgManager * mgr)
{
    while (mgr->a != NULL)
    {
        ArgMgr * a = mgr->a;

        < Perhaps —return— a character from the current arg 178d >
        < Perhaps start a new arg 179a >
        < Otherwise pop the current arg 179b >
    }

    return (pop(mgr->m));
}

```

178c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `argpop` 182, 183ab · uses `ArgManager` 178a, `ArgMgr` 177e, `pop` 138b

We separate individual arguments using spaces.

```

< Perhaps —return— a character from the current arg 178d > ≡
if (a->pv != NULL)
{
    char c = *a->pv++;

    if (c != '\0')
        return c;
    a->pv = NULL;
    return ' ';
}

```

178d · used in 178c

I'm pretty sure that only the first case is ever used. I don't think the others can occur in this context, which makes the whole —ArgManager— thing doubtful.

```

< Perhaps start a new arg 179a > ≡
if (a->arg) {
    Arglist * b = a->arg;

    a->arg = b->next;
    if (b->name == NULL) {
        a->bgn = a->pv = (char *)b->args;
    } else if (b->name == (Name *)1) {
        a->bgn = a->pv = "{Embedded Scrap}";
    } else {
        pushArglist(mgr, b->args);
    }
}

```

179a · used in 178c · uses `Arglist` 172d, `Name` 152b, `next` 129b, `pushArglist` 178b

```

< Otherwise pop the current arg 179b > ≡
} else {
    mgr->a = a->old;
    free(a);
}

```

179b · used in 178c

```

< scraps.c 179c > + ≡

```

```

static char
prev_char(ArgManager * mgr, int n)
{
    char c = '\0';
    ArgMgr * a = mgr->a;
    Manager * m = mgr->m;

    if (a != NULL) {
        ⟨ Get the nth previous character from an argument 179d ⟩
    } else {
        ⟨ Get the nth previous character from a scrap 180a ⟩
    }

    return c;
}

```

179c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `prev_char` 187a · uses `ArgManager` 178a, `ArgMgr` 177e, `Manager` 133c

```

⟨ Get the nth previous character from an argument 179d ⟩ ≡
if (a->pv && a->pv - n >= a->bgn)
    c = *a->pv;
else if (a->bgn) {
    int j = strlen(a->bgn) + 1;

    if (n >= j)
        c = a->bgn[j - n];
    else
        c = ' ';
}

```

179d · used in 179c · uses `strlen` 21b

When the offset k falls within the current slab we index it directly. When k is negative we must reach into the previous slab: since k is already negative, `SLAB_SIZE + k` yields the correct index (e.g. $k = -1$ gives `chars[1023]`, the last element).

```

⟨ Get the nth previous character from a scrap 180a ⟩ ≡
int k = m->index - n - 2;

if (k >= 0)
    c = m->scrap->chars[k];
else if (m->prev)
    c = m->prev->chars[SLAB_SIZE + k];

```

180a · used in 179c · uses `SLAB_SIZE` 129b

3.4.2 Building the Automata

```

⟨ Function prototypes 180b ⟩ + ≡
extern void search(void);

```

180b · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `search` 180c

```

⟨ scraps.c 180c ⟩ + ≡
static void build_gotos(Name *tree);
static int reject_match(Name *name, char post, ArgManager *reader);

```

```

void search(void)
{
    int i;
    for (i=0; i<256; i++)
        root[i] = NULL;
    max_depth = 10;
    depths = (Goto_Node **) arena_getmem(max_depth * sizeof(Goto_Node *));
    for (i=0; i<max_depth; i++)
        depths[i] = NULL;
    build_gotos(user_names);
    < Build failure functions 181b >
    < Search scraps 182 >
}

```

180c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines search 30, 42c, 180b, 192 · uses arena_getmem 190b, ArgManager 178a, build_gotos 180d, depths 177c, Goto_Node 177a, max_depth 177c, Name 152b, reject_match 187a, root 177c, user_names 152c

```

< scraps.c 180d > + ≡
static void build_gotos(Name *tree)
{
    while (tree) {
        < Extend goto graph with tree->spelling 181a >
        build_gotos(tree->rlink);
        tree = tree->llink;
    }
}

```

180d · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines build_gotos 180c · uses Name 152b

```

< Extend goto graph with tree->spelling 181a > ≡
{
    int depth = 2;
    char *p = tree->spelling;
    char c = *p++;
    Goto_Node *q = root[(unsigned char)c];
    Name_Node * last;
    if (!q) {
        q = (Goto_Node *) arena_getmem(sizeof(Goto_Node));
        root[(unsigned char)c] = q;
        q->moves = NULL;
        q->fail = NULL;
        q->output = NULL;
        q->next = depths[1];
        depths[1] = q;
    }
    while ((c = *p++)) {
        Goto_Node *new_node = goto_lookup(c, q);
        if (!new_node) {
            Move_Node *new_move = (Move_Node *) arena_getmem(sizeof(Move_Node));
            new_node = (Goto_Node *) arena_getmem(sizeof(Goto_Node));
            new_node->moves = NULL;
            new_node->fail = NULL;
            new_node->output = NULL;
            new_move->state = new_node;
            new_move->c = c;
            new_move->next = q->moves;
            q->moves = new_move;

```

```

    if (depth == max_depth) {
        int i;
        Goto_Node **new_depths =
            (Goto_Node **) arena_getmem(2*depth*sizeof(Goto_Node *));
        max_depth = 2 * depth;
        for (i=0; i<depth; i++)
            new_depths[i] = depths[i];
        depths = new_depths;
        for (i=depth; i<max_depth; i++)
            depths[i] = NULL;
    }
    new_node->next = depths[depth];
    depths[depth] = new_node;
}
q = new_node;
depth++;
}
last = q->output;
q->output = (Name_Node *) arena_getmem(sizeof(Name_Node));
q->output->next = last;
q->output->name = tree;
}

```

181a · used in 180d · uses [arena_getmem 190b](#), [depths 177c](#), [goto_lookup 177d](#), [Goto_Node 177a](#), [max_depth 177c](#), [Move_Node 177b](#), [Name_Node 176d](#), [next 129b](#), [root 177c](#)

⟨ *Build failure functions* 181b ⟩ ≡

```

{
    int depth;
    for (depth=1; depth<max_depth; depth++) {
        Goto_Node *r = depths[depth];
        while (r) {
            Move_Node *m = r->moves;
            while (m) {
                char a = m->c;
                Goto_Node *s = m->state;
                Goto_Node *state = r->fail;
                while (state && !goto_lookup(a, state))
                    state = state->fail;
                if (state)
                    s->fail = goto_lookup(a, state);
                else
                    s->fail = root[(unsigned char)a];
                if (s->fail) {
                    Name_Node *p = s->fail->output;
                    while (p) {
                        Name_Node *q = (Name_Node *) arena_getmem(sizeof(Name_Node));
                        q->name = p->name;
                        q->next = s->output;
                        s->output = q;
                        p = p->next;
                    }
                }
                m = m->next;
            }
            r = r->next;
        }
    }
}

```

181b · used in 180c · uses arena_getmem 190b, depths 177c, goto_lookup 177d, Goto_Node 177a, max_depth 177c, Move_Node 177b, Name_Node 176d, next 129b, root 177c

3.4.3 Searching the Scraps

⟨ Search scraps 182 ⟩ ≡

```
{
  for (i=1; i<scraps; i++) {
    char c, last = '\0';
    Manager rd;
    ArgManager reader;
    Goto_Node *state = NULL;
    rd.prev = NULL;
    rd.scrap = scrap_array(i).slab;
    rd.index = 0;
    reader.m = &rd;
    reader.a = NULL;
    c = argpop(&reader);
    while (c) {
      while (state && !goto_lookup(c, state))
        state = state->fail;
      if (state)
        state = goto_lookup(c, state);
      else
        state = root[(unsigned char)c];
      ⟨ Skip over at at 183a ⟩
      ⟨ Skip over a scrap use 183b ⟩
      ⟨ Skip over a block comment 50d ⟩
      last = c;
      c = argpop(&reader);
      if (state && state->output) {
        Name_Node *p = state->output;
        do {
          Name *name = p->name;
          if (!reject_match(name, c, &reader) &&
              scrap_array(i).sector == name->sector &&
              (!name->uses || name->uses->scrap != i)) {
            Scrap_Node *new_use =
              (Scrap_Node *) arena_getmem(sizeof(Scrap_Node));
            new_use->scrap = i;
            new_use->next = name->uses;
            name->uses = new_use;
            if (!scrap_is_in(name->defs, i))
              add_uses(&(scrap_array(i).uses), name);
          }
          p = p->next;
        } while (p);
      }
    }
  }
}
```

182 · used in 180c · uses add_uses 184c, arena_getmem 190b, ArgManager 178a, argpop 178c, defs 129c, goto_lookup 177d, Goto_Node 177a, Manager 133c, Name 152b, Name_Node 176d, next 129b, reject_match 187a, root 177c, scraps 130a, scrap_array 130a, scrap_is_in 184b, Scrap_Node 152a, sector 129c, slab 129c, uses 129c

⟨ Skip over at at 183a ⟩ ≡

```
if (last == nw_char && c == nw_char)
```

```

{
    last = '\0';
    c = argpop(&reader);
}

```

183a · used in 182 · uses [argpop 178c](#), [nw_char 28b](#), 29a

⟨ *Skip over a scrap use* 183b ⟩ ≡

```

if (last == nw_char && c == '<')
{
    char buf[MAX_NAME_LEN];
    char * p = buf;
    Arglist * args;
    int idx;

    c = argpop(&reader);
    while ((c = argpop(&reader)) != nw_char)
        *p++ = c;
    c = argpop(&reader);
    *p = '\0';
    if (sscanf(buf, "%d", &idx) != 1) {
        fprintf(stderr, "%s: found an internal problem (3)\n", command_name);
        exit(-1);
    }
    args = arglist_lookup(idx);
    pushArglist(&reader, args);
}

```

183b · used in 182 · uses [Arglist 172d](#), [arglist_lookup 142d](#), [argpop 178c](#), [command_name 29b](#), [exit 21b](#), [fprintf 21b](#), [MAX_NAME_LEN 23b](#), [nw_char 28b](#), 29a, [pushArglist 178b](#), [stderr 21b](#)

⟨ *Forward declarations for scraps.c* 184a ⟩ + ≡

```

static void add_uses(Uses **root, Name *name);
static int scrap_is_in(Scrap_Node *list, int i);

```

184a · defined at 142cd, 149b, 169b, 184a · used in 130a · uses [add_uses 184c](#), [Name 152b](#), [root 177c](#), [scrap_is_in 184b](#), [Scrap_Node 152a](#), [Uses 184d](#)

⟨ *scraps.c* 184b ⟩ + ≡

```

static int scrap_is_in(Scrap_Node * list, int i)
{
    while (list != NULL) {
        if (list->scrap == i)
            return TRUE;
        list = list->next;
    }
    return FALSE;
}

```

184b · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines [scrap_is_in 182](#), 184a · uses [FALSE 23a](#), [next 129b](#), [Scrap_Node 152a](#), [TRUE 23a](#)

⟨ *scraps.c* 184c ⟩ + ≡

```

static void add_uses(Uses * * root, Name *name)
{
    int cmp;

```

```

Uses *p, **q = root;

while ((p = *q, p != NULL)
      && (cmp = robs_strcmp(p->defn->spelling, name->spelling)) < 0)
    q = &(p->next);
if (p == NULL || cmp > 0)
{
    Uses *new_use = arena_getmem(sizeof(Uses));
    new_use->next = p;
    new_use->defn = name;
    *q = new_use;
}
}

184c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines add_uses 136b, 137a, 182, 184a · uses arena_getmem 190b,
Name 152b, next 129b, robs_strcmp 155c, root 177c, Uses 184d

< Type declarations 184d > + ≡
typedef struct uses {
    struct uses *next;
    Name *defn;
} Uses;

184d · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines Uses 98ab, 99c, 108, 110b, 130b,
131de, 184acf, 186a · uses Name 152b, next 129b, uses 129c

< Function prototypes 184e > + ≡
extern void format_uses_refs(FILE *, int);

184e · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b,
184e, 185d, 187d, 189d · used in 21a · uses FILE 21b, format_uses_refs 184f

< scraps.c 184f > + ≡

void
format_uses_refs(FILE * tex_file, int scrap)
{
    Uses * p = scrap_array(scrap).uses;
    if (p != NULL)
        < Write uses references 185a >
}

184f · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d,
177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines format_uses_refs 61ab, 184e · uses FILE 21b,
scrap_array 130a, Uses 184d, uses 129c

< Write uses references 185a > ≡
{
    char join = ' ';
    fputs("\\WEFTdot \\WEFTtxtIdentsUsed\\nobreak\\", tex_file);
    do {
        < Write one use reference 185b >
        join = ',';
        p = p->next;
    }while (p != NULL);
}

185a · used in 184f · uses fputs 21b, next 129b

```

⟨ *Write one use reference* 185b ⟩ ≡

```
Name * name = p->defn;
Scrap_Node *defs = name->defs;
int first = TRUE, page = -1;
fprintf(tex_file,
        "%c {\WEFTtint{weftkw}}\verb%c%s%c}\n\nobreak\n\n ",
        join, nw_char, name->spelling, nw_char);
if (defs)
{
    do {
        ⟨ Write one referenced scrap 185c ⟩
        first = FALSE;
        defs = defs->next;
    }while (defs!= NULL);
}
else
{
    fputs("\WEFTnotglobal", tex_file);
}
```

185b · used in 185a · uses [defs](#) 129c, [FALSE](#) 23a, [first](#) 190a, [fprintf](#) 21b, [fputs](#) 21b, [Name](#) 152b, [next](#) 129b, [nw_char](#) 28b, 29a, [page](#) 129c, [Scrap_Node](#) 152a, [TRUE](#) 23a

⟨ *Write one referenced scrap* 185c ⟩ ≡

```
fputs("\WEFTlink{weft", tex_file);
write_scrap_ref(tex_file, defs->scrap, -1, &page);
fputs("{}{", tex_file);
write_scrap_ref(tex_file, defs->scrap, first, &page);
fputs("}", tex_file);
```

185c · used in 185b, 186c · uses [defs](#) 129c, [first](#) 190a, [fputs](#) 21b, [page](#) 129c, [write_scrap_ref](#) 131b

⟨ *Function prototypes* 185d ⟩ + ≡

```
extern void format_defs_refs(FILE *, int);
```

185d · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses [FILE](#) 21b, [format_defs_refs](#) 186a

⟨ [scraps.c](#) 186a ⟩ + ≡

```
void
format_defs_refs(FILE * tex_file, int scrap)
{
    Uses * p = scrap_array(scrap).defs;
    if (p != NULL)
        ⟨ Write defs references 186b ⟩
}
```

186a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines [format_defs_refs](#) 61ab, 185d · uses [defs](#) 129c, [FILE](#) 21b, [scrap_array](#) 130a, [Uses](#) 184d

⟨ *Write defs references* 186b ⟩ ≡

```
{
    char join = ' ';
    fputs("\WEFTdot \WEFTtxtIdentsDefed\n\nobreak\n\n", tex_file);
    do {
        ⟨ Write one def reference 186c ⟩
        join = ',';
        p = p->next;
    }
```

```
    }while (p != NULL);
}
```

186b · used in 186a · uses fputs 21b, next 129b

⟨ *Write one def reference* 186c ⟩ ≡

```
Name * name = p->defn;
Scrap_Node *defs = name->uses;
int first = TRUE, page = -1;
fprintf(tex_file,
        "%c {\\WEFTtint{weftkw}\\verb%c%s%c}\\nobreak\\ ",
        join, nw_char, name->spelling, nw_char);
if (defs == NULL
    || (defs->scrap == scrap && defs->next == NULL)) {
    fputs("\\WEFTtxtIdentsNotUsed", tex_file);
}
else {
    do {
        if (defs->scrap != scrap) {
            ⟨ Write one referenced scrap 185c ⟩
            first = FALSE;
        }
        defs = defs->next;
    }while (defs!= NULL);
}
```

186c · used in 186b · uses defs 129c, FALSE 23a, first 190a, fprintf 21b, fputs 21b, Name 152b, next 129b, nw_char 28b, 29a, page 129c, Scrap_Node 152a, TRUE 23a, uses 129c

Rejecting Matches

A problem with simple substring matching is that the string “he” would match longer strings like “she” and “her.” Norman Ramsey suggested examining the characters occurring immediately before and after a match and rejecting the match if it appears to be part of a longer token. Of course, the concept of *token* is language-dependent, so we may be occasionally mistaken. For the present, we’ll consider the mechanism an experiment.

⟨ *scraps.c* 186d ⟩ + ≡

```
#define sym_char(c) (isalnum(c) || (c) == '_' )
```

```
static int op_char(char c)
{
    switch (c) {
        case '!':          case '#': case '%': case '$': case '^':
        case '&': case '*': case '-': case '+': case '=': case '/':
        case '|': case '~': case '<': case '>':
            return TRUE;
        default:
            return c==nw_char ? TRUE : FALSE;
    }
}
```

186d · defined at 25b, 129bc, 130a, 131abcde, 132bedef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines op_char 187a, sym_char 187a · uses FALSE 23a, nw_char 28b, 29a, TRUE 23a

⟨ *scraps.c* 187a ⟩ + ≡

```
static int reject_match(Name *name, char post, ArgManager *reader)
{
```

```

int len = strlen(name->spelling);
char first = name->spelling[0];
char last = name->spelling[len - 1];
char prev = prev_char(reader, len);
if (sym_char(last) && sym_char(post)) return TRUE;
if (sym_char(first) && sym_char(prev)) return TRUE;
if (op_char(last) && op_char(post)) return TRUE;
if (op_char(first) && op_char(prev)) return TRUE;
return FALSE; /* Here is 187a-01 */
}

```

187a · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `reject_match` 180c, 182 · uses `ArgManager` 178a, `FALSE` 23a, `first` 190a, `Name` 152b, `op_char` 186d, `prev_char` 179c, `strlen` 21b, `sym_char` 186d, `TRUE` 23a

3.5 Labels

Refer to 187b-01. And another one 187a-01.

```

⟨ Get label from 187b ⟩ ≡
char label_name[MAX_NAME_LEN];
char * p = label_name;
while (c = @1, c != nw_char) /* Here is 187b-01 */
    *p++ = c;
*p = '\0';
c = @1;

```

187b · used in 73a, 135a, 147a · uses `MAX_NAME_LEN` 23b, `nw_char` 28b, 29a

```

⟨ scraps.c 187c ⟩ + ≡
void
write_label(char label_name[], FILE * file)
⟨ Search for label(⟨ Write the label to file 188a ⟩, ⟨ Complain about missing label 188b ⟩) 188g ⟩

```

187c · defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac · defines `write_label` 73a, 147a, 187d · uses `FILE` 21b

```

⟨ Function prototypes 187d ⟩ + ≡
void write_label(char label_name[], FILE * file);
187d · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses FILE 21b, write_label 187c

```

```

⟨ Write the label to file 188a ⟩ ≡
write_single_scrap_ref(file, lbl->scrap);
fprintf(file, "-%02d", lbl->seq);
188a · used in 187c · uses fprintf 21b, write_single_scrap_ref 131c

```

```

⟨ Complain about missing label 188b ⟩ ≡
fprintf(stderr, "Can't find label %s.\n", label_name);
188b · used in 187c · uses fprintf 21b, stderr 21b

```

```

⟨ Save label to label store 188c ⟩ ≡
if (label_name[0])
⟨ Search for label(⟨ Complain about duplicate labels 188e ⟩, ⟨ Create a new label entry 188d ⟩) 188g ⟩
else
{
    ⟨ Complain about empty label 188f ⟩
}

```

```
}
```

188c · used in 135a

⟨ *Create a new label entry* 188d ⟩ ≡

```
lbl = (label_node *)arena_getmem(sizeof(label_node) + (p - label_name));
lbl->left = lbl->right = NULL;
strcpy(lbl->name, label_name);
lbl->scrap = current_scrap;
lbl->seq = ++lblseq;
*plbl = lbl;
```

188d · used in 188c · uses arena_getmem 190b, label_node 189b

⟨ *Complain about duplicate labels* 188e ⟩ ≡

```
fprintf(stderr, "Duplicate label %s.\n", label_name);
```

188e · used in 188c · uses fprintf 21b, stderr 21b

⟨ *Complain about empty label* 188f ⟩ ≡

```
fprintf(stderr, "Empty label.\n");
```

188f · used in 188c · uses fprintf 21b, stderr 21b

⟨ *Search for label(Found,Notfound)* 188g ⟩ ≡

```
{
    label_node * * plbl = &label_tab;
    for (;;)
    {
        label_node * lbl = *plbl;

        if (lbl)
        {
            int cmp = label_name[0] - lbl->name[0];

            if (cmp == 0)
                cmp = strcmp(label_name + 1, lbl->name + 1);
            if (cmp < 0)
                plbl = &lbl->left;
            else if (cmp > 0)
                plbl = &lbl->right;
            else
            {
                Found
                break;
            }
        }
        else
        {
            Notfound
            break;
        }
    }
}
```

188g · used in 187c, 188c · uses label_node 189b, label_tab 189a

⟨ *global.c* 189a ⟩ + ≡

```
label_node * label_tab = NULL;
```

189a · defined at 25e, 189a · defines label_tab 188g, 189c · uses label_node 189b

< Type declarations 189b > + ≡

```
typedef struct l_node
{
    struct l_node * left, * right;
    int scrap, seq;
    char name[1];
} label_node;
```

189b · defined at 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b · used in 21a · defines `label_node` 188dg, 189ac

< Global variable declarations 189c > + ≡

```
extern label_node * label_tab;
```

189c · defined at 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c · used in 21a · uses `label_node` 189b, `label_tab` 189a

3.6 Memory Management

I manage memory using a simple scheme inspired by Hanson’s idea of *arenas* [2]. Basically, I allocate all the storage required when processing a source file (primarily for names and scraps) using calls to `arena_getmem(n)`, where `n` specifies the number of bytes to be allocated. When the storage is no longer required, the entire arena is freed with a single call to `arena_free()`. Both operations are quite fast.

< Function prototypes 189d > + ≡

```
extern void *arena_getmem(size_t n);
extern void arena_free(void);
extern void arglist_reset(void);
```

189d · defined at 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d · used in 21a · uses `arena_free` 191, `arena_getmem` 190b, `arglist_reset` 142d, `size_t` 21b

< arena.c 189e > + ≡

```
typedef struct chunk {
    struct chunk *next;
    char *limit;
    char *avail;
} Chunk;
```

189e · defined at 25d, 189e, 190ab, 191 · defines `Chunk` 190acd · uses `next` 129b

We define an empty chunk called `first`. The variable `arena` points at the current chunk of memory; it’s initially pointed at `first`. As soon as some storage is required, a “real” chunk of memory will be allocated and attached to `first->next`; storage will be allocated from the new chunk (and later chunks if necessary).

< arena.c 190a > + ≡

```
static Chunk first = { NULL, NULL, NULL };
static Chunk *arena = &first;
```

190a · defined at 25d, 189e, 190ab, 191 · defines `arena` 190bcd, 191, `first` 30, 63e, 99ab, 100a, 108, 109, 112b, 130b, 131b, 185bc, 186c, 187a, 191, 192 · uses `Chunk` 189e

3.6.1 Allocating Memory

The routine `arena_getmem(n)` returns a pointer to (at least) `n` bytes of memory. Note that `n` is rounded up to ensure that returned pointers are always aligned. We align to the nearest 8-byte segment, since that’ll satisfy the more common 2-byte and 4-byte alignment restrictions too.

< arena.c 190b > + ≡

```

void *arena_getmem(size_t n)
{
    char *q;
    char *p = arena->avail;
    n = (n + 7) & ~7;           /* ensuring alignment to 8 bytes */
    q = p + n;
    if (q <= arena->limit) {
        arena->avail = q;
        return p;
    }
    < Find a new chunk of memory 190c >
}

```

190b · defined at 25d, 189e, 190ab, 191 · defines `arena_getmem` 35b, 39a, 46c, 53a, 131a, 133d, 134a, 136b, 137ad, 141bc, 154a, 157a, 172e, 175c, 180c, 181ab, 182, 184c, 188d, 189d · uses `arena` 190a, `size_t` 21b

If the current chunk doesn't have adequate space (at least `n` bytes) we examine the rest of the list of chunks (starting at `arena->next`) looking for a chunk with adequate space. If `n` is very large, we may not find it right away or we may not find a suitable chunk at all.

```

< Find a new chunk of memory 190c > ≡
{
    Chunk *ap = arena;
    Chunk *np = ap->next;
    while (np) {
        char *v = sizeof(Chunk) + (char *) np;
        if (v + n <= np->limit) {
            np->avail = v + n;
            arena = np;
            return v;
        }
        ap = np;
        np = ap->next;
    }
    < Allocate a new chunk of memory 190d >
}

```

190c · used in 190b · uses `arena` 190a, `Chunk` 189e, `next` 129b

If there isn't a suitable chunk of memory on the free list, then we need to allocate a new one.

```

< Allocate a new chunk of memory 190d > ≡
{
    size_t m = n + 10000;
    np = (Chunk *) malloc(m);
    np->limit = m + (char *) np;
    np->avail = n + sizeof(Chunk) + (char *) np;
    np->next = NULL;
    ap->next = np;
    arena = np;
    return sizeof(Chunk) + (char *) np;
}

```

190d · used in 190c · uses `arena` 190a, `Chunk` 189e, `malloc` 21b, `next` 129b, `size_t` 21b

3.6.2 Freeing Memory

To free all the memory in the arena, we need only point `arena` back to the first empty chunk.

⟨ arena.c 191 ⟩ + ≡

```
void arena_free(void)
{
    arena = &first;
    arglist_reset();
}
```

191 · defined at 25d, 189e, 190ab, 191 · defines arena_free 39b, 189d · uses arena 190a, arglist_reset 142d, first 190a

Chapter 4

Man page

Here is the UNIX man page for weft:

```
<weft.1 192> ≡
.TH WEFT 1 "March 2026"
.SH NAME
weft, a literate programming tool
.SH SYNOPSIS
.B weft
.br
\fBweft\fP [options] [file] ...
.SH DESCRIPTION
.I weft
is a literate programming tool like Knuth's
.I WEB,
only simpler.
A
.I weft
file contains program source code interleaved with documentation.
When
.I weft
is given a
.I weft
file, it writes the program file(s).
By default only tangling is performed. Weaving (documentation
generation) is requested with \fB-w\fP.
.SH COMMAND LINE OPTIONS
.br
\fB-t\fP Suppresses weave output (even if \fB-w\fP is given).
.br
\fB-o\fP Suppresses generation of the output files.
.br
\fB-d\fP List dangling identifier references in indexes.
.br
\fB-c\fP Forces output files to overwrite old files of the same
name without comparing for equality first.
.br
\fB-v\fP The verbose flag. Forces output of progress reports.
.br
\fB-n\fP Forces sequential numbering of scraps (instead of page
numbers).
.br
\fB-s\fP Doesn't print list of scraps making up file at end of
```

each scrap.

`\fB-p path`\fP Prepend path to the filenames for all the output files.

`\fB-V string`\fP Provide the string for replacement of the @v operation. This is intended as a means for including version information in generated output.

`\fB-x`\fP Include cross-references in comments in output files.

`\fB-h options`\fP Turn on hyperlinks using the hyperref package of LaTeX and provide the options to the package.

`\fB-r`\fP Turn on hyperlinks using the hyperref package of LaTeX, with the package options being in the text.

`\fB-I path`\fP Provide a directory to search for included files. This may appear several times.

.br

`\fB-m`\fP Emit a JSON map of all output files, fragments, their definitions, and dependency relationships to stdout. No weave or tangle is performed. Useful for AI tooling and project navigation.

.br

`\fB-e name`\fP Extract the named fragment and all its transitive dependencies as Markdown to stdout. Fragment names support prefix matching. No weave or tangle is performed.

.br

`\fB-R file[:line]`\fP Read section markers from a tangled output file and emit JSON mapping to stdout. If a line number is given, emit the single innermost scrap covering that line. Without a line number, emit all scrap regions. No .weft source files are needed.

.br

`\fB-w [format]`\fP Request weave output. Without a format token, the format is taken from the `\fB@W`\fP directive in the source file. With `\fBmd`\fP, a `\fB.md`\fP file is generated; with `\fBtex`\fP, a `\fB.tex`\fP file is generated. A CLI format overrides `\fB@W`\fP. If `\fB-w`\fP is given without a format and no `\fB@W`\fP exists, weft exits with an error.

.br

`\fB--bodies`\fP Include raw scrap body text in `\fB-m`\fP JSON output. Each definition entry gets a `\fB"body"`\fP field with the JSON-escaped source code of the scrap.

.br

`\fB--prose`\fP Include the documentation paragraph preceding each `\fB@d`\fP/\fB@o\fP directive as a `\fB"prose"`\fP field in `\fB-m`\fP JSON output.

.br

`\fB--diff`\fP Dry-run tangle: show a unified diff of what would change in each output file without writing anything. If a file does not yet exist, all lines are shown with a `\fB+`\fP prefix.

.br

`\fB--lint`\fP Static analysis of the literate source. Reports unused fragments (defined but never referenced), undefined references (used but never defined), and fuzzy-match suggestions for likely typos (Levenshtein distance ≤ 3).

.br

`\fB--callers`\fP When combined with `\fB-e name`\fP, show the upward call chain from `\fB@o`\fP roots down to the named fragment as an indented tree.

.br

`\fB--errors`\fP Read compiler or linter output from stdin and annotate `\fBfile:line`\fP patterns with `\fB.weft`\fP source locations by parsing section markers embedded in the tangled output. No `\fB.weft`\fP source files are needed. Example: `\fBmake 2>&1 | weft --errors`\fP.

.br

`\fB--help`\fP Print a comprehensive help message including all options,

usage examples, and an AI-assisted workflow guide, then exit.

.SH FORMAT OF WEFT FILES

A

.I weft

file contains mostly ordinary

.I LaTeX.

The file is read and copied to output (.tex file) unless a

.I weft

command is encountered. All

.I weft

commands start with an ‘‘at-sign’’ (@).

Files and fragments are defined with the following commands:

.PP

@o \fIfile-name flags scrap\fP where scrap is smaller than one page.

.br

@O \fIfile-name flags scrap\fP where scrap is bigger than one page.

.br

@d \fIfragment-name scrap\fP. Where scrap is smaller than one page.

.br

@D \fIfragment-name scrap\fP. Where scrap is bigger than one page.

.PP

@q \fIfragment-name scrap\fP. Where scrap is smaller than one page.

The scrap is not expanded in the output, allowing you to construct output files which can, perhaps after further processing, be input to

.I weft.

.br

@Q \fIfragment-name scrap\fP. Where scrap is bigger than one page.

Likewise.

.PP

Scraps have specific begin and end

markers;

which begin and end marker you use determines how the scrap will be typeset in the .tex file:

.br

\fB@{\fP...\fB@}\fP for code, typeset with the LaTeX \fBlistings\fP package (load \fB\usepackage{listings}\fP in your preamble).

.br

\fB@[\fP...\fB@]\fP for LaTeX paragraph mode formatting, and

.br

\fB@(\fP...\fB@)\fP for LaTeX math mode formatting.

.br

Any amount of whitespace

(including carriage returns) may appear between a name and the beginning of a scrap.

.PP

Several code/file scraps may have the same name;

.I weft

concatenates their definitions to produce a single scrap.

Code scrap definitions are like macro definitions;

.I weft

extracts a program by expanding one scrap.

The definition of that scrap contains references to other scraps, which are themselves expanded, and so on.

\fIweft\fP’s output is readable; it preserves the indentation of expanded scraps with respect to the scraps in which they appear.

.PP

.SH PER FILE OPTIONS

When defining an output file, the programmer has the option of using flags

to control the output.

.PP

\fB-d\fR option,

.I weft

will emit line number indications at scrap boundaries.

.br

\fB-i\fR option,

.I weft

supresses the indentation of fragments (useful for \fBFortran\fR).

.br

\fB-t\fP option makes \fIweft\fP

copy tabs untouched from input to output.

.br

\fB-c\fIx\fP Include comments in the output file.

\fIx\fP may be \fBc\fP for C-style comments, \fB+\fP for C++ and

\fBp\fP for perl and similar.

.br

\fB-s\fP Suppress section markers and \fB#line\fP directives in the

tangled output for this file. Language auto-detection still sets

the comment style and the \fBlang\fP tag (used for Markdown syntax

highlighting), but no markers appear in the output.

.PP

.SH MINOR COMMANDS

.br

@@ Causes a single ‘‘at-sign’’ to be copied into the output.

.br

@_ Causes the text between it and the next {\tt @_} to be made bold

(for keywords, etc.) in the formatted document

.br

%% Comments out a line so that it doesn’t appear in the output.

.br

@i \fBfilename\fR causes the file named to be included.

.br

@f Creates an index of output files.

.br

@m Creates an index of fragments.

.br

@u Creates an index of user-specified identifiers.

.br

@W \fIformat\fP Declares the weave output format (\fBmd\fP or \fBtex\fP).

Used by \fB-w\fP when no explicit format is given on the command line.

.br

@l \fItag\fP Sets the language for the next @o or @d directive.

The tag is looked up in the language table to determine the comment

style for section markers and the \fBclass\fP attribute for Markdown

syntax highlighting. Place it on the line before the directive it

applies to.

.br

@L \fIname style\fP [+d] Defines a custom language mapping.

\fIname\fP is the tag (e.g. \fBnim\fP, \fBgls\fP),

\fIstyle\fP is one of \fB/\fP, \fB#\fP, \fB--\fP, \fB/*\fP, or \fB<!--\fP.

The optional \fB+d\fP flag enables \fB#line\fP directive generation.

User-defined languages override built-in entries. Must appear in the

document preamble before any @o or @d that depends on it.

.PP

.SH SECTION MARKERS

By default, when a language is detected (via @l, @L, or file extension

of @o), weft wraps each scrap in the tangled output with opening and

closing section markers using the appropriate comment syntax. For example,

C output includes:
.nf

```
/* {1: source.w:10} */  
int main() { return 0; }  
/* {1} */
```

.fi
The opening marker shows the scrap number and source location; the closing marker shows only the scrap number. Nested fragment expansions produce nested markers.

.PP
Languages with \fB#line\fP support (C, C++, Objective-C, etc.) also emit \fB#line\fP directives at scrap boundaries.

.PP
The \fB-s\fP per-file flag suppresses both section markers and \fB#line\fP directives.

.PP
.SH LANGUAGE INFERENCE
When no @l tag precedes an @o directive, weft infers the language from the output file's extension. For example, \fB@o app.js\fP automatically detects JavaScript (// comments, no #line). The inferred language is used for section markers in the tangle and for the \fBclass="language-X"\fP attribute in the Markdown weave.

.PP
Over 250 file extensions are recognised, covering C, C++, Python, JavaScript, Go, Rust, Ruby, Lua, SQL, HTML, and many more. Extensions not in the table produce no markers.

.PP
To mark an identifier for inclusion in the index, it must be mentioned at the end of the scrap it was defined in. The line starts with @| and ends with the \fBend of scrap\fP mark \fB@}\fP.

.PP
.SH ERROR MESSAGES

.PP
.SH BUGS

.PP
.SH AUTHOR

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192 · uses [exit 21b](#), [FILE 21b](#), [first 190a](#), [main 25f](#), [next 129b](#), [page 129c](#), [prose 129c](#), [scraps 130a](#), [search 180c](#)

Chapter 5

Indices

Three sets of indices can be created automatically: an index of file names, an index of fragment names, and an index of user-specified identifiers. An index entry includes the name of the entry, where it was defined, and where it was referenced.

5.1 Files

`arena.c` defined at 25d, 189e, 190ab, 191
`global.c` defined at 25e, 189a
`global.h` defined at 21a
`html.c` defined at 24c, 100c, 101, 105c, 106ac
`input.c` defined at 25a, 125de, 126ac, 129a
`latex.c` defined at 24a, 54b, 55, 62b, 66c, 67ab, 68ab, 69bcd, 70, 71a, 75b, 76b, 78ab, 80bd
`main.c` defined at 23c, 25f
`markdown.c` defined at 24b, 82cd, 86c, 89ab, 92b, 93, 94b, 95bc, 97b, 98a, 99c
`names.c` defined at 25c, 153c, 154ab, 155ac, 156b, 157b, 164, 166, 167a, 170a, 172ce, 173, 176bc
`output.c` defined at 24d, 107, 108, 109, 110ab, 111abc, 112ab, 113ab, 114ab, 115ab, 116, 117, 119, 120, 121ce
`pass1.c` defined at 23d, 42bc
`scraps.c` defined at 25b, 129bc, 130a, 131abcde, 132bcdef, 133cdef, 137d, 138bc, 139, 140abcde, 141d, 142b, 150b, 176d, 177abcde, 178abc, 179c, 180cd, 184bcf, 186ad, 187ac
`weft.1` defined at 192

5.2 Fragments

< Accumulate prose character 43a > used in 42d
< Accumulate scrap and return scraps++ 134b > used in 133f
< Add a propagated argument 175b > used in 174
< Add an at character to the block or break 48b > used in 48a
< Add an inline scrap argument 175c > used in 174
< Add any other character to the block 48c > used in 48a
< Add buff to current arg list 176a > used in 175ab
< Add letters to scraps with duplicate page numbers 151a > used in 150b
< Add macro call argument 175e > used in 174
< Add more indentation char 144a > used in 143, 145e, 146a
< Add one char to the block buffer 48a > used in 47b
< Add plain string argument 175a > used in 174
< Add the source path to the include path list 39a > used in 38b
< Add user identifier definition 137a > used in 136a
< Add user identifier use 136b > used in 136a
< Add scrap to name's definition list 46c > used in 46ab
< Allocate a new chunk of memory 190d > used in 190c

< *Apply pending language if no explicit flags* 168b > used in 167a
 < *Apply pending language to macro* 169a > used in 170b, 172a
 < *Begin or end a block comment* 50e > used in 58c
 < *Begin the cross-reference environment* 65e > used in 61ab
 < *Begin the scrap environment* 63a > used in 61ab
 < *Bold Keyword* 73e > used in 71e
 < *Bold md Keyword* 91b > used in 90c
 < *Build a new arglist* 141c > used in 140e
 < *Build failure functions* 181b > used in 180c
 < *Build fragment definition* 46b > used in 44d
 < *Build output file definition* 46a > used in 44d
 < *Build source_name and tex_name* 38b > used in 38a
 < *Check at-sequence for end-of-scrap* 71e > used in 70
 < *Check for ambiguous prefix* 154c > used in 154b
 < *Check for cosmetic fence line* 83b > used in 83a
 < *Check for end of scrap name* 142a > used in 141d
 < *Check for long options* 29f > used in 29e
 < *Check for macro invocation in scrap* 146a > used in 143
 < *Check for macro parameters* 53a > used in 142a
 < *Check for terminating at-sequence and return name* 170b > used in 170a
 < *Check md at-sequence for end-of-scrap* 90c > used in 89b
 < *Cleanup and install name* 171d > used in 170b, 172a, 174
 < *Close the current sector* 45b > used in 44d, 58c, 84a, 103a
 < *Collect a block comment* 47b > used in 44d
 < *Collect include-file name* 128a > used in 127b
 < *Collect language apply in pass1* 43c > used in 44d
 < *Collect user language definition in pass1* 44a > used in 44d
 < *Collect user-specified index entries* 136a > used in 134c
 < *Collect weave format declaration in pass1* 44b > used in 44d
 < *Comment this macro use* 149a > used in 147d, 148ac
 < *Compare the temp file and the old file* 124a > used in 123a
 < *Complain about duplicate labels* 188e > used in 188c
 < *Complain about empty label* 188f > used in 188c
 < *Complain about missing label* 188b > used in 187c
 < *Copy block comment from scrap* 50a > used in 146a
 < *Copy file name into file* 147c > used in 146a
 < *Copy fragment title into file* 148c > used in 146a
 < *Copy label from scrap into file* 147a > used in 146a
 < *Copy label from source into* 73a > used in 58c, 71e, 84a, 90c, 103a
 < *Copy macro into file* 147d > used in 146a
 < *Copy source into html_file* 102c > used in 101
 < *Copy version info into file* 72b > used in 71e, 90c, 146a
 < *Copy version info into md file* 85a > used in 84a
 < *Copy version info into tex file* 59b > used in 58c
 < *Copy defs->scrap to file* 143 > used in 142b
 < *Copy source_file into md_file* 83a > used in 82d
 < *Copy source_file into tex_file* 58b > used in 55
 < *Create a new label entry* 188d > used in 188c
 < *Create new name entry* 157a > used in 154b, 156b
 < *Create new scrap, managed by writer* 134a > used in 133f
 < *Emit a quoted scrap argument inline* 75a > used in 75b
 < *End block* 63d > used in 50e, 63a, 65a
 < *Enter the next argument* 171a > used in 170b
 < *Expand macro into file* 59c > used in 58c
 < *Expand md macro into file* 85b > used in 84a
 < *Expand md tab into spaces* 90b > used in 89b
 < *Expand tab into spaces* 71c > used in 70, 145e
 < *Extend goto graph with tree->spelling* 181a > used in 180d

< *Fill in the html scrap 104b* > used in 103c, 104a
 < *Fill in the md scrap 87a* > used in 85c, 86a
 < *Fill in the middle of the scrap environment 63e* > used in 61ab
 < *Find a free temporary file 122b* > used in 122a
 < *Find a new chunk of memory 190c* > used in 190b
 < *Finish html scrap 104c* > used in 103c, 104a
 < *Finish md scrap 87b* > used in 85c, 86a
 < *Finish the cross-reference environment 65f* > used in 61ab
 < *Finish the scrap environment 65a* > used in 61ab
 < *For all remaining scraps 151c* > used in 151a
 < *Format a file index entry 77a* > used in 76b
 < *Format a md file index entry 94c* > used in 94b
 < *Format a md index entry 95d* > used in 95c
 < *Format a md user index entry 97c* > used in 97b
 < *Format a user index entry 80e* > used in 80d
 < *Format an index entry 79a* > used in 78b
 < *Format macro name 73g* > used in 71e
 < *Format macro parameters 53b* > used in 73g, 75b
 < *Format md macro name 92a* > used in 90c
 < *Forward declarations for scraps.c 142cd, 149b, 169b, 184a* > used in 130a
 < *Function prototypes 42a, 54a, 69a, 71b, 80a, 82ab, 100b, 121ab, 125a, 126b, 130b, 132a, 138a, 141a, 150a, 153b, 155b, 172b, 180b, 184e, 185d, 187d, 189d* > used in 21a
 < *Get comment delimiters 168a* > used in 167b
 < *Get label from 187b* > used in 73a, 135a, 147a
 < *Get label while collecting scrap 135a* > used in 134c
 < *Get the nth previous character from a scrap 180a* > used in 179c
 < *Get the nth previous character from an argument 179d* > used in 179c
 < *Global variable declarations 27, 28b, 29b, 45c, 49c, 65b, 125b, 133a, 152c, 189c* > used in 21a
 < *Global variable definitions 28a, 29ac, 45d, 49d, 65c, 125c, 133b, 153a* > used in 25e
 < *Handle an "at" character 127a* > used in 126a
 < *Handle at-sign during scrap accumulation 134c* > used in 134b
 < *Handle macro invocation in scrap 137b* > used in 134c
 < *Handle macro parameter substitution 51c* > used in 146a
 < *Handle optional per-file flags 167b* > used in 167a
 < *Handle reverse map mode 36d* > used in 25f
 < *Handle tab characters on output 145e* > used in 143
 < *Handle the file name in argv[arg] 38a* > used in 37
 < *Handle EOF 128b* > used in 126a
 < *HTML-escape character c 90a* > used in 89b
 < *If we break the line at this word 50c* > used in 50a
 < *Include a fragment use in comment 149d* > used in 149b
 < *Include an embedded scrap in comment 149c* > used in 149b
 < *Include block comment in a scrap 49f* > used in 134c
 < *Include files 21b* > used in 21a
 < *Indent suppressed 145d* > used in 145b
 < *Insert appropriate indentation 145b* > used in 143, 144b
 < *Insert debugging information if required 144b* > used in 143, 146b
 < *Insert section closing marker 145a* > used in 143
 < *Insert section opening marker 144c* > used in 143
 < *Interpret at-sequence 58c* > used in 58b
 < *Interpret command-line arguments 29de* > used in 25f
 < *Interpret html at-sequence 103a* > used in 102c
 < *Interpret md at-sequence 84a* > used in 83a
 < *Interpret the argument string s 33* > used in 29e
 < *Italic "whatever" 73f* > used in 71e
 < *Italic md "whatever" 91c* > used in 90c
 < *Limits 23b* > used in 21a
 < *Look for end of scrap name and return 174* > used in 173

< *Make this argument 171c* > used in 171a
 < *Move a word to the file 50b* > used in 50a
 < *Move the temporary file to the target, if required 123a* > used in 122a
 < *Note a delimiter collision 71d* > used in 70
 < *Open an include file 127b* > used in 127a
 < *Operating System Dependencies 26* > used in 21a, 25e
 < *Otherwise pop the current arg 179b* > used in 178c
 < *Perhaps add an include path 35b* > used in 29e
 < *Perhaps add letters to the page numbers 151d* > used in 151a
 < *Perhaps comment this macro 148a* > used in 51c, 147d
 < *Perhaps get the extract name 36b* > used in 29e
 < *Perhaps get the hyperref options 36a* > used in 29e
 < *Perhaps get the prepend path 35a* > used in 29e
 < *Perhaps get the reverse map argument 36c* > used in 29e
 < *Perhaps get the version info string 35c* > used in 29e
 < *Perhaps put a delayed indent 148b* > used in 50a, 148a
 < *Perhaps —return— a character from the current arg 178d* > used in 178c
 < *Perhaps skip white-space 49a* > used in 48c
 < *Perhaps start a new arg 179a* > used in 178c
 < *Print help message 30* > used in 29f
 < *Process a file 39b* > used in 38a
 < *Process the remaining arguments (file names) 37* > used in 25f
 < *Put out the indent 145c* > used in 51c, 145b
 < *Read line in .aux file 150c* > used in 150b
 < *Rename the temporary file to the target 124b* > used in 123a, 124a
 < *Resync the line number after a fragment reference 146b* > used in 146a
 < *Reverse cross-reference lists 51a* > used in 42c
 < *Rob's ordering 78c* > used in 78b, 95c
 < *Save label to label store 188c* > used in 135a
 < *Save macro name 137c* > used in 137b
 < *Save macro parameters 52* > used in 137b
 < *Save pending prose 43b* > used in 46ab
 < *Scan at-sequence 44d* > used in 42d
 < *Scan the source file, looking for at-sequences 42d* > used in 42c
 < *Search for label(Found,Notfound) 188g* > used in 187c, 188c
 < *Search scraps 182* > used in 180c
 < *Set locale information 36e* > used in 25f
 < *Set up name, args and next 141b* > used in 140e
 < *Show diff of temp file vs real file 123b* > used in 123a
 < *Show md presence of a block comment 91a* > used in 90c
 < *Show presence of a block comment 49e* > used in 71e
 < *Skip at-L language definition line 44c* > used in 58c, 84a, 103a
 < *Skip commented-out code 73d* > used in 71e, 90c, 134c
 < *Skip invisibles on p 156a* > used in 155c
 < *Skip over a block comment 50d* > used in 182
 < *Skip over a scrap use 183b* > used in 182
 < *Skip over an in-text scrap 46d* > used in 44d
 < *Skip over at at 183a* > used in 182
 < *Skip over at-sign or go to skipped 47a* > used in 46d
 < *Skip over index entries 73c* > used in 71e, 90c
 < *Skip to the next nw-char 49b* > used in 47b
 < *Skip until scrap begins, then return name 172a* > used in 170a
 < *Skip whitespace 47c* > used in 47b
 < *Sort key of size n for ordering 78d* > used in 78b, 95c
 < *Start block 63b* > used in 50e, 63a
 < *Step i to the next valid scrap 151b* > used in 151a
 < *Step to next sector 45a* > used in 44d, 58c, 84a, 103a
 < *Switch block 63c* > used in 63a

< *Type declarations* 23a, 51b, 121d, 152ab, 171b, 172d, 175d, 184d, 189b > used in 21a
 < *version 2* > used in 23b
 < *Warn (only once) about needing to rerun after LaTeX* 132g > used in 131b, 150c
 < *Write abbreviated definition list* 74 > used in 73g, 75b
 < *Write defining scrap numbers* 79b > used in 79a
 < *Write defs references* 186b > used in 186a
 < *Write file defs* 65g > used in 61a
 < *Write file's defining scrap numbers* 77b > used in 77a
 < *Write html file footer* 104d > used in 103c
 < *Write html index of file names* 105b > used in 103a
 < *Write html index of macro names* 105d > used in 103a
 < *Write html index of user names* 106b > used in 103a
 < *Write html macro definition* 104a > used in 103a
 < *Write html macro footer* 105a > used in 104a
 < *Write html output file definition* 103c > used in 103a
 < *Write in-text scrap* 65d > used in 58c
 < *Write index of file names* 76a > used in 58c
 < *Write index of macro names* 77c > used in 58c
 < *Write index of user-specified names* 80c > used in 58c
 < *Write LaTeX limbo definitions* 56a > used in 55
 < *Write macro definition* 61b > used in 58c
 < *Write macro defs* 66a > used in 61b
 < *Write macro refs* 66b > used in 61b
 < *Write md defining scrap numbers* 96a > used in 95d
 < *Write md defs references* 99d > used in 99c
 < *Write md file cross-refs* 87d > used in 85c
 < *Write md file defs* 87e > used in 87d
 < *Write md file's defining scrap numbers* 94d > used in 94c
 < *Write md in-text scrap* 87c > used in 84a
 < *Write md index of file names* 94a > used in 84a
 < *Write md index of macro names* 95a > used in 84a
 < *Write md index of user-specified names* 97a > used in 84a
 < *Write md macro cross-refs* 88a > used in 86a
 < *Write md macro definition* 86a > used in 84a
 < *Write md macro defs* 88b > used in 88a
 < *Write md macro refs* 88c > used in 88a
 < *Write md output file definition* 85c > used in 84a
 < *Write md referencing scrap numbers* 96b > used in 95d
 < *Write md uses references* 98b > used in 98a
 < *Write one def reference* 186c > used in 186b
 < *Write one md def reference* 100a > used in 99d
 < *Write one md referenced scrap* 99b > used in 99a, 100a
 < *Write one md use reference* 99a > used in 98b
 < *Write one referenced scrap* 185c > used in 185b, 186c
 < *Write one use reference* 185b > used in 185a
 < *Write out files->spelling* 122a > used in 121c
 < *Write output file definition* 61a > used in 58c
 < *Write referencing scrap numbers* 79c > used in 79a
 < *Write the AtBeginDocument setup* 58a > used in 56a
 < *Write the HTML document head* 102a > used in 101
 < *Write the HTML document tail* 102b > used in 101
 < *Write the hyperlink usage macro* 57a > used in 56a
 < *Write the label to file* 188a > used in 187c
 < *Write the macro's name* 62a > used in 61b, 79a
 < *Write the md macro's name* 86b > used in 86a, 95d
 < *Write the modern decoration macros* 57b > used in 56a
 < *Write the scrap formatting macros* 56b > used in 56a
 < *Write uses references* 185a > used in 184f

5.3 Identifiers

Knuth prints his index of identifiers in a two-column format. I could force this automatically by emitting the `\twocolumn` command; but this has the side effect of forcing a new page. Therefore, it seems better to leave it this up to the user.

add_to_use: 52, 137b, [137d](#), 138a, 175e.
add_uses: 136b, 137a, 182, 184a, [184c](#).
already_warned: 132g, [133a](#), 133b, 150b.
arena: [190a](#), 190bcd, 191.
arena_free: 39b, 189d, [191](#).
arena_getmem: 35b, 39a, 46c, 53a, 131a, 133d, 134a, 136b, 137ad, 141bc, 154a, 157a, 172e, 175c, 180c, 181ab, 182, 184c, 188d, 189d, [190b](#).
Arglist: 51c, 54b, 59c, 73g, 75b, 82c, 85b, 92a, 93, 103a, 130b, 137b, 138c, 140de, 141abcd, 142bd, 147d, 148c, 149b, 153b, [172d](#), 172e, 173, 175cd, 176a, 177e, 178b, 179a, 183b.
arglist_count: [142d](#).
arglist_lookup: 138c, 141d, [142d](#), 183b.
arglist_register: 137c, [142d](#).
arglist_reset: [142d](#), 189d, 191.
arglist_table: [142d](#).
ArgManager: [178a](#), 178bc, 179c, 180c, 182, 187a.
ArgMgr: [177e](#), 178abc, 179c.
argpop: [178c](#), 182, 183ab.
ARG_CHR: 59c, 62a, 73g, 75b, 86b, 92a, 93, 100c, 141b, 149ad, 171a, [171b](#), 174, 175b.
backup: [140c](#), 145b.
begin_tex_escape: 49e, [69d](#), 71e, 73efg.
blockBuff: 47b, [49c](#), 49df.
bodies_flag: [27](#), 28a, 29f, 108.
buildArglist: [172e](#), 174, 175c, 176a.
build_gotos: 180c, [180d](#).
callers_flag: [27](#), 28a, 29f, 39b.
Chunk: [189e](#), 190acd.
clear_marks: [111a](#), 112ab.
code_inline: 53b, [69d](#), 70.
collect_file_name: 46a, 61a, 85c, 103c, 153b, [167a](#).
collect_lang_def: 44a, 153b, [164](#).
collect_lint_names: [113b](#), 114b.
collect_macro_name: 46b, 61b, 86a, 104a, 153b, [170a](#).
collect_numbers: 39b, 150a, [150b](#).
collect_scrap: 46ab, 52, 130b, [133f](#), 175c.
collect_scrap_name: 59c, 73g, 85b, 92a, 103a, 137b, 153b, [173](#), 175e.
collect_weave_format: 44b, 153b, [166](#).
command_name: [29b](#), 29cd, 30, 33, 37, 39b, 44d, 46d, 47a, 55, 59c, 66b, 71d, 73g, 75b, 82d, 88c, 92a, 93, 101, 112ab, 122b, 124b, 127ab, 128a, 129a, 132g, 134bc, 136a, 141d, 142d, 147d, 154c, 164, 166, 167ab, 168a, 170ab, 171d, 172a, 173, 174, 183b.
comment_ArglistElement: 149a, [149b](#), 149d.
comment_begin: 50a, 144c, 145a, 148a, [169b](#).
comment_end: 50a, 144c, 145a, 148a, [169b](#).
comment_mid: 50a, [169b](#).
compare: [153c](#), 154b, 155a.
compare_flag: [27](#), 28a, 33, 123a.
copy_scrap: 53b, 54b, 63e, 65d, [70](#).
current_sector: 39b, 45ab, [45c](#), 45d, 77c, 80c, 95a, 97a, 105d, 106b, 134a, 136a, 170a, 173.
dangling_flag: [27](#), 28a, 33, 80e, 97c, 106c.
defs: 46c, 51c, 59c, 61ab, 65g, 66a, 73g, 74, 75b, 77b, 79b, 80e, 85b, 87e, 88b, 92a, 93, 94d, 96a, 97c, 99ab, 100ac, 103c, 104ad, 105ac, 106a, 108, 110b, 113b, 114a, 122a, [129c](#), 130b, 131e, 134a, 137a, 141b, 142b, 143, 144bc, 145a, 147cd, 148ac, 149ac, 152b, 157a, 175d, 176b, 182, 185bc, 186ac.
delayed_indent: 51c, [142c](#), 143, 145d, 147d, 148ab.
depths: [177c](#), 180c, 181ab.

diff_flag: [27](#), [28a](#), [29f](#), [123a](#).
double_at: [125d](#), [127a](#), [129a](#).
dump_scrap_json: [108](#), [132a](#), [139](#).
dump_scrap_text: [110b](#), [132a](#), [138c](#), [139](#).
Embed_Node: [51c](#), [141b](#), [149c](#), [175d](#).
emit_name_json: [108](#), [109](#).
end_line: [108](#), [129c](#), [132d](#), [134ac](#).
end_tex_escape: [49e](#), [69d](#), [71e](#), [73efg](#).
EQUAL: [153c](#), [154b](#), [155a](#).
errors_flag: [27](#), [28a](#), [29f](#), [36d](#).
exit: [21b](#), [25f](#), [29f](#), [36d](#), [37](#), [39b](#), [46d](#), [47a](#), [67b](#), [112ab](#), [117](#), [122b](#), [124b](#), [127ab](#), [128a](#), [129a](#), [134bc](#), [136a](#), [141d](#), [142d](#), [147d](#), [167a](#), [170ab](#), [171d](#), [172a](#), [173](#), [174](#), [178b](#), [183b](#), [192](#).
EXTENSION: [153c](#), [154b](#), [155a](#).
extract_flag: [27](#), [28a](#), [33](#), [36b](#).
extract_fragment: [110b](#), [112b](#).
extract_name: [27](#), [28a](#), [36b](#), [39b](#).
extra_scraps: [63e](#), [65a](#), [65b](#), [65c](#), [75a](#), [87ab](#), [93](#), [104bc](#).
FALLTHROUGH: [23a](#), [44d](#), [58c](#), [71e](#), [84a](#), [103a](#), [126a](#), [154b](#).
FALSE: [23a](#), [27](#), [28a](#), [33](#), [35abc](#), [36abc](#), [39b](#), [58bc](#), [63d](#), [65d](#), [66c](#), [69d](#), [70](#), [75a](#), [79b](#), [80be](#), [83ab](#), [84a](#), [85b](#), [87c](#), [89a](#), [93](#), [96a](#), [97c](#), [99a](#), [100a](#), [103a](#), [111a](#), [124b](#), [127a](#), [129a](#), [147d](#), [150b](#), [155a](#), [157ab](#), [164](#), [167b](#), [168b](#), [184b](#), [185b](#), [186cd](#), [187a](#).
fclose: [21b](#), [55](#), [75a](#), [82d](#), [101](#), [117](#), [119](#), [122ab](#), [123b](#), [124a](#), [128b](#), [139](#), [150b](#).
FILE: [21b](#), [30](#), [54b](#), [55](#), [62b](#), [66c](#), [68a](#), [69cd](#), [70](#), [75ab](#), [76b](#), [78b](#), [80d](#), [82bcd](#), [86c](#), [89ab](#), [92b](#), [93](#), [94b](#), [95c](#), [97b](#), [98a](#), [99c](#), [100c](#), [101](#), [105c](#), [106ac](#), [107](#), [108](#), [110b](#), [117](#), [119](#), [122a](#), [123b](#), [124a](#), [125de](#), [130b](#), [131bc](#), [132a](#), [138c](#), [139](#), [142b](#), [149b](#), [150b](#), [184ef](#), [185d](#), [186a](#), [187cd](#), [192](#).
file_line: [129c](#), [132c](#), [134ab](#), [143](#).
file_name: [42ac](#), [54a](#), [55](#), [82ad](#), [100b](#), [101](#), [129c](#), [132b](#), [134ab](#), [144bc](#).
file_names: [39b](#), [42c](#), [51a](#), [76a](#), [94a](#), [105b](#), [111c](#), [112b](#), [114b](#), [121b](#), [152c](#), [153a](#), [167a](#).
find_closest: [113b](#), [114a](#).
find_lang: [157b](#), [168b](#).
find_name: [110a](#), [112ab](#).
first: [30](#), [63e](#), [99ab](#), [100a](#), [108](#), [109](#), [112b](#), [130b](#), [131b](#), [185bc](#), [186c](#), [187a](#), [190a](#), [191](#), [192](#).
fopen: [21b](#), [55](#), [82d](#), [101](#), [117](#), [119](#), [122b](#), [123b](#), [124a](#), [127b](#), [129a](#), [150b](#).
format_defs_refs: [61ab](#), [185d](#), [186a](#).
format_entry: [54b](#), [77c](#), [78b](#).
format_file_entry: [54b](#), [76a](#), [76b](#).
format_user_entry: [54b](#), [80c](#), [80d](#).
format_uses_refs: [61ab](#), [184e](#), [184f](#).
fprintf: [21b](#), [33](#), [36e](#), [37](#), [39b](#), [42c](#), [44d](#), [46d](#), [47a](#), [49e](#), [53b](#), [55](#), [57a](#), [59c](#), [61a](#), [63ae](#), [65g](#), [66b](#), [67b](#), [71d](#), [73efg](#), [75b](#), [77a](#), [80e](#), [82d](#), [85c](#), [87ae](#), [88c](#), [91bc](#), [92a](#), [93](#), [94c](#), [97c](#), [99a](#), [100a](#), [101](#), [104b](#), [107](#), [108](#), [110b](#), [112ab](#), [114ab](#), [117](#), [122ab](#), [124b](#), [127ab](#), [128a](#), [129a](#), [131b](#), [132g](#), [134bc](#), [136a](#), [138c](#), [139](#), [141d](#), [142d](#), [144bc](#), [145a](#), [147cd](#), [149ab](#), [154c](#), [164](#), [166](#), [167ab](#), [168a](#), [170ab](#), [171d](#), [172a](#), [173](#), [174](#), [178b](#), [183b](#), [185b](#), [186c](#), [188abef](#).
fputs: [21b](#), [50a](#), [51c](#), [53b](#), [56ab](#), [57b](#), [58a](#), [59bc](#), [61ab](#), [62b](#), [63abcde](#), [65aefg](#), [66abc](#), [68a](#), [69cd](#), [70](#), [71e](#), [72b](#), [73g](#), [74](#), [75ab](#), [76a](#), [77abc](#), [79abc](#), [80ce](#), [85ac](#), [86ac](#), [87ae](#), [88bc](#), [89ab](#), [90ac](#), [91ab](#), [92ab](#), [93](#), [94acd](#), [95ad](#), [96ab](#), [97ac](#), [98b](#), [99abd](#), [100ac](#), [102ab](#), [103ac](#), [104abd](#), [105abcd](#), [106abc](#), [107](#), [108](#), [109](#), [110b](#), [111c](#), [117](#), [120](#), [123b](#), [131b](#), [139](#), [143](#), [147c](#), [148a](#), [185abc](#), [186bc](#).
getc: [21b](#), [124a](#), [126a](#), [127ab](#), [128ab](#), [129a](#).
getenv: [21b](#), [36e](#).
get_scrap_defs: [99c](#), [130b](#), [131e](#).
get_scrap_uses: [98a](#), [108](#), [110b](#), [130b](#), [131d](#).
goto_lookup: [177d](#), [181ab](#), [182](#).
Goto_Node: [177a](#), [177bcd](#), [180c](#), [181ab](#), [182](#).
GREATER: [153c](#), [154b](#), [155a](#).
has_sector: [77c](#), [80a](#), [80b](#), [80c](#), [95a](#), [97a](#), [105d](#), [106b](#).
html_emit_name: [100c](#), [104a](#), [106a](#).
html_esc: [100c](#).
html_format_entry: [101](#), [105d](#), [106a](#).
html_format_file_entry: [101](#), [105b](#), [105c](#).

[html_format_user_entry](#): 101, 106b, [106c](#).
[html_puts](#): [100c](#), 103c, 105c, 106c.
[html_ref](#): [100c](#), 103a.
[hyperoptions](#): [27](#), 28a, 36a, 56a, 57a.
[hyperopt_flag](#): [27](#), 28a, 33, 36a.
[hyperref_flag](#): [27](#), 28a, 33, 36a, 56a.
[incl](#): [27](#), 28a, 35b, 39a, 127b.
[includepath_flag](#): [27](#), 28a, 33, 35b.
[include_depth](#): [125d](#), 127b, 128b, 129a.
[init_scraps](#): 42c, 130b, [131a](#).
[instance](#): 85b, [140e](#), 141ab, 147d.
[isgraph](#): [21b](#), 128a, 167a, 173.
[islower](#): [21b](#), 155c.
[isspace](#): [21b](#), 65a, 87b, 104c, 136a, 167ab, 170a, 172a.
[is_file_name](#): [111b](#), 111c.
[json_puts](#): [107](#), 108, 117.
[label_node](#): 188dg, 189a, [189b](#), 189c.
[label_tab](#): 188g, [189a](#), 189c.
[LangEntry](#): [157b](#), 168b.
[lang_table](#): [157b](#).
[lang_to_flags](#): [157b](#), 168b, 169a.
[latex_code_putc](#): 54b, [68a](#), 69c, 70, 75a.
[latex_language](#): 54b, 63e, [68b](#).
[LESS](#): [153c](#), 154b, 155a.
[letter](#): 120, [129c](#), 131b, 134a, 151d.
[levenshtein](#): [113a](#), 113b.
[lint_flag](#): [27](#), 28a, 29f, 39b.
[lint_name](#): [114a](#), 114b.
[lint_names](#): [113b](#).
[lint_name_count](#): [113b](#), 114b.
[load_entry](#): [78a](#), 78b.
[lookup](#): 51c, [140d](#), 141b.
[macro_names](#): 39b, 42c, 51a, 77c, 95a, 105d, 112b, 114b, 121b, [152c](#), 153a, 171d.
[main](#): [25f](#), 192.
[make_parent_dirs](#): 121c, [121e](#), 122a.
[malloc](#): [21b](#), 78b, 95c, 178b, 190d.
[Manager](#): [133c](#), 133def, 138bc, 140c, 141d, 143, 178a, 179c, 182.
[map_flag](#): [27](#), 28a, 33, 39b.
[markdown_flag](#): [27](#), 28a, 33, 39b.
[MarkerFrame](#): [115a](#), 115b, 117, 119, 120.
[MAX_ARGLISTS](#): [142d](#).
[max_depth](#): [177c](#), 180c, 181ab.
[MAX_INDENT](#): 85b, [121d](#), 122a, 144a, 145c.
[MAX_LINT_NAMES](#): [113b](#).
[MAX_NAME_LEN](#): 23b, 136a, 138c, 141d, 167a, 170a, 173, 175a, 183b, 187b.
[md_copy_scrap](#): 82b, 87ac, [89b](#), 103a, 104b.
[md_format_defs_refs](#): 82c, 87d, 88a, [99c](#).
[md_format_entry](#): 82c, 95a, [95c](#).
[md_format_file_entry](#): 82c, 94a, [94b](#).
[md_format_user_entry](#): 82c, 97a, [97b](#).
[md_format_uses_refs](#): 82c, 87d, 88a, [98a](#).
[md_load_entry](#): [95b](#), 95c.
[md_print_scrap_numbers](#): 82b, 87e, 88bc, [89a](#), 94d, 96b, 104d, 105ac, 106ac.
[md_scrap_type](#): [89b](#), 90c.
[md_write_arg](#): 82c, 86b, [86c](#), 90c, 93.
[md_write_ArglistElement](#): 82c, 92a, [93](#).
[md_write_literal](#): 82c, 92a, [92b](#), 93.
[mkdirs_flag](#): [27](#), 28a, 29f, 122a, 124b.

Move_Node: [177b](#), [177d](#), [181ab](#).
 Name: [30](#), [46ab](#), [51c](#), [54b](#), [59c](#), [61ab](#), [70](#), [73g](#), [75b](#), [76b](#), [78abd](#), [80abd](#), [82bc](#), [85bc](#), [86a](#), [89b](#), [92a](#), [93](#), [94b](#), [95bc](#), [97b](#), [99a](#), [100ac](#), [101](#), [103c](#), [104a](#), [105c](#), [106ac](#), [108](#), [109](#), [110ab](#), [111abc](#), [112ab](#), [113b](#), [114ab](#), [121abc](#), [132af](#), [136a](#), [137bd](#), [138a](#), [140bde](#), [141b](#), [147d](#), [149b](#), [152b](#), [152c](#), [153ab](#), [154b](#), [155a](#), [156b](#), [157a](#), [167a](#), [170a](#), [172bcde](#), [174](#), [175c](#), [176bd](#), [179a](#), [180cd](#), [182](#), [184acd](#), [185b](#), [186c](#), [187a](#).
 name_add: [136b](#), [137a](#), [153b](#), [156b](#), [167a](#).
 Name_Node: [176d](#), [177a](#), [181ab](#), [182](#).
 next: [27](#), [30](#), [33](#), [35b](#), [39a](#), [46c](#), [51c](#), [59c](#), [65g](#), [66abc](#), [73g](#), [74](#), [75b](#), [77b](#), [79bc](#), [80e](#), [87e](#), [88bc](#), [89a](#), [92a](#), [93](#), [94d](#), [96ab](#), [97c](#), [98b](#), [99ad](#), [100ac](#), [104d](#), [105a](#), [108](#), [110b](#), [111c](#), [112a](#), [125a](#), [127b](#), [129b](#), [133d](#), [136b](#), [137ad](#), [138b](#), [140de](#), [141bc](#), [142b](#), [149ad](#), [152a](#), [172de](#), [175ce](#), [176acd](#), [177abd](#), [179a](#), [181ab](#), [182](#), [184bcd](#), [185ab](#), [186bc](#), [189e](#), [190cd](#), [192](#).
 number_flag: [27](#), [28a](#), [33](#), [39b](#), [150b](#).
 num_scraps: [78b](#), [95c](#), [130a](#), [130b](#).
 nw_char: [28b](#), [29a](#), [30](#), [39b](#), [42d](#), [44d](#), [46d](#), [47a](#), [48a](#), [49bf](#), [50d](#), [52](#), [53ab](#), [58bc](#), [61a](#), [67a](#), [69c](#), [70](#), [71ae](#), [73ce](#), [77a](#), [80e](#), [83a](#), [84a](#), [89b](#), [90c](#), [91b](#), [102c](#), [103a](#), [126a](#), [127a](#), [129a](#), [134bc](#), [135a](#), [136a](#), [137bc](#), [138c](#), [141d](#), [143](#), [145d](#), [146a](#), [147cd](#), [149abd](#), [154c](#), [164](#), [166](#), [167a](#), [170ab](#), [171a](#), [172a](#), [173](#), [174](#), [175a](#), [183ab](#), [185b](#), [186cd](#), [187b](#).
 op_char: [186d](#), [187a](#).
 output_flag: [27](#), [28a](#), [33](#), [39b](#).
 owner: [111c](#), [112a](#), [129c](#), [132f](#), [134a](#), [140b](#).
 page: [66c](#), [79b](#), [80e](#), [89a](#), [96a](#), [97c](#), [99ab](#), [100a](#), [129c](#), [130b](#), [131bc](#), [134a](#), [150bc](#), [151bd](#), [185bc](#), [186c](#), [192](#).
 Parameters: [51b](#), [53a](#), [85b](#), [130b](#), [141d](#), [142b](#), [143](#).
 parse_close_marker: [116](#), [117](#), [119](#).
 parse_open_marker: [115b](#), [117](#), [119](#).
 pass1: [39b](#), [42a](#), [42c](#), [71e](#), [90c](#).
 pending_lang: [43c](#), [152c](#), [153a](#), [168b](#), [169a](#).
 pending_prose: [42b](#), [42c](#), [43b](#), [46ab](#).
 pop: [50abd](#), [53a](#), [138b](#), [138c](#), [141d](#), [142a](#), [143](#), [145b](#), [146a](#), [147b](#), [178c](#).
 pop_scrap_name: [141d](#), [147d](#).
 PREFIX: [153c](#), [154b](#), [155a](#).
 prefix_add: [153b](#), [154b](#), [171d](#).
 prev_char: [179c](#), [187a](#).
 prev_sector: [39b](#), [45a](#), [45c](#), [45d](#).
 print_callers: [111c](#), [112a](#).
 print_scrap_numbers: [54b](#), [65g](#), [66ab](#), [66c](#), [77b](#), [79c](#).
 prose: [27](#), [29f](#), [30](#), [63e](#), [108](#), [129c](#), [132e](#), [134a](#), [140a](#), [192](#).
 prose_buf: [42b](#), [43ab](#).
 prose_flag: [27](#), [28a](#), [29f](#), [42d](#), [43b](#), [108](#).
 prose_len: [42b](#), [42c](#), [43ab](#).
 push: [49f](#), [52](#), [133d](#), [133e](#), [134bc](#), [135a](#), [137bc](#).
 pushArglist: [178b](#), [179a](#), [183b](#).
 pushes: [52](#), [133e](#), [135a](#), [137c](#).
 putc: [21b](#), [50ab](#), [58bc](#), [59c](#), [68a](#), [70](#), [71c](#), [73g](#), [75ab](#), [77ab](#), [79abc](#), [80e](#), [83ab](#), [84a](#), [89b](#), [90abc](#), [92a](#), [93](#), [94c](#), [95d](#), [96a](#), [100c](#), [102c](#), [103a](#), [106a](#), [131b](#), [143](#), [145ce](#), [146a](#), [147d](#), [148abc](#).
 reject_match: [180c](#), [182](#), [187a](#).
 remove: [21b](#), [123ab](#), [124ab](#).
 reverse: [27](#), [30](#), [176b](#), [176c](#).
 reverse_lists: [51a](#), [153b](#), [176b](#).
 reverse_lookup_in_file: [119](#), [120](#).
 reverse_map_arg: [27](#), [28a](#), [36cd](#).
 reverse_map_flag: [27](#), [28a](#), [33](#), [36c](#).
 robs_strcmp: [78c](#), [110a](#), [155b](#), [155c](#), [156b](#), [157a](#), [184c](#).
 root: [102a](#), [111b](#), [177c](#), [180c](#), [181ab](#), [182](#), [184ac](#).
 save_string: [35b](#), [39a](#), [43b](#), [127b](#), [134a](#), [153b](#), [154a](#), [154b](#), [157a](#), [172c](#), [176a](#).
 SCRAP: [130a](#), [131a](#), [134a](#).
 ScrapEntry: [129c](#), [130a](#), [131a](#), [134a](#).
 scraps: [24abc](#), [27](#), [30](#), [53b](#), [54b](#), [61ab](#), [63a](#), [65ae](#), [66c](#), [82b](#), [85c](#), [86a](#), [87bd](#), [88a](#), [89a](#), [103c](#), [104acd](#), [105a](#), [108](#), [130a](#), [131a](#), [134a](#), [150bc](#), [151bc](#), [182](#), [192](#).
 scrap_array: [130a](#), [131bde](#), [132bcdef](#), [134abc](#), [136b](#), [137a](#), [138c](#), [140ab](#), [143](#), [144bc](#), [150bc](#), [151bd](#), [182](#), [184f](#), [186a](#).

SCRAP_BITS: [130a](#).
 scrap_end_line: [108](#), [132a](#), [132d](#).
 scrap_file_line: [108](#), [110b](#), [114a](#), [132a](#), [132c](#).
 scrap_file_name: [108](#), [110b](#), [114a](#), [132a](#), [132b](#).
 scrap_flag: [27](#), [28a](#), [33](#), [61a](#), [87d](#), [104d](#).
 scrap_is_in: [182](#), [184a](#), [184b](#).
 SCRAP_MASK: [130a](#), [134a](#).
 Scrap_Node: [46c](#), [51c](#), [54b](#), [59c](#), [66c](#), [74](#), [75a](#), [77b](#), [79bc](#), [80e](#), [82b](#), [89a](#), [92a](#), [93](#), [94d](#), [96ab](#), [97c](#), [99a](#), [100a](#), [108](#), [110b](#), [111c](#), [112a](#), [130b](#), [136b](#), [137ad](#), [141b](#), [142b](#), [152a](#), [152b](#), [175cd](#), [176bc](#), [182](#), [184ab](#), [185b](#), [186c](#).
 scrap_owner: [111c](#), [112a](#), [132a](#), [132f](#).
 scrap_prose: [108](#), [132a](#), [132e](#).
 SCRAP_SHIFT: [130a](#), [134a](#).
 SCRAP_SIZE: [130a](#), [131a](#), [134a](#).
 scrap_type: [69b](#), [69d](#), [70](#), [71de](#), [73g](#).
 search: [30](#), [42c](#), [180b](#), [180c](#), [192](#).
 sector: [30](#), [54ab](#), [55](#), [77c](#), [78b](#), [79a](#), [80bcde](#), [82acd](#), [95acd](#), [97abc](#), [100b](#), [101](#), [105d](#), [106abc](#), [108](#), [129c](#), [134a](#), [136ab](#), [137ac](#), [138c](#), [147d](#), [149d](#), [152b](#), [153b](#), [154bc](#), [155a](#), [156b](#), [157a](#), [170a](#), [171d](#), [173](#), [182](#).
 setlocale: [21b](#), [36e](#).
 set_scrap_owner: [46c](#), [132a](#), [140b](#).
 set_scrap_prose: [46ab](#), [132a](#), [140a](#).
 size_t: [21b](#), [75a](#), [115b](#), [120](#), [189d](#), [190bd](#).
 Slab: [129b](#), [129c](#), [133cd](#), [134a](#), [138b](#).
 slab: [129b](#), [129c](#), [134a](#), [138c](#), [143](#), [182](#).
 SLAB_SIZE: [129b](#), [133d](#), [138b](#), [140c](#), [180a](#).
 source_file: [125d](#), [126ac](#), [127ab](#), [128ab](#), [129a](#).
 source_get: [42d](#), [43c](#), [44cd](#), [45ab](#), [46d](#), [47ac](#), [48ab](#), [49ab](#), [52](#), [53b](#), [58bc](#), [59bc](#), [65ad](#), [70](#), [71e](#), [73bcde](#), [76a](#), [77c](#), [80c](#), [83ab](#), [84a](#), [85ab](#), [87bc](#), [89b](#), [90c](#), [91b](#), [94a](#), [95a](#), [97a](#), [102c](#), [103a](#), [104c](#), [105bd](#), [106b](#), [125a](#), [126a](#), [127b](#), [128b](#), [134bc](#), [135b](#), [136a](#), [137b](#), [164](#), [166](#), [167ab](#), [168a](#), [170ab](#), [171a](#), [172a](#), [173](#), [174](#), [175a](#).
 source_last: [53b](#), [63e](#), [70](#), [89b](#), [125a](#), [126a](#).
 source_line: [44d](#), [46d](#), [47a](#), [125b](#), [125c](#), [126ac](#), [127ab](#), [128ab](#), [129a](#), [134ac](#), [136a](#), [154c](#), [164](#), [166](#), [167ab](#), [168a](#), [170a](#), [171d](#), [173](#), [174](#).
 source_name: [38ab](#), [39ab](#), [44d](#), [46d](#), [47a](#), [125b](#), [125c](#), [127ab](#), [128ab](#), [129a](#), [134ac](#), [136a](#), [154c](#), [164](#), [166](#), [167ab](#), [168a](#), [170ab](#), [171d](#), [172a](#), [173](#), [174](#).
 source_open: [42c](#), [55](#), [82d](#), [101](#), [125a](#), [129a](#).
 source_peek: [47c](#), [48b](#), [49a](#), [125a](#), [126a](#), [126c](#), [127ab](#), [128b](#), [129a](#).
 source_ungetc: [48ab](#), [49b](#), [77c](#), [95a](#), [105d](#), [126b](#), [126c](#).
 stack: [117](#), [119](#), [125e](#), [127b](#), [128b](#).
 stderr: [21b](#), [33](#), [36e](#), [37](#), [39b](#), [42c](#), [44d](#), [46d](#), [47a](#), [55](#), [59c](#), [65g](#), [66b](#), [67b](#), [71d](#), [73g](#), [75b](#), [82d](#), [87e](#), [88c](#), [92a](#), [93](#), [101](#), [112ab](#), [114ab](#), [117](#), [122ab](#), [124b](#), [127ab](#), [128a](#), [129a](#), [132g](#), [134bc](#), [136a](#), [141d](#), [142d](#), [147d](#), [154c](#), [164](#), [166](#), [167ab](#), [168a](#), [170ab](#), [171d](#), [172a](#), [173](#), [174](#), [178b](#), [183b](#), [188bef](#).
 strlen: [21b](#), [50a](#), [113a](#), [117](#), [121e](#), [150c](#), [153c](#), [154a](#), [179d](#), [187a](#).
 sym_char: [186d](#), [187a](#).
 tex_flag: [27](#), [28a](#), [33](#), [39b](#), [42c](#), [147d](#).
 toupper: [21b](#), [155c](#).
 TRUE: [23a](#), [27](#), [28a](#), [29f](#), [33](#), [36a](#), [39b](#), [53b](#), [58c](#), [63be](#), [66c](#), [71d](#), [75a](#), [79b](#), [80be](#), [83ab](#), [84a](#), [85b](#), [87ab](#), [89a](#), [93](#), [96a](#), [97c](#), [99a](#), [100a](#), [103a](#), [104b](#), [110b](#), [111c](#), [127a](#), [131c](#), [132g](#), [147d](#), [155a](#), [157ab](#), [164](#), [167b](#), [184b](#), [185b](#), [186cd](#), [187a](#).
 update_delimit_scrap: [44d](#), [58c](#), [71a](#), [71b](#), [84a](#), [103a](#).
 user_langs: [157b](#), [164](#).
 user_lang_count: [157b](#), [164](#).
 user_lang_names: [157b](#), [164](#).
 user_names: [42c](#), [51a](#), [80c](#), [97a](#), [106b](#), [136b](#), [137a](#), [152c](#), [153a](#), [180c](#).
 Uses: [98ab](#), [99c](#), [108](#), [110b](#), [130b](#), [131de](#), [184ac](#), [184d](#), [184f](#), [186a](#).
 uses: [30](#), [56a](#), [66b](#), [79c](#), [80e](#), [88c](#), [96b](#), [97c](#), [100a](#), [105a](#), [106c](#), [108](#), [110b](#), [111c](#), [112a](#), [114a](#), [129c](#), [131d](#), [134a](#), [136b](#), [137d](#), [152b](#), [157ab](#), [176b](#), [182](#), [184df](#), [186c](#).
 verbose_flag: [27](#), [28a](#), [33](#), [42c](#), [55](#), [82d](#), [101](#), [122a](#).
 version_info_flag: [27](#), [28a](#), [33](#), [35c](#).
 version_string: [27](#), [28a](#), [35c](#), [59b](#), [72b](#), [85a](#), [103a](#).
 weave_flag: [27](#), [28a](#), [33](#), [39b](#).

weave_format: [27](#), [28a](#), [33](#), [39b](#), [166](#).
weave_format_from_cli: [27](#), [28a](#), [33](#), [39b](#), [166](#).
WEFT_VERSION: [23b](#), [28a](#), [29f](#), [30](#).
write_arg: [54b](#), [59c](#), [62a](#), [62b](#), [71e](#), [75b](#).
write_ArglistElement: [54b](#), [73g](#), [75b](#).
write_callers: [39b](#), [112a](#), [121b](#).
write_error_annotator: [36d](#), [120](#), [121b](#).
write_extract: [39b](#), [112b](#), [121b](#).
write_files: [39b](#), [121a](#), [121c](#).
write_html: [39b](#), [100b](#), [101](#).
write_label: [73a](#), [147a](#), [187c](#), [187d](#).
write_lint: [39b](#), [114b](#), [121b](#).
write_literal: [54b](#), [69c](#), [73g](#), [75b](#).
write_map: [39b](#), [109](#), [121b](#).
write_md: [39b](#), [82a](#), [82d](#).
write_reverse_map: [36d](#), [117](#), [121b](#).
write_scraps: [51c](#), [75a](#), [85b](#), [93](#), [122a](#), [130b](#), [142b](#), [147d](#), [149c](#).
write_scrap_ref: [66c](#), [79b](#), [80e](#), [89a](#), [96a](#), [97c](#), [99b](#), [130b](#), [131b](#), [131c](#), [185c](#).
write_single_scrap_ref: [53b](#), [59c](#), [61ab](#), [65e](#), [66b](#), [74](#), [77b](#), [79c](#), [80e](#), [85c](#), [86a](#), [88c](#), [92a](#), [93](#), [94d](#), [96b](#), [97c](#), [100c](#),
 [103c](#), [104ad](#), [105a](#), [130b](#), [131c](#), [148ac](#), [188a](#).
write_tex: [39b](#), [54a](#), [55](#).
xref_flag: [27](#), [28a](#), [33](#), [148ac](#).

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